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STABILITY THEORY BY LIAPUNOV'S SECOND METHOD

by T. Yoshizawa

THE MATHEMATICAL SOCIETY OF JAPAN

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PREFACE

The true creator of stability theory is A. M. Liapunov and the starting point for modern theories is his famous memoir published in Russian in 1892 and in French translation in 1907 "Problème général de la stabilité du mouvement". Although Liapunov introduced a method, which is called the second or direct method, and used it only to establish stability theorems, this second method has been recognized to be very general and powerful in the qualitative theory of ordinary differential equations, for example stability, asymptotic behavior of solutions, and boundedness which is the same concept as stability in a certain sense. Modifications and refinements of the concepts of stability and boundedness have been introduced and converse theorems have made it necessary to distinguish various types of stability and boundedness.

In this monograph, I shall discuss the stability and boundedness of solutions of differential equations and related topics; the underlying theme and connective thread being Liapunov's second method. I have attempted to give an introduction to Liapunov's second method which incorporates recent modifications and illustrates the scope and power of this method.

There is a vast literature on the theory and applications of Liapunov's second method, and due to the nature of this series and the resultant restrictions in size, I have emphasized the derivation and application of stability criteria for ordinary differential equations. As in any monograph of this nature, the selection of topics has also been dictated by the interests of the author.

Liapunov's second method is also an important tool in the theory of control systems, dynamical systems and functional-differential equations. Since an excellent book on stability theory in control systems has been published recently by Lefschetz [79], I have omitted all statements on control systems. For the stability in control systems, see [72], [74], [78]-[80], [153]. For dynamical systems, there are many interesting investigations [8]-[10], [15], [76], [103], [152], but dynamical systems are briefly treated in Section 22. Functional-differential equations are considered in Chapter VIII where a Liapunov function is generalized to a Liapunov functional and similar results are discussed.

There are two excellent English language books on this subject; an introductory one by LaSalle and Lefschetz [74], and one by Hahn [37]. Also, the outstanding books by Krasovskii [62] and Zubov [152] are now available in English translations.

The first chapter gives background material and introduces Liapunov's second method. In Chapter II the stability and boundedness of solutions are discussed. Positive limiting sets and the semi-invariant set are used to discuss the asymptotic behavior of solutions (an extension of stability theory) in Chapter III. Then, in Chapter IV extreme stability and stability of a set are discussed where sufficient conditions are established. In Chapter V converse theorems on stability and boundedness are discussed and utilized in Chapter VI to derive properties of solutions of perturbed systems and the asymptotic behavior of solutions near integral manifolds. Next, using fixed point theorems and Liapunov functions, existence of periodic and almost periodic solutions is discussed in Chapter VII. The concluding Chapter VIII shows how Liapunov's second method may be generalized to functional-differential equations to obtain similar results to those for ordinary differential equations.

Finally, I would like to express my sincere gratitude to Professor M. Hukuhara who suggested writing this monograph and to Professors W. A. Harris, Jr. and J. K. Hale who read the manuscript and made many helpful comments.

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CHAPTER I. PRELIMINARIES

The fundamental properties of solutions will be stated and some of them will be discussed by using Liapunov functions.

§ 1. Existence, uniqueness and Liapunov function.

Throughout this monograph, real systems of differential equations will be considered and the following notations will be used. The real intervals $a<t<b$, $a\leqq t\leqq b$, $a\leqq t<b$ and $a<t\leqq b$ will be denoted by (a, b), $[a, b]$, $[a, b)$ and $(a, b]$, respectively. Let I denote the interval $0\leqq t<\infty$ and R^n denote Euclidean n-space. For $x\in R^n$, let $\|x\|$ be the Euclidean norm of x. For an $n\times n$ matrix $A=(a_{ij})$, define the norm $\|A\|$ of A by $\|A\|=\sum_{i,j=1}^{n}|a_{ij}|$. It is easily seen that for two $n\times n$ matrices A, B and an n-vector x, we have

$$\|A+B\|\leqq\|A\|+\|B\|, \qquad \|AB\|\leqq\|A\|\,\|B\|$$

and

$$\|Ax\|\leqq\|A\|\,\|x\|.$$

The closure of a set S will be denoted by $\bar{S}$, and $N(\varepsilon, S)$ represents the ε-neighborhood of S. We denote by $d(p, S)$ the distance between a point p and a set S, and we shall sometimes denote by S_α the set of x such that $\|x\|\leqq\alpha$.

We shall denote by $C_0(x)$ the family of functions which satisfy locally a Lipschitz condition with respect to x, and if for any compact set $K\subset R^n$, there exists a constant $L(K)>0$ such that $\|f(t, x)-f(t, x')\|\leqq L(K)\,\|x-x'\|$ for $x\in K$, $x'\in K$, we shall write $f(t, x)\in\bar{C}_0(x)$. For brevity, CI and CIP will denote the families of continuous increasing functions and continuous increasing, positive definite functions, respectively.

Consider a system of differential equations

$$\frac{dx}{dt} = F(t, x), \tag{1.1}$$

where x is an n-vector and $F(t, x)$ is an n-vector function which is defined on a region in $I \times R^n$ and is continuous in (t, x). Throughout this monograph, a solution through a point (t_0, x_0) in $I \times R^n$ will be denoted by such forms as $x(t; x_0, t_0)$. We shall also sometimes write $\dot{x}$ for $\dfrac{dx}{dt}$.

There are many well known theorems of existence for (1.1). We mention here only one, see [25], [77]. As is well known, the solution $x(t)$ of (1.1) through (t_0, x_0) satisfies an integral equation

$$x(t) = x_0 + \int_{t_0}^{t} F(s, x(s))ds \tag{1.2}$$

on the interval on which $x(t)$ is defined. Conversely, a continuous function $x(t)$ satisfying (1.2) on some interval J is a solution of (1.1) defined on J such that $x(t_0) = x_0$.

THEOREM 1.1. *If $F(t, x)$ is continuous on $D: |t - t_0| \leqq a$, $\|x - x_0\| \leqq b$, and if $\|F(t, x)\| \leqq M$ on D, then there exists a solution $x(t)$ of (1.1) on $|t - t_0| \leqq \alpha$ for which $\|x(t) - x_0\| \leqq b$ and $x(t_0) = x_0$, where* $\alpha = \min\left(a, \dfrac{b}{M}\right)$.

This theorem can be proved by applying the following theorem (cf. [25]).

ASCOLI'S THEOREM. *On a bounded interval J, let $F = \{f\}$ be an infinite, uniformly bounded, equicontinuous set of functions. Then F contains a sequence $\{f_k\}$, $k = 1, 2, \cdots$, which is uniformly convergent on J.*

For other fundamental theorems, see [25], [77].

For the uniqueness of solutions, many sufficient conditions are well known (cf. [25], [52], [123]). Here, we shall discuss the uniqueness by using a Liapunov function. Such functions play an important role in Liapunov's second method.

Throughout this monograph, a Liapunov function will be assumed to be a scalar continuous function which satisfies locally a Lipschitz condition with respect to x.

Let $V(t, x)$ be a continuous scalar function defined on an open set S and let $V(t, x) \in C_0(x)$. Corresponding to $V(t, x)$, we define the function

$$(1.3) \qquad V'_{(1.1)}(t, x) = \overline{\lim_{h \to 0^+}} \frac{1}{h} \{V(t+h, x+hF(t, x)) - V(t, x)\} .$$

Let $x(t)$ be a solution of (1.1) which stays in S, and denote by $V'(t, x(t))$ the upper right-hand derivative of $V(t, x(t))$, i.e.,

$$(1.4) \qquad V'(t, x(t)) = \overline{\lim_{h \to 0^+}} \frac{1}{h} \{V(t+h, x(t+h)) - V(t, x(t))\} .$$

For a point $(t, x) \in S$ and small h, there exists a neighborhood U of (t, x) such that $\bar{U} \subset S$, $(t+h, x+hF(t, x)) \in U$ and $(t+h, x(t+h)) \in U$. Let L be the Lipschitz constant of $V(t, x)$ with respect to x in $\bar{U}$, and write

$$(1.5) \qquad \begin{aligned} &V(t+h, x(t+h)) - V(t, x(t)) \\ &\quad = V(t+h, x+hF(t, x)+h\varepsilon) - V(t, x) \\ &\quad \leqq V(t+h, x+hF(t, x)) + Lh\|\varepsilon\| - V(t, x) , \end{aligned}$$

where ε tends to zero with h. From (1.5), it follows that

$$(1.6) \qquad \begin{aligned} &\overline{\lim_{h \to 0^+}} \frac{1}{h} \{V(t+h, x(t+h)) - V(t, x(t))\} \\ &\quad \leqq \overline{\lim_{h \to 0^+}} \frac{1}{h} \{V(t+h, x+hF(t, x)) - V(t, x)\} . \end{aligned}$$

On the other hand, we have

$$\begin{aligned} &V(t+h, x(t+h)) - V(t, x(t)) \\ &\quad \geqq V(t+h, x+hF(t, x)) - Lh\|\varepsilon\| - V(t, x) , \end{aligned}$$

which implies that $V'_{(1.1)}(t, x) \leqq V'(t, x(t))$. Thus, from this and (1.6), we obtain

$$(1.7) \qquad V'_{(1.1)}(t, x) = V'(t, x(t)) .$$

By the same calculation, we obtain the relation

$$
\begin{aligned}
(1.8) \qquad & \varliminf_{h\to 0^+} \frac{1}{h}\{V(t+h,\, x(t+h)) - V(t,\, x(t))\} \\
& = \varliminf_{h\to 0^+} \frac{1}{h}\{V(t+h,\, x+hF(t,\, x)) - V(t,\, x)\}\,.
\end{aligned}
$$

In case $V(t, x)$ has continuous partial derivatives of the first order, it is evident that

$$
(1.9) \qquad V'_{(1.1)}(t,\, x) = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \cdot F(t,\, x)\,,
$$

where "$\cdot$" denotes the scalar product.

As is known (cf. [97]), if $V'_{(1.1)}(t, x) \leqq 0$ and consequently $V'(t, x(t)) \leqq 0$, the function $V(t, x(t))$ is a non-increasing function of t, which implies that $V(t, x)$ is non-increasing along a solution of (1.1). Conversely, if $V(t, x)$ is non-increasing along a solution of (1.1), we have $V'_{(1.1)}(t, x) \leqq 0$. Further, if

$$
\varliminf_{h\to 0^+} \frac{1}{h}\{V(t+h,\, x+hF(t,\, x)) - V(t,\, x)\} \geqq 0\,,
$$

the function $V(t, x)$ is non-decreasing along a solution of (1.1), and conversely.

The following property of a Liapunov function $V(t, x)$ is important, especially in studying the behavior of solutions of perturbed systems.

Let $x(s)$ and $y(s)$ be continuous and differentiable functions defined for $s \geqq t$ such that $x(t) = y(t) = x$. Then, by the definition,

$$
V'(t,\, x(t)) = \varlimsup_{h\to 0^+} \frac{1}{h}\{V(t+h,\, x(t+h)) - V(t,\, x(t))\}\,,
$$

$$
V'(t,\, y(t)) = \varlimsup_{h\to 0^+} \frac{1}{h}\{V(t+h,\, y(t+h)) - V(t,\, y(t))\}\,.
$$

Let L be a Lipschitz constant of $V(t, x)$ in a neighborhood of the point (t, x). Then, for sufficiently small h,

$$
\begin{aligned}
V'(t,\, y(t)) \leqq & \varlimsup_{h\to 0^+} \frac{1}{h}\{V(t+h,\, x(t+h)) - V(t,\, y(t))\} \\
& + \varlimsup_{h\to 0^+} \frac{1}{h}\{V(t+h,\, y(t+h)) - V(t+h,\, x(t+h))\}
\end{aligned}
$$

$$\leqq \varlimsup_{h\to 0^+} \frac{1}{h}\{V(t+h, x(t+h))-V(t, x(t))\}$$

$$+\varlimsup_{h\to 0^+} \frac{1}{h} L \| y(t+h)-x(t+h) \| .$$

Thus, we have

$$V'(t, y(t)) \leqq V'(t, x(t)) + L \| \dot{y}(t)-\dot{x}(t) \| .$$

First of all, let us consider the uniqueness of a given solution of (1.1). In this case, by changing variables, we can assume that $F(t, 0) \equiv 0$.

THEOREM 1.2. *Suppose that* $F(t, x)$ *of the system* (1.1) *is continuous on* $D: 0 \leqq t \leqq a,\ \|x\| \leqq b$ *and that* $F(t, 0) \equiv 0$. *If there exists a Liapunov function* $V(t, x)$ *defined on* D *which satisfies the conditions* (i) $V(t, 0) \equiv 0$, (ii) $V(t, x) > 0$ *for* $\|x\| \neq 0$ *and* (iii) $V'_{(1.1)}(t, x) \leqq 0$ *in the interior of* D, *then the solution through* $(0, 0)$ *of* (1.1) *is unique to the right. If* $\varliminf_{h\to 0^+} \frac{1}{h}\{V(t+h, x+hF(t, x))-V(t, x)\} \geqq 0$, *the solution through* $(a, 0)$ *is unique to the left.*

PROOF. Suppose that there is a solution $x(t)$ through $(0, 0)$ of (1.1) which is different from the zero solution. Then there are $t_0, t_1, 0 \leqq t_0 < t_1 \leqq a$, such that $x(t_0)=0$, $x(t_1) \neq 0$ and that $\|x(t)\| > 0$ for $t \in (t_0, t_1)$. By (ii), $V(t_1, x(t_1)) > 0$, and by (iii), we have $V(t_1, x(t_1)) \leqq V(t_0, x(t_0))$. As $V(t_0, x(t_0))=0$ by (i), there arises a contradiction. Thus, the uniqueness of the solution is proved.

Now we shall see that the existence of such a function $V(t, x)$ as above is necessary for uniqueness of the solution. Let $p_1(t_1, x_1)$ and $p_2(t_2, x_2)$ be two points in D. For p_1 and p_2, $t_1 < t_2$, denote by $X_{p_1 p_2}$ the family of all functions $x(t)$, which are continuous on $[t_1, t_2]$, with the properties that their derivatives are continuous except for at most one value of t and that $x(t_1)=x_1$, $x(t_2)=x_2$ and $(t, x(t)) \in D$ for $t \in [t_1, t_2]$. Let $v(p_1, p_2)$ be defined for $t_1 < t_2$ by

$$v(p_1, p_2) = \inf_{x(t) \in X_{p_1 p_2}} \int_{t_1}^{t_2} \| \dot{x}(t) - F(t, x(t)) \| \, dt . \tag{1.10}$$

If $t_1=t_2$, we set $v(p_1, p_2)=\| x_1-x_2 \|$ and if $t_1>t_2$, we set $v(p_1, p_2)=v(p_2, p_1)$.

LEMMA 1.1. *In order that p_1 and p_2 should be on a solution of* (1.1), *it is necessary and sufficient that $v(p_1, p_2)=0$.*

PROOF. It is clear that the condition is necessary. Now suppose that $v(p_1, p_2)=0$, and there exists a sequence of functions $\{x_k(t)\}$, $x_k(t)\in X_{p_1p_2}$, such that

$$\lim_{k\to\infty}\int_{t_1}^{t_2}\| \dot{x}_k(t)-F(t, x_k(t))\| dt=0\,.$$

If we set $\varphi_k(t)=x_k(t)-x_1-\int_{t_1}^{t} F(s, x_k(s))ds$, clearly $\lim_{k\to\infty}\varphi_k(t)=0$. For $t_1\leqq t'\leqq t''\leqq t_2$ and $y_k(t)=x_k(t)-\varphi_k(t)$, we have $y_k(t'')-y_k(t')=\int_{t'}^{t''} F(s, x_k(s))ds$, and hence $\| y_k(t'')-y_k(t')\| \leqq M(t''-t')$ for a constant $M>0$ such that $\| F(t, x)\| \leqq M$ on D. Therefore, $\{y_k(t)\}$ is equicontinuous and by Ascoli's Theorem, there exists a uniformly convergent subsequence, which we denote by $\{y_k(t)\}$ again. Let $y(t)$ be the limit function. Clearly $y(t_1)=x_1$, $y(t_2)=x_2$ and

$$y(t)=x_1+\int_{t_1}^{t} F(s, y(s))ds\,,$$

because $x_k(t)\to y(t)$ as $\varphi_k(t)\to 0$. Thus, there exists a solution of (1.1) through p_1 and p_2.

LEMMA 1.2. *For two points p_1 and p_2, $t_1<t_2$, we have*

$$|v(p_1, p_2)-\| x_1-x_2\| |\leqq M(t_2-t_1)\,. \tag{1.11}$$

PROOF. For any $x(t)\in X_{p_1p_2}$,

$$\begin{aligned}\int_{t_1}^{t_2}\| \dot{x}(t)-F(t, x(t))\| dt &\geqq \int_{t_1}^{t_2}\| \dot{x}(t)\| dt-\int_{t_1}^{t_2}\| F(t, x(t))\| dt\\ &\geqq \| x_2-x_1\| -M(t_2-t_1)\,,\end{aligned}$$

which implies that $v(p_1, p_2)\geqq \| x_2-x_1\| -M(t_2-t_1)$. Next, let $x(t)$ be the function which represents the segment $\overline{p_1\,p_2}$, which is a function in $X_{p_1p_2}$. Since

$$\int_{t_1}^{t_2} \| \dot{x}(t) \| dt = \| \int_{t_1}^{t_2} \dot{x}(t) dt \| = \| x_2 - x_1 \| ,$$

we have

$$\int_{t_1}^{t_2} \| \dot{x}(t) - F(t, x(t)) \| dt \leqq \| x_2 - x_1 \| + M(t_2 - t_1) ,$$

whence $v(p_1, p_2) \leqq \| x_2 - x_1 \| + M(t_2 - t_1)$. Thus, (1.11) is proved.

LEMMA 1.3. *For points $p_i(t_i, x_i)$, $i = 1, 2, 3$, such that $t_1 \leqq t_2 \leqq t_3$,*

$$v(p_1, p_3) \leqq v(p_1, p_2) + v(p_2, p_3) . \tag{1.12}$$

PROOF. It is clear that (1.12) is valid in case $t_1 < t_2 < t_3$. Now assume that $t_1 < t_2 = t_3$. Let $\{x_k(t)\}$ be the sequence such that

$$v(p_1, p_2) = \lim_{k \to \infty} \int_{t_1}^{t_2} \| \dot{x}_k(t) - F(t, x_k(t)) \| dt$$

and take points $Q_k = (t', x_k(t'))$, $t_1 < t' < t_2 = t_3$. Then, for p_1, Q_k and p_3, we have $v(p_1, p_3) \leqq v(p_1, Q_k) + v(Q_k, p_3)$, and hence

$$\begin{aligned} &\int_{t_1}^{t_2} \| \dot{x}_k(t) - F(t, x_k(t)) \| dt \\ &\quad = \int_{t_1}^{t'} \| \dot{x}_k(t) - F(t, x_k(t)) \| dt + \int_{t'}^{t_2} \| \dot{x}_k(t) - F(t, x_k(t)) \| dt \\ &\quad \geqq v(p_1, Q_k) + v(Q_k, p_2) \geqq v(p_1, p_3) - v(Q_k, p_3) + v(Q_k, p_2) . \end{aligned}$$

From this and (1.11), it follows that

$$\int_{t_1}^{t_2} \| \dot{x}_k(t) - F(t, x_k(t)) \| dt \geqq v(p_1, p_3) - \| x_2 - x_3 \| - 2M(t_2 - t') ,$$

and by letting $k \to \infty$ and $t' \to t_2$, we have $v(p_1, p_2) \geqq v(p_1, p_3) - \| x_2 - x_3 \|$, i. e., (1.12), because $v(p_2, p_3) = \| x_2 - x_3 \|$.

LEMMA 1.4. *For $p_i(t_i, x_i)$, $i = 1, 2, 3$, such that $t_1 \leqq t_2 \leqq t_3$,*

$$\left\{ \begin{aligned} v(p_1, p_3) &\geqq v(p_1, p_2) - \| x_3 - x_2 \| - M(t_3 - t_2) \\ v(p_1, p_3) &\geqq v(p_2, p_3) - \| x_2 - x_1 \| - M(t_2 - t_1) . \end{aligned} \right. \tag{1.13}$$

PROOF. In case $t_1 < t_2 < t_3$, consider the sequence $\{x_k(t)\}$ such that $v(p_1, p_3) = \lim_{k\to\infty} \int_{t_1}^{t_3} \| \dot{x}_k(t) - F(t, x_k(t)) \| \, dt$. For points $Q_k = (t_2, x_k(t_2))$,

$$\begin{aligned}
\int_{t_1}^{t_3} \| \dot{x}_k(t) - F(t, x_k(t)) \| \, dt &\geqq v(p_1, Q_k) + v(Q_k, p_3) \\
&\geqq v(p_1, p_2) - v(Q_k, p_2) + v(Q_k, p_3) \quad \text{(by (1.12))} \\
&\geqq v(p_1, p_2) - \| x_k(t_2) - x_2 \| + \| x_3 - x_k(t_2) \| - M(t_3 - t_2) \quad \text{(by (1.11))} \\
&\geqq v(p_1, p_2) - \| x_3 - x_2 \| - M(t_3 - t_2),
\end{aligned}$$

which implies that $v(p_1, p_3) \geqq v(p_1, p_2) - \| x_3 - x_2 \| - M(t_3 - t_2)$. The other cases can be easily proved in a similar fashion.

From the lemmas above, it follows that $v(p_1, p_2)$ is a non-negative continuous function of (p_1, p_2) and that

$$(1.14) \qquad | v(p_1, p_2) - v(p_1, p_3) | \leqq \| x_2 - x_3 \| + M | t_3 - t_2 | .$$

THEOREM 1.3. *Suppose that* $F(t, x)$ *of* (1.1) *is continuous on* $D: 0 \leqq t \leqq a$, $\| x \| \leqq b$, *and that* $F(t, 0) \equiv 0$. *If the solution through* (0, 0) *of* (1.1) *is unique to the right, there exists a Liapunov function* $V(t, x)$ *defined on* D *which satisfies the conditions* (i), (ii) *and* (iii) *in Theorem* 1.2. *In case the solution through* $(a, 0)$ *of* (1.1) *is unique to the left, the condition* $V'_{(1.1)}(t, x) \leqq 0$ *is replaced by*

$$\varliminf_{h\to 0+} \frac{1}{h} \{ V(t+h, x+hF(t, x)) - V(t, x) \} \geqq 0 .$$

PROOF. For $p_1 = (0, 0)$ and $p_2 = (t, x) \in D$, we set $V(t, x) = v(p_1, p_2)$. Since $x(t) \equiv 0$ is a solution of (1.1) through (0, 0), clearly $V(t, 0) \equiv 0$. For any point $(t, x) \in D$ such that $x \neq 0$, there is no solution of (1.1) through (0, 0) and (t, x), and hence, by Lemma 1.1, we have $V(t, x) > 0$. From (1.14), it follows that

$$(1.15) \qquad | V(t, x) - V(t', x') | \leqq \| x - x' \| + M | t - t' | .$$

Moreover, Lemma 1.1 and Lemma 1.3 show that $V(t, x)$ is non-increasing along the solution of (1.1), which implies $V'_{(1.1)}(t, x) \leqq 0$, because of (1.15).

THEOREM 1.4. *Suppose that* $F(t, x)$ *of* (1.1) *is continuous on* $D_1: 0 \leqq t < a,\ \|x\| < b$. *In order that every solution of* (1.1) *starting from a point in* D_1 *is unique to the right, it is necessary and sufficient that for any point* $(t_0, x_0) \in D_1$, *there exists a neighborhood* U *of* (t_0, x_0) *which has the following property: Let* W *be a set of* (t, x, y) *such that* $(t, x) \in U$, $(t, y) \in U$. *Then there exists a Liapunov function* $V(t, x, y)$ *defined on* W, *which satisfies the conditions* (i) $V(t, x, y) \equiv 0$ *if* $x = y$, (ii) $V(t, x, y) > 0$ *if* $x \neq y$ *and* (iii) $V(t, x, y) \in C_0(x, y)$ *and*

$$(1.16)\quad V'(t, x, y) = \overline{\lim_{h \to 0+}} \frac{1}{h}\{V(t+h, x+hF(t, x), y+hF(t, y)) - V(t, x, y)\} \leqq 0 .$$

PROOF. The sufficiency can be easily proved. We shall only prove that the condition is necessary. Consider the following system associated with (1.1)

$$(1.17)\qquad \dot{x} = F(t, x), \qquad \dot{y} = F(t, y).$$

For each point $(t_0, x_0) \in D_1$, let U be a neighborhood of (t_0, x_0) which is contained in the set $S = \{(t, x);\ |t - t_0| \leqq \alpha,\ \|x - x_0\| \leqq \beta\} \subset D_1$. For two points $p(t, x, y)$ and $Q(\tau, \xi, \eta)$, $(t, x) \in S$, $(t, y) \in S$, $(\tau, \xi) \in S$, $(\tau, \eta) \in S$, define the function $v(p, Q)$. Let $v^*(p)$ be

$$v^*(p) = \min_{\substack{0 \leqq \tau \leqq t \\ \xi = \eta}} v(p, Q) .$$

Since $v(p, Q)$ is continuous in Q and Q varies on a compact set, $\min v(p, Q) = v^*(p)$ exists and is continuous in p, and we have $v^*(p) = v(p, Q^*)$ for some Q^*. Let $V(t, x, y)$ be such that $V(t, x, y) = v^*(p)$. If p is on $x = y$, $v^*(p) = 0$ or $V(t, x, y) = 0$, and if $x \neq y$, $v^*(p) > 0$ or $V(t, x, y) > 0$, because of the uniqueness of solutions. For two points $p(t, x, y)$, $p'(t, x', y')$, for which we can assume that $v^*(p') \geqq v^*(p)$, $0 \leqq v^*(p') - v^*(p) \leqq v(p', Q) - v(p, Q) \leqq \overline{p'p}$, where $v^*(p) = v(p, Q)$. From this, it follows that $|V(t, x, y) - V(t, x', y')| \leqq \|x - x'\| + \|y - y'\|$.

Finally, suppose that $p(t, x, y)$, $R(t', x', y')$, $t' < t$, are on a solution of (1.17). There exists a $Q(t_1, x_1, y_1)$ on $x = y$ such that

$v^*(R)=v(R, Q)$. By Lemmas 1.1 and 1.3, $v(p, Q)\leqq v(p, R)+v(R, Q) \leqq v(R, Q)$. As we have $v^*(p)\leqq v(p, Q)$, $v^*(p)\leqq v^*(R)$, which means that $V(t, x, y)$ is non-increasing along the solution of (1.17).

REMARK. Under the conditions of Theorems 1.3 and 1.4, it is known that there exist Liapunov functions which have continuous first order partial derivatives (cf. [104]).

EXAMPLE. In the case when $F(t, x)$ of (1.1) satisfies a Lipschitz condition

$$\| F(t, x)-F(t, x')\| \leqq L \| x-x' \| ,$$

where $L>0$ is a constant, the function $V(t, x, y)=e^{-2Lt}\| x-y \|^2$ satisfies

$$V'_{(1.17)}(t, x, y)\leqq e^{-2Lt}\{-2L \| x-y \|^2+2 \| x-y \| \| F(t, x)-F(t, y) \| \}\leqq 0 .$$

Thus, it can be seen that $V(t, x, y)$ satisfies the conditions in Theorem 1.4 and hence, every solution is unique to the right.

§ 2. Differential equations of Carathéodory type.

A system of differential equations defined on a region of (t, x)

$$\frac{dx}{dt}=F(t, x) \tag{2.1}$$

is said to be of Carathéodory type, if $F(t, x)$ is finite, a measurable function of t for fixed x and a continuous function of x for fixed t. It is known that if there exists a summable function $M(t)$ such that $\| F(t, x)\| \leqq M(t)$, the system admits a solution for initial value problem (cf. [25], [97]).

For a system of Carathéodory type, the same problems as in an ordinary system can be discussed, and quite similar results are obtained. However, if we want to use a Liapunov function, an additional condition on the Liapunov function will be required. It is the condition that the Liapunov function $V(t, x)$ is absolutely continuous on t uniformly at a point (t_0, x_0), that is, for a certain number $q>0$ and for any number $\varepsilon>0$, there exists a $\delta(\varepsilon)>0$ such that

$$\sum_{k=1}^{m} | V(t'_k, x_k) - V(t_k, x_k) | < \varepsilon ,$$

provided $\sum_{k=1}^{m} (t'_k - t_k) < \delta$, $t_0 - q \leqq t_1 \leqq t'_1 \leqq \cdots \leqq t_m \leqq t'_m \leqq t_0 + q$, where x_k is an arbitrary vector satisfying $\| x_k - x_0 \| \leqq q$ and δ is independent of the choice of t_k, t'_k, x_k and m.

If $V(t, x)$ has the property above and $V(t, x) \in C_0(x)$ and if $x(t)$ is an absolutely continuous vector function, then $V(t, x(t))$ also is an absolutely continuous function in a neighborhood of t_0. The relations (1.7) and (1.8) hold good almost everywhere with respect to t and hence, inequalities including $V'_{(2.1)}(t, x)$ should be understood to hold almost everywhere with respect to t (cf. [137]).

We often use the condition $F(t, x) \in C_0(x)$. This condition will be replaced by a generalized Lipschitz condition, i. e., for $x, x' \in S_\alpha$ (α : arbitrary) there is a summable function $L(t)$ such that $\| F(t, x) - F(t, x') \| \leqq L(t) \| x - x' \|$, where $L(t)$ may depend on α.

§ 3. Continuation of solutions.

In the system (1.1), suppose that $F(t, x)$ is defined and continuous on a region D. Let $x(t; x_0, t_0)$ be a solution of (1.1) which is defined on $[t_0, a)$, where $(t_0, x_0) \in D$. If there exists a solution $x^*(t; x_0, t_0)$ of (1.1) defined on $[t_0, T]$, $a \leqq T$, and if $x(t; x_0, t_0) = x^*(t; x_0, t_0)$ on $[t_0, a)$, the solution $x(t; x_0, t_0)$ is said to be continuable to $t = T$. For a solution $x(t)$ of (1.1) defined on $[a, b]$, we understand that $\dot{x}(a)$, $\dot{x}(b)$ are the right-hand and the left-hand derivative, respectively.

THEOREM 3.1. *Suppose that $F(t, x)$ of (1.1) is continuous on a domain $D \subset I \times R^n$ and that (1.1) has a solution $x(t)$ which exists on a finite interval (a, b) and $x(t_0) = x_0$, $(t_0, x_0) \in D$, $a < t_0 < b$. If $F(t, x)$ is bounded on D, the limits $x(a+0)$ and $x(b-0)$ exist. If $(a, x(a+0))$ (or $(b, x(b-0))$) is in D, then the solution $x(t)$ is continuable to $t = a$ (or $t = b$).*

For the proof, see [25].

THEOREM 3.2. *If $F(t, x)$ of (1.1) is continuous on $[0, T] \times R^n$*

and if there exists a constant $M>0$ such that $\|F(t,x)\| \leqq M$ for all $t \in [0,T]$, $x \in R^n$, then every solution of (1.1) *is continuable to* $t=T$.

PROOF. Let $x(t)$ be a solution through (t_0, x_0), $0 \leqq t_0 < T$, $x_0 \in R^n$, which is defined on $[t_0, a)$, $a \leqq T$. By Theorem 3.1, this solution is defined on $[t_0, a]$. In case $a=T$, $x(t)$ is continuable to $t=T$. In case $a<T$, from Theorem 1.1, it follows that there exists a solution $x^*(t)$ of (1.1) which is defined on $[a, T]$ and passes through the point $(a, x(a))$. Joining $x(t)$ and $x^*(t)$, we find a solution of (1.1) defined on $[t_0, T]$, which shows that $x(t)$ is continuable to $t=T$.

THEOREM 3.3. *Suppose that $F(t,x)$ of* (1.1) *is defined and continuous on $0 \leqq t \leqq T$, $\|x\| \leqq H$, $H>0$ (or $\|x\|<\infty$). If every solution $x(t; x_0, t_0)$ of* (1.1) *through (t_0, x_0) is strictly bounded by a constant $\beta>0$, $\beta \leqq H$ (or $\beta<\infty$), as long as $x(t; x_0, t_0)$ is defined, then all solutions of* (1.1) *through (t_0, x_0) are continuable to $t=T$. If $F(t,x)$ is defined for all $t \in I$, $x(t; x_0, t_0)$ is continuable to any t. In this case, $x(t; x_0, t_0)$ is said to exist in the future.*

PROOF. Let $x(t)$ be a solution of (1.1) through (t_0, x_0) defined on $[t_0, a)$, $a \leqq T$. Since $F(t,x)$ is continuous on the compact set $S=[0,T]\times S_\beta$, there exists a continuous bounded functon $F^*(t,x)$ which is defined on $[0,T]\times R^n$ and is the same as $F(t,x)$ on S. By Theorem 3.2, every solution of

$$\frac{dx}{dt}=F^*(t,x) \tag{3.1}$$

is continuable to $t=T$. Clearly, $x(t)$ is a solution of (3.1) through (t_0, x_0) for $t \in [t_0, a)$, because $\|x(t)\|<\beta$, and hence, there exists a solution $x^*(t)$ of (3.1) through (t_0, x_0) defined on $[t_0, T]$ for which $x(t)=x^*(t)$ on $[t_0, a)$. Suppose that $\|x^*(t)\|=\beta$ at some t. Then there is a t_1 such that $\|x^*(t_1)\|=\beta$ and that $\|x^*(t)\|<\beta$ for $t \in [t_0, t_1)$. Thus, $x^*(t)$ is a solution of (1.1) on $[t_0, t_1]$, which is bounded strictly by β. This is a contradiction, and hence $\|x^*(t)\|<\beta$ for all $t \in$

$[t_0, T]$, which shows that $x^*(t)$ is a solution of (1.1) through (t_0, x_0) defined on $[t_0, T]$, that is, $x(t)$ is continuable to $t=T$.

Now we shall discuss the continuation of solutions by using a Liapunov function.

THEOREM 3.4. *Suppose that $F(t, x)$ of (1.1) is defined and continuous on $[0, T]\times R^n$. If there exists a Liapunov function $V(t, x)$ defined on $D: 0\leqq t\leqq T$, $\|x\|\geqq R$ (R may be large) which satisfies the conditions*

(i) *$a(\|x\|)\leqq V(t, x)$, where $a(r)$ is continuous for $r\geqq R$ and $a(r)\to\infty$ as $r\to\infty$,*

(ii) *$V'_{(1.1)}(t, x)\leqq 0$ in the interior of D,*

then every solution of (1.1) is continuable to $t=T$.

PROOF. Let $x(t; x_0, t_0)$ be a solution of (1.1) such that $0\leqq t_0<T$, $\|x_0\|\leqq\alpha$, where we can assume that $\alpha>R$, and let $K(\alpha)>0$ be such that $V(t_0, x_0)\leqq K(\alpha)$. By (i), there exists $\beta(\alpha)>0$, $\beta>\alpha$, such that $K(\alpha)<V(t, x)$ for $t\in[0, T]$, $\|x\|=\beta$. Suppose that $\|x(t; x_0, t_0)\|=\beta$ at some t. Then there exist t_1, t_2, $t_0\leqq t_1<t_2\leqq T$, such that $\|x(t_1; x_0, t_0)\|=\alpha$, $\|x(t_2; x_0, t_0)\|=\beta$ and that $\alpha<\|x(t; x_0, t_0)\|<\beta$ for $t\in(t_1, t_2)$. By (ii), we have

$$V(t_2, x(t_2; x_0, t_0))\leqq V(t_1, x(t_1; x_0, t_0)). \tag{3.2}$$

On the other hand, $V(t_1, x(t_1; x_0, t_0))\leqq K(\alpha)$ and $V(t_2, x(t_2; x_0, t_0))>K(\alpha)$, which contradicts (3.2). Therefore, as long as $x(t; x_0, t_0)$ is defined, we have $\|x(t; x_0, t_0)\|<\beta$. Thus, by Theorem 3.3, every solution of (1.1) is continuable to $t=T$.

EXAMPLE 3.1. Suppose that $F(t, x)$ of (1.1) is defined and continuous on $I\times R^n$ and that $F(t, x)$ satisfies $\|F(t, x)\|\leqq\lambda(t)\,\varphi(\|x\|)$ for $\|x\|\geqq R$, where $\lambda(t)$ is continuous on I and $\varphi(u)$ is continuous on $R\leqq u<\infty$ and $\int_R^\infty\frac{d(u)}{\varphi(u)}=\infty$.

In this case, a continuous scalar function

$$V(t, x)=-\int_0^t\lambda(s)ds+\int_R^r\frac{d(u)}{\varphi(u)}, \qquad r=\|x\|,$$

satisfies the condition (i) in Theorem 3.4 for any fixed T. As is easily seen, $V(t, x) \in C_0(x)$ and

$$V'_{(1.1)}(t, x) \leqq -\lambda(t) + \frac{1}{\varphi(r)} \cdot \frac{\|x\| \|F(t, x)\|}{r} \leqq 0,$$

from which the condition (ii) in Theorem 3.4 follows. Therefore, every solution of (1.1) is continuable to arbitrary T, that is, every solution exists in the future.

From Example 3.1, it follows that every solution of a linear system

$$\frac{dx}{dt} = A(t)x + p(t)$$

exists in the future, where $A(t)$ is a continuous $n \times n$ matrix and $p(t)$ is a continuous function on I.

Now we shall see that the existence of such a Liapunov function as in Theorem 3.4 is a necessary condition for continuation of solutions. This will be seen by reducing the problem to the uniqueness at infinity.

Let $F(t, x)$ of (1.1) be continuous on $[0, T] \times R^n$. Setting $\|x\| = r$, let $F^*(r)$ be

$$F^*(r) = r + \max_{\substack{\|\xi\| \leqq r \\ 0 \leqq t \leqq T}} \|F(t, \xi)\|, \qquad r \geqq R > 0.$$

Then, $F^*(r) \in CI$ (see Section 1) and $F^*(r) \to \infty$ as $r \to \infty$. If we set $\varphi(r) = \int_r^{r+1} F^*(s) ds$, $\varphi(r) \in CI$ and $F^*(r) \leqq \varphi(r)$, and hence $\varphi(r) \to \infty$ as $r \to \infty$. Consider a function $w(r) = \int_r^\infty \frac{ds}{s^2 \varphi(s)}$. Then $w(r)$ is defined for $r \geqq R$, because $w(r) \leqq \frac{1}{\varphi(r)} \int_r^\infty \frac{ds}{s^2}$. Clearly, $w(r)$ is decreasing and is continuous with its derivative, and $w(r) > 0$, $\frac{dw}{dr} < 0$ and $w(r) \to 0$, $\frac{dw}{dr} \to 0$ as $r \to \infty$. Consider a transformation $y = \frac{w(r)}{r} x$. Then the domain $0 \leqq t \leqq T$, $\|x\| \geqq R$ is transformed into a domain $0 \leqq t \leqq T$, $\|y\| \leqq w(R)$. $\|y\| = w(r)$ implies

$x=\frac{w^{-1}(\|y\|)}{\|y\|}y$. Thus, the system (1.1) is transformed into a system

$$\frac{dy}{dt}=G(t, y), \tag{3.3}$$

where

$$G(t, y)=\frac{dw(r)}{dr}\frac{dr}{dt}\frac{x}{r}-\frac{w(r)}{r^2}\frac{dr}{dt}x+\frac{w(r)}{r}F(t, x),$$

$$r=w^{-1}(\|y\|), \qquad x=\frac{w^{-1}(\|y\|)}{\|y\|}y.$$

On the other hand, $\|G(t, y)\|\leqq\left|\frac{dw}{dr}\right|\|F(t, x)\|+\frac{2w(r)}{r}\|F(t, x)\|$, which implies that $G(t, y)\to 0$ uniformly in t as $y\to 0$, because

$$\left|\frac{dw}{dr}\right|\|F(t, x)\|\leqq\frac{1}{r^2\varphi(r)}\varphi(r)=\frac{1}{r^2}\to 0 \quad \text{as} \quad r\to\infty,$$

$$\frac{w(r)}{r}\|F(t, x)\|\leqq\frac{1}{r}\frac{1}{r\varphi(r)}\varphi(r)=\frac{1}{r^2}\to 0 \quad \text{as} \quad r\to\infty.$$

Therefore, if we define $G(t, 0)\equiv 0$, $G(t, y)$ is defined and continuous on $0\leqq t\leqq T$, $\|y\|\leqq w(R)$.

Suppose that every solution of (1.1) is continuable to $t=T$. Let $x(t)$ be a solution of (1.1) defined on $[t_0, a)$, $0\leqq t_0<T$, $a<T$. Then there exists a solution $x^*(t)$ of (1.1) defined on $[t_0, T]$ such that $x^*(t)=x(t)$ on $[t_0, a)$. Hence, there is a constant $\alpha>0$ such that $\|x(t)\|\leqq\alpha$ on $[t_0, a)$, which implies that the zero solution of (3.3) is unique to the left. Then, by Theorem 1.3, there exists a continuous scalar function $W(t, y)$ defined on $0\leqq t\leqq T$, $\|y\|\leqq w(R)$ which satisfies the conditions that $W(t, 0)\equiv 0$, $W(t, y)>0$ if $y\neq 0$ and that $W(t, y)\in C_0(y)$ and $\lim_{h\to 0+}\frac{1}{h}\{W(t+h, y+hG(t, y))-W(t, y)\}\geqq 0$. If we set

$$V(t, x)=\frac{1}{W\left(t, \frac{w(r)}{r}x\right)},$$

$V(t, x)$ is defined on $0\leqq t\leqq T$, $\|x\|\geqq R$ and satisfies the conditions (i), (ii) in Theorem 3.4. Thus, we have the following theorem.

THEOREM 3.5. *Suppose that $F(t, x)$ of* (1.1) *is continuous on* $[0, T]\times R^n$. *If every solution of* (1.1) *is continuable to $t=T$, there exists a Liapunov function $V(t, x)$ defined on $0\leqq t\leqq T$, $\|x\|\geqq R$ (R may be large) which satisfies the conditions* (i) *and* (ii) *in Theorem* 3.4.

The following theorem can be obtained immediately from Theorem 3.5.

THEOREM 3.6. *Suppose that $F(t, x)$ of* (1.1) *is continuous on* $[0, T]\times R^n$. *If every solution of* (1.1) *is continuable to $t=T$, corresponding to each $\alpha>0$, there exists a $\beta(\alpha)>0$ such that if $x_0\in S_\alpha$ and $t_0\in[0, T)$, $\|x(t; x_0, t_0)\|<\beta(\alpha)$ for all $t\in[t_0, T]$.*

By the same argument as above, the theorem on continuation to the left can be obtained, and by combining these two theorems, we can obtain a theorem on continuation in both directions.

REMARK. When $F(t, x)\in C_0(x)$, a Liapunov function $V(t, x)$ can be easily constructed by setting $V(t, x)=\sup_\tau\{\|x(\tau; x, t)\|;\ t\leqq\tau\leqq T\}$. In this case, clearly $\|x\|\leqq V(t, x)$. For the continuation to the left, $V(t, x)=\sup_\tau\{\|x(\tau; x, t)\|;\ 0\leqq\tau\leqq t\}$.

THEOREM 3.7. *Suppose that $F(t, x)$ of* (1.1) *is continuous on $[0, T]\times R^n$ and let K be a compact set in $[0, T]\times R^n$. If every solution $x(t; x_0, t_0)$ of* (1.1) *through $(t_0, x_0)\in K$ is continuable to $t=T$, then there exists a $\beta(K)>0$ such that $\|x(t; x_0, t_0)\|<\beta(K)$ for all $t\in[t_0, T]$.*

PROOF. Suppose that there is no such β. Then, there exist sequences $\{x_k\}$, $\{t_k\}$ and $t_0<T$ such that $(t_0, x_k)\to(t_0, x_0)\in K$, $\|x(t_k; x_k, t_0)\|\to\infty$ as $k\to\infty$. Let N be a sufficiently large integer. If k is large, there is τ_k such that $\|x(\tau_k; x_k, t_0)\|=N+1$ and that $\|x(t; x_k, t_0)\|<N+1$ for $t\in[t_0, \tau_k)$. Consider a system

$$\frac{dx}{dt}=F^*(t, x), \tag{3.4}$$

where $F^*(t, x)$ is a bounded continuous function on $[0, T]\times R^n$ and $F^*(t, x)=F(t, x)$ on $0\leqq t\leqq T$, $\|x\|\leqq N+1$. Let $y(t; x_k, t_0)$ be a solution of (3.4) such that $y(t; x_k, t_0)=x(t; x_k, t_0)$ for $t\in[t_0, \tau_k]$. As the sequence $\{y(t; x_k, t_0)\}$ is uniformly bounded and equicontinuous on $[0, T]$, it has a uniformly convergent subsequence, which we denote by $\{y(t; x_k, t_0)\}$ again. Let $y(t; x_0, t_0)$ be its limit function which is a solution of (3.4) through $(t_0, x_0)\in K$. If $\|y(t; x_0, t_0)\|<N$ on $[t_0, T]$, there arises a contradiction, and hence, there exists a t_1 such that $\|y(t_1; x_0, t_0)\|=N$ and $\|y(t; x_0, t_0)\|<N$ for $t\in[t_0, t_1)$. Thus, on $[t_0, t_1]$, $y(t; x_0, t_0)$ is a solution of (1.1).

Next, consider a function $F^{**}(t, x)$ which is bounded, continuous on $[0, T]\times R^n$ and $F^{**}(t, x)=F(t, x)$ on $0\leqq t\leqq T$, $\|x\|\leqq 2N+1$. Let $z(t; x_k, t_0)$ be a solution of

$$\frac{dx}{dt}=F^{**}(t, x), \tag{3.5}$$

such that $z(t; x_k, t_0)=y(t; x_k, t_0)$ on $[t_0, t_1]$ and that $z(t; x_k, t_0)=x(t; x_k, t_0)$ on $t_1<t\leqq\tau_k'$, where τ_k' is the first time such that $\|x(\tau_k'; x_k, t_0)\|=2N+1$. Then, by taking a subsequence, we can find a solution $z(t; x_0, t_0)$ of (3.5) which is a solution of (1.1) on $[t_0, t_2]$, where t_2 is the first time such that $\|z(t_2; x_0, t_0)\|=2N$, and which is equal to $y(t; x_0, t_0)$ on $[t_0, t_1]$.

By repeating this process, we can find a solution $x(t; x_0, t_0)$ of (1.1) defined on $[t_0, \sigma)$, $\sigma\leqq T$, such that for σ_k tending to σ as $k\to\infty$, $\|x(\sigma_k; x_0, t_0)\|=kN$. On the other hand, $x(t; x_0, t_0)$ is continuable to $t=T$, because $(t_0, x_0)\in K$. Therefore, $\|x(t; x_0, t_0)\|<B$ for some $B>0$, which contradicts the above. Thus, the existence of a suitable constant $\beta(K)$ has been demonstrated.

§ 4. Comparison principle.

Consider the system (1.1) in which $F(t, x)$ is defined and continuous on $I\times R^n$. Let $V(t, x)$ be a Liapunov function, and suppose that there exists a real valued continuous function $\omega(t, u)$ defined on $0\leqq t<\infty$, $|u|<\infty$ such that for all $(t, x)\in I\times R^n$

$$V'_{(1.1)}(t, x)\leqq\omega(t, V(t, x)). \tag{4.1}$$

Let $u(t; u_0, t_0)$ be the maximal solution of

$$\frac{du}{dt} = \omega(t, u), \qquad u_0 = V(t_0, x_0). \tag{4.2}$$

Then, as a consequence of (4.1), the solution $x(t; x_0, t_0)$ of (1.1) and $u(t; u_0, t_0)$ are related by the inequality

$$V(t, x(t; x_0, t_0)) \leqq u(t; u_0, t_0) \tag{4.3}$$

which holds for all $t \geqq t_0$ for which $x(t; x_0, t_0)$ and $u(t; u_0, t_0)$ are defined.

This is the simplest form of a very general comparison principle. In case $F(t, x)$ is defined on $0 \leqq t < \infty$, $\|x\| \leqq H$, $H > 0$, we also have a similar principle. The comparison principle has been widely used in dealing with a variety of qualitative problems, see for example [6], [7], [19], [27], [28], [43], [66]. It is a very important tool in applications, because it reduces the problem of determining the behavior of solutions of (1.1) to the solution of a scalar equation (4.2) and the properties of the Liapunov function V.

The comparison principle can be verified by the following theorem [105]. Consider a scalar differential equation

$$\frac{du}{dt} = f(t, u), \tag{4.4}$$

where $f(t, u)$ is continuous on $D: 0 \leqq t \leqq T$, $|u| < A$, $A > 0$.

THEOREM 4.1. *Suppose that the maximal solution $u(t)$ of* (4.4) *such that $u(t_0) = u_0$, $(t_0, u_0) \in D$, stays in D for $t \in [t_0, T]$. If a continuous function $x(t)$ with $x(t_0) = u_0$ satisfies*

$$x'(t) = \overline{\lim_{h \to 0+}} \frac{1}{h} \{x(t+h) - x(t)\} \leqq f(t, x(t)) \tag{4.5}$$

on $[t_0, T]$, $|u| < A$, then we have $x(t) \leqq u(t)$ for $t \in [t_0, T]$.

PROOF. Let $F(t, u)$ be such that

$$F(t, u) = \begin{cases} f(t, u) & (u \geqq x(t)) \\ f(t, x(t)) & (u < x(t)). \end{cases}$$

and let $u^*(t)$ be a solution of $\dot{u}=F(t, u)$ with $u^*(t_0)=u_0$. Suppose that $u^*(t)<x(t)$ at some t. Then there exists a t_1 such that $u^*(t_1) \leqq x(t_1)$ and $\dot{u}^*(t_1)<x'(t_1)$. Thus,

$$x'(t_1) \leqq f(t_1, x(t_1)) = F(t_1, u^*(t_1)) = \dot{u}^*(t_1),$$

which contradicts $\dot{u}^*(t_1)<x'(t_1)$. Therefore, $u^*(t) \geqq x(t)$, which implies that $u^*(t)$ is a solution of (4.4) with $u^*(t_0)=u_0$. Since $u(t)$ is the maximal solution of (4.4) such that $u(t_0)=u_0$, we have $u^*(t) \leqq u(t)$. From this, it follows that $x(t) \leqq u(t)$ for all $t \in [t_0, T]$.

By the same argument, the following theorem can be proved.

THEOREM 4.2. *Suppose that the minimal solution $u(t)$ of (4.4) such that $u(t_0)=u_0$, $(t_0, u_0) \in D$, stays in D for $t \in [t_0, T]$. If a continuous function $x(t)$ with $x(t_0)=u_0$ satisfies*

$$\varliminf_{h \to 0+} \frac{1}{h}\{x(t+h)-x(t)\} \geqq f(t, x(t))$$

on $[t_0, T]$, $|u|<A$, then we have $x(t) \geqq u(t)$ for $t \in [t_0, T]$.

REMARK. In Theorems 4.1. and 4.2, $h \to 0^+$ can be replaced by $h \to 0^-$.

As an application of the comparison principle, we shall discuss the uniqueness of solution of the system (1.1), where $F(t, x)$ is assumed to be continuous on $0 \leqq t < T$, $\|x\| < b$ (cf. [6], [19]).

THEOREM 4.3. *Let $\omega(t, r)$ be a continuous non-negative function on $(0, T)$, $r \geqq 0$. Suppose that the only solution of*

$$\frac{dr}{dt} = \omega(t, r), \tag{4.6}$$

which satisfies $r(0)=\dot{r}(0)=0$, on $[0, \alpha]$, for any α in $0<\alpha<T$, is the identically zero solution. Moreover, suppose that there exists a Liapunov function $V(t, x, y)$ defined on $0 \leqq t < T$, $\|x\|<b$, $\|y\|<b$ and satisfying the conditions;

(i) *$a(\|x-y\|) \leqq V(t, x, y) \leqq c\|x-y\|$, where $a(r) \in CIP$ and $c>0$ is a constant,*

(ii) *$V(t, x, y) \in C_0(x, y)$ and for the system*

(4.7) $$\dot{x}=F(t, x), \qquad \dot{y}=F(t, y),$$

we have

(4.8) $$V'_{(4.7)}(t, x, y) \leqq \omega(t, V(t, x, y)).$$

Then, there exists at most one solution of (1.1) *through* (0, 0) *on* $[0, T)$.

PROOF. Suppose $x(t)$ and $y(t)$ are two solutions of (1.1) on $[0, T)$ which pass through (0, 0). Consider the function $v(t) = V(t, x(t), y(t))$. Then by (4.8),

(4.9) $$v'(t)=\overline{\lim_{h\to 0^+}}\frac{1}{h}\{v(t+h)-v(t)\} \leqq \omega(t, v(t)).$$

Suppose that there exists a σ, $0<\sigma<T$, such that $v(\sigma)>0$. There exists a minimal solution $r(t)$ of (4.6) through the point $(\sigma, v(\sigma))$, existing on some interval to the left of σ. If $r(t)$ vanishes at a point, it can be continued to the interval $0<t\leqq\sigma$ as a solution of (4.6) such that $r(0)=\dot{r}(0)=0$, and hence $r(t)$ must vanish identically by hypothesis. From this, it follows that $0<r(t)$ as long as $r(t)$ exists. By applying Theorem 4.2, we have

(4.10) $$0<r(t)\leqq v(t)$$

as far to the left of σ as $r(t)$ exists, because of (4.9). As $v(t)$ is defined on $[0, \sigma]$ and is continuous, $r(t)$ can be continued to the whole interval $0<t\leqq\sigma$ as a solution of (4.6). Since $v(t)\to 0$ as $t\to 0^+$, we have $\lim_{t\to 0^+} r(t)=0$ and hence, we may define $r(0)=0$.

On the other hand, corresponding to any $\eta>0$, there exists a $\delta>0$ such that $\|x(t)-y(t)\|<\eta t$ for $t\in[0, \delta)$, because

$$\|x(t)-y(t)\| \leqq \int_0^t \|F(s, x(s))-F(s, y(s))\|\, ds,$$

and thus, for any $\varepsilon>0$ there exists a $\gamma>0$ such that $v(t)\leqq c\|x(t)-y(t)\|<\varepsilon t$ for $t\in[0, \gamma)$ by (i). This implies that $0<\frac{v(t)}{t}<\varepsilon$ for $t\in(0, \delta)$, and hence, $\frac{v(t)}{t}\to 0$ as $t\to 0^+$, which proves $\dot{r}(0)=0$. Thus, $r(t)$ is a solution of (4.6) with $r(0)=\dot{r}(0)=0$, and hence $r(t)$

$\equiv 0$ on $[0, \sigma]$, which contradicts (4.10). This proves that $x(t) \equiv y(t)$ on $[0, T)$.

EXAMPLE 4.1. If we consider the special function $V(t, x, y) = \|x-y\|$, it satisfies condition (i) in Theorem 4.3 and $V(t, x, y) \in C_0(x, y)$. Since we have $V'_{(4.7)}(t, x, y) \leqq \|F(t, x)-F(t, y)\|$, if we assume that

$$\|F(t, x)-F(t, y)\| \leqq \omega(t, \|x-y\|),$$

$V(t, x, y)$ satisfies the condition (4.8).

EXAMPLE 4.2. If $F(t, x)$ satisfies an inequality

$$\|F(t, x)-F(t, y)\| \leqq \lambda(t)\varphi(\|x-y\|),$$

where $\int_0 \frac{dr}{\varphi(r)} = \infty$ and $\int_0^T \lambda(t)dt < \infty$, then there is at most one solution of (1.1) through (0, 0). This is Osgood's uniqueness theorem (cf. [52]).

EXAMPLE 4.3. Letting $\omega(t, r) = \frac{r}{t}$ and $V(t, x, y) = \|x-y\|$, Perron's theorem can be obtained (cf. [52]).

§ 5. Dependence of solutions on initial values.

The dependence of solutions on initial values which generalizes Opial's result [107] will be discussed (cf. [144]). As a special case, we obtain the continuity property with respect to initial values.

LEMMA 5.1. *In the system* (1.1), *suppose that* $F(t, x)$ *is continuous on* $D: t_0 \leqq t \leqq T$, $\|x\| < \infty$, *and* $F(t, x)$ *is bounded by* $M > 0$ *on* D. *If* E *is the set of all points consisting of solution curves through* (t_0, x_0), $x_0 \in R^n$, *then* E *is a compact set.*

PROOF. It is clear that E is bounded. To show that E is a closed set, let (t', x') be a cluster point of E. Then there exists a sequence of points $\{(t_k, x_k)\}$, $k = 1, 2, \cdots$, of E such that $(t_k, x_k) \to (t', x')$ as $k \to \infty$. By the definition of E, there exists a solution

$x_k(t; x_0, t_0)$ of (1.1) through (t_k, x_k) which is defined on $[t_0, T]$ by Theorem 3.2. Since we have

$$x_k(t; x_0, t_0) = x_0 + \int_{t_0}^{t} F(s, x_k(s; x_0, t_0))ds$$

and $\|F(t, x)\| \leqq M$, we have a uniformly convergent subsequence by Ascoli's Theorem. Let $x(t)$ be its limit function. Then by letting $k \to \infty$ we obtain

$$x(t) = x_0 + \int_{t_0}^{t} F(s, x(s))ds\,.$$

Therefore, $x(t)$ is a solution of (1.1) through (t_0, x_0), and from $x_k = x_k(t_k; x_0, t_0) = [x_k(t_k; x_0, t_0) - x(t_k)] + x(t_k)$, it follows that $x' = x(t')$. This shows that (t', x') is on the solution curve $x = x(t)$, i.e., $(t', x') \in E$. Thus, E is a closed set.

THEOREM 5.1. *Suppose that $F(t, x)$ of* (1.1) *is continuous on an open set D in $I \times Q$, Q an open set in R^n, and that every solution issuing from a point $(t_0, x_0) \in D$ to the right is continuable to $t = t_1$. Let E be the set of all points consisting of the solution curves for $J = [t_0, t_1]$, starting from (t_0, x_0), and assume that E is contained in a compact set in D.*

Then, corresponding to each $\varepsilon > 0$, there exists a $\delta > 0$ such that every solution $x^*(t; x_0^*, t_0^*)$, $t_0^* \in J$, of

$$\frac{dx}{dt} = F(t, x) + g(t)\,, \tag{5.1}$$

where $g(t)$ is a continuous function such that $\int_J \|g(t)\|dt \leqq \delta$, passing through $P^*(t_0^*, x_0^*)$ such that $d(P^*, E) \leqq \delta$, exists on $[t_0^*, t_1]$ and satisfies

$$\|x^*(t; x_0^*, t_0^*) - x(t; x_0, t_0)\| < \varepsilon\,,$$

where $x(t; x_0, t_0)$ is a solution of (1.1) *contained in E and $x(t; x_0, t_0)$ may depend on $x^*(t; x_0^*, t_0^*)$.*

PROOF. By Lemma 5.1, it is clear that E is a closed set. Suppose that for some $\varepsilon > 0$ there exists no δ that satisfies the condition in Theorem 5.1. We can assume that $\overline{N(\varepsilon, E)} \subset D$. Since

$\overline{N(\varepsilon, E)}$ is a compact set, there exists a function $F^*(t, x)$ which is continuous and bounded on $|t| < \infty$, $x \in R^n$ and is equal to $F(t, x)$ on $\overline{N(\varepsilon, E)}$. A solution of (1.1) remaining in $\overline{N(\varepsilon, E)}$ is a solution of

$$\frac{dx}{dt} = F^*(t, x) \tag{5.2}$$

and the set of all points consisting of the solution curves for $t \in J$ of (5.2) through (t_0, x_0) coincides with E. Therefore, we can assume that for $\varepsilon > 0$ and the equation

$$\frac{dx}{dt} = F^*(t, x) + g(t), \tag{5.3}$$

the conclusion of Theorem 5.1 is not verified. Every solution of (5.3) exists for all t. From our hypothesis, there are a sequence of points $\{P_k(t_k, x_k)\}$ and a sequence of functions $\{g_k(t)\}$ such that $d(P_k, E) \to 0$, $\int_J \| g_k(t) \| dt \to 0$ as $k \to \infty$ and moreover, there exists a solution $\varphi_k(t)$ of $\dot{x} = F^*(t, x) + g_k(t)$ through P_k such that there is no solution curve of (5.2) lying in E with the property that the distances of all points on the arc of the former to the latter are smaller than ε. Since we have

$$\varphi_k(t) = \varphi_k(t_0) + \int_{t_0}^{t} F^*(s, \varphi_k(s))ds + \int_{t_0}^{t} g_k(s)ds, \tag{5.4}$$

$\{\varphi_k(t)\}$ is uniformly bounded and equicontinuous on J. Hence, by Ascoli's Theorem, there exists a uniformly convergent subsequence, the suffix of which we denote by k again. Let $\varphi(t)$ be its limit function. Then from (5.4), it follows that

$$\varphi(t) = \varphi(t_0) + \int_{t_0}^{t} F^*(s, \varphi(s))ds,$$

which shows that $\varphi(t)$ is a solution of (5.2). If we let t' be a cluster point of $\{t_k\}$, clearly $(t', \varphi(t')) \in E$, because E is a closed set. By $\varphi(t)$ and a solution joining (t_0, x_0) and $(t', \varphi(t'))$, we have a solution $x = \varphi^*(t)$ of (5.2) through (t_0, x_0). Therefore,

$$\varphi^*(t) \subset E \quad \text{and} \quad \varphi^*(t) = \varphi(t) \quad \text{for} \quad t \geqq t'.$$

If k is sufficiently large and t_k is sufficiently close to t', the distance between $\varphi_k(t)$ and $\varphi(t)$ is smaller than ε, because $\varphi_k(t)$ is uniformly convergent to $\varphi(t)$. This contradicts our hypothesis. Thus, the theorem is proved.

As a special case, if the solution of (1.1) is unique, we obtain the theorem on continuous dependence upon initial values.

THEOREM 5.2. *Suppose that* $F(t, x)$ *of* (1.1) *is continuous on an open set* D *in* $I\times R^n$ *and that a solution* $x(t\,;x_0, t_0)$ *of* (1.1) *through* $(t_0, x_0)\in D$ *is defined on* $[t_0, t_1]$ *and is unique. Then, corresponding to each* $\varepsilon>0$, *there exists a* $\delta>0$ *such that every solution* $x^*(t\,;x_0^*, t_0^*)$, $t_0\leqq t_0^*<t_1$, *of* $\dot{x}=F(t, x)+g(t)$, *where* $g(t)$ *is a continuous function satisfying* $\int_{t_0}^{t}\|g(t)\|\,dt\leqq\delta$, *passing through* $P^*(t_0^*, x_0^*)$ *such that its distance from the solution* $x(t\,;x_0, t_0)$ *is less than* δ, *exists on* $[t_0^*, t_1]$ *and satisfies*

$$\|x^*(t\,;x_0^*, t_0^*)-x(t\,;x_0, t_0)\|<\varepsilon\,.$$

The case where $g(t)\equiv 0$ is the ordinary continuous dependence theorem.

THEOREM 5.3. *Suppose that* $F(t, x)$ *of* (1.1) *is continuous on* $a\leqq t\leqq b$, $\|x-x_0(t)\|\leqq r$, $r>0$, *where* $x_0(t)$ *is a solution defined on* $a\leqq t\leqq b$, *and that* $x_0(t)$ *is unique to the right. Then, given any* $\varepsilon>0$, *there exists a* $\delta>0$ *such that if* $a\leqq t_0\leqq b$ *and* $\|x_0-x_0(t)\|<\delta$, $\|x(t\,;x_0, t_0)-x_0(t)\|<\varepsilon$ *for* $t\in[t_0, b]$.

REMARK. Theorem 5.3 can also be proved directly by Theorem 1.3.

§6. Almost periodic function.

Since almost periodic functions appear in this monograph, we summarize here some of their properties which we shall need later. The considerations for an almost periodic vector function $f(t, x)$ defined on $-\infty<t<\infty$, $x\in X$ are the same as for an almost periodic scalar function $f(t)$ defined on $-\infty<t<\infty$, and hence we shall state some theorems without proofs. The proofs can be supplied by following the respective proof in [14] and [35].

Further, for ordinary differential equations $X \subset R^n$ but for functional-differential equations $X \subset C^n$, where for a given $h > 0$, C^n denotes the space of continuous functions mapping the interval $[-h, 0]$ into R^n with the norm $|\varphi| = \sup_{-h \leq \theta \leq 0} \|\varphi(\theta)\|$ for $\varphi \in C^n$.

In the following, let X be a subset of R^n or of C^n.

DEFINITION 6.1. A function $f(t, x)$, where f is an m-vector, t is a real scalar and x is an n-vector, is said to be almost periodic in t uniformly with respect to $x \in X$, if $f(t, x)$ is continuous in (t, x) for $t \in (-\infty, \infty)$, $x \in X$ and if for any $\varepsilon > 0$, it is possible to find an $l(\varepsilon) > 0$ such that, in any interval of length $l(\varepsilon)$, there exists a τ suth that the inequality

$$\|f(t+\tau, x) - f(t, x)\| \leq \varepsilon$$

is satisfied for all $t \in (-\infty, \infty)$, $x \in X$. This number τ is called an ε-translation number of $f(t, x)$.

If $f(t, x)$ is almost periodic in t uniformly with respect to $x \in X$, then let $\{\lambda_j\}$ be the set of all real numbers such that

$$\lim_{T \to \infty} \frac{1}{T} \int_0^T f(t, x) e^{-i\lambda_j t} dt, \qquad i = \sqrt{-1},$$

is not identically zero for $x \in X$. As X is separable, $\{\lambda_j\}$ is countable.

DEFINITION 6.2. The set $\{\sum_{j=1}^{N} n_j \lambda_j\}$ for all integers N and integers n_j is called the module of $f(t, x)$.

In the following theorems except Theorem 6.4, we assume that $f(t, x)$ and $g(t, x)$ are almost periodic in t uniformly with respect to $x \in X$.

THEOREM 6.1. *If X is compact, then $f(t, x)$ is bounded.*

THEOREM 6.2. *If X is compact, then $f(t, x)$ is uniformly continuous in (t, x) for $t \in (-\infty, \infty)$, $x \in X$.*

THEOREM 6.3. *If X is compact, then $f(t, x)$ is almost periodic in t uniformly with respect to $x \in X$, if and only if, for every sequence of real numbers $\{\tau_k\}$, the sequence $\{f(t+\tau_k, x)\}$ has a subsequence $\{f(t+\tau_{k_j}, x)\}$ which is convergent uniformly with respect to $t \in (-\infty, \infty)$, $x \in X$.*

THEOREM 6.4. *Let C_H^n be a subset of C^n such that $|\varphi| \leqq H$, $\varphi \in C^n$, and let $f(t, x)$ be almost periodic in t uniformly with respect to $x \in C_H^n$ (or $x \in C^n$). If corresponding to any $\alpha > 0$, there exists an $L(t, \alpha) > 0$, where $L(t, \alpha)$ is continuous in t, such that for $x \in C_\alpha^n$ we have $\| f(t, x) \| \leqq L(t, \alpha)$, then there exists a $B(\alpha) > 0$ such that $\| f(t, x) \| \leqq B(\alpha)$ for $x \in C_\alpha^n$.*

THEOREM 6.5. *If X is compact, $f(t, x)+g(t, x)$ also is almost periodic in t uniformly with respect to $x \in X$.*

THEOREM 6.6. *If X is compact and if for any sequence of real numbers $\{\tau_k\}$ having its limit (including infinity) for which $\{f(t+\tau_k, x)\}$ is uniformly convergent, $\{g(t+\tau_k, x)\}$ also is uniformly convergent, then the module of $g(t, x)$ is contained in the module of $f(t, x)$.*

Finally, the following should be noticed: Let $f(t, x)$ be almost periodic in t uniformly with respect to $x \in X$, where X is a compact set, and be Lipschitzian in x. In this case, it is not necessarily possible to choose a Lipschitz constant which is independent of t. For example, a function $f(t, x)$ defined for $-\infty < t < +\infty$, $x \geqq 0$ such that

$$f(t, x) = \begin{cases} \displaystyle\sum_{n=1}^{\infty} \frac{x^{n^4}}{n^2}\left(\sin \frac{t}{n}\right)^4 & (0 \leqq x \leqq 1) \\ \displaystyle\sum_{n=1}^{\infty} \frac{2-x^{-n^4}}{n^2}\left(\sin \frac{t}{n}\right)^4 & (1 < x) \end{cases}$$

shows that in a neighborhood of $x=1$, we cannot have a Lipschitz constant which is independent of t.

CHAPTER II. LIAPUNOV STABILITY AND BOUNDEDNESS OF SOLUTIONS

Sufficient conditions on stability and boundedness of solutions will be discussed by Liapunov's second method.

§ 7. Definitions of stability.

Consider a system of differential equations

$$\frac{dx}{dt} = F(t, x), \tag{7.1}$$

where x is an n-vector. Suppose that $F(t, x)$ is continuous in (t, x) on $I \times D$, where D is a connected open set in R^n. Let C be a class of solutions of (7.1) which remain in D and let $x_0(t)$ be an element of C. Setting $x = y + x_0(t)$, the system (7.1) is transformed into

$$\frac{dy}{dt} = F(t, y + x_0(t)) - F(t, x_0(t)). \tag{7.2}$$

If we denote by $G(t, y)$ the right-hand side of (7.2), clearly $G(t, 0) \equiv 0$ and the zero solution $y(t) \equiv 0$ of (7.2) corresponds to $x_0(t)$. Therefore, it is sufficient to discuss the stability of $y(t) \equiv 0$ of (7.2) in place of $x_0(t)$. For this reason, we can assume without loss of generality that $F(t, 0) \equiv 0$ and that D is a domain such that $\| x \| < H$, $H > 0$. Here, we give the definitions of stability in the sense of Liapunov (cf. [5], [37], [85], [92], [94]).

DEFINITION 7.1. The solution $x(t) \equiv 0$ of (7.1) is stable, if for any $\varepsilon > 0$ and any $t_0 \in I$, there exists a $\delta(t_0, \varepsilon) > 0$ such that if $\| x_0 \| < \delta(t_0, \varepsilon)$, we have $\| x(t; x_0, t_0) \| < \varepsilon$ for all $t \geqq t_0$ [85].

DEFINITION 7.2. The zero solution of (7.1) is uniform-stable, if the δ above is independent of t_0 [112].

DEFINITION 7.3. The zero solution of (7.1) is asymptotically stable, if $x(t)\equiv 0$ is stable and if there exists a $\delta_0(t_0)>0$ such that if $\|x_0\|<\delta_0(t_0)$, $x(t;x_0,t_0)\to 0$ as $t\to\infty$ [85].

DEFINITION 7.4. The zero solution of (7.1) is quasi-equiasymptotically stable, if given any $\varepsilon>0$ and any $t_0\in I$, there exist a $\delta_0(t_0)>0$ and a $T(t_0,\varepsilon)>0$ such that if $\|x_0\|<\delta_0(t_0)$, $\|x(t;x_0,t_0)\|<\varepsilon$ for all $t\geqq t_0+T(t_0,\varepsilon)$ [5].

DEFINITION 7.5. The zero solution of (7.1) is equiasymptotically stable, if it is stable and is quasi-equiasymptotically stable [92].

DEFINITION 7.6. The zero solution of (7.1) is quasi-uniform-asymptotically stable, if the δ_0 and the T in Definition 7.4 are independent of t_0 [5].

DEFINITION 7.7. The zero solution of (7.1) is uniform-asymptotically stable, if it is uniform-stable and is quasi-uniform-asymptotically stable [88].

DEFINITION 7.8. The zero solution of (7.1) is exponential-asymptotically stable, if there exists a $\lambda>0$ and, given any $\varepsilon>0$, there exists a $\delta(\varepsilon)>0$ such that if $\|x_0\|<\delta(\varepsilon)$,

$$\|x(t;x_0,t_0)\|\leqq\varepsilon\exp[-\lambda(t-t_0)]$$

for all $t\geqq t_0$ [87].

In [92], we can find several examples which show that these concepts are not equivalent. In the definitions above, all solutions of (7.1) near $x(t)\equiv 0$ are considered. However, we can also define the stability with respect to some class of solutions or the stability regarding the behavior of solutions for $-t$ large (cf. [77]).

Here, it is noticed that if $x(t)\equiv 0$ is the unique solution of (7.1) through (0, 0) quasi-equiasymptotic stability of $x(t)\equiv 0$ implies the stability of $x(t)\equiv 0$. This can be easily seen from Theorem 5.3. (cf. [136]).

It is clear that if the zero solution of (7.1) is stable, the solution of (7.1) through the point $(t_0, 0)$, $t_0 \in I$, is unique to the right.

Now we observe some relationships between the definitions above. First of all, we consider a linear system

$$\frac{dx}{dt} = A(t)x, \tag{7.3}$$

where $A(t)$ is an $n \times n$ matrix of continuous functions defined on I. Let $X(t)$ be a non-singular solution of the matrix equation associated with (7.3). Then the solution of (7.3) through (t_0, x_0) has the form $x(t; x_0, t_0) = X(t)X^{-1}(t_0)x_0$.

THEOREM 7.1. *If the solution $x(t) \equiv 0$ of (7.3) is asymptotically stable, then it is equiasymptotically stable.*

PROOF. Since the asymptotic stability implies that all elements of $X(t)$ tend to zero as $t \to \infty$, there exists a $T(t_0, \varepsilon) > 0$ such that if $t \geqq T(t_0, \varepsilon)$, $\| X(t)X^{-1}(t_0) \| < \dfrac{\varepsilon}{\delta_0}$, where ε is given and $\delta_0 > 0$ is a constant. Therefore, if $\| x_0 \| < \delta_0$ and $t \geqq t_0 + T(t_0, \varepsilon)$,

$$\| x(t; x_0, t_0) \| \leqq \| X(t)X^{-1}(t_0) \| \, \| x_0 \| < \varepsilon,$$

which shows that $x(t) \equiv 0$ of (7.3) is equiasymptotically stable.

THEOREM 7.2. *If the zero solution of (7.3) is uniform-asymptotically stable, then it is exponential-asymptotically stable.*

PROOF. For a fixed t_0, $cX(t)X^{-1}(t_0)$ is a matrix solution, where $c > 0$ is a constant. From uniform-stability, it follows that $\| cX(t)X^{-1}(t_0) \| < 1$ for small c, which implies the existence of a constant $N \geqq 1$ such that $\| X(t)X^{-1}(t_0) \| \leqq N$ for all $t \geqq t_0 \geqq 0$. Since $x(t) \equiv 0$ is quasi-uniform-asymptotically stable, there exists a $T > 0$ such that $\| X(t)X^{-1}(t_0) \| < \dfrac{1}{2}$ for all $t \geqq t_0 + T$, and hence, $\| X(t)X^{-1}(t_0) \| < 2^{-k}N$ for $t \geqq t_0 + kT$, where k is a positive integer. Thus, we have

$$\| X(t)X^{-1}(t_0) \| < 2Ne^{-\lambda(t-t_0)}, \qquad \lambda = \frac{\log 2}{T},$$

for all $t \geqq t_0 \geqq 0$, which implies

$$\| x(t\,;\, x_0,\, t_0) \| = \| X(t)X^{-1}(t_0)x_0 \| < 2Ne^{-\lambda(t-t_0)} \| x_0 \| \,.$$

Therefore, the zero solution is exponential-asymptotically stable.

Next, we consider the stability in a periodic system

$$\frac{dx}{dt} = F(t, x)\,, \tag{7.4}$$

where $F(t, x)$ is continuous on $I \times D$, $F(t, 0) \equiv 0$ and $F(t, x)$ is periodic in t of period $\omega > 0$, i. e., $F(t+\omega, x) = F(t, x)$. This system has the following properties.

THEOREM 7.3. *If the zero solution of* (7.4) *is stable, then it is uniform-stable.*

PROOF. By the stability of $x(t) \equiv 0$, for any $\varepsilon > 0$, there exists a $\delta_1(\omega, \varepsilon) > 0$ such that if $\| x_0 \| < \delta_1(\omega, \varepsilon)$, $\| x(t\,;\, x_0,\, \omega) \| < \varepsilon$ for all $t \geqq \omega$. Since the stability of $x(t) \equiv 0$ implies the uniqueness of $x(t) \equiv 0$, by Theorem 5.3 there exists a $\delta(\delta_1) > 0$ for which $\| x(\omega\,;\, x_0,\, t_0) \| < \delta_1$ if $t_0 \in [0, \omega)$ and $\| x_0 \| < \delta$. Thus, if $t_0 \in [0, \omega)$ and $\| x_0 \| < \delta$, $\| x(t\,;\, x_0,\, t_0) \| < \varepsilon$ for all $t \geqq t_0$. From the periodicity of $F(t, x)$, it follows that the behavior of a solution $x(t\,;\, x_0,\, t_0)$ such that $k\omega \leqq t_0 < (k+1)\omega$, $k = 1, 2, \cdots$, is the same as one of the solution $x(t\,;\, x_0,\, t_0 - k\omega)$. Therefore, if $\| x_0 \| < \delta$ and $t_0 \in I$, $\| x(t\,;\, x_0,\, t_0) \| < \varepsilon$ for all $t \geqq t_0$, where clearly δ depends only on ε, which proves the uniform-stability of $x(t) \equiv 0$.

THEOREM 7.4. *If the zero solution of* (7.4) *is asymptotically stable, it is uniform-asymptotically stable.*

PROOF. First of all, we prove the equiasymptotic stability of $x(t) \equiv 0$. Suppose that it is not so. Since $x(t) \equiv 0$ is asymptotically stable, there exists a $\delta_0(t_0) > 0$ such that $\| x_0 \| \leqq \delta_0(t_0)$ implies $\| x(t\,;\, x_0,\, t_0) \| \to 0$ as $t \to \infty$. By our hypothesis, for some $t_0 \in I$ and some $\varepsilon > 0$, there exist sequences $\{x_k\}$, $\{\tau_k\}$ such that $\| x_k \| \leqq \delta_0(t_0)$, $\tau_k \to \infty$ as $k \to \infty$ and that $\| x(\tau_k\,;\, x_k,\, t_0) \| \geqq \varepsilon$. Let x_0 be a cluster point of $\{x_k\}$, and then $\| x_0 \| \leqq \delta_0(t_0)$. On any finite interval, the sequence $\{x(t; x_k, t_0)\}$ is uniformly bounded and equicontinuous and hence, by Ascoli's Theorem, there exists a subsequence which converges to a solution $x(t; x_0, t_0)$ uniformly on any finite interval. We denote the subsequence by $\{x(t; x_k, t_0)\}$ again. Since $\|x_0\| \leqq \delta_0(t_0)$, $x(t; x_0, t_0) \to 0$ as $t \to \infty$. Therefore, corresponding to any $\eta > 0$, if m is large enough, we have $\|x(m\omega; x_0, t_0)\| < \eta$, where $m > 0$ is an integer. As $\{x(t; x_k, t_0)\}$ is uniformly convergent to $x(t; x_0, t_0)$, $\|x(m\omega; x_k, t_0)\| < \eta$ for some k large enough, and besides $\|x(\tau_k; x_k, t_0)\| \geqq \varepsilon$ for $\tau_k > m\omega$. Thus, there exists a solution $x(t; x(m\omega; x_k, t_0), 0)$ such that $\|x(\tau_k - m\omega; x(m\omega; x_k, t_0), 0)\| \geqq \varepsilon$, which contradicts the stability of $x(t) \equiv 0$. Therefore, the zero solution is equiasymptotically stable.

The uniform-asymptotic stability can be proved by the same idea as used in the proof of Theorem 7.3.

Some relationships between stability and uniform-stability in an almost periodic system have been discussed by Seifert [127], [128] and by Conley and Miller (J. Diff. Eqs., **1**, (1965), 333-336).

§ 8. **Theorems on Liapunov stability.**

Liapunov [85] gave sufficient conditions for stability and asymptotic stability. Here, we shall consider a more general case and give other sufficient conditions. We shall consider the differential system (7.1) under the assumption that $F(t, x)$ is continuous on $0 \leqq t < \infty$, $\|x\| < H$, $H > 0$, and $F(t, 0) \equiv 0$. Whereas Liapunov considered a Liapunov function with continuous first order partial derivatives, we shall not assume this and generalize the Liapunov function by considering $V'_{(7.1)}(t, x)$ defined in Section 1.

THEOREM 8.1. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $0 \leqq t < \infty$, $\|x\| < H$ which satisfies the following conditions;*

(i) $V(t, 0) \equiv 0$,

(ii) $a(\|x\|) \leqq V(t, x)$, *where* $a(r) \in CIP$ (*see Section* 1),

(iii) $V'_{(7.1)}(t, x) \leqq 0$.

Then, the solution $x(t) \equiv 0$ of the system (7.1) *is stable.*

PROOF. Corresponding to any $\varepsilon>0$, $\varepsilon<H$, we have $a(\varepsilon) \leqq V(t, x)$ for $t\in I$ and x such that $\|x\|=\varepsilon$. For a fixed $t_0\in I$, we can choose a $\delta(t_0, \varepsilon)>0$ such that $\|x_0\|<\delta$ implies $V(t_0, x_0)<a(\varepsilon)$, because $V(t_0, 0)=0$ and $V(t, x)$ is continuous. Suppose that a solution $x(t; x_0, t_0)$ of (7.1) such that $\|x_0\|<\delta$ satisfies $\|x(t_1; x_0, t_0)\| =\varepsilon$ at some t_1. From (iii), it follows that $V(t_1, x(t_1; x_0, t_0))\leqq V(t_0, x_0)$, and hence $a(\varepsilon)\leqq V(t_1, x(t_1; x_0, t_0))\leqq V(t_0, x_0)<a(\varepsilon)$. This is a contradiction, and hence, if $\|x_0\|<\delta(t_0, \varepsilon)$, then $\|x(t; x_0, t_0)\| <\varepsilon$ for all $t\geqq t_0$ by Theorem 3.3, that is, $x(t)\equiv 0$ is stable.

THEOREM 8.2. *If condition* (ii) *in Theorem* 8.1 *is replaced by*

(ii)′ $a(\|x\|)\leqq V(t, x)\leqq b(\|x\|)$, *where* $a(r)\in CIP$ *and* $b(r)\in CIP$, *the solution* $x(t)\equiv 0$ *of* (7.1) *is uniform-stable.*

By choosing a $\delta(\varepsilon)>0$ so that $b(\delta)<a(\varepsilon)$ and by the same argument as in Theorem 8.1, it can be proved that if $\|x_0\|<\delta(\varepsilon)$ and $t_0\in I$, $\|x(t; x_0, t_0)\|<\varepsilon$ for all $t\geqq t_0$, which shows the uniform-stability of $x(t)\equiv 0$.

Next, we shall discuss some theorems on asymptotic stability of the solution $x(t)\equiv 0$ of (7.1).

THEOREM 8.3. *Under the same assumptions as in Theorem* 8.2, *if* $V'_{(7.1)}(t, x)\leqq -c(\|x\|)$, *where* $c(r)$ *is continuous on* $[0, H]$ *and is positive definite, then the solution* $x(t)\equiv 0$ *of* (7.1) *is uniform-asymptotically stable.*

PROOF. By Theorem 8.2, the zero solution is uniform-stable, and hence, there exists a $\delta_0>0$ such that if $t_0\in I$ and $\|x_0\|<\delta_0$, $\|x(t; x_0, t_0)\|<H$ for all $t\geqq t_0$. Moreover, for any $\varepsilon>0$, there exists a $\delta(\varepsilon)>0$ such that if $t_0\in I$ and $\|x_0\|<\delta$, $\|x(t; x_0, t_0)\|<\varepsilon$ for all $t\geqq t_0$. It will be shown that every solution $x(t; x_0, t_0)$ of (7.1) such that $t_0\in I$, $\|x_0\|<\delta_0$ satisfies $\|x(t; x_0, t_0)\|<\delta(\varepsilon)$ at some t. Suppose that $\delta(\varepsilon)\leqq \|x(t; x_0, t_0)\|<H$ for all $t\geqq t_0$. Since there exists a $\gamma>0$ such that $V'_{(7.1)}(t, x)\leqq -\gamma$ on $\delta\leqq\|x\|<H$, we have

$$V(t, x(t; x_0, t_0))\leqq V(t_0, x(t_0; x_0, t_0))-\gamma(t-t_0)\,. \tag{8.1}$$

If $t>t_0+T$, $T=\dfrac{b(\delta_0)-a(\delta)}{\gamma}$, we have $V(t_0, x_0)-\gamma(t-t_0)<a(\delta)$,

because $V(t_0, x_0) \leqq b(\delta_0)$. By (8.1), $V(t, x(t; x_0, t_0)) < a(\delta)$, which contradicts $V(t, x(t; x_0, t_0)) \geqq a(\delta)$. Thus, at some t_1 such that $t_0 \leqq t_1 \leqq t_0 + T$, we must have $\| x(t_1; x_0, t_0) \| < \delta(\varepsilon)$. Therefore, if $t \geqq t_0 + T$, we have $\| x(t; x_0, t_0) \| < \varepsilon$, where clearly T depends only on ε. This shows that $x(t) \equiv 0$ is quasi-uniform-asymptotically stable. Thus, the proof is completed.

COROLLARY. *Under the same assumption as in Theorem* 8.2, *if* $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c > 0$ *is a constant, then the solution* $x(t) \equiv 0$ *of* (7.1) *is uniform-asymptotically stable.*

This is an immediate consequence of Theorem 8.3. However, applying Theorem 4.1,

$$V(t, x(t; x_0, t_0)) \leqq V(t_0, x_0) e^{-c(t-t_0)},$$

and hence, the uniform-asymptotic stability can be easily proved. As we shall see later, the existence of $V(t, x)$ satisfying the condition in Corollary is a necessary condition for the uniform-asymptotic stability and it is very useful in discussing the behavior of solutions of perturbed systems.

THEOREM 8.4. *Suppose that there exists a Liapunov function* $V(t, x)$ *defined on* $t \in I$, $\| x \| < H$, *which satisfies the following conditions;*

(i) $a(\| x \|) \leqq V(t, x) \leqq b(\| x \|)$, *where* $a(r) \in CIP$ *and* $b(r) \in CIP$,

(ii) $V'_{(7.1)}(t, x) + V^*(t, x) \to 0$ *uniformly on* $0 < \lambda \leqq \| x \| < H$, *for any* λ, *as* $t \to \infty$, *where* $V^*(t, x)$ *is continuous and there exists a continuous function* $c(r) > 0$ *for* $0 < r < H$ *such that* $c(\| x \|) \leqq V^*(t, x)$.

Then, if the solution $x(t) \equiv 0$ *of* (7.1) *is unique to the right, it is uniform-asymptotically stable.*

PROOF. For any $\varepsilon > 0$, $\varepsilon \leqq H^* < H$, choose a $\delta(\varepsilon) > 0$ so that $b(\delta) < a(\varepsilon)$, $\delta < \varepsilon$. On the domain $\delta \leqq \| x \| \leqq H^*$, there exist a $\gamma(\varepsilon) > 0$ and a $T(\varepsilon) > 0$ such that $V^*(t, x) \geqq \gamma(\varepsilon)$ and that if $t \geqq T(\varepsilon)$, we have $V'_{(7.1)}(t, x) + V^*(t, x) \leqq \dfrac{\gamma(\varepsilon)}{2}$. This follows from condition (ii), and hence

$$V'_{(7.1)}(t, x) \leqq -\frac{\gamma(\varepsilon)}{2} \tag{8.2}$$

on the domain D_ε; $t \geqq T(\varepsilon)$, $\delta(\varepsilon) \leqq \|x\| \leqq H^*$. Suppose that a solution $x(t; x_0, t_0)$ of (7.1) such that $t_0 \geqq T(\varepsilon)$, $\|x_0\| < \delta(\varepsilon)$, satisfies $\|x(t; x_0, t_0)\| = \varepsilon$ at some t. Then there exist t_1, t_2, $t_0 \leqq t_1 < t_2$, such that $\|x(t_1; x_0, t_0)\| = \delta$, $\|x(t_2; x_0, t_0)\| = \varepsilon$ and that $\delta < \|x(t; x_0, t_0)\| < \varepsilon$ for $t \in (t_1, t_2)$. From (i) and (8.2), it follows that $a(\varepsilon) \leqq V(t_2, x(t_2; x_0, t_0)) \leqq V(t_1, x(t_1; x_0, t_0)) \leqq b(\delta)$, which contradicts $b(\delta) < a(\varepsilon)$. Therefore, we have $\|x(t; x_0, t_0)\| < \varepsilon$ for all $t \geqq t_0$. Since $x(t) \equiv 0$ is unique to the right, by Theorem 5.3, there exists a $\delta_1(\varepsilon) > 0$ such that if $0 \leqq t_0 \leqq T(\varepsilon)$ and $\|x_0\| < \delta_1(\varepsilon)$, we have $\|x(T(\varepsilon); x_0, t_0)\| < \delta(\varepsilon)$. Thus, if $t_0 \in I$ and $\|x_0\| < \delta_1(\varepsilon)$, $\|x(t; x_0, t_0)\| < \varepsilon$ for all $t \geqq t_0$, which proves the uniform-stability of $x(t) \equiv 0$.

By uniform-stability, there exists a $\delta_0 > 0$ for which $\|x(t; x_0, t_0)\| < H^* < H$ for all $t \geqq t_0$ if $t_0 \in I$ and $\|x_0\| < \delta_0$. For any $\varepsilon > 0$, we can assume that $\delta(\varepsilon) \leqq \delta_0$. Consider a solution $x(t; x_0, t_0)$ of (7.1) such that $t_0 \geqq T(\varepsilon)$ and $\|x_0\| < \delta_0$. On the domain D_ε, we have (8.2) and hence, by the same argument in the proof of Theorem 8.3, it can be seen that if $t \geqq t_0 + T_1(\varepsilon)$, $T_1(\varepsilon) = \dfrac{2}{\gamma(\varepsilon)}\{b(H^*) - a(\delta)\}$, we have $\|x(t; x_0, t_0)\| < \varepsilon$. For a solution $x(t; x_0, t_0)$ such that $0 \leqq t_0 < T(\varepsilon)$ and $\|x_0\| < \delta_0$, clearly $\|x(t; x_0, t_0)\| < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon) + T_1(\varepsilon)$, because $\|x(T(\varepsilon); x_0, t_0)\| < H^*$. Thus, we can find a $T_2(\varepsilon) > 0$ such that if $t_0 \in I$ and $\|x_0\| < \delta_0$, $\|x(t; x_0, t_0)\| < \varepsilon$ for all $t \geqq t_0 + T_2(\varepsilon)$, which shows that $x(t) \equiv 0$ is quasi-uniform-asymptotically stable. This completes the proof (cf. [92], [150]).

The following theorem due to Marachkoff [90] is a sufficient condition for asymptotic stability, which does not necessarily imply uniform-asymptotic stability (cf. [92]).

THEOREM 8.5. *Under the assumption in Theorem* 8.1, *if* $V'_{(7.1)}(t, x) \leqq -c(\|x\|)$, *where* $c(r)$ *is continuous on* $[0, H]$ *and positive definite, and if* $F(t, x)$ *is bounded, then the solution* $x(t) \equiv 0$ *of* (7.1) *is asymptotically stable.*

PROOF. By Theorem 8.1, $x(t) \equiv 0$ is stable. Suppose that $x(t) \equiv 0$ is not asymptotically stable. Then for some $\varepsilon > 0$ there exist a solution $x(t; x_0, t_0)$ and a divergent sequence $\{t_k\}$ for which $\| x(t_k; x_0, t_0) \| \geqq \varepsilon$. Since $F(t, x)$ is bounded, there exists a $K > 0$ such that $\left| \frac{d\| x \|}{dt} \right| < K$. Therefore, on the intervals

$$t_k - \frac{\varepsilon}{2K} \leqq t \leqq t_k + \frac{\varepsilon}{2K}, \tag{8.3}$$

we have $\| x(t; x_0, t_0) \| \geqq \frac{\varepsilon}{2}$. We can assume that these intervals are disjoint and $t_1 - \frac{\varepsilon}{2K} > t_0$ by taking, if necessary, a subsequence of $\{t_k\}$.

Since $V'_{(7.1)}(t, x) \leqq -c(\| x \|)$, there exists a constant $\gamma > 0$ such that $V'_{(7.1)}(t, x) \leqq -\gamma$ on the intervals (8.3), and $V'_{(7.1)}(t, x) \leqq 0$ elsewhere. Therefore,

$$V\left(t_k + \frac{\varepsilon}{2K}, x\left(t_k + \frac{\varepsilon}{2K}; x_0, t_0\right)\right) - V(t_0, x_0) < -\gamma \frac{\varepsilon}{K} k \to -\infty$$

as $k \to \infty$, which contradicts $V(t, x) \geqq 0$. Thus, we see that $x(t) \equiv 0$ is asymptotically stable.

COROLLARY. *If the system* (7.1) *is periodic in t or autonomous, and if there exists a Liapunov function which satisfies the conditions in Theorem* 8.5, *the solution* $x(t) \equiv 0$ *is uniform-asymptotically stable.*

The following theorem can be easily proved by the comparison principle [7], [28].

THEOREM 8.6. *Suppose that there exists a Liapunov function* $V(t, x)$ *defined on* $t \in I$, $\| x \| < H$, *which satisfies the following conditions;*

(i) $V(t, 0) \equiv 0$ *and* $a(\| x \|) \leqq V(t, x)$, *where* $a(r) \in CIP$,

(ii) $V'_{(7.1)}(t, x) \leqq \omega(t, V(t, x))$, *where* $\omega(t, r)$ *is a continuous function on* $I \times R^1$ *and* $\omega(t, 0) \equiv 0$.

Then, the stability or the asymptotic stability of the zero solution of

$$\frac{dr}{dt} = \omega(t, r) \tag{8.4}$$

implies the same type of stability for the solution $x(t) \equiv 0$ *of* (7.1). *Moreover, in addition, if* $V(t, x) \to 0$ *as* $\| x \| \to 0$ *uniformly on* I, *the uniform-stability or the uniform-asymptotic stability of the zero solution of* (8.4) *implies the same type of stability for the zero solution of* (7.1).

By applying Theorem 4.2, we obtain a similar theorem which is concerned with instability properties, see [6].

The comparison principle has other important applications to questions of conditional stability and the boundedness of some but not necessarily all solutions. This is clearly brought out by the inequality (4.1).

§ 9. Definitions of boundedness.

Consider the system (7.1), and suppose that $F(t, x)$ is defined and continuous on $I \times R^n$. First of all, the definitions of boundedness of solutions will be given (cf. [5], [37], [74], [137]).

DEFINITION 9.1. A solution $x(t; x_0, t_0)$ of (7.1) is bounded, if there exists a $\beta > 0$ such that $\| x(t; x_0, t_0) \| < \beta$ for all $t \geqq t_0$, where β may depend on each solution.

DEFINITION 9.2. The solutions of (7.1) are equi-bounded, if for any $\alpha > 0$ and $t_0 \in I$, there exists a $\beta(t_0, \alpha) > 0$ such that if $x_0 \in S_\alpha$, $\| x(t; x_0, t_0) \| < \beta(t_0, \alpha)$ for all $t \geqq t_0$.

DEFINITION 9.3. The solutions of (7.1) are uniform-bounded, if the β in Definition 9.2 is independent of t_0.

DEFINITION 9.4. The solutions of (7.1) are ultimately bounded for bound B, if there exist a $B > 0$ and a $T > 0$ such that for every solution $x(t; x_0, t_0)$ of (7.1), $\| x(t; x_0, t_0) \| < B$ for all $t \geqq t_0 + T$, where B is independent of the particular solution while T may depend on each solution.

DEFINITION 9.5. The solutions of (7.1) are equiultimately bounded for bound B, if there exists a $B>0$ and if corresponding to any $\alpha>0$ and $t_0\in I$, there exists a $T(t_0, \alpha)>0$ such that $x_0\in S_\alpha$ implies that $\| x(t; x_0, t_0)\| < B$ for all $t\geqq t_0+T(t_0, \alpha)$.

DEFINITION 9.6. The solutions of (7.1) are uniform-ultimately bounded for bound B, if the T in Definition 9.5 is independent of t_0.

These concepts are actually different concepts, and we can find several examples showing this in [137].

It is evident that a linear transformation of coordinates does not affect the boundedness properties as well as the stability properties. However, a general transformation of coordinates will affect those properties.

It also is evident that if the solutions of (7.1) are equiultimately bounded, they are equi-bounded.

Now we consider the linear system (7.3) and the periodic system (7.4) in which $F(t, x)$ is continuous on $I\times R^n$. For the linear system (7.3), the following properties can be easily demonstrated (cf. [137]).

THEOREM 9.1. *If all solutions of* (7.3) *are bounded, then they are equi-bounded. Moreover, if the solutions of* (7.3) *are ultimately bounded, then they are equiultimately bounded.*

For the periodic system (7.4), we have the following properties.

THEOREM 9.2. *If the solutions of* (7.4) *are equi-bounded, then they are uniform-bounded.*

PROOF. For a given $\alpha>0$, consider solutions starting from (t_0, x_0) such that $0\leqq t_0<\omega$, $x_0\in S_\alpha$. By Theorem 3.6, there exists a $\beta(\alpha)>0$ by which the solutions considered are bounded on $[t_0, \omega]$, because every solution is continuable to $t=\omega$. Since the solutions are equi-bounded, there exists a $\gamma(\beta)>0$ such that if $x_0\in S_\beta$, $\| x(t; x_0, t_0)\| <\gamma$ for all $t\geqq\omega$, which implies that if $0\leqq t_0<\omega$ and $x_0\in S_\alpha$, $\| x(t; x_0, t_0)\| <\gamma$ for all $t\geqq t_0$. From the periodicity of

$F(t, x)$, it follows that if $t_0 \in I$ and $x_0 \in S_\alpha$, $\|x(t; x_0, t_0)\| < \gamma$ for all $t \geqq t_0$. This proves the uniform-boundedness.

THEOREM 9.3. *If the solutions of* (7.4) *are equiultimately bounded, then they are uniform-ultimately bounded.*

The proof can be given by the same idea as in the proof of Theorem 9.2.

§ 10. Theorems on boundedness.

In this section, we shall apply Liapunov's second method to show boundedness of solutions of the system (7.1), where $F(t, x)$ in (7.1) is continuous on $I \times R^n$ (cf. [5], [37], [74], [118], [123], [137], [138]).

THEOREM 10.1. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the following conditions;*

(i) $a(\|x\|) \leqq V(t, x)$, *where* $a(r) \in CI$ *and* $a(r) \to \infty$ *as* $r \to \infty$,

(ii) $V'_{(7.1)}(t, x) \leqq 0$.

Then, the solutions of (7.1) *are equi-bounded.*

PROOF. Let $x(t; x_0, t_0)$ be a solution of (7.1) such that $t_0 \in I$ and $x_0 \in S_\alpha$. Since $V(t, x)$ is continuous, there exists a $K(t_0, \alpha) > 0$ such that if $x_0 \in S_\alpha$, $V(t_0, x_0) \leqq K(t_0, \alpha)$. By (i), we can choose a $\beta(t_0, \alpha) > 0$ so large that $a(\beta) > K(t_0, \alpha)$. Suppose that $\|x(t_1; x_0, t_0)\| = \beta$ at some t_1, $t_1 > t_0$. By (ii), $V(t_1, x(t_1; x_0, t_0)) \leqq V(t_0, x(t_0; x_0, t_0))$, which implies $a(\beta) \leqq K(t_0, \alpha)$. This contradicts the choice of $\beta(t_0, \alpha)$. Thus $\|x(t; x_0, t_0)\| < \beta(t_0, \alpha)$ for all $t \geqq t_0$ by Theorem 3.2. This shows the equi-boundedness of solutions of (7.1).

THEOREM 10.2. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $0 \leqq t < \infty$, $\|x\| \geqq R$, where R may be large, which satisfies the following conditions;*

(i) $a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, *where* $a(r) \in CI$, $a(r) \to \infty$ *as* $r \to \infty$ *and* $b(r) \in CI$,

(ii) $V'_{(7.1)}(t, x) \leqq 0$.

Then, the solutions of (7.1) *are uniform-bounded.*

By the same idea used in the proof of Theorem 3.4, this theorem can be proved by choosing $\beta(\alpha)$ such that $b(\alpha) < a(\beta)$.

EXAMPLE 10.1. [3]. Consider the equation

$$\ddot{x} + \phi(x, \dot{x})\dot{x} + h(x) = e(t), \tag{10.1}$$

where we assume that the following conditions are satisfied: $\phi(x, y)$, $h(x)$ are continuous for all values of their variables, $e(t)$ is continuous on I and $\int_0^\infty |e(t)|dt < \infty$. $\phi(x, y) \geqq 0$ for all x, y, $H(x) = \int_0^x h(u)du > 0$ for all $x \neq 0$ and $H(x) \to \infty$ as $|x| \to \infty$. Then, every solution $x(t)$ of (10.1) satisfies $|x(t)| < c$, $|\dot{x}(t)| < c$, where c may depend on the solution.

In this case, consider the system

$$\dot{x} = y, \qquad \dot{y} = -\phi(x, y)y - h(x) + e(t) \tag{10.2}$$

and set

$$V(t, x, y) = \sqrt{y^2 + 2H(x)} - \int_0^t |e(s)|ds$$

for $x^2 + y^2 \geqq R^2$, and note that $V(t, x, y)$ satisfies the conditions in Theorem 10.2. In fact,

$$V'_{(10.2)}(t, x, y) = \frac{1}{\sqrt{y^2 + 2H(x)}} \{h(x)y + y(-\phi(x, y)y - h(x) + e(t))\} - |e(t)|$$
$$\leqq 0 .$$

Therefore, the solutions of (10.2) are uniform-bounded, and thus we have $|x(t)| < c$, $|\dot{x}(t)| < c$.

EXAMPLE 10.2. In Example 3.1, if we assume that $\int_0^\infty \lambda(t)dt < \infty$, the solutions are uniform-bounded. In fact, the Liapunov function $V(t, x) = -\int_0^t \lambda(s)ds + \int_R^r \frac{du}{\varphi(u)}$, $r = \|x\|$, satisfies the conditions in Theorem 10.2.

In some cases, the following theorem on boundedness is more convenient to apply. We consider a system

$$(10.3)\qquad \begin{cases} \dfrac{dx}{dt}=F(t,x,y) \\[2ex] \dfrac{dy}{dt}=G(t,x,y), \end{cases}$$

where x, y are n and m-vectors, respectively, and $F(t,x,y)$, $G(t,x,y)$ are continuous on $I\times R^n\times R^m$.

THEOREM 10.3. *Suppose that there exists a Liapunov function $V(t,x,y)$ defined on $t\in I$, $\|x\|^2+\|y\|^2\geqq R^2$, where R may be large, which satisfies the following conditions;*

(i) *$V(t,x,y)$ tends to infinity uniformly in (t,x) as $\|y\|\to\infty$,*

(ii) *$V(t,x,y)\leqq b(\|x\|, \|y\|)$, where $b(r,s)$ is a continuous function,*

(iii) *$V'_{(10.3)}(t,x,y)\leqq 0$.*

Moreover, suppose that corresponding to each $K>0$, there exists a Liapunov function $W(t,x,y)$ defined on $t\in I$, $\|x\|\geqq R_1(K)$, $\|y\|\leqq K$ (R_1 may be large), which satisfies the following conditions;

(iv) *$W(t,x,y)$ tends to infinity uniformly in (t,y) as $\|x\|\to\infty$,*

(v) *$W(t,x,y)\leqq c(\|x\|)$, where $c(r)$ is a continuous function,*

(vi) *$W'_{(10.3)}(t,x,y)\leqq 0$.*

Then, the solutions of (10.3) *are uniform-bounded.*

PROOF. Let $\{x(t;x_0,y_0,t_0), y(t;x_0,y_0,t_0)\}$ be a solution of (10.3) through (t_0,x_0,y_0) such that $\|x_0\|^2+\|y_0\|^2\leqq\alpha^2$, $\alpha>R$. Choose a $\beta(\alpha)>0$ so large that $\sup\limits_{\|x\|^2+\|y\|^2=\alpha^2} V(t,x,y)<\inf\limits_{\|y\|=\beta} V(t,x,y)$. This is possible by (i), (ii). Then, it can be seen that if $\|x_0\|^2+\|y_0\|^2\leqq\alpha^2$, $\|y(t;x_0,y_0,t_0)\|<\beta(\alpha)$ as long as the solution exists. Next, consider $W(t,x,y)$ defined on $0\leqq t<\infty$, $\|x\|\geqq R_1(\beta)$, $\|y\|\leqq\beta(\alpha)$. Let α^* be $\max(\alpha, R_1(\beta))$, where α^* can be assumed to depend only on α. Choose $\gamma(\alpha)$ so large that

$$\sup\{W(t,x,y);\ t\in I,\ \|x\|=\alpha^*,\ \|y\|\leqq\beta\}$$
$$<\inf\{W(t,x,y);\ t\in I,\ \|x\|=\gamma,\ \|y\|\leqq\beta\}\,.$$

Then, by (vi), it can be shown that $\|x(t;x_0,y_0,t_0)\|<\gamma(\alpha)$, as long as the solution exists. Thus, we obtain the uniform-boundedness of solutions of (10.3).

EXAMPLE 10.3. [101]. In the equation

$$\ddot{x}+f(x)\dot{x}+g(x)=p(t), \tag{10.4}$$

suppose that $f(x)$, $g(x)$, $p(t)$ are continuous, $F(x)=\int_0^x f(u)du \to \pm\infty$ as $x\to\pm\infty$, respectively, $g(x)\ \mathrm{sgn}\ x \geqq 0$ for $|x|>q$ and $P(t) =\int_0^t p(s)ds$ is bounded. Then, every solution of (10.4) is bounded with its derivative.

To see this, consider a system equivalent to (10.4)

$$\dot{x}=y-F(x)+P(t), \qquad \dot{y}=-g(x), \tag{10.5}$$

and choosing positive constants a, b suitably, define a Liapunov function $V(t, x, y)$ in the following way; Setting $G(x)=\int_0^x g(u)du$,

$$V(t,x,y)=\begin{cases} G(x)+\dfrac{y^2}{2} & (x\geqq a,\ |y|<\infty) \\[2ex] G(x)+\dfrac{y^2}{2}-x+a & (|x|\leqq a,\ y\geqq b) \\[2ex] G(x)+\dfrac{y^2}{2}+2a & (x\leqq -a,\ y\geqq b) \\[2ex] G(x)+\dfrac{y^2}{2}+\dfrac{2a}{b}y & (x\leqq -a,\ |y|\leqq b) \\[2ex] G(x)+\dfrac{y^2}{2}-2a & (x\leqq -a,\ y\leqq -b) \\[2ex] G(x)+\dfrac{y^2}{2}+x-a & (|x|\leqq a,\ y\leqq -b). \end{cases}$$

Then, we can see easily that $V(t, x, y)$ satisfies the conditions in Theorem 10.3. Moreover, choosing $c>0$ suitably, $W(t, x, y)=|x|$ defined for $|x|\geqq c$ satisfies the conditions in Theorem 10.3. Therefore, $x(t)$ and $y(t)$ of (10.5) are bounded and consequently $\dot{x}(t)$ also is bounded, because $y(t)$, $F(x(t))$ and $P(t)$ are bounded.

By the same argument used in the proof of Theorem 8.3, we can prove the following theorem which is concerned with uniform-ultimate boundedness of solutions of the system (7.1).

THEOREM 10.4. *Under the assumption in Theorem* 10.2, *if* $V'_{(7.1)}(t, x) \leqq -c(\|x\|)$, *where* $c(r)$ *is positive and continuous, then the solutions of* (7.1) *are uniform-ultimately bounded.*

COROLLARY. *Under the assumption in Theorem* 10.2, *if* $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c>0$ *is a constant, then the solutions of* (7.1) *are uniform-ultimately bounded.*

EXAMPLE 10.4. [22], [120]. In the equation

$$\ddot{x}+kf(x)\dot{x}+g(x)=kp(t), \qquad k>0, \tag{10.6}$$

the following assumptions will be made: $f(x)$ is continuous and $F(x)=\int_0^x f(u)du \to \pm\infty$ as $x\to\pm\infty$, respectively. $g(x)$ is continuous, $xg(x)>0$ for $|x|\geqq q>0$ and $G(x)=\int_0^x g(u)du\to\infty$ as $|x|\to\infty$. Moreover, $p(t)$ is continuous and $P(t)=\int_0^t p(s)ds$ is bounded.

Considering an equivalent system

$$\dot{x}=y-kF(x)+kP(t), \qquad \dot{y}=-g(x) \tag{10.7}$$

and a Liapunov function

$$V(t, x, y)=\begin{cases} G(x)+\dfrac{y^2}{2} & (x\geqq a,\ |y|<\infty) \\[2ex] G(x)+\dfrac{y^2}{2}-x+a & (|x|\leqq a,\ y\geqq b) \\[2ex] G(x)+\dfrac{y^2}{2}+2a & (x\leqq -a,\ y\geqq b) \\[2ex] G(x)+\dfrac{y^2}{2}+\dfrac{2a}{b}y & (x\leqq -a,\ |y|\leqq b) \\[2ex] G(x)+\dfrac{y^2}{2}-2a & (x\leqq -a,\ y\leqq -b) \\[2ex] G(x)+\dfrac{y^2}{2}+x-a & (|x|\leqq a,\ y\leqq -b). \end{cases}$$

where a, b are suitable large constants, we can apply Theorem 10.4 and see that the solutions of (10.7) are uniform-ultimately bounded. Thus, we can find $B>0$ independent of the solutions, for which $|x(t)|<B$, $|\dot{x}(t)|<B$ for t large.

THEOREM 10.5. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $I\times R^n$ which satisfies the following conditions;*

(i) $a(\|x\|)\leqq V(t, x)$ for $\|x\|\geqq B$, where $a(r)\in CI$, $a(r)>0$ for $r\geqq B$ and $a(r)\to\infty$ as $r\to\infty$,

(ii) $V'_{(7.1)}(t, x)\leqq -cV(t, x)$, where $c>0$ is a constant.

Then, the solutions of (7.1) are equiultimately bounded for bound B.

PROOF. Since $V(t, x)$ is continuous, there exists a $K(t_0, \alpha)>0$ such that if $x_0\in S_\alpha$, $V(t_0, x_0)\leqq K(t_0, \alpha)$. Let $x(t; x_0, t_0)$ be a solution of (7.1) such that $x_0\in S_\alpha$. It is bounded for all $t\geqq t_0$. Suppose that for $t>t_0+\dfrac{1}{c}\log\dfrac{K(t_0, \alpha)}{a(B)}$, $\|x(t; x_0, t_0)\|\geqq B$. From (ii), it follows that $V(t, x(t; x_0, t_0))\leqq V(t_0, x_0)e^{-c(t-t_0)}$, and hence,

$$a(B)<K(t_0, \alpha)e^{-\log\frac{K(t_0,\alpha)}{a(B)}}=a(B).$$

This is a contradiction. Therefore, if $t>t_0+\dfrac{1}{c}\log\dfrac{K(t_0, \alpha)}{a(B)}$, we have $\|x(t; x_0, t_0)\|<B$. Thus, the solutions of (7.1) are equiultimately bounded for bound B.

There are many interesting results concerning the boundedness of solutions of differential equations of second order or of higher order, see for example [32]- [34], [114], [115], [117], etc.. Many excellent references can be found in [118], [123].

§ 11. Asymptotic stability in the large.

In the system (7.1), we assume that $F(t, x)$ is continuous on $I\times R^n$ and that $F(t, 0)\equiv 0$.

DEFINITION 11.1. The zero solution of (7.1) is asymptotically stable in the large, if it is stable and if every solution of (7.1) tends to zero as $t \to \infty$ [12].

DEFINITION 11.2. The zero solution of (7.1) is quasi-equi-asymptotically stable in the large, if for any $\alpha > 0$, any $\varepsilon > 0$ and $t_0 \in I$, there exists a $T(t_0, \varepsilon, \alpha) > 0$ such that if $x_0 \in S_\alpha$, $\| x(t; x_0, t_0) \| < \varepsilon$ for all $t \geqq t_0 + T(t_0, \varepsilon, \alpha)$.

DEFINITION 11.3. The zero solution of (7.1) is equiasymptotically stable in the large, if it is stable and is quasi-equiasymptotically stable in the large.

DEFINITION 11.4. The zero solution of (7.1) is quasi-uniform-asymptotically stable in the large, if the T in Definition 11.2 is independent of t_0.

DEFINITION 11.5. The zero solution of (7.1) is uniform-asymptotically stable in the large, if it is uniform-stable and is quasi-uniform-asymptotically stable in the large and if the solutions of (7.1) are uniform-bounded [13].

By Theorem 5.3, it can be proved that if $x(t) \equiv 0$ is the unique solution of (7.1) through (0, 0), Definition 11.2 implies Definition 11.3 (cf. [136]).

DEFINITION 11.6. The zero solution of (7.1) is exponential-asymptotically stable in the large, if there exists a $c > 0$ and for any $\alpha > 0$, there exists a $K(\alpha) > 0$ such that if $x_0 \in S_\alpha$,

$$\| x(t; x_0, t_0) \| \leqq K(\alpha) e^{-c(t-t_0)} \| x_0 \| \tag{11.1}$$

for all $t \geqq t_0$.

For the linear system (7.3), the following properties can be easily seen.

THEOREM 11.1. *If the zero solution of* (7.3) *is asymptotically stable, it is asymptotically stable in the large and it is also equi-asymptotically stable in the large. Moreover, if the zero solution* *of* (7.3) *is uniform-asymptotically stable, it is exponential-asymptotically stable in the large. In this case, we can find a* $K>0$ *independent of* α *in* (11.1).

A more interesting property is the following. Roughly speaking, stability of $x(t)\equiv 0$ of (7.3) and boundedness of solutions of (7.3) are equivalent.

THEOREM 11.2. *For the system* (7.3),

(a) *stability and boundedness (consequently equi-boundedness) are equivalent,*

(b) *uniform-stability and uniform-boundedness are equivalent,*

(c) *asymptotic stability and ultimate-boundedness are equivalent, and consequently equiasymptotic stability and equiultimate boundedness are equivalent,*

(d) *quasi-uniform-asymptotic stability in the large and uniform-ultimate boundedness are equivalent.*

PROOF. For the proofs of (a) and (b), refer to [77]. (c) Let $X(t)$ be a non-singular solution of the matrix equation associated with (7.3). Suppose that the solutions are equiultimately bounded. Then $x(t)\equiv 0$ is stable, because all solutions are bounded. Moreover, for some constant $B>0$, there exists a $T(t_0, \alpha)>0$ such that if $x_0\in S_\alpha$, $\|X(t)X^{-1}(t_0)x_0\|<B$ for all $t\geqq t_0+T(t_0, \alpha)$, and hence $\|X(t)X^{-1}(t_0)\|<\dfrac{n\sqrt{n}B}{\alpha}$ for all $t\geqq t_0+T(t_0, \alpha)$. Therefore, if for any $\varepsilon>0$, we choose $\alpha(\varepsilon)$ so that $\dfrac{n\sqrt{n}B}{\alpha}<\varepsilon$, $\|X(t)X^{-1}(t_0)\|<\varepsilon$ for all $t\geqq t_0+T(t_0, \alpha(\varepsilon))$. This shows that the zero solution is equiasymptotically stable. It is clear that equiasymptotic stability implies equiultimate boundedness. (d) can be proved in the same way as (c).

For a periodic system, the following property can be easily seen.

THEOREM 11.3. *If the system* (7.1) *is a periodic system in* t *and if the zero solution of* (7.1) *is asymptotically stable in the large,* *then it is uniform-asymptotically stable in the large.*

THEOREM 11.4. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the following conditions;*

(i) $V(t, 0) \equiv 0$,

(ii) $a(\| x \|) \leqq V(t, x)$, *where* $a(r) \in CIP$ *and* $a(r) \to \infty$ *as* $r \to \infty$,

(iii) $V'_{(7.1)}(t, x) \leqq -cV(t, x)$ *where* $c > 0$ *is a constant.*

Then, the solution $x(t) \equiv 0$ *of* (7.1) *is equiasymptotically stable in the large.*

PROOF. By Theorem 8.1, the solution $x(t) \equiv 0$ is stable. Moreover, by Theorem 10.1, the solutions of (7.1) are equi-bounded and hence, every solution exists in the future. Let $x(t; x_0, t_0)$ be a solution such that $\| x_0 \| \leqq \alpha$. Applying Theorem 4.1, by (iii)

$$V(t, x(t; x_0, t_0)) \leqq V(t_0, x_0) e^{-c(t-t_0)}. \tag{11.2}$$

Let $M(t_0, \alpha) = \max_{\| x_0 \| \leqq \alpha} V(t_0, x_0)$, and let $T(t_0, \varepsilon, \alpha)$ be such that $T(t_0, \varepsilon, \alpha) = \dfrac{1}{c} \log \dfrac{M(t_0, \alpha)}{a(\varepsilon)}$. Then, from (11.2), it follows that for $t > t_0 + T(t_0, \varepsilon, \alpha)$

$$V(t, x(t; x_0, t_0)) < M(t_0, \alpha) \frac{a(\varepsilon)}{M(t_0, \alpha)} = a(\varepsilon).$$

Since $a(r)$ is increasing and $a(\| x \|) \leqq V(t, x)$, we have $\| x(t; x_0, t_0) \| < \varepsilon$ for $t > t_0 + T(t_0, \varepsilon, \alpha)$, which proves quasi-equiasymptotic stability in the large of $x(t) \equiv 0$. This completes the proof.

THEOREM 11.5. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the following conditions;*

(i) $a(\| x \|) \leqq V(t, x) \leqq b(\| x \|)$, *where* $a(r) \in CIP$, $a(r) \to \infty$ *as* $r \to \infty$ *and* $b(r) \in CIP$,

(ii) $V'_{(7.1)}(t, x) \leqq -c(\| x \|)$, *where* $c(r)$ *is continuous and positive definite.*

Then, the solution $x(t) \equiv 0$ *of* (7.1) *is uniform-asymptotically stable in the large.*

This theorem can be proved by the same arguments used in the proof of Theorem 8.3. The uniform-boundedness of solutions of (7.1) follows immediately from Theorem 10.2 (cf. [13]).

COROLLARY. *Under the assumption in Theorem* 11.5, *if* $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c>0$ *is a constant, the zero solution of* (7.1) *is uniform-asymptotically stable in the large.*

Let us consider Liénard's equation

$$\ddot{x}+f(x)\dot{x}+g(x)=0, \tag{11.3}$$

where $f(x)$, $g(x)$ are continuous on $x \in R^1$. Suppose that $g(x)F(x) > 0$ for $x \neq 0$, where $F(x)=\int_0^x f(u)du$, and that $xg(x)>0$ for $x \neq 0$ and $G(x)=\int_0^x g(u)du \to \infty$ as $|x| \to \infty$. Consider an equivalent system

$$\dot{x}=y-F(x), \qquad \dot{y}=-g(x) \tag{11.4}$$

and a Liapunov function $V(t, x, y)=G(x)+\frac{y^2}{2}$. Clearly $V(t, x, y) \to \infty$ uniformly as $x^2+y^2 \to \infty$ and $V(t, 0, 0) \equiv 0$. Since we have

$$V'_{(11.4)}(t, x, y)=-g(x)F(x) \leqq 0,$$

by Theorem 8.2, the zero solution of (11.4) is uniform-stable, and by Theorem 10.2, the solutions of (11.4) are uniform-bounded. However, $V'_{(11.4)}(t, x, y)$ does not satisfy condition (ii) in Theorem 11.5, and hence, Theorem 11.5 cannot be applied to this case. In fact, the zero solution of (11.4) is uniform-asymptotically stable in the large. For this reason we shall discuss some extension of stability theory in the following chapter.

THEOREM 11.6. *Suppose that there exists a Liapunov function* $V(t, x)$ *defined on* $I \times R^n$ *satisfying the following conditions;*

(i) $\|x\| \leqq V(t, x) \leqq K(\alpha)\|x\|$ *for* $x \in S_\alpha$,

(ii) $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c>0$ *is a constant.*

Then, the solution $x(t) \equiv 0$ *of* (7.1) *is exponential-asymptotically stable in the large, that is,* $\|x(t; x_0, t_0)\| \leqq K(\alpha)e^{-c(t-t_0)}\|x_0\|$ *for* $x_0 \in S_\alpha$.

PROOF. By Theorem 10.2, the solutions of (7.1) are uniform-bounded, and hence, they exist in the future. Consider the function $V(t, x(t; x_0, t_0))$. Then we have $V(t, x(t; x_0, t_0)) \leqq V(t_0, x_0)e^{-c(t-t_0)}$, which implies that $\| x(t; x_0, t_0) \| \leqq K(\alpha)e^{-c(t-t_0)} \| x_0 \|$ if $x_0 \in S_\alpha$, because of (i).

As an application of the comparison principle, we have the following theorem [17].

THEOREM 11.7. *Suppose that there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ such that $a(\| x \|) \leqq V(t, x)$, where $a(r) \in CIP$ and $a(r) \to \infty$ as $r \to \infty$. Moreover, suppose that there exists a continuous scalar function $\omega(t, r)$ defined on $0 \leqq t < \infty$, $r \geqq 0$ such that $V'_{(7.1)}(t, x) \leqq \omega(t, V(t, x))$.*

Then, if all solutions of

$$\frac{dr}{dt} = \omega(t, r) \tag{11.5}$$

tend to zero as $t \to \infty$, all solutions of (7.1) *tend to zero.*

PROOF. Let $x(t; x_0, t_0)$ be a solution of (7.1) and let $r(t)$ be the maximal solution of (11.5) such that $r(t_0) = V(t_0, x_0)$. Since $r(t) \to 0$ as $t \to \infty$, $V(t, x(t; x_0, t_0)) \to 0$ as $t \to \infty$, which implies that $x(t; x_0, t_0) \to 0$ as $t \to \infty$.

REMARK. For Theorem 11.7, the assumption that $F(t, 0) \equiv 0$ is not necessary.

Now we consider the case where $F(t, x)$ in the system (7.1) is almost periodic in t. Suppose that $F(t, x)$ in (7.1) is defined and continuous on $-\infty < t < \infty$, $x \in R^n$ and that $F(t, 0) \equiv 0$. When $F(t, x)$ is almost periodic in t uniformly with respect to $x \in S$ for any compact set in R^n, we shall say that $F(t, x)$ is almost periodic in t uniformly with respect to $x \in R^n$.

DEFINITION 11.7. The zero solution of (7.1) is perfectly uniform-asymptotically stable in the large, if the δ in Definition 7.1, the β in Definition 9.2 and the T in Definition 11.2 are independent of t_0

for all $t_0 \in (-\infty, \infty)$.

Other perfect properties can be defined in the similar way (cf. [149]).

THEOREM 11.8. *Suppose that $F(t, x)$ is almost periodic in t uniformly with respect to $x \in R^n$ and that there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the following conditions;*

(i) $a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, *where* $a(r) \in CIP$, $a(r) \to \infty$ *as* $r \to \infty$ *and* $b(r) \in CIP$,

(ii) *for any* $\alpha > 0$, *if* $x \in S_\alpha$ *and* $x' \in S_\alpha$, $|V(t, x) - V(t, x')| \leqq h(\alpha)\|x - x'\|$,

(iii) $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c > 0$ *is a constant.*

Then, the zero solution of (7.1) *is perfectly uniform-asymptotically stable in the large.*

PROOF. For a given $\varepsilon > 0$, $\varepsilon < 1$, choose a $\delta(\varepsilon) > 0$ so that $a(\varepsilon) > 2b(\delta)$. Let $x(t; x_0, t_0)$ be a solution of (7.1) such that $-\infty < t_0 < \infty$, $\|x_0\| < \delta(\varepsilon)$, and suppose that at some t, $\|x(t; x_0, t_0)\| = \varepsilon$. Then, there exists a t_1, $t_0 < t_1$, such that $\|x(t_1; x_0, t_0)\| = \varepsilon$ and that for $t_0 < t < t_1$, $\|x(t; x_0, t_0)\| < \varepsilon$. Clearly, there is a compact set S for which $x(t; x_0, t_0) \in S$ for $t \in [t_0, t_1]$. Let K denote $h(\alpha)$ in (ii) for an α such that the set $\|x\| < \alpha$ contains S, and let τ be an $\dfrac{a(\delta)c}{2K}$-translation number of $F(t, x)$ for $x \in S$ such that $t_0 + \tau \geqq 0$, that is,

$$\|F(s+\tau, x) - F(s, x)\| \leqq \frac{a(\delta)c}{2K} \quad \text{for } s \in (-\infty, \infty), \quad x \in S. \tag{11.6}$$

Consider the function $V(s+\tau, x(s))$ for $s \in [t_0, t_1]$, where $x(s) = x(s; x_0, t_0)$. We have

$$\begin{aligned} V'(s+\tau, x(s)) &= \overline{\lim_{h \to 0+}} \frac{1}{h}\{V(s+\tau+h, x(s+h)) - V(s+\tau, x(s))\} \\ &\leqq \overline{\lim_{h \to 0+}} \frac{1}{h}\{V(s+\tau+h, x^*(s+\tau+h)) - V(s+\tau, x(s))\} \\ &\quad + \overline{\lim_{h \to 0+}} \frac{1}{h}\{V(s+\tau+h, x(s+h)) - V(s+\tau+h, x^*(s+\tau+h))\}, \end{aligned}$$

where $x^*(t)$ is the solution of (7.1) such that $x^*(s+\tau)=x(s)$. By condition (iii),

$$\overline{\lim_{h\to 0+}} \frac{1}{h}\{V(s+\tau+h, x^*(s+\tau+h))-V(s+\tau, x(s))\} \leqq -cV(s+\tau, x(s)),$$

and by (ii),

$$\begin{aligned} &\overline{\lim_{h\to 0+}} \frac{1}{h}\{V(s+\tau+h, x(s+h))-V(s+\tau+h, x^*(s+\tau+h))\} \\ &\leqq \overline{\lim_{h\to 0+}} \frac{1}{h} K\| x(s+h)-x^*(s+\tau+h)\| \\ &\leqq K\| F(s, x(s))-F(s+\tau, x(s))\| \\ &\leqq \frac{a(\delta)c}{2}, \end{aligned}$$

because of (11.6). Thus,

$$V'(s+\tau, x(s)) \leqq -cV(s+\tau, x(s))+\frac{a(\delta)c}{2}. \tag{11.7}$$

Applying Theorem 4.1, we have

$$\begin{aligned} V(t_1+\tau, x(t_1)) &\leqq e^{-c(t-t_0)}V(t_0+\tau, x(t_0))+\frac{a(\delta)}{2} \\ &\leqq b(\delta)+\frac{a(\delta)}{2} \leqq 2b(\delta). \end{aligned}$$

Since $V(t_1+\tau, x(t_1)) \geqq a(\varepsilon)$, there arises a contradiction, because $a(\varepsilon)>2b(\delta)$. Thus, for any $\varepsilon>0$ there exists a $\delta(\varepsilon)>0$ such that if $t_0\in(-\infty, \infty)$ and $\| x_0\| <\delta$, $\| x(t; x_0, t_0)\| <\varepsilon$ for all $t\geqq t_0$, which shows that the zero solution is perfectly uniform-stable.

For a given $\alpha>0$, by choosing $\beta(\alpha)>0$ so that $2b(\alpha)<a(\beta)$ and applying the same argument as above, it can be verified that if $t_0\in(-\infty, \infty)$ and $x_0\in S_\alpha$, $\| x(t; x_0, t_0)\| <\beta(\alpha)$ for all $t\geqq t_0$.

Let $x(t; x_0, t_0)$ be a solution such that $t_0\in(-\infty, \infty)$, $x_0\in S_\alpha$. Then there exists a $\beta(\alpha)$ for which $\| x(t; x_0, t_0)\| <\beta(\alpha)$ for all $t\geqq t_0$. Let S be a compact set in R^n such that $\| x\| \leqq\beta(\alpha)$. As was seen above, for a given $\varepsilon>0$, there exists a $\delta(\varepsilon)>0$ such that $a(\varepsilon)>2b(\delta)$ which proves uniform-stability. Let K be the Lipschitz constant of $V(t, x)$ for $x\in S_\beta$ and let τ be an $\frac{a(\delta)c}{2K}$-translation number of

$F(t, x)$ for $x \in S$ such that $t_0+\tau \geqq 0$. By the same argument as in the first part of this proof, we have (11.7) for all $s \geqq t_0$. Suppose that $\|x(t; x_0, t_0)\| \geqq \delta(\varepsilon)$ for all $t \geqq t_0$. Then, from (11.7), if $t > t_0+T$, $T = \frac{1}{c} \log \frac{2b(\alpha)}{a(\delta)}$,

$$V(t+\tau, x(t)) \leqq e^{-c(t-t_0)} V(t_0+\tau, x(t_0)) + \frac{a(\delta)}{2}$$
$$< \frac{a(\delta)}{2b(\alpha)} b(\alpha) + \frac{a(\delta)}{2} = a(\delta),$$

which contradicts $a(\delta) \leqq V(t+\tau, x(t))$. Therefore, at some t_1 such that $t_0 \leqq t_1 \leqq t_0+T$, we have $\|x(t_1; x_0, t_0)\| < \delta(\varepsilon)$, which implies that if $t \geqq t_0+T$, $\|x(t; x_0, t_0)\| < \varepsilon$. Since T depends only on α and ε, the theorem is proved.

In the case where $F(t, x)$ of the system (7.1) is defined and continuous on $-\infty < t < \infty$, $\|x\| \leqq H$, the following theorem can be proved by the same argument used in Theorem 11.8 (cf. [149]).

THEOREM 11.9. *Suppose that $F(t, x)$ of (7.1) is continuous on $-\infty < t < \infty$, $\|x\| \leqq H$, $H > 0$, and that $F(t, x)$ is almost periodic in t uniformly with respect to $x \in S_H$. Moreover, suppose that there exists a Liapunov function $V(t, x)$ defined on $t \in I$, $\|x\| < H$ which satisfies the following conditions;*

(i) $a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, *where* $a(r) \in CIP$ *and* $b(r) \in CIP$,

(ii) $|V(t, x) - V(t, x')| \leqq K\|x - x'\|$, *where* $K > 0$ *is a constant,*

(iii) $V'_{(7.1)}(t, x) \leqq -cV(t, x)$, *where* $c > 0$ *is a constant.*

Then, the zero solution of (7.1) is perfectly uniform-asymptotically stable.

CHAPTER III. EXTENSION OF STABILITY THEORY

Considering positive limiting sets of solutions and the semi-invariant set, we shall discuss the asymptotic behavior of solutions which is an extension of the stability theory.

§ 12. Positive limiting set.

Consider a system of differential equations

$$\frac{dx}{dt} = F(t, x) + G(t, x). \tag{12.1}$$

Let Q be an open set in R^n and suppose that $F(t, x)$, $G(t, x)$ are continuous on $I \times Q$. Moreover, suppose that if $x(t)$ is continuous and bounded on $[t_0, \infty)$, that is, for a compact set $Q^* \subset Q$, $x(t) \subset Q^*$ for all $t \in [t_0, \infty)$, then we have

$$\int_{t_0}^{\infty} \| G(s, x(s)) \| \, ds < \infty. \tag{12.2}$$

Let $x(t; x_0, t_0)$ be a solution of (12.1) through (t_0, x_0) and let Γ^+ be a set in R^n such that

$$\Gamma^+ = \bigcap_{s<\tau} \overline{\bigcup_{s \leqq t} x(t; x_0, t_0)},$$

where the interval $[t_0, \tau)$ is the maximum interval on which the solution $x(t; x_0, t_0)$ is defined. Then, Γ^+ is a closed subset in Q, and if $x(t; x_0, t_0)$ is bounded, Γ^+ is compact and non-empty (cf. [25], [77]).

THEOREM 12.1. *Suppose that $F(t, x)$ and $G(t, x)$ are continuons on $I \times Q$. If Γ^+ is non-empty, then we have $\tau = +\infty$.*

PROOF. Suppose that $\tau < \infty$ and $\omega \in \Gamma^+$. Let B_1, B_2 be two spheres centered at ω, lying in Q and $B_1 \subset B_2$. If $\Gamma^+ \subset B_2$, then $\tau = +\infty$, because $x(t; x_0, t_0)$ does not approach the boundary of Q.

Otherwise, $x(t; x_0, t_0)$ must traverse the set $B_2 - B_1$ an infinite number of times. Since $\|F(t, x)+G(t, x)\|$ is bounded for $t \in [t_0, \tau]$ and $x \in B_2$, there arises a contradiction. Therefore, we have $\tau = +\infty$.

As is well known, Γ^+ is called the positive limiting set of the solution $x(t; x_0, t_0)$. If Γ^+ is non-empty, $x(t) = x(t; x_0, t_0)$ is defined on $[t_0, \infty)$ by Theorem 12.1. In this case, the existence of a point ω in Γ^+ is equivalent to saying that there exists a sequence $\{t_k\}$ such that $x(t_k) \to \omega$, $t_k \to \infty$ as $k \to \infty$.

When for each $\varepsilon > 0$ there is a $T > 0$ with the property that for all $t > T$ the points $x(t)$ are contained in $N(\varepsilon, S)$, where S is a set in R^n, we say that $x(t)$ approaches S as $t \to \infty$ and write $x(t) \to S$.

LEMMA 12.1. *Let $x(t)$ be a solution of* (12.1) *bounded for $t \geqq t_0$ and let Γ^+ be the positive limiting set of $x(t)$. Then, $x(t) \to \Gamma^+$ as $t \to \infty$. Moreover, if $x(t)$ is bounded for $t \geqq t_0$ and if M contains Γ^+, then $x(t) \to M$ as $t \to \infty$.*

LEMMA 12.2. *Let Ω be a closed set in the space Q. Suppose that a solution $x(t)$ approaches Ω as $t \to \infty$. Then, the positive limiting set Γ^+ of $x(t)$ satisfies $\Gamma^+ \subset \Omega$.*

These lemmas can be easily verified.

Relationships between the positive limiting sets of solutions of a perturbed system and solutions of the unperturbed system have been discussed by Markus [91] and Opial [107] concerning a system of form

$$\frac{dx}{dt} = H(x) + G(t, x), \tag{12.3}$$

where $G(t, x)$ is a perturbation term. Markus has discussed the case where $G(t, x)$ tends to zero as $t \to \infty$ uniformly in any compact set in R^n. Opial has discussed the case where $G(t, x)$ is integrable in the sense of (12.2). In the following section, we shall obtain a more general result.

§ 13. Semi-invariant set.

In this section, under the assumption that a solution $x(t)$ is bounded and approaches a closed set Ω, it will be shown that the positive limiting set of $x(t)$ is composed of solutions of some system defined on Ω which is related to the unperturbed system (cf. [144]).

DEFINITION 13.1. For a system defined on a set D

$$\frac{dx}{dt}=H(x), \qquad x\in D, \tag{13.1}$$

and for a subset M of D, M is said to be a semi-invariant set of (13.1), if for each point of M there exists at least one solution of (13.1) which remains in M for all future time.

If we assume the uniqueness of every solution of (13.1) for the initial value problem, the semi-invariant set is an invariant set of (13.1).

Now let Ω be a closed set in the space Q, and suppose that $F(t, x)$ of (12.1) satisfies the following conditions:

(a) *$F(t, x)$ tends to a function $H(x)$ for $x\in\Omega$ as $t\to\infty$ and on any compact set in Ω this convergence is uniform. Consequently, $H(x)$ is a continuous function on Ω.*

(b) *Corresponding to each $\varepsilon>0$ and each $y\in\Omega$, there exists a $\delta(\varepsilon, y)>0$ and a $T(\varepsilon, y)>0$ such that if $\|x-y\|<\delta(\varepsilon, y)$ and $t\geqq T(\varepsilon, y)$, we have $\|F(t, x)-F(t, y)\|<\varepsilon$.*

REMARK. Here, since $F(t, x)$ is defined on I, we can choose $\delta(\varepsilon, y)$ so that $\|F(t, x)-F(t, y)\|<\varepsilon$ for all $t\geqq 0$, if we have condition (b). However, in case $F(t, x)$ is defined on $0<t<\infty$, we require the existence of $T(\varepsilon, y)$ to obtain our results.

The following lemma can be proved in the same manner as in the standard proof of uniform continuity of a continuous function on a compact set.

LEMMA 13.1. *In case $F(t, x)$ satisfies condition (b), for y which* *belongs to a compact set Ω_1 in Ω, we can choose δ and T which are independent of y and depend only on Ω_1.*

Markus discussed relationships between solutions of

$$\frac{dx}{dt}=F(t, x) \tag{13.2}$$

and

$$\frac{dx}{dt}=H(x), \tag{13.3}$$

where $F(t, x)$, $H(x)$ are continuous in (t, x) for $x\in Q$ and for all $t>t_0$. The system (13.2) is said to be asymptotic to the system (13.3) in case for each compact set $K\subset Q$ and for each $\varepsilon>0$, there exists a $T(K, \varepsilon)>t_0$ such that $\| F(t, x)-H(x)\| <\varepsilon$ for all $x\in K$ and all $t>T$.

Any $F(t, x)$ with this property satisfies conditions (a), (b). In fact, if Ω is an arbitrary closed set in Q, it is clear that there is such a function $H(x)$ as in (a), because $F(t, x)$ tends to $H(x)$ uniformly on any compact set in Ω. Next, consider a neighborhood $N(y)$ of $y\in\Omega$. We can assume that $\overline{N(y)}\subset Q$. If $x\in\overline{N(y)}$, corresponding to each $\varepsilon>0$ there exists a $T(\varepsilon, y)$ such that for $t\geqq T(\varepsilon, y)$,

$$\begin{gathered} \| F(t, x)-H(x)\| <\frac{\varepsilon}{3}, \\ \| F(t, y)-H(y)\| <\frac{\varepsilon}{3}, \end{gathered} \tag{13.4}$$

because of the uniform convergence. Since $H(x)$ is continuous on $\overline{N(y)}$, there is a $\delta(\varepsilon, y)>0$ such that if $\| x-y\| <\delta(\varepsilon, y)$, then $\| H(x)-H(y)\| <\frac{\varepsilon}{3}$. From this and (13.4), it follows that if $t\geqq T(\varepsilon, y)$ and $\| x-y\| <\delta(\varepsilon, y)$, then $\| F(t, x)-F(t, y)\| <\varepsilon$, which shows that $F(t, x)$ satisfies (*b*).

Thus, Markus' condition is the special case where $Q=\Omega$. In particular, the system (12.3) discussed by Opial is the special case of the above, where $\Omega=Q=R^n$.

THEOREM 13.1. *Suppose that a solution $x(t; x_0, t_0)$ of* (12.1) *is bounded and approaches a closed set Ω in the space Q. If $F(t, x)$ satisfies conditions* (a), (b), *then the positive limiting set Γ^+ of $x(t; x_0, t_0)$ is a semi-invariant set contained in Ω of the equation*

$$\frac{dx}{dt} = H(x), \qquad x \in \Omega. \tag{13.5}$$

PROOF. Since $x(t) = x(t; x_0, t_0)$ is bounded in Q, there exists a compact set Q^* in Q such that for all $t \geqq t_0$, $x(t) \in Q^*$. By Lemma 12.2, $\Gamma^+ \subset \Omega \cap Q^* = \Omega_1$. Since Ω_1 is a compact set in R^n, there exists a continuous, bounded function $H^*(x)$ on R^n such that $H^*(x) = H(x)$ on Ω_1. Let ω be a point of Γ^+. Then $\omega \in \Omega_1$ and there exists a sequence $\{t_k\}$ such that

$$x(t_k) \to \omega, \ t_k \to \infty \quad \text{as} \quad k \to \infty. \tag{13.6}$$

Now, consider the systems

$$\frac{dx}{dt} = H^*(x), \tag{13.7}$$

$$\frac{dx}{dt} = H^*(x) + F(t+t_k, x(t+t_k)) - H^*(x(t+t_k)) + G(t+t_k, x(t+t_k)). \tag{13.8}$$

Since $H^*(x)$ is bounded, for any $\lambda > 0$, all solutions of (13.7) are defined on $0 \leqq t \leqq \lambda$. It is clear that $x = x(t+t_k)$ is a solution of (13.8) through $(0, x(t_k))$. As $x(t)$ is bounded, by condition (12.2), $\int_{t_0}^{\infty} \| G(s, x(s)) \| ds < \infty$, and hence, if k is sufficiently large, say $k \geqq k_1$, for a given $\delta > 0$

$$\int_{t_k}^{t_k+\lambda} \| G(s, x(s)) \| ds < \frac{\delta}{2} \quad \text{or} \quad \int_0^{\lambda} \| G(s+t_k, x(s+t_k)) \| ds < \frac{\delta}{2}. \tag{13.9}$$

As Ω_1 is a compact set, for every point $x(t+t_k)$ there is a point $y(t+t_k) \in \Omega_1$ for which $d(x(t+t_k), \Omega_1) = \| x(t+t_k) - y(t+t_k) \|$.

From condition (b) and Lemma 13.1, it follows that corresponding to $\frac{\delta}{6\lambda}$, there are $\delta_1 > 0$ and $T > 0$ such that if $y \in \Omega_1$, $\| x - y \| < \delta_1$ and $t \geqq T$, we have $\| F(t, x) - F(t, y) \| < \frac{\delta}{6\lambda}$. On the other hand, for sufficiently large t, $x(t+t_k)\subset N(\delta_1, \Omega_1)\cap Q^*$, because $x(t)\to\Omega_1$ as $t\to\infty$. Therefore, if k is greater than $k_2>0$, on $0\leqq t\leqq\lambda$

$$\|F(t+t_k, x(t+t_k))-F(t+t_k, y(t+t_k))\|<\frac{\delta}{6\lambda}. \tag{13.10}$$

By condition (a), $F(t, x)\to H(x)$ as $t\to\infty$ uniformly in $x\in\Omega_1$, and hence $\|F(t, x)-H(x)\|<\frac{\delta}{6\lambda}$ for sufficiently large t and for $x\in\Omega_1$. Therefore, if $k\geqq k_3$ for sufficiently large $k_3>0$, on $0\leqq t\leqq\lambda$

$$\|F(t+t_k, y(t+t_k))-H(y(t+t_k))\|<\frac{\delta}{6\lambda}. \tag{13.11}$$

Moreover, since $H^*(x)$ is continuous on Q^*, there exists a $\delta_2>0$ such that if $\|x-y\|<\delta_2$, we have $\|H^*(x)-H^*(y)\|<\frac{\delta}{6\lambda}$. Thus, if $k\geqq k_4$ for sufficiently large $k_4>0$, on $0\leqq t\leqq\lambda$

$$\|H^*(x(t+t_k))-H^*(y(t+t_k))\|<\frac{\delta}{6\lambda}. \tag{13.12}$$

Since $H^*(y(t+t_k))=H(y(t+t_k))$, from (13.10), (13.11) and (13.12), it follows that if $k>\max k_i$ $(i=1, 2, 3, 4)$,

$$\|F(t+t_k, x(t+t_k))-H^*(x(t+t_k))\|<\frac{\delta}{2\lambda} \tag{13.13}$$

on $0\leqq t\leqq\lambda$. From (13.9) and (13.13),

$$\int_0^\lambda \|F(s+t_k, x(s+t_k))-H^*(x(s+t_k))+G(s+t_k, x(s+t_k))\|\,ds<\delta.$$

On the other hand, by (13.6), we have $\|x(t_k)-\omega\|<\delta$ for sufficiently large k. Therefore, applying Theorem 5.1, there exists a solution $\varphi_k(t)$ defined on $0\leqq t\leqq\lambda$ of (13.7) through $(0, \omega)$ such that for a given $\varepsilon>0$, $\|x(t+t_k)-\varphi_k(t)\|<\varepsilon$ for $t\in[0, \lambda]$.

Since $\varphi_k(t)$ is a solution of (13.7) through $(0, \omega)$, we have

$$\varphi_k(t)=\omega+\int_0^t H^*(\varphi_k(s))ds$$

on $0\leqq t\leqq\lambda$. Thus, for a sequence $\{\varepsilon_k\}$ approaching zero as $k\to\infty$, there exist solutions $\varphi_k(t)$ of (13.7) such that for $t\in[0, \lambda]$

$$\text{(13.14)} \qquad \begin{cases} \varphi_k(t) = \omega + \int_0^t H^*(\varphi_k(s))ds \\ \varphi_k(t) \subset N(\varepsilon_k, \Gamma^+). \end{cases}$$

Since $\{\varphi_k(t)\}$ is uniformly bounded and equicontinuous, it has a uniformly convergent subsequence. Let $\varphi(t)$ be its limit function. Then by (13.14)

$$\varphi(t) = \omega + \int_0^t H^*(\varphi(s))ds\,, \qquad \varphi(t) \subset \Gamma^+$$

for $t \in [0, \lambda]$. On the other hand, $\Gamma^+ \subset \Omega_1$, and hence, $H^*(\varphi(t)) = H(\varphi(t))$, which implies that $\varphi(t) = \omega + \int_0^t H(\varphi(s))ds$ on $[0, \lambda]$, that is, $\varphi(t)$ is a solution of the system (13.5) which passes through $(0, \omega)$ and remains in Γ^+. Since λ is arbitrary, we can find a solution of (13.5) defined on I which passes through $(0, \omega)$ and remains in Γ^+. Thus, Γ^+ is a semi-invariant set of (13.5).

REMARK 1. As we can see from the proof above, in the case where $F(t, x)$ and $G(t, x)$ are defined for $t \in (0, \infty)$, $x \in Q$, the conclusion of Theorem 13.1 is also true for a solution $x(t\,;\, x_0, t_0)$ such that $t_0 > 0$, $x_0 \in Q$.

REMARK 2. As we can see from the proof above, if we take k so large that $[t_k - \lambda, t_k]$ is contained in I, the same argument can be applied to showing that there is a solution of (13.5) defined on $-\infty < t \leqq 0$ which passes through $(0, \omega)$ and remains in Γ^+. Therefore, Γ^+ is a semi-invariant set in both directions.

REMARK 3. In Theorem 13.1, if $H(x) \in C_0(x)$, the positive limiting set Γ^+ is an invariant set of (13.5) contained in Ω.

From Theorem 13.1, the following corollaries can be obtained immediately (cf. [91], [107]). We assume that $F(t, x)$ of the system (12.1) satisfies conditions (a), (b) for a closed set in the space Q.

COROLLARY 1. *If for a solution* $x(t)$ *of* (12.1) *approaching* Ω, *we have* $\lim_{t \to \infty} x(t) = x_0$, *then the point* x_0 *is a critical point of* (13.5), *that is,* $H(x_0)=0$.

COROLLARY 2. *If every solution of* (13.5) *is unbounded, a solution of* (12.1) *approaching* Ω *is unbounded.*

We stated Theorem 13.1 in a special form which is convenient for applications, but the proof of Theorem 13.1 is easily modified so as to be applied to a more general case, and we can prove the following theorem (cf. [141]).

THEOREM 13.2. *Suppose that the positive limiting set* Γ^+ *of* $x(t;x_0,t_0)$ *of* (12.1) *is non-empty and* $x(t;x_0,t_0)\to\Omega$ *as* $t\to\infty$, *where* Ω *is a closed set in the space* Q *and that* $F(t,x)$ *satisfies conditions* (a), (b). *Then,* Γ^+ *is the union of solutions of* (13.5).

With this generalization, as a special case, we have the following theorem which includes the results of both Markus and Opial.

THEOREM 13.3. *Suppose that the positive limiting set* Γ^+ *of* $x(t;x_0,t_0)$ *of* (12.1) *is non-empty and that* $F(t,x)$ *satisfies Markus' condition given for the system* (13.2). *Then,* Γ^+ *is the union of solutions of* $\dot{x}=H(x)$ *on* Q.

For Markus' equation, $G(t,x)\equiv 0$ and $\Omega=Q$, and for Opial's equation, $F(t,x)=H(x)$ and $\Omega=Q$.

§ 14. Asymptotic behavior of solutions.

Using the result in Section 13 and a Liapunov function, we shall obtain some results concerning the asymptotic behavior of solutions which can be applied to the equation (11.3) and implies uniform-asymptotic stability in the large [144]. As special cases, we shall obtain some results due to LaSalle [69], [70], [74], and Levin and Nohel [81], [82].

DEFINITION 14.1. A scalar function $W(x)$ defined for $x\in Q$ is said to be positive definite with respect to a set S, if $W(x)=0$ for $x\in S$ and if corresponding to each $\varepsilon>0$ and each compact set Q^* in Q, there exists a positive number $\delta(\varepsilon,Q^*)$ such that

$$W(x) \geqq \delta(\varepsilon, Q^*) \quad \text{for} \quad x \in Q^* - N(\varepsilon, S).$$

THEOREM 14.1. *Suppose that $F(t, x)$ of the system* (12.1) *is bounded for all t when x belongs to an arbitrary compact set in Q. Moreover, suppose that there exists a non-negative Liapunov function $V(t, x)$ such that*

$$V'_{(12.1)}(t, x) \leqq -W(x), \tag{14.1}$$

where $W(x)$ is positive definite with respect to a closed set Ω in the space Q. Then, every bounded solution of (12.1) *approaches Ω as $t \to \infty$.*

PROOF. Consider a bounded solution $x(t) = x(t; x_0, t_0)$ of (12.1). Then there exists a compact set Q^* in Q such that $x(t) \in Q^*$ for all $t \geqq t_0$. Suppose that $x(t)$ does not approach Ω. Then, for some $\varepsilon > 0$ there exists a sequence $\{t_k\}$ tending to infinity with k such that

$$x(t_k) \notin N(\varepsilon, \Omega), \qquad x(t_k) \in Q^*.$$

By assumption, there exists a $K > 0$ for which $\| F(t, x) \| < K$ for all $t \geqq 0$ and $x \in Q^*$. Here, t_1 can be assumed to be sufficiently large so that, by (12.2),

$$\int_{t_k}^{t_k + \frac{\varepsilon}{4K}} \| G(s, x(s)) \| \, ds < \frac{\varepsilon}{4}$$

on the intervals

$$t_k \leqq t \leqq t_k + \frac{\varepsilon}{4K}. \tag{14.2}$$

Therefore, on the intervals (14.2), we have $\| x(t) - x(t_k) \| < \frac{\varepsilon}{2}$. From $d(x(t), \Omega) \geqq d(x(t_k), \Omega) - \| x(t) - x(t_k) \|$, it follows that $d(x(t), \Omega) > \frac{\varepsilon}{2}$ or $x(t) \notin N\left(\frac{\varepsilon}{2}, \Omega\right)$. We can assume that these intervals are disjoint by taking, if necessary, a subsequence of $\{t_k\}$. By (14.1), there exists a $\delta\left(\frac{\varepsilon}{2}\right) > 0$ such that $V'_{(12.1)}(t, x) \leqq -\delta\left(\frac{\varepsilon}{2}\right)$ on the intervals (14.2), and $V'_{(12.1)}(t, x) \leqq 0$ elsewhere, so that

$$V\left(t_k+\frac{\varepsilon}{4K}, x\left(t_k+\frac{\varepsilon}{4K}\right)\right)-V(t_0, x_0) \leqq -\delta\left(\frac{\varepsilon}{2}\right)\frac{\varepsilon}{4K}k \to -\infty$$

as $k \to \infty$, which is absurd because $V(t, x) \geqq 0$. Thus, it can be seen that $x(t) \to \Omega$ as $t \to \infty$.

REMARK. In case F and G are defined for $t \in (0, \infty)$, $x \in Q$, the theorem is also true for a bounded solution $x(t; x_0, t_0)$ such that $t_0 > 0$, $x_0 \in Q$, if $F(t, x)$ is bounded for all $t \geqq t_0$ when $x \in Q^*$, a compact set.

By Theorem 13.1 and Theorem 14.1, a sufficient condition for asymptotic behavior of solutions of (12.1) will be obtained.

THEOREM 14.2. *Suppose that there exists a non-negative Liapunov function $V(t, x)$ on $I \times Q$ such that $V'_{(12.1)}(t, x) \leqq -W(x)$, where $W(x)$ is positive definite with respect to a closed set Ω in the space Q. Moreover, suppose that $F(t, x)$ of the system* (12.1) *is bounded for all t when x belongs to an arbitrary compact set in Q and that $F(t, x)$ satisfies conditions* (a), (b) *in Section* 13 *with respect to Ω. Then, every bounded solution of* (12.1) *approaches the largest semi-invariant set of the system* (13.5) *contained in Ω as $t \to \infty$. In particular, if all solutions of* (12.1) *are bounded, every solution of* (12.1) *approaches the largest semi-invariant set of* (13.5) *contained in Ω as $t \to \infty$.*

As can be seen from the proof, Theorem 14.2 is verified for a family of solutions which are in some region. Thus, one can determine something about the size of the region of asymptotic stability. LaSalle [69], [70], [74] has established propositions leading to criteria to determine the extent of asymptotic stability. These results follow from our theorem.

Let Q_1 be a set in $I \times Q^*$, where Q^* is a compact set in Q, and let Ω be a closed set in R^n such that $I \times \Omega \subset Q_1$.

THEOREM 14.3. *Suppose that every solution of* (12.1) *starting from Q_1 remains in Q_1 for all future time and that $F(t, x)$ satisfies the same conditions as in Theorem* 14.2. *If there exists a non-*

negative Liapunov function $V(t, x)$ *on* Q_1 *and if we have* (14.1) *on* Q_1, *then every solution of* (12.1) *starting from* Q_1 *approaches* Ω.

As a special case, consider an autonomous system

$$\frac{dx}{dt} = F(x), \tag{14.3}$$

which LaSalle has considered, under the assumptions that $F(x)$ is continuous on an open set Q in R^n and every solution is unique. Then, LaSalle's results can be obtained from Theorems 14.2 and 14.3.

THEOREM 14.4. *Suppose that* $F(x)$ *of* (14.3) *is continuous on* R^n *and that there exists a Liapunov function* $V(x)$ *with continuous partial derivatives of the first order such that* $V(x) > 0$ *for all* $x \neq 0$ *and* $V'_{(14.3)}(x) \leqq 0$. *Let* E *be the set of all points where* $V'_{(14.3)}(x) = 0$ *and let* M *be the largest invariant set contained in* E. *Then, every bounded solution of* (14.3) *approaches* M *as* $t \to \infty$.

THEOREM 14.5. *Let* Ω *be a compact set with the property that every solution of* (14.3) *starting in* Ω *remains in* Ω *for all future time. Suppose that there exists a Liapunov function* $V(x) \geqq 0$ *with continuous partial derivatives of the first order and that* $V'_{(14.3)}(x) \leqq 0$ *in* Ω. *Let* E *be the set of all points in* Ω *where* $V'_{(14.3)}(x) = 0$ *and let* M *be the largest invariant set in* E. *Then, every solution starting in* Ω *approaches* M *as* $t \to \infty$.

In some applications, the construction of Liapunov function $V(x)$ in Theorem 14.4 will guarantee the existence of a set Ω.

THEOREM 14.6. *Let* Ω *be the closed region defined by* $V(x) \leqq l$ *and suppose that* $V(x)$ *has continuous partial derivatives of the first order in* Ω. *If* Ω *is bounded and* $V'_{(14.3)}(x) \leqq 0$ *in* Ω, *then every solution of* (14.3) *starting in* Ω *approaches* M *as* $t \to \infty$, *where* M *is as defined in Theorem* 14.5.

We can find good simple examples in [70], [74]. Applying Theorem 14.4, we can see that every solution of Liénard's equation (11.3) approaches the origin. Since the system (11.4) is autonomous, the zero solution is uniform-asymptotically stable in the large.

Here, let us consider another example which has been discussed by Levin and Nohel [82].

EXAMPLE 14.1. In the equation

$$\ddot{x}+h(t, x, \dot{x})\dot{x}+f(x)=e(t), \tag{14.4}$$

we assume that

(i) $h(t, x, y)$ is continuous, non-negative in $I\times R^1\times R^1$ and is bounded when x^2+y^2 is bounded, and moreover, $h(t, x, y)$ satisfies $h(t, x, y)\geqq k(x, y)>0$ for $y\neq 0$, where $k(x, y)$ is a continuous function,

(ii) $f(x)$ is continuous in R^1, $xf(x)>0$ for $x\neq 0$ and $F(x) =\int_0^x f(u)du\rightarrow+\infty$ as $|x|\rightarrow\infty$,

(iii) $e(t)$ is continuous on I and $E(t)=\int_0^t |e(s)|ds<\infty$.

Then, every solution $x(t)$ of (14.4) exists on I and $x(t)\rightarrow 0$, $\dot{x}(t)\rightarrow 0$ as $t\rightarrow\infty$.

To prove this, consider a system equivalent to (14.4)

$$\dot{x}=y, \qquad \dot{y}=-h(t, x, y)y-f(x)+e(t) \tag{14.5}$$

and the Liapunov function

$$V(t, x, y)=e^{-2E(t)}\left\{F(x)+\frac{y^2}{2}+1\right\}.$$

Then, we have

$$e^{-2E(\infty)}\left\{F(x)+\frac{y^2}{2}+1\right\}\leqq V(t, x, y)\leqq F(x)+\frac{y^2}{2}+1$$

and

$$V'_{(14.5)}(t, x, y)\leqq -e^{-2E(\infty)}h(t, x, y)y^2, \tag{14.6}$$

where $E(\infty)=\int_0^\infty |e(s)|ds$.

Then, by Theorem 10.2, the solutions of (14.5) are uniform-bounded. By the condition on $h(t, x, y)$ and (14.6), the set Ω in Theorem 14.2 is the set of all points where $y=0$, i.e., the x-axis. The other conditions in Theorem 14.2 can be easily verified. On the set Ω,

$$\dot{x}=0, \quad \dot{y}=-f(x), \tag{14.7}$$

which corresponds to the system (13.5). From the condition on $f(x)$, it follows that the largest semi-invariant set contained in Ω is only the origin. Thus, by Theorem 14.2, $x(t)\to 0$, $y(t)\to 0$ as $t\to\infty$.

Finally, we shall note some extensions of the above results. LaSalle [71] has extended his results so that they apply to a periodic system

$$\frac{dx}{dt}=F(t, x), \tag{14.8}$$

where $F(t+\omega, x)=F(t, x)$ for all t, x.

THEOREM 14.7. *Suppose that every solution of* (14.8) *is unique. Let $V(t, x)$ be a Liapunov function with continuous partial derivatives of the first order, and suppose that $V(t+\omega, x)=V(t, x)\geqq 0$ and $V'_{(14.8)}(t, x)\leqq 0$ for all t, x. Define $E=\{(t_0, x_0);\ V'_{(14.8)}(t_0, x_0)=0\}$ and let M be the union over all $(t_0, x_0)\in E$ of all trajectories $x(t; x_0, t_0)$ such that $(t, x(t; x_0, t_0))\in E$ for all t. Then, all solutions of* (14.8) *bounded in the future approach M as $t\to\infty$.*

THEOREM 14.8. *Suppose that $V(t, x)$ satisfies all conditions of Theorem* 14.7 *for all t and all x in a compact set Ω. The sets E and M are defined as before relative to Ω. If Ω_0 is a set with the property that all solutions of* (14.8) *starting in Ω_0 remain in the future in Ω, then every solution of* (14.8) *starting in Ω_0 approaches M as $t\to\infty$. In particular, if M is an equilibrium state of* (14.8), *then Ω_0 is in the region of asymptotic stability of this equilibrium state.*

For the proofs of these theorems, see [71].

For a system of form

$$\frac{dx}{dt}=A(t, x)+F(t, x)+G(t, x),$$

where $A(t, x)$ is almost periodic in t, $F(t, x)\to 0$ uniformly on any compact set in R^n as $t\to\infty$ and $G(t, x)$ is integrable in the sense of (12.2), Miller [98] has extended some of our results.

CHAPTER IV. EXTREME STABILITY AND STABILITY OF SETS

By using Liapunov functions, extreme stability and stability of a set will be discussed, and some sufficient conditions will be given.

§ 15. Stability and boundedness of systems.

In this section, some stronger types of stability and boundedness will be discussed. Some of these results are applicable in showing the existence and uniqueness of periodic or almost periodic solutions, which will be discussed in Chapter VII.

Consider a system of differential equations

$$\frac{dx}{dt} = F(t, x). \tag{15.1}$$

If for each pair of solutions $x(t)$ and $y(t)$ of (15.1), $x(t)-y(t) \to 0$ as $t \to \infty$, the system (15.1) is said to be extremely stable (cf. [68], [74]). In studying the behavior of a pair of solutions, it is natural to introduce the product system

$$\dot{x} = F(t, x), \qquad \dot{y} = F(t, y). \tag{15.2}$$

The following theorem can be proved by the same argument used in the proof of Theorem 8.3 (cf. [137]).

THEOREM 15.1. *Suppose that* $F(t, x)$ *of* (15.1) *is continuous on* $I \times R^n$ *and that the solutions of* (15.1) *are ultimately bounded for bound* $B > 0$. *Moreover, suppose that there exists a Liapunov function* $V(t, x, y)$ *defined on* $I \times S_B \times S_B$ *which satisfies the following conditions;*

(i) $a(\| x-y \|) \leqq V(t, x, y) \leqq b(\| x-y \|)$, *where* $a(r) \in CIP$ *and* $b(r) \in CIP$ *(for CI and CIP, see Section* 1),

(ii) *for any* $\lambda > 0$, *there exists a* $c(\lambda) > 0$ *such that if* $\| x-y \|$

$\geqq \lambda$, $V'_{(15.2)}(t, x, y) \leqq -c(\lambda)$ *in the interior of* $I \times S_B \times S_B$.

Then, the system (15.1) *is extremely stable.*

As an example, consider the equation (10.6) [114]. Under the previous assumption on (10.6) given before, there exist two numbers $A>0$, $B>0$ such that $|x(t)|<A$, $|\dot{x}(t)|<B$ ultimately, where $x(t)$ is a solution of (10.6). Here, we assume that $f(x)>0$, $\dfrac{dg}{dx}>0$ and $\dfrac{d^2g}{dx^2}$ exists and is bounded for $|x| \leqq A$. Then, there exist positive constants a_1, a_2, a_3, a_4 and $\gamma(A)$, independent of k, such that for $|x| \leqq A$

$$a_1 \leqq f(x) \leqq a_2\,, \qquad a_3 \leqq \frac{dg}{dx} \leqq a_4\,, \qquad \left|\frac{d^2g}{dx^2}\right| \leqq \gamma(A).$$

In this case, it will be concluded that if $k > \dfrac{\gamma(A)B}{a_1a_3}$, for any two solutions $x(t)$ and $u(t)$ of (10.6), $x(t)-u(t) \to 0$, $\dot{x}(t)-\dot{u}(t) \to 0$ as $t \to \infty$.

To see this, consider the equivalent system

$$\dot{x} = y - kF(x)\,, \qquad \dot{y} = -g(x) + kp(t) \tag{15.3}$$

and the associated system

$$\begin{cases} \dot{x} = y - kF(x)\,, & \dot{y} = -g(x) + kp(t) \\ \dot{u} = v - kF(u)\,, & \dot{v} = -g(u) + kp(t)\,, \end{cases} \tag{15.4}$$

where $F(x) = \int_0^x f(\xi)d\xi$. Let $V(x, y, u, v)$ be defined by

$$V(x, y, u, v) = \varphi(x, u)(x-u)^2 + (y-v)^2 - 2c(x-u)(y-v)\,,$$

where $\varphi(x, u) = \dfrac{g(x)-g(u)}{x-u}$ and $c>0$ is a constant. Since $a_3 \leqq \varphi(x, u)$, if we choose c sufficiently small, $V(x, y, u, v)$ is positive definite with respect to $(x-u)$ and $(y-v)$.

Setting $\dfrac{\partial\varphi}{\partial x}\dot{x} + \dfrac{\partial\varphi}{\partial u}\dot{u} = \Phi$, $k\dfrac{F(x)-F(u)}{x-u} = H$, $x-u=X$ and $y-v=Y$,

$$V'_{(15.4)}(x, y, u, v) = -(2H\varphi - \Phi - 2c\varphi)X^2 - 2cY^2 + 2cHXY. \tag{15.5}$$

From $|\dot{x}-\dot{u}| < 2B$ and $\left|\dfrac{d^2g}{dx^2}\right| \leqq \gamma(A)$, it follows that $|\Phi| \leqq 2B\,\gamma(A)$.

Moreover, $a_1 \leqq f(x)$ and $a_3 \leqq \frac{dg}{dx}$ imply that $2ka_1a_3 - 2B\ \gamma(A) \leqq 2H\varphi - \Phi$, and hence $0 < M \leqq 2H\varphi - \Phi$, because $\frac{\gamma(A)B}{a_1a_3} < k$. If we choose c so that

$$(15.6) \qquad c^2H^2 < 2c(M - 2c\varphi),$$

the right hand side of (15.5) is negative for $X \neq 0$, $Y \neq 0$. It is possible to choose $c > 0$ which satisfies (15.6), because $ka_1 \leqq H \leqq ka_2$ and $a_3 \leqq \varphi \leqq a_4$.

Thus, $V(x, y, u, v)$ satisfies the conditions in Theorem 15.1, if c is chosen suitably. Therefore, the system (15.3) is extremely stable, i.e., $x(t) - u(t) \to 0$ and $y(t) - v(t) \to 0$ as $t \to \infty$. From $\dot{x}(t) - \dot{u}(t) = y(t) - v(t) - k\{F(x(t)) - F(u(t))\}$, it follows that $\dot{x}(t) - \dot{u}(t) \to 0$ as $t \to \infty$.

Now we shall discuss more general cases. Consider the system (15.1) and suppose that $F(t, x)$ is continuous on $I \times S_H$, $H > 0$.

Definition 15.1. The system (15.1) is uniform-stable with respect to (H^*, H), $H^* \leqq H$, if for every $\varepsilon > 0$ and every $t_0 \geqq 0$, there exists a $\delta(\varepsilon) > 0$ such that if $x_0 \in S_{H^*}$, $x_0' \in S_{H^*}$ and $\| x_0 - x_0' \| < \delta(\varepsilon)$, then $\| x(t; x_0, t_0) - x(t; x_0', t_0) \| < \varepsilon$ for all $t \geqq t_0$.

Definition 15.2. The system (15.1) is quasi-uniform-asymptotically stable with respect to (H^*, H), if there exist a $\delta_0 > 0$ and a $T(\varepsilon) > 0$ for every $\varepsilon > 0$ such that if $x_0 \in S_{H^*}$, $x_0' \in S_{H^*}$ and $\| x_0 - x_0' \| < \delta_0$, then $\| x(t; x_0, t_0) - x(t; x_0', t_0) \| < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon)$.

Definition 15.3. The system (15.1) is uniform-asymptotically stable with respect to (H^*, H), if the conditions in Definitions 15.1 and 15.2 are satisfied.

In the case where $F(t, x)$ of (15.1) is defined and continuous on $I \times R^n$, similar definitions apply for stability with respect to (H^*, ∞), $H^* \leqq \infty$. For example,

Definition 15.4. The system (15.1) is quasi-uniform-asymptotically stable with respect to (∞, ∞), if there exist a $\delta_0 > 0$ and a $T(\varepsilon)>0$ for every $\varepsilon>0$ such that if $\|x_0-x_0'\|<\delta_0$, then $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<\varepsilon$ for all $t\geqq t_0+T(\varepsilon)$.

The following definition is equivalent to Definition 15.4.

DEFINITION 15.5. The system (15.1) is quasi-uniform-asymptotically stable with respect to (∞,∞), if for every $\varepsilon>0$ and every $\alpha>0$ there exists a $T(\varepsilon,\alpha)>0$ such that if $\|x_0-x_0'\|\leqq\alpha$, then $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<\varepsilon$ for all $t\geqq t_0+T(\varepsilon,\alpha)$.

It is clear that Definition 15.5 implies Definition 15.4. For any $\alpha>0$, there exists a smallest integer N such that $\frac{\alpha}{\delta_0}<N$. For any $\varepsilon>0$, we can find a number $T\left(\frac{\varepsilon}{N}\right)>0$ such that $\|x_0-x_0'\|<\delta_0$ and $t\geqq t_0+T\left(\frac{\varepsilon}{N}\right)$ imply that $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<\frac{\varepsilon}{N}$. Thus, if $\|x_0-x_0'\|\leqq\alpha$, by subdividing the line segment joining x_0 and x_0', we see that $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<\varepsilon$ for all $t\geqq t_0+T\left(\frac{\varepsilon}{N}\right)$, which shows that Definition 15.4 implies Definition 15.5.

For the solutions of (15.1), where $F(t,x)$ is defined and continuous on $I\times R^n$, we shall consider some types of boundedness.

DEFINITION 15.6. The system (15.1) is uniform-distance-bounded, if for each $\alpha>0$ there exists a $\beta(\alpha)>0$ such that if $\|x_0-x_0'\|\leqq\alpha$, then $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<\beta(\alpha)$ for all $t\geqq t_0$.

DEFINITION 15.7. The system (15.1) is uniform-ultimately distance-bounded for bound B, if there exists a $B>0$ and for any $\alpha>0$, there exists a $T(\alpha)>0$ such that if $\|x_0-x_0'\|\leqq\alpha$, then $\|x(t;x_0,t_0)-x(t;x_0',t_0)\|<B$ for all $t\geqq t_0+T(\alpha)$, where B is independent of the particular solution.

Similar definitions apply for stability and boundedness corresponding to the definitions in Sections 7 and 9 (cf. [139]).

For a linear system

$$\frac{dx}{dt}=A(t)x+f(t)\,, \tag{15.7}$$

where $A(t)$ is an $n\times n$ matrix of continuons functions on I and

$f(t)$ is continuous on I, by the transformation $x=y+\varphi(t)$, (15.7) is transformed into

$$\frac{dy}{dt}=A(t)y\,, \tag{15.8}$$

where $\varphi(t)$ is a solution of (15.7). Therefore, uniform-stability of $y(t)\equiv 0$ of (15.8), uniform-stability of a solution of (15.7) and uniform-stability of the system (15.7) are equivalent. Moreover, uniform-asymptotic stability of the system (15.7) is equivalent to uniform-asymptotic stability in the large of $y(t)\equiv 0$ of the system (15.8).

First of all, we shall prove the following theorem which shows a relationship between uniform-asymptotic stability in the large of a bounded solution of (15.1) and uniform-asymptotic stability of the system (15.1).

THEOREM 15.2. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $I\times R^n$ and $F(t, x)\in \bar{C}_0(x)$. If there exists a solution $x(t\,;x_0, 0)$ of* (15.1) *which is bounded by H_1 for $t\geqq 0$ and is uniform-asymptotically stable in the large, then the system* (15.1) *is uniform-asymptotically stable with respect to (H^*, H), where $H^*>0$ is an arbitrary number such that $H_1<H^*$ and $H>0$ is a number depending on H^* and H_1.*

PROOF. Since $x(t)=x(t\,;x_0, 0)$ is uniform-asymptotically stable in the large, there exists a $\beta(2H^*)>0$ such that if $\|x(t_0)-x_0'\|\leqq 2H^*$, then $\|x(t)-x(t\,;x_0', t_0)\|\leqq\beta(2H^*)$ for all $t\geqq t_0$, and hence, if $x_0'\in S_{H^*}$, we have $\|x(t\,;x_0', t_0)\|\leqq H$, where $H=\beta(2H^*)+H_1$. Moreover, for $\xi_1\in S_{H^*}$, $\xi_2\in S_{H^*}$, corresponding to each $\varepsilon>0$, each $t_0\geqq 0$ and $2H^*$, there exists a $T(\varepsilon)>0$ such that if $t\geqq t_0+T(\varepsilon)$, then $\|x(t)-x(t\,;\xi_1, t_0)\|<\frac{\varepsilon}{2}$ and $\|x(t)-x(t\,;\xi_2, t_0)\|<\frac{\varepsilon}{2}$. Therefore, if $\xi_1\in S_{H^*}$ and $\xi_2\in S_{H^*}$, then $\|x(t\,;\xi_1, t_0)-x(t\,;\xi_2, t_0)\|<\varepsilon$ for all $t\geqq t_0+T(\varepsilon)$, which shows that the system (15.1) is quasi-uniform-asymptotically stable with respect to (H^*, H).

If $x\in S_H$ and $x'\in S_H$, there exists an $L(H)$ such that $\|F(t, x)-F(t, x')\|\leqq L(H)\,\|x-x'\|$. For a given $\varepsilon>0$, choose a $\delta(\varepsilon)>0$ so that $\delta e^{L(H)T(\varepsilon)} < \varepsilon$. Then, if $\xi_1 \in S_{H^*}$, $\xi_2 \in S_{H^*}$ and $\|\xi_1 - \xi_2\| < \delta(\varepsilon)$, then $\|x(t; \xi_1, t_0) - x(t; \xi_2, t_0)\| < \varepsilon$ for $t \geqq t_0 + T(\varepsilon)$ and for $t \in [t_0, t_0 + T(\varepsilon)]$,

$$\|x(t; \xi_1, t_0) - x(t; \xi_2, t_0)\| \leqq e^{L(H)T(\varepsilon)} \|\xi_1 - \xi_2\| < \varepsilon,$$

which shows that (15.1) is uniform-stable with respect to (H^*, H). Thus, the theorem is proved.

THEOREM 15.3. *Suppose that* $F(t, x)$ *of* (15.1) *is continuous on* $I \times R^n$ *and that there exists a Liapunov function* $V(t, x, y)$ *defined on* $0 \leqq t < \infty$, $\|x - y\| \leqq H$, $H > 0$, *in* $I \times R^n \times R^n$ *and satisfying the following conditions;*

(i) $a(\|x-y\|) \leqq V(t, x, y) \leqq b(\|x-y\|)$, *where* $a(r) \in CIP$ *and* $b(r) \in CIP$,

(ii) $V'_{(15.2)}(t, x, y) \leqq 0$ *in the interior of the domain.*

Then, the system (15.1) *is uniform-stable with respect to* (∞, ∞).

PROOF. First of all, we must establish that every solution of (15.1) exists in the future. Suppose that a solution $x(t; x_0, t_0)$ of (15.1) satisfies $\|x(t; x_0, t_0)\| \to \infty$ as $t \to \sigma - 0$, where $t_0 < \sigma < \infty$. Let $x(t; x_0', \sigma)$ be a solution of (15.1) through (σ, x_0'). Then there is an interval $[\sigma - h, \sigma + h]$, $h > 0$, on which this solution is defined. For an $\varepsilon > 0$, $\varepsilon < H$, let δ be such that $b(\delta) < a(\varepsilon)$. Then, by considering the function $V(t, x(t), y(t))$, where $x(t)$, $y(t)$ are two solutions of (15.1), it can be easily seen that if $\|x(\sigma-h) - y(\sigma-h)\| < \delta$, then $\|x(t) - y(t)\| < \varepsilon$ for $t \in [\sigma-h, \sigma+h]$. Let N be the smallest integer such that $\|x(\sigma-h; x_0, t_0) - x(\sigma-h; x_0', \sigma)\|/\delta < N$. By subdividing the line segment joining $x(\sigma-h; x_0, t_0)$ and $x(\sigma-h; x_0', \sigma)$, we see that $\|x(t; x_0, t_0) - x(t; x_0', \sigma)\| < \varepsilon N$ for $t \in [\sigma-h, \sigma+h]$. For $t \in [\sigma-h, \sigma+h]$, $x(t; x_0', \sigma)$ is bounded, and hence, there arises a contradiction. Thus, every solution exists in the future. The remainder of the proof can be accomplished by the same method used in the theory of stability (in the sense of Liapunov), by choosing a $\delta > 0$ for each $\varepsilon > 0$ so that $b(\delta) < a(\varepsilon)$.

Corresponding to the theorems in Sections 8, 10 and 11, we have the following theorems (cf. [139]).

THEOREM 15.4. *Suppose that $F(t, x)$ of (15.1) is continuous on $I\times R^n$ and that there exists a Liapunov function $V(t, x, y)$ defined on $t\in I$, $\|x-y\|\geqq R$ in $I\times R^n\times R^n$, where $R>0$ may be large, and satisfying the following conditions;*

(i) *$a(\|x-y\|)\leqq V(t, x, y)\leqq b(\|x-y\|)$, where $a(r)$, $b(r)$ are non-negative continuous functions and $a(r)\to\infty$ as $r\to\infty$,*

(ii) *$V'_{(15.2)}(t, x, y)\leqq 0$.*

Then, the system (15.1) is uniform-distance-bounded.

As an example, consider the equation

$$\ddot{x}+f(\dot{x})+x=e(t, \dot{x}), \tag{15.9}$$

where $f(y)$ is continuous, non-decreasing and $e(t, y)$ is continuous. If $(y-v)\{e(t, y)-e(t, v)\}\leqq 0$, the equation is uniform-distance-bounded. To see this, consider the system

$$\begin{cases} \dot{x}=y, & \dot{y}=-x-f(y)+e(t, y) \\ \dot{u}=v, & \dot{v}=-u-f(v)+e(t, v) \end{cases} \tag{15.10}$$

and a Liapunov function $V(x, y, u, v)=(x-u)^2+(y-v)^2$. Then,

$$\begin{aligned} V'_{(15.10)}(x, y, u, v)= &-2(y-v)\{f(y)-f(v)\} \\ &+2(y-v)\{e(t, y)-e(t, v)\}\leqq 0, \end{aligned}$$

and hence, the uniform-distance-boundedness follows from Theorem 15.4.

A special equation of type (15.9) is $\ddot{x}+\dot{x}^3+x=1+t$. This equation has a solution $x=t$, and hence all solutions are unbounded.

THEOREM 15.5. *Suppose that $F(t, x)$ of (15.1) is continuous on $I\times R^n$ and that there exists a Liapunov function $V(t, x, y)$ defined on $t\in I$, $\|x-y\|<H$, $H>0$, in $I\times R^n\times R^n$ and satisfying the following conditions;*

(i) *$a(\|x-y\|)\leqq V(t, x, y)\leqq b(\|x-y\|)$, where $a(r)\in CIP$ and $b(r)\in CIP$,*

(ii) *$V'_{(15.2)}(t, x, y)\leqq -c(\|x-y\|)$, where $c(r)$ is continuous, positive definite.*

Then, the system (15.1) is uniform-asymptotically stable with respect to (∞, ∞).

THEOREM 15.6. *Suppose that* $F(t, x)$ *of* (15.1) *is continuous on* $t \in I$, $\|x\| \leqq H$ *and that if* $x_0 \in S_{H^*}$, $H^* \leqq H$, *we have* $\|x(t; x_0, t_0)\| \leqq H$ *for all* $t \geqq t_0$. *Moreover, suppose that there exists a Liapunov function* $V(t, x, y)$ *on* $t \in I$, $x \in S_H$, $y \in S_H$ *and* $\|x-y\| < H_0$, *where* $H_0 > 0$ *is a constant, which satisfies the conditions* (i), (ii) *in Theorem* 15.5. *Then, the system* (15.1) *is uniform-asymptotically stable with respect to* (H^*, H).

THEOREM 15.7. *Suppose that* $F(t, x)$ *of* (15.1) *is continuous on* $I \times R^n$ *and that there exists a Liapunov function* $V(t, x, y)$ *satisfying the following conditions in* $0 \leqq t < \infty$, $\|x-y\| \geqq R$, *where* R *may be large;*

(i) $a(\|x-y\|) \leqq V(t, x, y) \leqq b(\|x-y\|)$, *where* $a(r)$, $b(r)$ *are positive continuous increasing functions and* $a(r) \to \infty$ *as* $r \to \infty$,

(ii) $V'_{(15.2)}(t, x, y) \leqq -c(\|x-y\|)$, *where* $c(r)$ *is positive, continuous.*

Then, the system (15.1) *is uniform-distance-bounded and uniform-ultimately distance-bounded.*

§ 16. Stability of sets.

In Chapter III, we considered a Liapunov function whose derivative along the solution is negative definite with respect to a set and discussed the asymptotic behavior of solutions. These considerations are some types of stability of a set. The stability of a set has been discussed by Lefschetz [76], Hale and Stokes [51], Seibert [124], Zubov [152] etc..

In this section, we shall discuss stability of a set which includes, as special cases, stability in the sense of Liapunov, orbital stability, and ultimate boundedness of solutions.

Consider the system (15.1) and a set M in $I \times R^n$ and suppose that $F(t, x)$ of (15.1) is defined and continuous on $I \times R^n$ or in a neighborhood of M. Here, the following notations will be used: π_σ is the hyperplane $t = \sigma$. $M(\sigma)$ represents the set $M \cap \pi_\sigma$ and $M(\sigma, \varepsilon)$ is the ε-neighborhood of $M(\sigma)$ in R^n. If there is a compact set Q in R^n such that $M(t) \subset Q$ for all $t \in I$, M is said to be bounded. Throughout this monograph, we shall denote by 0-set the set of points (t, x) such that $t \in I$, $x = 0$.

DEFINITION 16.1. M is a stable set of (15.1), if for any $\varepsilon > 0$, any $\alpha > 0$ and $t_0 \in I$, there exists a $\delta(t_0, \varepsilon, \alpha) > 0$ such that if $d(x_0, M(t_0)) < \delta$ and $x_0 \in S_\alpha$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0$.

DEFINITION 16.2. M is a uniform-stable set of (15.1) with respect to t, if the δ above is independent of t_0. In case δ is independent of α, M is a uniform-stable set of (15.1) with respect to α. When δ depends only on ε, we say that M is a uniform-stable set of (15.1).

DEFINITION 16.3. M is a uniform-asymptotically stable set of (15.1) with respect to t, if it is a uniform-stable set with respect to t and if for any $\varepsilon > 0$, any $\alpha > 0$, there exist a $\delta_0(\alpha) > 0$ and a $T(\varepsilon, \alpha) > 0$ such that if $d(x_0, M(t_0)) < \delta_0$ and $x_0 \in S_\alpha$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon, \alpha)$.

DEFINITION 16.4. The solutions of (15.1) are equi-bounded with respect to M, if for any $\eta > 0$, any $\alpha > 0$ and $t_0 \in I$, there exists a $\beta(t_0, \eta, \alpha) > 0$ such that if $d(x_0, M(t_0)) \leqq \eta$ and $x_0 \in S_\alpha$, then $d(x(t; x_0, t_0), M(t)) < \beta(t_0, \eta, \alpha)$ for all $t \geqq t_0$. The solutions are uniform-bounded with respect to M and t, if β is independent of t_0. When β depends only on η, we say that the solutions are uniform-bounded with respect to M.

DEFINITION 16.5. M is a quasi-asymptotically stable set of (15.1) in the large, if every solution of (15.1) approaches M as $t \to \infty$.

DEFINITION 16.6. M is a quasi-equiasymptotically stable set of (15.1) in the large, if for any $\varepsilon > 0$, any $\eta > 0$, any $\alpha > 0$ and $t_0 \in I$, there exists a $T(t_0, \varepsilon, \eta, \alpha) > 0$ such that if $d(x_0, M(t_0)) \leqq \eta$ and $x_0 \in S_\alpha$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0 + T(t_0, \varepsilon, \eta, \alpha)$. If T is independent of t_0, M is a quasi-uniform-asymptotically stable set of (15.1) in the large with respect to t. When T is independent of α, we say that M is quasi-uniform-asymptotically stable set in the large with respect to α. If T is independent of both t_0 and α, M is said to be a quasi-uniform-asymptotically stable set in the large.

DEFINITION 16.7. M is an equiasymptotically stable set of (15.1) in the large, if M is a stable set of (15.1) and a quasi-equiasymptotically stable set of (15.1) in the large and if the solutions of (15.1) are equi-bounded with respect to M. If the solutions are uniform-bounded with respect to M and t and if M is a uniform-stable set with respect to t and is a quasi-uniform-asymptotically stable set in the large with respect to t, M is said to be a uniform-asymptotically stable set of (15.1) in the large with respect to t.

In particular, if M is bounded, the stability and boundedness above are clearly uniform with respect to α, and the boundedness is what was defined in Section 9. Moreover, orbital stability of a closed orbit C in R^n is equivalent to the stability of the set $I \times C$.

We can deal with ultimate boundedness of solutions of (15.1) defined in Section 9 as a special case of quasi-asymptotic stability of a set in the large.

THEOREM 16.1. *If the solutions of* (15.1) *are ultimately bounded for bound* B*, the set* $I \times S_B$ *is a quasi-asymptotically stable set of* (15.1) *in the large. Conversely, if the set* $I \times S_B$ *is a quasi-asymptotically stable set of* (15.1) *in the large, the solutions of* (15.1) *are ultimately bounded for bound* B' *such that* $B' > B$.

THEOREM 16.2. *If the solutions of* (15.1) *are equiultimately bounded for bound* B*, the set* $I \times S_B$ *is a quasi-uniform-asymptotically stable set of* (15.1) *in the large with respect to* α*. Conversely, if* $I \times S_B$ *is a quasi-uniform-asymptotically stable set in the large with respect to* α*, the solutions are equiultimately bounded for bound* B', $B' > B$.

THEOREM 16.3. *If the solutions of* (15.1) *are uniform-ultimately bounded for bound B, the set $I\times S_B$ is a quasi-uniform-asymptotically stable set of* (15.1) *in the large. Conversely, if $I\times S_B$ is a quasi-uniform-asymptotically stable set in the large, the solutions are uniform-ultimately bounded for bound B', $B'>B$.*

Now we assume that $M(t)$ is non-empty for any $t\in I$ and that if (t, x), (t', x) belong to a compact set Q in $I\times R^n$, there exists a $K>0$ depending on Q such that

$$(16.1) \qquad |d(x, M(t))-d(x, M(t'))|\leqq K|t-t'|.$$

Throughout this section and the next section, these conditions on M will be assumed, and hence, we always have $\overline{M\cap\pi_t}=\overline{M}\cap\pi_t$.

DEFINITION 16.8. M is an invariant set of (15.1), if for any point $(t_0, x_0)\in M$, every solution $x(t; x_0, t_0)$ exists in the future and $(t, x(t; x_0, t_0))\in M$ for all $t\geqq t_0$.

From the definitions and the conditions on M, it follows that each kind of stability of M is equivalent to one of $\overline{M}$. Further it is clear that if M is a stable set of (15.1), $\overline{M}$ is an invariant set of (15.1). There are several relationships between various types of stability which imply relationships in stability in the sense of Liapunov and in boundedness of solutions, see [143]. For this purpose, we require the following property.

THEOREM 16.4. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $0\leqq t\leqq T$, $d(x, M(t))\leqq H$, $H>0$. If $\overline{M}$ is an invariant set of* (15.1) *on $[0, T]$, then for each $\varepsilon>0$ and each $\alpha>0$, there exists a $\delta(\varepsilon, \alpha)>0$ such that if $t_0\in[0, T]$, $x_0\in S_\alpha$ and $(t_0, x_0)\in N(\delta, M)$, then $(t, x(t; x_0, t_0))\in N(\varepsilon, M)$ for $t\geqq t_0$.*

PROOF. Since a solution $x(t; x_0, t_0)$, $x_0\in S_\alpha$, $(t_0, x_0)\in\overline{M}$, is continuable to $t=T$, by Theorem 3.7, there exists a $\gamma(\alpha)>0$ for which $\|x(t; x_0, t_0)\|\leqq\gamma(\alpha)$ for $t\in[t_0, T]$. For a suitable γ', $\gamma'>\gamma$, let $F^*(t, x)$ be a continuous function which coincides with $F(t, x)$ on $[0, T]\times S_{\gamma'}\cap\overline{N(\varepsilon, M)}$ and is bounded in the whole space. Then, all solutions of

$$(16.2) \qquad \frac{dx}{dt} = F^*(t, x)$$

are continuable to $t = T$.

Suppose that there is a sequence of points $\{(t_k, x_k)\}$ such that $(t_k, x_k) \in N(\delta_k, M)$, $t_k \in [0, T)$, $x_k \in S_\alpha$, where $\delta_k \to 0$ as $k \to \infty$, and that the solutions $x^*(t; x_k, t_k)$ of (16.2) do not remain in $[0, T] \times S_{\gamma'} \cap N(\varepsilon, M)$. By uniform boundedness and equicontinuity of $x^*(t; x_k, t_k)$, we can choose a subsequence which converges uniformly to a solution $x^*(t; x_0, t_0)$ of (16.2), where $(t_0, x_0) \in \overline{M}$ and $(t_k, x_k) \to (t_0, x_0)$ as $k \to \infty$. Since $(t, x^*(t; x_0, t_0)) \in \overline{M}$ and $\| x^*(t; x_0, t_0) \| < \gamma'$, there arises a contradiction. Therefore, for a suitable $\delta > 0$, if $t_0 \in [0, T]$, $x_0 \in S_\alpha$ and $(t_0, x_0) \in N(\delta, M)$, we have $(t, x^*(t; x_0, t_0)) \in [0, T] \times S_{\gamma'} \cap N(\varepsilon, M)$ and hence, $x^*(t; x_0, t_0)$ is a solution of (15.1). Thus, the theorem is proved.

THEOREM 16.5. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $I \times R^n$ and that every solution of* (15.1) *is unique and the solutions of* (15.1) *are uniform-bounded with respect to M. If M is a quasi-asymptotically stable set of* (15.1) *in the large, then there exists a set M^* which is a quasi-equiasymptotically stable set of* (15.1) *in the large.*

REMARK. If equi-boundedness of solutions is assumed, the assumption that every solution is unique is not required.

THEOREM 16.6. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $I \times R^n$ and that $F(t, x)$, $M(t)$ are periodic in t with the same period. If the solutions of* (15.1) *are equi-bounded and if M is an asymptotically stable set of* (15.1) *in the large, then M is a uniform-asymptotically stable set of* (15.1) *in the large with respect to t.*

THEOREM 16.7. *Suppose that $F(t, x)$, $M(t)$ satisfy the assumption in Theorem* 16.6. *Moreover, suppose that every solution of* (15.1) *is unique and that the solutions of* (15.1) *are uniform-bounded with respect to M and α. If M is a quasi-asymptotically stable set of*

(15.1) *in the large, then there exists a set which is quasi-uniform-asymptotically stable set of* (15.1) *in the large with respect to t.*

The remark for Theorem 16.5 is also valid for Theorem 16.7. Thus, from Theorem 16.7, we obtain the following theorem which is concerned with ultimate boundedness.

THEOREM 16.8. *Suppose that $F(t,x)$ is continuous on $I\times R^n$ and is periodic in t. If the solutions of* (15.1) *are equi-bounded and ultimately bounded, then they are uniform-ultimately bounded.*

Here, we consider a special case. Consider a system

$$(16.3)\qquad \begin{cases} \dfrac{dx}{dt}=F(t,x,y)+H(t,x,y) \\[2ex] \dfrac{dy}{dt}=G(t,x,y) \end{cases}$$

in the product space, where x, y are n-vector and m-vector, respectively, and $F, G, H,$ are continuous on $I\times R^n\times R^m$. Let N be a non-empty set in R^n and let $M=I\times N\times R^m$. We assume that for any continous bounded function $\{x(t), y(t)\}$, $t\geqq t_0$, $\int_{t_0}^{\infty}\|H(t, x(t), y(t))\|\,dt<\infty$.

THEOREM 16.9. *Let $\{x(t;x_0,y_0,t_0), y(t;x_0,y_0,t_0)\}$ be a solution of* (16.3) *through (t_0,x_0,y_0) which is bounded by $\alpha>0$ for all $t\geqq t_0$. Suppose that $F(t,x,y)$ is bounded for bounded x, y and that there exists a non-negative Liapunov function $V(t,x,y)$ defined on $t_0\leqq t<\infty$, $\|x\|^2+\|y\|^2\leqq\alpha^2$, which satisfies the condition that $V'_{(16.3)}(t,x,y)\leqq -W(x,y)$, where $W(x,y)$ is continuous and is positive definite with respect to the set $N\times R^m$. Then $x(t;x_0,y_0,t_0)\to N$ as $t\to\infty$.*

PROOF. Suppose that $x(t)=x(t;x_0,y_0,t_0)$ does not approach N as $t\to\infty$. Then there exist an $\varepsilon>0$ and a sequence $\{t_k\}$ such that $t_k\to\infty$ as $k\to\infty$ and that $d(x(t_k),N)>\varepsilon$. Since the solution is bounded, there is an $L>0$ by which $\|\dot{x}(t)\|$ is bounded for all $t\geqq t_0$. Then, on the intervals $t_k\leqq t\leqq t_k+\dfrac{\varepsilon}{4L}$, we have $d(x(t),N)$

$>\frac{\varepsilon}{2}$ if t_1 is sufficiently large, and hence, by the same argument as in the proof of Theorem 14.1, there arises a contradiction. Thus, it can be shown that $x(t)\to N$ as $t\to\infty$.

COROLLARY. *Suppose that there exists a non-negative Liapunov function $V(t, x, y)$ defined on $I\times R^n\times R^m$ which satisfies the condition that $V'_{(16.3)}(t, x, y)\leqq -W(x, y)$, where $W(x, y)$ is continuous and is positive definite with respect to $N\times R^m$. If $F(t, x, y)$ is bounded for bounded x, y, then every bounded solution of* (16.3) *approaches $N\times R^m$ as $t\to\infty$. Moreover, if every solution of* (16.3) *is bounded, the set M is a quasi-asymptotically stable set of* (16.3) *in the large.*

REMARK. In Corollary, if $V(t, x, y)\to\infty$ uniformly as $\|x\|^2+\|y\|^2\to\infty$, every solution of (16.3) is bounded, by Theorem 10.1.

§ 17. Eventual properties.

Let us consider a simple scalar equation

$$\frac{dx}{dt}=-x+2e^{-t}x^2. \tag{17.1}$$

Clearly the zero solution of $\dot{x}=-x$ is uniform-asymptotically stable in the large. However, the zero solution of (17.1) is not uniform-asymptotically stable in the large, because the solutions are not uniform-bounded, while the zero solution is uniform-stable. Clearly, (17.1) has a solution $x=e^t$. Though the solutions of (17.1) are not uniform-bounded, if t_0 is large enough, for a given $\alpha>0$ there exists a $\beta(\alpha)>0$ such that if $|x_0|\leqq\alpha$, then $|x(t;x_0,t_0)|<\beta(\alpha)$, and such a solution tends to zero as $t\to\infty$. We call such properties the eventual properties. Here, we shall give difinitions which we need later.

For more general cases, we give definitions of eventual properties for the system (15.1) and the set M stated in Section 16.

DEFINITION 17.1. *M is an eventually stable set of* (15.1), *if for any $\varepsilon>0$ and any $\alpha>0$, there exists an $S(\varepsilon, \alpha)\geqq 0$ such that*

Definition 16.1 is satisfied for all $t_0 \geqq S(\varepsilon, \alpha)$. In particular, when M is the 0-set, i.e., the t-axis, we say that $x=0$ is eventually stable.

DEFINITION 17.2. M is an eventually uniform-stable set of (15.1) with respect to t, if for any $\varepsilon > 0$ and any $\alpha > 0$, there exist an $S(\varepsilon. \alpha) \geqq 0$ and a $\delta(\varepsilon, \alpha) > 0$ such that if $x_0 \in S_\alpha$, $d(x_0, M(t_0)) < \delta(\varepsilon, \alpha)$ and $t_0 \geqq S(\varepsilon, \alpha)$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0$.

DEFINITION 17.3. M is an eventually uniform-asymptotically stable set of (15.1) with respect to t, if M is an eventually uniform-stable set of (15.1) with respect to t and if for any $\varepsilon > 0$ and any $\alpha > 0$, there exist a $\delta_0(\alpha) > 0$, a $T(\varepsilon, \alpha) > 0$ and an $S_1(\varepsilon, \alpha) \geqq 0$ such that if $t_0 \geqq S_1(\varepsilon, \alpha)$, $x_0 \in S_\alpha$ and $d(x_0, M(t_0)) < \delta_0(\alpha)$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon, \alpha)$.

DEFINITION 17.4. The solutions of (15.1) are eventually uniform-bounded with respect to M and t, if for any $\eta > 0$, any $\alpha > 0$, there exist an $S_2(\eta, \alpha) \geqq 0$ and a $\beta(\eta, \alpha) > 0$ such that if $t_0 \geqq S_2(\eta, \alpha)$, $x_0 \in S_\alpha$ and $d(x_0, M(t_0)) \leqq \eta$, then $d(x(t; x_0, t_0), M(t)) < \beta(\eta, \alpha)$ for all $t \geqq t_0$.

DEFINITION 17.5. M is an eventually quasi-uniform-asymptotically stable set of (15.1) in the large with respect to t, if for any $\varepsilon > 0$, any $\alpha > 0$ and any $\eta > 0$, there exist an $S_3(\eta, \alpha) \geqq 0$ and a $T(\varepsilon, \eta, \alpha) > 0$ such that if $t_0 \geqq S_3(\eta, \alpha)$, $x_0 \in S_\alpha$ and $d(x_0, M(t_0)) \leqq \eta$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon, \eta, \alpha)$.

DEFINITION 17.6. M is an eventually uniform-asymptotically stable set of (15.1) in the large with respect to t, if the conditions in Definitions 17.2, 17.4 and 17.5 are satisfied.

Similar definitions apply for eventual stability and boundedness with respect to M.

THEOREM 17.1. *Suppose that* $F(t, x)$ *of the system* (15.1) *is continuous on* D: $0 \leqq t < \infty$, $d(x, M(t)) \leqq H$, $H > 0$. *If* M *is an even-*

tually uniform-stable set of (15.1) *with respect to t and $\overline{M}$ is an invariant set of* (15.1) *and if the solutions of* (15.1) *are uniform-bounded, then M is a uniform-stable set of* (15.1) *with respect to t.*

PROOF. By the uniform-boundedness of solutions, corresponding to each $\alpha > 0$, there exists a $\gamma(\alpha) > 0$ such that $x_0 \in S_\alpha$ implies $\| x(t; x_0, t_0) \| \leqq \gamma(\alpha)$ for all $t \geqq t_0$. Since M is an eventually uniform-stable set with respect to t, for any $\varepsilon > 0$ there are $S(\varepsilon, \alpha) \geqq 0$ and $\delta_1(\varepsilon, \alpha) > 0$ such that if $t_0 \geqq S(\varepsilon, \alpha)$, $x_0 \in S_\gamma$ and $d(x_0, M(t_0)) < \delta_1(\varepsilon, \alpha)$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0$. By Theorem 16.3, there is a $\delta(\varepsilon, \alpha) > 0$, $\delta < \delta_1$, such that if $t_0 \in [0, S]$, $x_0 \in S_\alpha$ and $d(x_0, M(t_0)) < \delta(\varepsilon, \alpha)$, then $d(x(S; x_0, t_0), M(S)) < \delta_1(\varepsilon, \alpha)$. Thus, if $t_0 \in I$, $x_0 \in S_\alpha$ and $d(x_0, M(t_0)) < \delta(\varepsilon, \alpha)$, then $d(x(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0$, which shows that M is a uniform-stable set with respect to t.

In a similar manner, we have the following theorem.

THEOREM 17.2. *Suppose that $F(t, x)$ of* (15.1) *is continuous on D in Theorem* 17.1. *If M is an eventually uniform-stable set of* (15.1) *and $\overline{M}$ is an invariant set of* (15.1), *then M is a uniform-stable set of* (15.1) *with respect to t. Moreover, if M is bounded, M is a uniform-stable set of* (15.1).

Now we shall discuss eventual properties in simple cases.

THEOREM 17.3. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $I \times R^n$ and that there exists a Liapunov function $V(t, x)$ defined on $t \in I$, $\| x \| \geqq R$, where R may be large, which satisfies the following conditions;*

(i) $a(\| x \|) \leqq V(t, x) \leqq b(\| x \|)$, *where $a(r) \in CI$, positive, $a(r) \to \infty$ as $r \to \infty$ and $b(r) \in CI$ (see, Section* 1),

(ii) $V'_{(15.1)}(t, x) \leqq h(t) q(t, x)$, *where* $\int_0^\infty |h(t)| dt < \infty$, *and $q(t, x)$ is continuous on $t \in I$, $\| x \| \geqq R$ and is bounded for bounded x.*

Then, the solutions of (15.1) *are eventually uniform-bounded.*

PROOF. Corresponding to any $\alpha > R$, choose a $\beta(\alpha) > 0$ so that

$2b(\alpha) < a(\beta)$, $\alpha < \beta$, and let $K(\beta)$ be a positive number such that $x \in S_\beta$ implies $|q(t, x)| \leqq K(\beta)$. By the condition on $h(t)$, there exists a $T(\alpha) > 0$ such that if $t_0 \geqq T(\alpha)$,

$$\int_{t_0}^{\infty} |h(t)|\, dt < \frac{b(\alpha)}{K(\beta)}. \tag{17.2}$$

Let $x(t; x_0, t_0)$ be a solution of (15.1) such that $t_0 \geqq T(\alpha)$, $x_0 \in S_\alpha$, and suppose that $\|x(t_1; x_0, t_0)\| = \beta$ at some t_1. Then, there exist t_2, t_3 such that $t_0 \leqq t_2 < t_3 \leqq t_1$, $\|x(t_2; x_0, t_0)\| = \alpha$, $\|x(t_3; x_0, t_0)\| = \beta$ and that for $t \in (t_2, t_3)$, $\alpha < \|x(t; x_0, t_0)\| < \beta$. Since $V'_{(15.1)}(t, x) \leqq K(\beta)|h(t)|$ in the domain $0 \leqq t < \infty$, $\alpha \leqq \|x\| \leqq \beta$, we have

$$V(t_3, x(t_3; x_0, t_0)) \leqq V(t_2, x(t_2; x_0, t_0)) + \int_{t_2}^{t_3} K(\beta)|h(t)|\, dt.$$

From this and (17.2), it follows that $V(t_3, x(t_3; x_0, t_0)) \leqq 2b(\alpha) < a(\beta)$, and hence by (i), $\|x(t_3; x_0, t_0)\| < \beta$, which contradicts $\|x(t_3; x_0, t_0)\| = \beta$. Therefore, if $t_0 \geqq T(\alpha)$ and $x_0 \in S_\alpha$, then $\|x(t; x_0, t_0)\| < \beta(\alpha)$ for all $t \geqq t_0$. This proves the theorem.

THEOREM 17.4. *Suppose that $F(t, x)$ of (15.1) is continuous on $I \times S_H$, $H > 0$. Suppose that there exists a Liapunov function $V(t, x)$ defined on $0 \leqq t < \infty$, $\|x\| < H$, which satisfies the following conditions;*

(i) *$a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, where $a(r) \in CIP$ and $b(r) \in CIP$,*

(ii) *$V'_{(15.1)}(t, x) \leqq h(t)q(t; x)$, where $\int_0^\infty |h(t)|\, dt < \infty$ and $q(t, x)$ is bounded.*

Then, $x = 0$ is eventually uniform-stable. Moreover, if the 0-set is an invariant set of (15.1), then the zero solution is uniform-stable.

PROOF. For a given $\varepsilon > 0$, $\varepsilon < H$, choose a $\delta(\varepsilon) > 0$ so that $2b(\delta) < a(\varepsilon)$, $\delta < \varepsilon$. Since $q(t, x)$ is bounded, there is an $L > 0$ such that $|q(t, x)| \leqq L$. Let $T(\varepsilon) > 0$ be such that if $t_0 \geqq T(\varepsilon)$, then $\int_{t_0}^{\infty} |h(t)|\, dt < \frac{b(\delta)}{L}$. Then, by the same argument used in the proof of Theorem 17.3, it can be proved that if $t_0 \geqq T(\varepsilon)$ and $\|x_0\| < \delta$, then $\|x(t; x_0, t_0)\| < \varepsilon$ for all $t \geqq t_0$, which shows that $x = 0$ is eventually uniform-stable.

If the 0-set is an invariant set of (15.1), from Theorem 17.2, it follows that the zero solution is uniform-stable.

THEOREM 17.5. *Suppose that $F(t, x)$ of* (15.1) *is continuous on $I\times R^n$ and let M be a set such that $M=I\times N$, where N is a non-empty set in R^n. Moreover, suppose that the solutions of* (15.1) *are uniform-bounded and that there exists a Liapunov function $V(t, x)$ defined on $I\times R^n$ which satisfies the following conditions;*

(i) *$a(d(x, N))\leqq V(t, x)\leqq b(d(x, N), \|x\|)$, where $a(r)$ is continuous, $a(r)>0$ for $r\neq 0$ and $b(r, s)\in CI$, $b(r, s)\to 0$ as $r\to 0$,*

(ii) *$V'_{(15.1)}(t, x)+V^*(t, x)\to 0$ uniformly on $0<\lambda\leqq d(x, N)\leqq \mu$, $x\in S_\alpha$ for any λ, μ, α as $t\to\infty$, where $V^*(t, x)$ is a continuous function and there exists a function $W(x)$ which is positive definite with respect to N and satisfies $W(x)\leqq V^*(t, x)$.*

Then, the set M is an eventually uniform-asymptotically stable set of (15.1) *in the large with respect to t. If $\overline{M}$ is invariant, M is a uniform-asymptotically stable set in the large with respect to t.*

PROOF. First of all, we shall see that the solutions of (15.1) are uniform-bounded with respect to M and t. The uniform-boundedness of solutions implies that for a given $\alpha>0$ there exists a $\beta(\alpha)>0$ such that if $t_0\in I$ and $x_0\in S_\alpha$, then $\|x(t; x_0, t_0)\|<\beta(\alpha)$ for all $t\geqq t_0$. Since N is non-empty, if we choose $\beta(\alpha)$ large enough, the region S_β contains a point of N. Therefore, we have $d(x(t; x_0, t_0), N)\leqq 2\beta(\alpha)$, which shows that the solutions are uniform-bounded with respect to M and t.

Consider a solution $x(t; x_0, t_0)$ of (15.1) such that $x_0\in S_\alpha$. By the condition on $b(r, s)$, for a given $\varepsilon>0$, there exists a $\delta(\varepsilon, \alpha)>0$ such that $b(\delta, \beta(\alpha))<a(\varepsilon)$. By (ii), for x such that $x\in S_\beta$ and $\delta(\varepsilon, \alpha)\leqq d(x, N)\leqq\varepsilon$, there are $\gamma_1(\varepsilon, \alpha)>0$ and $T_1(\varepsilon, \alpha)>0$ for which $V^*(t, x)\geqq\gamma_1(\varepsilon, \alpha)$ and $V'_{(15.1)}(t, x)\leqq -\dfrac{\gamma_1(\varepsilon, \alpha)}{2}$ for $t\geqq T_1(\varepsilon, \alpha)$. From this, it follows that if $t_0\geqq T_1(\varepsilon, \alpha)$ and $d(x_0, N)<\delta(\varepsilon, \alpha)$, then $d(x(t; x_0, t_0), N)<\varepsilon$ for all $t\geqq t_0$, which shows that M is an eventually uniform-stable set of (15.1) with respect to t.

Now, let $x(t; x_0, t_0)$ be a solution such that $t_0 \in I$, $x_0 \in S_\alpha$. For x such that $x \in S_\beta$, $\delta(\varepsilon, \alpha) \leqq d(x, N) \leqq 2\beta(\alpha)$, there are $\gamma(\varepsilon, \alpha) > 0$ and $T_2(\varepsilon, \alpha) > 0$ such that if $t \geqq T_2(\varepsilon, \alpha)$, we have $V'_{(15.1)}(t, x) \leqq -\dfrac{\gamma(\varepsilon, \alpha)}{2}$. By the same argument used in the proof of Theorem 8.4, we can find a $T(\varepsilon, \alpha) > 0$ such that $d(x(t; x_0, t_0), N) < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon, \alpha)$. In fact, we can set

$$T(\varepsilon, \alpha) = \max(T_1(\varepsilon, \alpha), T_2(\varepsilon, \alpha)) + \frac{2}{\gamma(\varepsilon, \alpha)} \{b(2\beta(\alpha), \beta(\alpha)) - a^*(\varepsilon, \alpha)\},$$

where $a^*(\varepsilon, \alpha) = \min\{a(r); \delta(\varepsilon, \alpha) \leqq r \leqq 2\beta(\alpha)\}$. This shows that M is a quasi-uniform-asymptotically stable set of (15.1) in the large with respect to t.

If $\overline{M}$ is invariant, by Theorem 17.1 M is a uniform-stable set with respect to t and hence, M is a uniform-asymptotically stable set of (15.1) in the large with respect to t.

Now we shall observe the following (cf. [73], [150]). Let us consider the system (16.3) under the same assumptions on F, G and H. Moreover, the following assumptions will be made:

(a) $F(t, x, y)$ is bounded for bounded x, y and all $t \geqq 0$.

(b) There exists a Liapunov function $V(t, x, y)$ defined on $I \times R^n \times R^m$ which satisfies the following conditions;

(i) $a(\|x\|^2 + \|y\|^2) \leqq V(t, x, y) \leqq b(\|x\|^2 + \|y\|^2)$, where $a(r) \in CIP$, $a(r) \to \infty$ as $r \to \infty$ and $b(r) \in CIP$,

(ii) $V'_{(16.3)}(t, x, y) \leqq -W(x) + h(t)q(t, x, y)$, where $W(x)$ is continuous, positive definite, $\int_0^\infty |h(t)|\,dt < \infty$ and $q(t, x, y)$ is continuous and is bounded for bounded x, y and all $t \geqq 0$.

Then, $\{x = 0, y = 0\}$ is eventually uniform-stable, and for any $\alpha > 0$, there exists a $T(\alpha) > 0$ such that $\|x(t_0)\|^2 + \|y(t_0)\|^2 < \alpha^2$ for some $t_0 \geqq T(\alpha)$ implies that $y(t)$ is bounded and $x(t) \to 0$ as $t \to \infty$.

In addition, if we have the following condition

(c) for some $R > 0$, if $\|x\|^2 + \|y\|^2 \geqq R^2$, we have $|q(t, x, y)| \leqq \varphi(V(t, x, y))$, where $\varphi(u)$ is a continuous positive function and $\int^\infty \dfrac{du}{\varphi(u)} = \infty$, then all solutions $y(t)$ are bounded and all $x(t) \to 0$

as $t \to \infty$.

By Theorem 17.4, it is clear that $\{x=0, y=0\}$ is eventually uniform-stable. By Theorem 17.3, the solutions are eventually uniform-bounded and hence, for any $\alpha > 0$ there exist $T(\alpha) > 0$ and $\beta(\alpha) > 0$ such that if $t_0 \geqq T(\alpha)$ and $\|x(t_0)\|^2 + \|y(t_0)\|^2 \leqq \alpha^2$, we have $\|x(t)\|^2 + \|y(t)\|^2 < \beta^2$. By the condition on $q(t, x, y)$, there exists an $L > 0$ such that if $\|x\|^2 + \|y\|^2 \leqq \beta^2$, then $|q(t, x, y)| \leqq L$. On the domain $0 \leqq t < \infty$, $\|x\|^2 + \|y\|^2 \leqq \beta^2$, consider a Liapunov function

$$U(t, x, y) = e^{-\int_0^t |h(s)| ds} \{V(t, x, y) + L\} .$$

Then we have

$$U'_{(16.3)}(t, x, y) \leqq e^{-\int_0^\infty |h(s)| ds} W(x) .$$

Thus, from Theorem 16.8, it follows that $x(t) \to 0$ as $t \to \infty$.

In case we have condition (c), on the domain $0 \leqq t < \infty$, $\|x\|^2 + \|y\|^2 \geqq R^2$, consider a Liapunov function

$$U(t, x, y) = \exp\left\{-\int_0^t |h(s)| ds + \int_{V_0}^V \frac{du}{\varphi(u)}\right\} .$$

Then, we have

$$\begin{aligned} &U'_{(16.3)}(t, x, y) \\ &\quad \leqq U(t, x, y)\left\{-|h(t)| + \frac{1}{\varphi(V)}(-W(x) + |h(t)|\,|q(t, x, y)|)\right\} \\ &\quad \leqq 0 , \end{aligned}$$

and hence, by Theorem 10.2, the solutions are uniform-bounded. Thus, by the same consideration as above, we can see that $x(t) \to 0$ as $t \to \infty$.

CHAPTER V. CONVERSE THEOREMS ON STABILITY AND BOUNDEDNESS

The converse theorems on stability and boundedness will be discussed. The converse theorems are also important in studying the properties of solutions of perturbed systems.

§ 18. Converse theorems on stability.

Let $f(x)=\sum_{i,j=1}^{n} p_{ij}x_i x_j$, $p_{ij}=p_{ji}$, be a quadratic form and let P be an $n\times n$ matrix such that $P=(p_{ij})$. We denote by x the column vector and denote by $Q^{\top}$ the transpose of a matrix Q. Then, $f(x)=x^{\top}Px$. If $f(x)$ is positive definite, we say that P is positive and denote it by $P>0$. Liapunov [85] proved the following.

THEOREM 18.1. *Suppose that all characteristic roots of an $n\times n$ constant matrix A have negative real parts. For a given $C>0$, there exists one and only one $B>0$ which satisfies $A^{\top}B+BA=-C$.*

Actually, B is defined by $B=\int_{-\infty}^{0} Y^{\top}(t)CY(t)dt$, where $Y(t)$ is the matrix solution of $\dot{Y}=-YA$ such that $Y(0)=E$, where E is the unit matrix.

As a corollary, Liapunov's theorem on asymptotic stability admits a reciprocal theorem in the case of linear systems with constant coefficients. In a linear system

$$\frac{dx}{dt}=Ax, \tag{18.1}$$

suppose that all characteristic roots of A have negative real parts, where A is an $n\times n$ constant matrix. Then, clearly every solution of (18.1) tends to zero as $t\to\infty$ and hence, every solution is bounded, which implies that the zero solution is stable, by Theorem 11.2. Therefore, the zero solution is asymptotically stable. As the system (18.1) is autonomous, by Theorem 7.4, the zero solution is uniform-asymptotically stable. Conversely, if the zero solution is asymptotically stable, all characteristic roots of A have negative real parts.

For the linear system (18.1), consider a quadratic form $V(x) = x^{\top}Bx$. Then its derivative along the solution of (18.1) is $V'_{(18.1)}(x) = x^{\top}(A^{\top}B+BA)x$. By Theorem 18.1, for a given $C>0$, there exists a $B>0$ such that $A^{\top}B+BA=-C$. Therefore, by choosing a $B>0$, we have a positive definite quadratic form $V(x)$ such that $V'_{(18.1)}(x)$ is negative definite. Thus, we have the following theorem.

THEOREM 18.2. *If the zero solution of* (18.1) *is asymptotically stable, there exists a positive definite quadratic form* $V(x)$ *such that* $V'_{(18.1)}(x)$ *is negative definite.*

Corresponding to this theorem, Massera [94] proved a converse theorem in the case of stability (non-asymptotic).

THEOREM 18.3. *If the zero solution of* (18.1) *is stable, there exists a positive definite Liapunov function* $V(x)$ *such that* $V'_{(18.1)}(x)$ *is negative semidefinite.* $V(x)$ *may be taken as an algebraic form of any given even degree.*

Since Persidskii [111]-[113] and Malkin [87]-[89] discussed converse theorems, there have appeared many papers concerning converse theorems. A good survey of these papers can be found in [5], [37], [123]. Converse theorems are very important for studying the behavior of solutions of a perturbed system.

Let us begin with a converse theorem on uniform stability (cf. [60], [63], [135]). Consider a system of differential equations

$$\frac{dx}{dt} = F(t, x). \tag{18.2}$$

In this section, D will denote the domain such that $I \times S_H$, $H>0$, and we assume that $F(t, 0) \equiv 0$.

THEOREM 18.4. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* D *and that* $F(t, x) \in C_0(x)$. *If the zero solution of* (18.2) *is uniform-*

stable, there exists a Liapunov function $V(t, x) \in C_0(t, x)$ *satisfying the conditions in Theorem* 8.2 *on the domain* $0 \leqq t < \infty$, $\| x \| < H_1$, *where* $H_1 > 0$ *is some constant such that* $H_1 < H$.

PROOF. By uniform-stability, for any $\varepsilon > 0$ there exists a $\delta(\varepsilon) > 0$, $\delta < \varepsilon$, such that $x_0 \in S_\delta$ implies that $\| x(t; x_0, t_0) \| < \varepsilon$ for all $t \geqq t_0$. This $\delta(\varepsilon)$ can be assumed to be continuous, strictly monotone in ε. Let H^* be a positive constant such that $H^* < H$ and let D^* be the domain $0 \leqq t < \infty$, $\| x \| < \delta(\delta(H^*))$. Consider a solution $x(t; x_0, t_0)$, $(t_0, x_0) \in D^*$. Then, $\| x(t; x_0, t_0) \| < H$ for all $t \in [0, t_0]$ or at some t_1, $t_1 < t_0$, $\| x(t_1; x_0, t_0) \| = H$.

If we set

$$V(t, x) = \min_\tau \{ \| x(\tau; x, t) \| ; \ \tau \in [0, t] \cap I^* \}, \tag{18.3}$$

where I^* is the largest interval to the left of t on which $x(\tau; x, t)$ is defined, then clearly $V(t, x)$ is defined for each point $(t, x) \in D^*$ and we have $V(t, x) \leqq \| x \|$. Let $\varepsilon(\delta)$ be the inverse function of $\delta(\varepsilon)$ above. Then $\varepsilon(\delta)$ is continuous, strictly monotone increasing in δ, for which we have $\varepsilon(\| x \|) \leqq V(t, x)$.

Now we shall show that $V(t, x) \in C_0(t, x)$. When, the solution $x(\tau; x_0, t_0)$ through $P(t_0, x_0) \in D^*$ satisfies $\| x(\tau; x_0, t_0) \| < H$ for all $\tau \in [0, t_0]$, there exists a neighborhood $N(P)$ of P such that solutions starting in $N(P)$ stay in a neighborhood of $x(\tau; x_0, t_0)$ in D for all $\tau \in [0, t]$, because every solution is unique. Therefore, for the solutions starting in $N(P)$, I^* in (18.3) is the interval $[0, t]$. On the other hand, for a solution $x(\tau; x_0, t_0)$ such that $\| x(t_1; x_0, t_0) \| = H$ at some t_1, $t_1 < t_0$, let t^* be the value of t for which $\| x(t^*; x_0, t_0) \| = H^*$ at the first time to the left of t_0, $t^* < t_0$. If ε is small enough and $H^* - \varepsilon > \delta(H^*)$, the solutions starting in a suitable neighborhood $N(P)$ of $P(t_0, x_0)$ stay in the ε-neighborhood of $x(\tau; x_0, t_0)$, $t^* \leqq \tau \leqq t_0$, and hence, they are continuable to t^*. Suppose that one such solution $x(\tau; x, t)$ satisfies $\| x(t'; x, t) \| < \delta(\delta(H^*))$ at t', $0 \leqq t' < t^*$. Since $\| x(t'; x, t) \| < \delta(\delta(H^*))$, we have $\| x(t^*; x, t) \| < \delta(H^*)$, which contradicts $\| x(t^*; x, t) \| > H^* - \varepsilon$. Thus, $\min \| x(\tau; x, t) \|$, $(t, x) \in N(P)$, takes place at some τ such that $t^* \leqq \tau \leqq t$.

To show that $V(t, x) \in C_0(t, x)$, by the preceding it is sufficient to consider (t, x) and (t', x') which are close, and proceed in the following manner. Let $K(t) = \max\{\| F(\tau, x)\|\,;\ 0 \leqq \tau \leqq t,\ \|x\| \leqq H\}$ and let $A(t) = e^{K(t)t}$. If we set $V(t, x') = \|x(\tau'\,; x', t)\|$, we have $V(t, x) - V(t, x') \leqq \|x(\tau'\,; x, t)\| - \|x(\tau'\,; x', t)\| \leqq A(t)\|x - x'\|$. If we set $V(t, x) = \|x(\tau\,; x, t)\|$, we have $V(t, x) - V(t, x') \geqq \|x(\tau\,; x, t)\| - \|x(\tau\,; x', t)\| \geqq -A(t)\|x - x'\|$. Thus, we have

$$|V(t, x) - V(t, x')| \leqq A(t)\|x - x'\|\,. \tag{18.4}$$

Next, consider $V(t, x) - V(t', x)$, where $t < t'$. If $\|x(\tau\,; x, t')\|$ assumes its minimum at τ' where $t^* \leqq \tau' \leqq t$, $t^* \geqq 0$, then

$$\begin{aligned} V(t, x) - V(t', x) &\leqq \|x(\tau'\,; x, t)\| - \|x(\tau'\,; x, t')\| \\ &\leqq \|x(\tau'\,; x, t)\| - \|x(\tau'\,; X, t)\| \\ &\leqq A(t)\|X - x\|\,, \end{aligned}$$

where $X = x(t\,; x, t')$, and hence, $V(t, x) - V(t', x) \leqq A(t)K(t')(t' - t)$. In the same way, we have $V(t, x) - V(t', x) \geqq -A(t)K(t')(t' - t)$. If $V(t, x) = \|x(\tau\,; x, t)\|$ and $V(t', x) = \|x(\tau'\,; x, t')\|$, $t < \tau' \leqq t'$, we have

$$\begin{aligned} V(t, x) - V(t', x) &= \|x(\tau\,; x, t)\| - \|x(\tau'\,; x, t')\| \\ &\leqq \|x(t\,; x, t)\| - \|x(t\,; x, t')\| + \|x(t\,; x, t')\| \\ &\quad - \|x(\tau'\,; x, t')\| \\ &\leqq \|x - X\| + \|x(t\,; x, t') - x(\tau'\,; x, t')\| \\ &\leqq 2K(t')(t' - t)\,, \end{aligned}$$

and in the same way, we have $V(t, x) - V(t', x) \geqq -A(t)K(t')(t' - t)$. Thus, from these results and (18.4), it follows that $V(t, x) \in C_0(t, x)$.

Since $V(t, x(t\,; x_0, t_0))$ is clearly a non-increasing function of t, we have $V'_{(18.2)}(t, x) \leqq 0$. This completes the proof.

REMARK. In case $F(t, x) \in C_0(t, x)$, we can find a Liapunov function which has continuous partial derivatives of all orders.

Here, it should be noticed that in an autonomous system

$$\frac{dx}{dt} = F(x)\,, \qquad F(0) = 0\,,$$

an autonomous Liapunov function $V(x)$ does not necessarily exist, though the zero solution is uniform-stable. If such a Liapunov function exists, the stability property of the zero solution is stronger than uniform stability, that is, it has the integrable stability property as introduced by Vrkoč [134] which is equivalent to strong stability as introduced by Okamura (cf. [134], [135]). For example, in a scalar equation

$$\frac{dx}{dt}=\begin{cases} x\sin^2\dfrac{1}{x} & (x \neq 0) \\ 0 & (x=0), \end{cases}$$

the zero solution is uniform-stable, but not strongly stable. Of course, strong stability implies uniform-stability (cf. [135]).

If we only assume that $F(t, x)$ of (18.2) is continuous on D, we have the following theorem.

THEOREM 18.5. *In order that the zero solution of* (18.2) *is stable, it is necessary and sufficient that there exists a scalar function $V(t, x)$ defined on D which satisfies the following conditions;*

(i) $V(t, 0)\equiv 0$ *and* $V(t, x)>0$ *for* $x \neq 0$,

(ii) *corresponding to each* $\varepsilon>0$, $\varepsilon\leqq H$, *there exists a* $\lambda(\varepsilon)>0$ *such that* $\lambda(\varepsilon)\leqq V(t, x)$, *whenever* $0\leqq t<\infty$, $\|x\|=\varepsilon$,

(iii) $V(t, x)$ *is continuous at* $x=0$,

(iv) *for any solution* $x(t)$ *of* (18.2), *the function* $V(t, x(t))$ *is a non-increasing function of* t.

PROOF. The sufficiency will be proved easily. We shall see that the condition is necessary. Consider solutions of (18.2) starting from a point $P(t, x)\in D$ and denote by X_P this family. We define a function $v(P)$ in the following way: If P is on $\|x\|=H$ or if there exists a solution in X_P which reaches $\|x\|=H$, $v(P)=H$, and if otherwise,

$$v(P)=\sup\{\|x(\tau\,;\,x,t)\|\,;\ t\leqq\tau<\infty,\ x(\tau\,;\,x,t)\in X_P\}\,.$$

If we set $V(t, x)=v(P)$, $P=(t, x)$, then $V(t, x)$ is the desired function. Since $x(t)\equiv 0$ is stable, the solution through a point

$(t, 0)$ is unique to the right, and hence, $v(P)=0$ for $P(t, 0)$ and consequently $V(t, 0)\equiv 0$. For any point $P(t, x)$, it is clear that $\|x\|\leqq v(P)$, which implies (ii). By the stability, for any $\varepsilon>0$, there exists a $\delta(t, \varepsilon)>0$ such that if $\|x\|<\delta(t, \varepsilon)$, then $\|x(\tau ; x, t)\|<\varepsilon$ for all $\tau\geqq t$. From this, it follows that $V(t, x)$ is continuous at $x=0$. If $P(t, x)$ and $Q(t', x')$ are on the same solution and $t\leqq t'$, clearly $v(P)\geqq v(Q)$, which implies (iv).

REMARK. From the proof above, it can be easily seen that for uniform stability, condition (iii) is replaced by the condition

(iii)′ $V(t, x)$ is continuous at $x=0$ uniformly for all $t\in I$.

The result above is not an exact converse of Liapunov's Theorem. Kurzweil and Vrkoč [65] showed that, under only the assumption that $F(t, x)$ is continuous, uniformity of stability alone is not sufficient to assure the existence of a continuous Liapunov function, but additional conditions on $F(t, x)$ are required. To obtain a converse theorem in the case where F is continuous, they introduced a new concept of stability, which differs from the usual one. It is shown that, if uniqueness holds, both concepts coincide. They proved the existence of a Liapunov function which has continuous partial derivatives of all orders.

Kurzweil [63] proved the following theorem.

THEOREM 18.6. *Suppose that $F(t, x)$ of* (18.2) *is continuous with its partial derivatives of the first order with respect to x in D. If the zero solution of* (18.2) *is stable, there exists a Liapunov function $V(t, x)$ defined on D which satisfies the conditions in Theorem* 8.1 *and which has continuous partial derivatives of the first order with respect to all variables. In this case, $V'_{(18.2)}(t, x)$ is represented by* (1.9).

PROOF. From the stability, it follows that there exists a $\delta_0>0$ such that if $\|x_0\|<\delta_0$, then $\|x(t ; x_0, 0)\|<\frac{H}{2}$ for all $t\geqq 0$. Let G_1 and G_2 be the sets of those points $(t, x)\in D$ for which $\|x(0 ; x, t)\|$

$<\frac{\delta_0}{2}$ and $\| x(0; x, t) \| < \delta_0$, respectively, and let G_1^c be the complement of G_1 in D. Consider a function $\omega(r)$, which is continuously differentiable for $r \geqq 0$, such that $\omega(r) = 0$ for $r \leqq 0$, $\omega(r) > 0$ for $r > 0$ and $\omega(r) = \delta_0^2$ for $r \geqq \frac{\delta_0}{2}$. Now we define a Liapunov function $V(t, x)$ in the following way:

(18.5) $$V(t, x) = \delta_0^2 \quad \text{for} \quad (t, x) \in G_1^c,$$

(18.6) $$V(t, x) = \omega(\| x(0; x, t) \|) \quad \text{for} \quad (t, x) \in G_2 .$$

The two definitions (18.5), (18.6) agree in $G_1^c \cap G_2$.

Clearly $V(t, x)$ is non-negative and $V(t, 0) \equiv 0$. It can be easily verified that $V(t, x)$ has continuous partial derivatives with respect to all variables, because $\frac{\partial F}{\partial x}$ is continuous and G_1, G_2 are the open sets. Since $V(t, x)$ is a constant along every solution, we have $V'_{(18.2)}(t, x) = 0$.

Finally, we shall see the existence of a function $a(r)$. By the stability, for any $\varepsilon > 0$, $\varepsilon < H$, there exists a $\delta(\varepsilon) > 0$ such that if $\| x_0 \| < \delta$, $\| x(t; x_0, 0) \| < \varepsilon$ for all $t \geqq 0$. Therefore, if $\varepsilon \leqq \| x \| \leqq H$ and $(t, x) \in G_2$, we have $\| x(0; x, t) \| \geqq \delta$, which implies that $V(t, x) \geqq \min(\delta_0^2, \omega(\delta))$. Thus, we can find such a function $a(r)$.

Krasovskii [60] and Kurzweil [63] proved the converse of Theorem 8.2, by assuming that $\frac{\partial F}{\partial x}$ is continuous, where $V(t, x)$ also has continuous partial derivatives of the first order with respect to all variables. In [65], Kurzweil and Vrkoč showed that the assumption of continuity of the partial derivatives of F may be replaced by other weaker conditions.

§ 19. Converse theorems on asymptotic stability.

Let us begin with converse theorems on asymptotic stability in the large. First of all, we consider the linear system

(19.1) $$\frac{dx}{dt} = A(t)x ,$$

where $A(t)$ is a continuous $n\times n$ matrix on I.

THEOREM 19.1. *Suppose that there exist a $K>0$ and a c such that*

$$\| x(t\,;x_0,t_0)\| \leqq Ke^{-c(t-t_0)}\| x_0\|, \tag{19.2}$$

where $x(t\,;x_0,t_0)$ is a solution of (19.1) *and c is a constant* ($\gtreqless 0$). *Then, there exists a Liapunov function $V(t,x)$ which satisfies the following conditions;*

(i) $\| x\| \leqq V(t,x)\leqq K\| x\|$,

(ii) $|V(t,x)-V(t,x')|\leqq K\| x-x'\|$,

(iii) $V'_{(19.1)}(t,x)\leqq -cV(t,x)$.

PROOF. Let $V(t,x)$ be defined by

$$V(t,x)=\sup_{\tau\geqq 0}\| x(t+\tau\,;x,t)\| e^{c\tau}. \tag{19.3}$$

Then clearly $\| x\|\leqq V(t,x)$ and by (19.2),

$$V(t,x)\leqq \sup_{\tau\geqq 0} Ke^{-c\tau}\| x\| e^{c\tau}=K\| x\|.$$

Since the system is linear, we have the relation $x(t+\tau\,;x,t)-x(t+\tau\,;x',t)=x(t+\tau\,;x-x',t)$, and hence,

$$\begin{aligned}|V(t,x)-V(t,x')| &\leqq \sup_{\tau\geqq 0}\| x(t+\tau\,;x,t)-x(t+\tau\,;x',t)\| e^{c\tau}\\ &\leqq \sup_{\tau\geqq 0} Ke^{-c\tau}\| x-x'\| e^{c\tau}=K\| x-x'\|.\end{aligned}$$

Now we shall prove the continuity of $V(t,x)$. Take a $\delta\geqq 0$. We have

$$\begin{aligned}|V(t+\delta,x')-V(t,x)| \leqq\, &|V(t+\delta,x')-V(t+\delta,x)|\\ &+|V(t+\delta,x)-V(t+\delta,x(t+\delta\,;x,t))|\\ &+|V(t+\delta,x(t+\delta\,;x,t))-V(t,x)|.\end{aligned}$$

Since $V(t,x)$ is Lipschitzian in x and $x(t+\delta\,;x,t)$ is continuous in δ, the first two terms are small when $\| x-x'\|$ and δ are small. Let us consider the third term. Since $x(t+\delta+\tau\,;x(t+\delta\,;x,t),t+\delta)=x(t+\delta+\tau\,;x,t)$, we have

$$
\begin{aligned}
&| V(t+\delta, x(t+\delta\,; x, t)) - V(t, x)| \\
&= \Big| \sup_{\tau \geqq 0} \| x(t+\delta+\tau\,; x(t+\delta\,; x, t), t+\delta) \| e^{c\tau} - \sup_{\tau \geqq 0} \| x(t+\tau\,; x, t) \| e^{c\tau} \Big| \\
&= \Big| \sup_{\tau \geqq \delta} \| x(t+\tau\,; x, t) \| e^{c\tau} e^{-c\delta} - \sup_{\tau \geqq 0} \| x(t+\tau\,; x, t) \| e^{c\tau} \Big| .
\end{aligned}
$$

Set $a(\delta) = \sup_{\tau \geqq \delta} \| x(t+\tau\,; x, t) \| e^{c\tau}$. Then $a(\delta)$ is non-increasing and $a(\delta) \to a(0)$ as $\delta \to 0$, because $\| x(t+\tau\,; x, t) \| e^{c\tau}$ is a bounded continuous function for all $\tau \geqq 0$. Thus,

$$
| V(t+\delta, x(t+\delta\,; x, t)) - V(t, x)| = | a(\delta) e^{-c\delta} - a(0)|
$$

implies that the third term tends to zero as $\delta \to 0$. Therefore, the continuity of $V(t, x)$ is verified.

Finally, we shall establish condition (iii). Let $x' = x(t+h\,; x, t)$, $h > 0$. Then,

$$
\begin{aligned}
V(t+h, x') &= \sup_{\tau \geqq 0} \| x(t+h+\tau\,; x', t+h) \| e^{c\tau} \\
&= \sup_{\tau \geqq h} \| x(t+\tau\,; x, t) \| e^{c\tau} e^{-ch} \leqq V(t, x) e^{-ch} ,
\end{aligned}
$$

which implies

$$
\frac{V(t+h, x') - V(t, x)}{h} \leqq V(t, x) \frac{e^{-ch} - 1}{h} .
$$

From this, we obtain $V'_{(19.1)}(t, x) \leqq -cV(t, x)$.

By the same argument as above, we can obtain a converse theorem on exponential-asymptotic stability for a general system.

THEOREM 19.2. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* $I \times R^n$ *and that* $F(t, 0) \equiv 0$. *If* $F(t, x) \in C_0(x)$ *and the zero solution of* (18.2) *is exponential-asymptotically stable in the large, i. e., there exists a* $c > 0$ *and for any* $\alpha > 0$, *there exists a* $K(\alpha) > 0$ *such that if* $x_0 \in S_\alpha$,

$$
\| x(t\,; x_0, t_0) \| \leqq K(\alpha) e^{-c(t-t_0)} \| x_0 \| \quad \text{for all } t \geqq t_0 , \tag{19.4}
$$

there exists a Liapunov function $V(t, x)$ *defined on* $I \times R^n$ *which satisfies the following conditions;*

(i) $\| x \| \leqq V(t, x) \leqq K(\alpha) \| x \|$ *for* $x \in S_\alpha$,

(ii) $| V(t, x) - V(t, x')| \leqq L(t, \alpha) \| x - x' \|$ *for* $x \in S_\alpha$, $x' \in S_\alpha$,

(iii) $V'_{(18.2)}(t, x) \leqq -qcV(t, x)$, *where* $0 < q < 1$.

PROOF. For a constant q such that $0<q<1$, define

$$V(t, x)=\sup_{\tau \geqq 0} \|x(t+\tau ; x, t)\| e^{qc\tau} . \tag{19.5}$$

Then $\|x\| \leqq V(t, x)$ and for $x \in S_\alpha$

$$\|x\| \leqq V(t, x) \leqq \sup_{\tau \geqq 0} K(\alpha) e^{-(1-q)c\tau}\|x\| \leqq K(\alpha)\|x\| . \tag{19.6}$$

Now we shall prove (ii). In the course of the proof, we shall determine $L(t, \alpha)$ explicitly. Let $T(\alpha)$ be such that $K(\alpha)=e^{(1-q)cT(\alpha)}$. If $\tau \geqq T(\alpha)$, then $K(\alpha)e^{-(1-q)c\tau}\|x\| \leqq \|x\|$ and hence, from (19.6), it follows that $V(t, x)$ must be defined for τ such that $0 \leqq \tau \leqq T(\alpha)$. Therefore, for $x \in S_\alpha$ and $x' \in S_\alpha$,

$$\begin{aligned}|V(t, x)-V(t, x')| &\leqq \sup_{0 \leqq \tau \leqq T(\alpha)} \|x(t+\tau ; x, t)-x(t+\tau ; x', t)\| e^{qc\tau} \\ &\leqq e^{qcT(\alpha)} \exp\left\{\int_t^{t+T(\alpha)} M(s, \alpha) ds\right\}\|x-x'\| ,\end{aligned}$$

where $M(s, \alpha)$ is a continuous function of s such that $\|F(s, x)-F(s, x')\| \leqq M(s, \alpha)\|x-x'\|$ for $x \in S_{\alpha K(\alpha)}$ and $x' \in S_{\alpha K(\alpha)}$.

The remainder of the proof can be verified by the same argument as in Theorem 19.1.

COROLLARY. *Suppose that $F(t, x)$ of* (18.2) *is defined and continuous on $I \times S_H$, $H>0$, and that $F(t, x) \in \bar{C}_0(x)$. If there exist a $\delta_0>0$, a $c>0$ and a $K>0$ such that $x_0 \in S_{\delta_0}$ implies that $\|x(t; x_0, t_0)\| \leqq Ke^{-c(t-t_0)}\|x_0\|$ for all $t \geqq t_0$, then there exists a Liapunov function $V(t, x)$ defined on $I \times S_{\delta_0}$ which satisfies the conditions* (i) $\|x\| \leqq V(t, x) \leqq K\|x\|$, (ii) $|V(t, x)-V(t, x')| \leqq L\|x-x'\|$, *where $L>0$ is a constant, and* (iii) $V'_{(18.2)}(t, x) \leqq -qcV(t, x)$, *where $0<q<1$.*

Next, we shall prove a converse theorem on equiasymptotic stability in the large. First of all, we state a simple form of a lemma due to Massera [94].

LEMMA 19.1. *Given any real function $A(r, t, \varepsilon)$ of real variables, defined, continuous and positive in $Q: r \geqq 0$, $t \geqq 0$, $\varepsilon>0$, there are three continuous functions $h(r)$, $f(t)$, $g(\varepsilon)$ such that $h(r)>0$, $f(t)>0$, $g(\varepsilon)>0$ for $\varepsilon>0$, $g(0)=0$ and that $h(r)f(t)g(\varepsilon) \leqq A(r, t, \varepsilon)$ in Q.*

PROOF. Let $A^*(r, t, \varepsilon)$ be such that

$$A^*(r, t, \varepsilon) = \min\left\{A(r, t, \varepsilon),\ A\left(r, t, \frac{1}{\varepsilon}\right),\ \varepsilon,\ \frac{1}{\varepsilon}\right\}.$$

Then $A^*(r, t, \varepsilon)$ is continuous and positive in Q and clearly $A^*(r, t, \varepsilon) = A^*\left(r, t, \frac{1}{\varepsilon}\right)$. If we set $g(\varepsilon) = \inf\left\{A^*(r, t, \varepsilon);\ 0 \leqq r, t \leqq \max\left(\varepsilon, \frac{1}{\varepsilon}\right)\right\}$, $g(\varepsilon)$ is continuous, $g(\varepsilon) > 0$ for $\varepsilon > 0$, $g(0) = 0$ and $g\left(\frac{1}{\varepsilon}\right) = g(\varepsilon)$. Consider a function $B(r, t)$ such that $B(r, t) = \min[1, \inf\{A^*(r, t, \varepsilon)/g(\varepsilon);\ \varepsilon > 0\}]$. Then clearly $B(r, t)$ is positive. Since $A^*(r, t, \varepsilon) = A^*\left(r, t, \frac{1}{\varepsilon}\right)$ and $g(\varepsilon) = g\left(\frac{1}{\varepsilon}\right)$, we have $\inf\{A^*(r, t, \varepsilon)/g(\varepsilon);\ \varepsilon > 0\} = \inf\{A^*(r, t, \varepsilon)/g(\varepsilon);\ 0 < \varepsilon \leqq 1\}$. For an $\varepsilon > 0$ such that $\max(r, t) \leqq \frac{1}{\varepsilon}$, we have $\inf\{A^*(r, t, \varepsilon)/g(\varepsilon)\} \geqq 1$ and hence, the infimum can be taken for ε such that $\frac{1}{\max(r, t)} \leqq \varepsilon \leqq 1$ if $\max(r, t) > 1$. If $\max(r, t) \leqq 1$, for any ε, $0 < \varepsilon \leqq 1$, we have $\max(r, t) \leqq \frac{1}{\varepsilon}$ and hence, $B(r, t) = 1$. Thus, we have established the continuity of $B(r, t)$.

Moreover, if we set $h(r) = \inf\{B(r, t);\ 0 \leqq t \leqq r\}$, $h(r)$ is positive and continuous on $[0, \infty)$. Let $f(t)$ be $f(t) = \min[1, \inf\{B(r, t)/h(r);\ r \geqq 0\}]$. Since $\inf B(r, t)/h(r) \geqq 1$ if $r \geqq t$, $f(t) = \min[1, \inf\{B(r, t)/h(r);\ t \geqq r \geqq 0\}]$. Therefore, $f(t)$ is positive and continuous on $[0, \infty)$.

Thus, from the definitions of $A^*(r, t, \varepsilon)$, $B(r, t)$ and $f(t)$, it follows that $h(r)f(t)g(\varepsilon) \leqq A(r, t, \varepsilon)$.

REMARK. To simplify the proof, we have considered $A(r, t, \varepsilon)$. However, as can be seen easily from the proof, this lemma is valid for positive continuous functions of any number of variables.

THEOREM 19.3. *Suppose that $F(t, x)$ of (18.2) is continuous on $I \times R^n$ and that $F(t, 0) \equiv 0$. If $F(t, x) \in C_0(x)$ and if the zero solution of (18.2) is equiasymptotically stable in the large, there exists a*

Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the conditions in Theorem 11.4 and

$$|V(t, x) - V(t, x')| \leqq h(\alpha) f(t) \| x - x' \| \tag{19.7}$$

for $x \in S_\alpha$, $x' \in S_\alpha$, where $h(\alpha)$, $f(t)$ are suitable functions.

PROOF. Let $\Omega_{\sigma,\alpha}$ be a domain such that $t \in [0, \sigma]$, $x \in S_\alpha$. By equiasymptotic stability in the large, the solutions are equi-bounded, which implies the existence a $\beta(\sigma, \alpha) > 0$ such that if $(t_0, x_0) \in \Omega_{\sigma,\alpha}$, then $\| x(t; x_0, t_0) \| \leqq \beta(\sigma, \alpha)$ for all $t \geqq t_0$. Moreover, there exists a $T(\sigma, \alpha, \varepsilon) > 0$ such that $(t_0, x_0) \in \Omega_{\sigma,\alpha}$ implies that $\| x(t; x_0, t_0) \| < \varepsilon$ for all $t \geqq t_0 + T(\sigma, \alpha, \varepsilon)$. If $\varepsilon > 1$, we set $T(\sigma, \alpha, \varepsilon) = T(\sigma, \alpha, 1)$. Since $F(t, x) \in C_0(x)$, there exists an $L(\sigma, \alpha, \varepsilon) > 0$ such that if $0 \leqq t \leqq \sigma + T(\sigma, \alpha, \varepsilon)$, $\| x \| \leqq \beta(\sigma, \alpha)$, $\| x' \| \leqq \beta(\sigma, \alpha)$, we have $\| F(t, x) - F(t, x') \| \leqq L(\sigma, \alpha, \varepsilon) \| x - x' \|$. Let $F^*(\sigma, \alpha, \varepsilon)$ be $1 + \max \| F(t, x) \|$ for $0 \leqq t \leqq \sigma + T(\sigma, \alpha, \varepsilon)$, $\| x \| \leqq \beta(\sigma, \alpha)$. Further, β, T, L and F^* can be assumed to be continuous (cf. [136]).

For a $c > 0$, let $A(\sigma, \alpha, \varepsilon)$ be such that

$$A(\sigma, \alpha, \varepsilon) = e^{cT(\sigma,\alpha,\varepsilon)} \{ 2F^*(\sigma, \alpha, \varepsilon) e^{L(\sigma,\alpha,\varepsilon) T(\sigma,\alpha,\varepsilon)} + \beta(\sigma, \alpha) \} . \tag{19.7}$$

By Lemma 19.1, there exist three functions $h(\alpha)$, $f(\sigma)$ and $g(\varepsilon)$ such that $h(\alpha) > 0$, $f(\sigma) > 0$, $g(\varepsilon) > 0$ for $\varepsilon > 0$, $g(0) = 0$ and that

$$g(\varepsilon) A(\sigma, \alpha, \varepsilon) \leqq h(\alpha) f(\sigma) . \tag{19.8}$$

For $k = 1, 2, \cdots$, we define $V_k(t, x)$ as follows:

$$V_k(t, x) = g\left(\frac{1}{k}\right) \sup_{\tau \geqq 0} G_k(\| x(t+\tau; x, t) \|) e^{c\tau} , \tag{19.9}$$

where $G_k(z)$ is a function such that

$$G_k(z) = \begin{cases} z - \dfrac{1}{k} & \left(z \geqq \dfrac{1}{k}\right) \\ 0 & \left(0 \leqq z < \dfrac{1}{k}\right) . \end{cases}$$

Clearly $G_k(z) \to \infty$ as $z \to \infty$ for each k and $|G_k(z) - G_k(z')| \leqq |z - z'|$. From the definition of $V_k(t, x)$, it is clear that

$$g\left(\frac{1}{k}\right) G_k(\| x \|) \leqq V_k(t, x) \quad \text{and} \quad V_k(t, 0) \equiv 0 \tag{19.10}$$

and for $(t, x) \in \Omega_{\sigma,\alpha}$,

$$V_k(t, x) \leqq g\left(\frac{1}{k}\right) G_k(\beta(\sigma, \alpha)) e^{cT\left(\sigma, \alpha, \frac{1}{k}\right)} \tag{19.11}$$

$$\leqq g\left(\frac{1}{k}\right) \beta(\sigma, \alpha) e^{cT\left(\sigma, \alpha, \frac{1}{k}\right)},$$

and hence, by (19.7) and (19.8),

$$V_k(t, x) \leqq h(\alpha) f(\sigma) \quad \text{for} \quad (t, x) \in \Omega_{\sigma,\alpha}. \tag{19.12}$$

Now we shall see that $V_k(t, x) \in C_0(t, x)$. For $(t, x) \in \Omega_{\sigma,\alpha}$ and $(t', x') \in \Omega_{\sigma,\alpha}$, $t < t'$,

$$\begin{aligned}
&| V_k(t, x) - V_k(t', x') | \\
&\leqq g\left(\frac{1}{k}\right) \sup_{\tau \geqq 0} | G_k(\| x(t+\tau\,; x, t) \|) - G_k(\| x(t'+\tau\,; x', t') \|) | e^{c\tau} \\
&\leqq g\left(\frac{1}{k}\right) \sup_{T\left(\sigma, \alpha, \frac{1}{k}\right) \geqq \tau \geqq 0} e^{c\tau} \| x(t+\tau\,; x, t) - x(t'+\tau\,; x', t') \| \\
&\leqq g\left(\frac{1}{k}\right) \sup_{\tau} e^{c\tau} \{ \| x(t+\tau\,; x, t) - x(t'+\tau\,; x, t) \| \\
&\qquad + \| x(t'+\tau\,; x, t) - x(t'+\tau\,; x', t') \| \} \\
&\leqq g\left(\frac{1}{k}\right) \sup_{\tau} e^{c\tau} \left\{ F^*\left(\sigma, \alpha, \frac{1}{k}\right)(t'-t) \right. \\
&\qquad \left. + \| x(t'+\tau\,; X, t') - x(t'+\tau\,; x', t') \| \right\},
\end{aligned}$$

where $X = x(t'\,; x, t)$. On the other hand, for $0 \leqq \tau \leqq T\left(\sigma, \alpha, \frac{1}{k}\right)$

$$\begin{aligned}
&\| x(t'+\tau\,; X, t') - x(t'+\tau\,; x', t') \| \\
&\leqq \| X - x' \| \exp\left\{ L\left(\sigma, \alpha, \frac{1}{k}\right) T\left(\sigma, \alpha, \frac{1}{k}\right) \right\} \\
&\leqq \left\{ \| X - x \| + \| x - x' \| \right\} \exp\left\{ L\left(\sigma, \alpha, \frac{1}{k}\right) T\left(\sigma, \alpha, \frac{1}{k}\right) \right\} \\
&\leqq \left\{ F^*\left(\sigma, \alpha, \frac{1}{k}\right)(t'-t) + \| x - x' \| \right\} \exp\left\{ L\left(\sigma, \alpha, \frac{1}{k}\right) T\left(\sigma, \alpha, \frac{1}{k}\right) \right\}.
\end{aligned}$$

Thus, we have

$$(19.13)\quad |V_k(t,x)-V_k(t',x')|\leqq g\Big(\frac{1}{k}\Big)2F^*\Big(\sigma,\alpha,\frac{1}{k}\Big)$$
$$\times\exp\Big\{\Big(c+L\Big(\sigma,\alpha,\frac{1}{k}\Big)\Big)T\Big(\sigma,\alpha,\frac{1}{k}\Big)\Big\}[(t'-t)+\|x-x'\|]$$
$$\leqq h(\alpha)f(\sigma)\{|t-t'|+\|x-x'\|\}\qquad \text{(by (19.7) and (19.8))}.$$

Next, we shall prove that $V'_{k(18.2)}(t,x)\leqq -cV_k(t,x)$. For $h>0$ and $x'=x(t+h;x,t)$,

$$V_k(t+h,x')=g\Big(\frac{1}{k}\Big)\sup_{\tau\geqq 0}G_k(\|x(t+h+\tau;x',t+h)\|)e^{c\tau}$$
$$=g\Big(\frac{1}{k}\Big)\sup_{\tau\geqq 0}G_k(\|x(t+h+\tau;x,t)\|)e^{c\tau}$$
$$=g\Big(\frac{1}{k}\Big)\sup_{\tau\geqq h}G_k(\|x(t+\tau;x,t)\|)e^{c\tau}e^{-ch}$$
$$\leqq V_k(t,x)e^{-ch},$$

which implies

$$(19.14)\qquad V'_{k(18.2)}(t,x)\leqq -cV_k(t,x).$$

Now we define the desired function $V(t,x)$ by setting

$$(19.15)\qquad V(t,x)=\sum_{k=1}^{\infty}\frac{1}{2^k}V_k(t,x).$$

Since (19.12) implies the uniform convergence of the series of (19.15) in $\Omega_{\sigma,\alpha}$ and σ, α are arbitrary, $V(t,x)$ is defined and continuous on $I\times R^n$. Clearly $V(t,0)\equiv 0$. For x such that $\|x\|\geqq 1$, by (19.10) and (19.15),

$$V(t,x)>\frac{1}{2}V_1(t,x)\geqq\frac{1}{2}g(1)G_1(\|x\|)\geqq\frac{1}{2}g(1)(\|x\|-1)$$

and for x such that $\frac{1}{k}\leqq\|x\|<\frac{1}{k-1}$,

$$V(t,x)\geqq\frac{1}{2^{k+1}}V_{k+1}(t,x)\geqq\frac{1}{2^{k+1}}g\Big(\frac{1}{k+1}\Big)G_{k+1}(\|x\|)$$
$$\geqq\frac{1}{2^{k+1}}g\Big(\frac{1}{k+1}\Big)\Big(\|x\|-\frac{1}{k+1}\Big)\geqq\frac{1}{2^{k+1}}g\Big(\frac{1}{k+1}\Big)\frac{1}{k(k+1)}.$$

Therefore, we can find an $a(r)\in CIP$ such that $a(r)\to\infty$ as $r\to\infty$ and that $a(\|x\|)\leqq V(t,x)$.

For $(t, x) \in \Omega_{\sigma,\alpha}$ and $(t', x') \in \Omega_{\sigma,\alpha}$,

$$
\begin{aligned}
& | V(t, x) - V(t', x') | \\
& = \Big| \sum_{k=1}^{\infty} \frac{1}{2^k} \{ V_k(t, x) - V_k(t', x') \} \Big| \\
& \leqq \sum_{k=1}^{\infty} \frac{1}{2^k} | V_k(t, x) - V_k(t', x') | \\
& \leqq \sum_{k=1}^{\infty} \frac{1}{2^k} h(\alpha) f(\sigma) \{ |t-t'| + \| x-x' \| \} \quad \text{(by (19.13))} \\
& \leqq h(\alpha) f(\sigma) \{ |t-t'| + \| x-x' \| \} ,
\end{aligned}
$$

which implies (19.7).

Finally, we shall prove that $V'_{(18.2)}(t, x) \leqq -cV(t, x)$. In fact, we have

$$
\begin{aligned}
V'_{(18.2)}(t, x) & = \overline{\lim_{h \to 0+}} \frac{1}{h} \Big\{ \sum_{k=1}^{\infty} \frac{1}{2^k} V_k(t+h, x+hF(t, x)) - \sum_{k=1}^{\infty} \frac{1}{2^k} V_k(t, x) \Big\} \\
& \leqq \sum_{k=1}^{\infty} \overline{\lim_{h \to 0+}} \frac{1}{2^k} \frac{1}{h} \{ V_k(t+h, x+hF(t, x)) - V_k(t, x) \} \\
& \leqq \sum_{k=1}^{\infty} \frac{1}{2^k} (-cV_k(t, x)) = -cV(t, x) .
\end{aligned}
$$

As we can see easily from the theorem above, for the system (18.2) defined on $I \times R^n$, where $x(t) \equiv 0$ is not necessarily the solution of (18.2), we obtain a necessary condition in order that the 0-set (see Section 16) is quasi-equiasymptotically stable set of (18.2) in the large.

THEOREM 19.4. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* $I \times R^n$ *and that* $F(t, x) \in C_0(x)$. *If the* 0-*set is a quasi-equiasymptotically stable set of* (18.2) *in the large, there exists a Liapunov function* $V(t, x)$ *defined on* $I \times R^n$ *which satisfies the conditions in Theorem* 19.3, *except* $V(t, 0) \equiv 0$.

From Theorem 19.3, a converse theorem on uniform-asymptotic stability in the large is obtained. In this case, β and T in the proof of Theorem 19.3 can be replaced by $\beta(\alpha)$ and $T(\alpha, \varepsilon)$, respectively, and we can assume that $\beta(\alpha) \to 0$ as $\alpha \to 0$, because of uniform stability of the zero solution. From (19.11) and $e^{cT(\alpha,\varepsilon)} \leqq A(0, \alpha, \varepsilon)$, it follows that $V_k(t, x) \leqq g\left(\frac{1}{k}\right) A\left(0, \alpha, \frac{1}{k}\right) \beta(\alpha) \leqq h(\alpha) f(0) \beta(\alpha)$, and hence, $V(t, x) \leqq h(\alpha) f(0) \beta(\alpha)$ for all $t \in I$ and $x \in S_\alpha$. This implies the existence of a function $b(r) \in CI$ such that $b(r) \to 0$ as $r \to 0$ and that $V(t, x) \leqq b(\|x\|)$. Thus, we obtain the following theorem.

THEOREM 19.5. *Suppose that $F(t, x)$ of* (18.2) *is continuous on $I \times R^n$ and that $F(t, 0) \equiv 0$. If $F(t, x) \in C_0(x)$ and if the zero solution of* (18.2) *is uniform-asymptotically stable in the large, there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the conditions in Theorem* 11.5 *and* (19.7). *In particular, condition* (ii) *in Theorem* 11.5 *can be replaced by $V'_{(18.2)}(t, x) \leqq -cV(t, x)$.*

Barbašin and Krasovskii [12], [13] were the first to formulate the converse theorem on uniform-asymptotic stability in the large. For differential equations whose solutions are not uniquely determined by the initial values, Kurzweil [64] defined a concept which comprises uniform-asymptotic stability in the large. Kurzweil [64] and Massera [94], [96] showed that there exists a Liapunov function which has continuous partial derivatives of all orders under the assumption that $F(t, x) \in C_0(t, x)$.

Here we give some notes on Theorem 19.5. As is clear from the proof of Theorem 19.3, if $F(t, x) \in \bar{C}_0(x)$, L and F^* also depend only on α and hence, for a suitable continuous function $h(\alpha)$, if $x \in S_\alpha$ and $x' \in S_\alpha$,

$$(19.16) \qquad |V(t, x) - V(t, x')| \leqq h(\alpha) \|x - x'\| .$$

THEOREM 19.6. *Under the assumption in Theorem* 19.5, *if $F(t, x) \in \bar{C}_0(x)$, there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the conditions in Theorem* 11.5 *and* (19.16).

Now, let us consider the case where $F(t, x)$ is almost periodic in t. The following assumptions will be made:

(a) $F(t, x)$ of the system (18.2) is defined and continuous on $-\infty < t < \infty$, $x \in R^n$, $F(t, x) \in \bar{C}_0(x)$ and $F(t, 0) \equiv 0$.

(b) $F(t, x)$ is almost periodic in t uniformly with respect to

$x \in R^n$, i.e., uniformly with respect to $x \in S$ for any compact set S in R^n.

THEOREM 19.7. *Under the assumptions* (a), (b), *if the zero solution of* (18.2) *is unifom-asymptotically stable in the large for* $t_0 \geqq 0$, *then it is perfectly uniform-asymptotically stable in the large.*

This theorem is an immediate consequence of Theorems 19.6 and 11.8.

The following theorem shows the existence of an almost periodic Liapunov function (cf. [99], [149]).

THEOREM 19.8. *Under the assumptions* (a), (b), *if the zero solution of* (18.2) *is uniform-asymptotically stable in the large for* $t_0 \geqq 0$, *then there exists a Liapunov function* $V(t, x)$ *defined on* $-\infty < t < \infty$, $x \in R^n$ *which is almost periodic in t uniformly with respect to* $x \in R^n$ *and which satisfies* (19.16) *and the following conditions;*

(i) $a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, *where* $a(r)$ *and* $b(r)$ *are the same as in Theorem* 11.5,

(ii) $V'_{(18.2)}(t, x) \leqq -cV(t, x)$, *where* $c > 0$ *is a constant.*

In particular, if $F(t, x)$ *is periodic in t, then* $V(t, x)$ *also is periodic in t with the same period. In case F is autonomous, we can find an autonomous Liapunov function.*

PROOF. The construction of the desired Liapunov function $V(t, x)$ is done in the same way as in Theorem 19.3. In this case, $\Omega_{\sigma,\alpha}$ is the domain $[-\sigma, \sigma] \times S_\alpha$. By Theorem 19.7, the zero solution is perfectly uniform-asymptotically stable in the large and hence, β, T and F^* in Theorem 19.3 are independent of σ and consequently L depends only on α. For a $c > 0$, let $A(\alpha, \varepsilon)$ be such that

$$A(\alpha, \varepsilon) = e^{cT(\alpha, \varepsilon)}\{(2F^*(\alpha) + T(\alpha, \varepsilon))e^{L(\alpha)T(\alpha, \varepsilon)} + \beta(\alpha)\},$$

and let the two functions $h(\alpha)$, $g(\varepsilon)$ be such that $g(\varepsilon)A(\alpha, \varepsilon) \leqq h(\alpha)$ by Lemma 19.1.

Define $V_k(t, x)$ by (19.9) with $g(\varepsilon)$ above. Here, we shall show that $V_k(t, x)$ is almost periodic in t uniformly with respect to $x \in S$ for any compact set S in R^n. Without loss of generality, we can assume S to be a set S_α. If $x \in S$, then $\|x(\tau; x, t)\| < \beta(\alpha)$ for all $\tau \geqq t$. By almost periodicity of $F(t, x)$ and Theorem 6.3, for any sequence there exists a subsequence $\{\tau_p\}$ for which $F(t+\tau_p, x)$ is uniformly convergent for $t \in (-\infty, \infty)$, $x \in S_\beta$.

Let us consider $V_k(t+\tau_p, x) - V_k(t+\tau_q, x)$. For $x \in S$, we have

$$\begin{aligned}
&|V_k(t+\tau_p, x) - V_k(t+\tau_q, x)| \\
&\quad \leqq \sup_{\tau \geqq 0} g\left(\frac{1}{k}\right) e^{c\tau} |G_k(\|x(t+\tau_p+\tau; x, t+\tau_p)\|) \\
&\quad\quad -G_k(\|x(t+\tau_q+\tau; x, t+\tau_q)\|)| \\
&\quad \leqq g\left(\frac{1}{k}\right) \sup_{T\left(\alpha, \frac{1}{k}\right) \geqq \tau \geqq 0} e^{c\tau} \|x(t+\tau_p+\tau; x, t+\tau_p) \\
&\quad\quad -x(t+\tau_q+\tau; x, t+\tau_q)\|.
\end{aligned}$$

On the other hand, we have

$$x(t+\tau_p+\tau; x, t+\tau_p) = x + \int_t^{t+\tau} F(\sigma+\tau_p, x(\sigma+\tau_p; x, t+\tau_p)) d\sigma,$$

$$x(t+\tau_q+\tau; x, t+\tau_q) = x + \int_t^{t+\tau} F(\sigma+\tau_q, x(\sigma+\tau_q; x, t+\tau_q)) d\sigma$$

and hence,

$$\begin{aligned}
&\|x(t+\tau_p+\tau; x, t+\tau_p) - x(t+\tau_q+\tau; x, t+\tau_q)\| \\
&\quad \leqq \int_t^{t+\tau} \|F(\sigma+\tau_p, x(\sigma+\tau_p; x, t+\tau_p)) \\
&\quad\quad -F(\sigma+\tau_q, x(\sigma+\tau_p; x, t+\tau_p))\| d\sigma \\
&\quad\quad +\int_t^{t+\tau} \|F(\sigma+\tau_q, x(\sigma+\tau_p; x, t+\tau_p)) \\
&\quad\quad -F(\sigma+\tau_q, x(\sigma+\tau_q; x, t+\tau_q))\| d\sigma.
\end{aligned}$$

Since $x(\sigma+\tau_p; x, t+\tau_p) \in S_\beta$, there is an integer $n_0(\varepsilon) > 0$ such that for all $p, q \geqq n_0(\varepsilon)$, $\|F(\sigma+\tau_p, x(\sigma+\tau_p; x, t+\tau_p)) - F(\sigma+\tau_q, x(\sigma+\tau_p; x, t+\tau_p))\| < \varepsilon$. Thus, for $p, q \geqq n_0(\varepsilon)$,

$$\|x(t+\tau_p+\tau\,;\,x,\,t+\tau_p)-x(t+\tau_q+\tau\,;\,x,\,t+\tau_q)\|$$

$$\leqq \tau\varepsilon+L(\alpha)\int_t^{t+\tau}\|x(\sigma+\tau_p\,;\,x,\,t+\tau_p)-x(\sigma+\tau_q\,;\,x,\,t+\tau_q)\|\,d\sigma\,.$$

By Gronwall's inequality

$$\|x(t+\tau_p+\tau\,;\,x,\,t+\tau_p)-x(t+\tau_q+\tau\,;\,x,\,t+\tau_q)\|\leqq\tau\varepsilon e^{L\tau}\,,$$

which implies that if $p,\,q\geqq n_0(\varepsilon)$,

$$|V_k(t+\tau_p,\,x)-V_k(t+\tau_q,\,x)|$$

$$\leqq g\Big(\frac{1}{k}\Big)T\Big(\alpha,\,\frac{1}{k}\Big)\exp\Big\{(L(\alpha)+c)T\Big(\alpha,\,\frac{1}{k}\Big)\Big\}\,\varepsilon\leqq h(\alpha)\varepsilon\,.$$

This shows that $V_k(t+\tau_p,\,x)$ is uniformly convergent for $t\in(-\infty,\infty)$, $x\in S$ and hence, by Theorem 6.3, $V_k(t,\,x)$ is almost periodic in t uniformly with respect to $x\in R^n$. If we define $V(t,\,x)$ by (19.15), for all $p,\,q\geqq n_0(\varepsilon)$ and $x\in S$

$$|V(t+\tau_p,\,x)-V(t+\tau_q,\,x)|\leqq h(\alpha)\varepsilon\,,$$

which implies the almost periodicity of $V(t,\,x)$.

From the proof of Theorem 19.3, converse theorems on equi-asymptotic stability and uniform-asymptotic stability follow immediately. Massera [92] was the first to formulate such a theorem and to give an indirect proof for equiasymptotic stability. Under the assumption that $F(t,\,x)$ has continuous partial derivatives of the first order with respect to x, Malkin [88] gave a converse theorem on uniform-asymptotic stability. Kurzweil [64] also considered a converse theorem by assuming only that $F(t,\,x)$ is continuous, but he replaced the requirement of uniform-asymptotic stability by strong stability. Barbašin [11] considered the converse problem by means of completely different techniques, and used methods of the theory of dynamical systems. Another proof of Barbašin's result was given by Krasovskii [61].

Let $F(t,\,x)$ of the system (18.2) be assumed to be continuous on $I\times S_H$, $H>0$, and let H' be a positive number such that $H'<\delta_0$, where $\delta_0>0$ is the number in Definition 7.6. By considering only α such that $\alpha=H'$ in Theorem 19.5, the following theorem is obtained.

THEOREM 19.9. *Suppose that $F(t, x)$ of* (18.2) *is continuous on $I\times S_H$ and that $F(t, 0)\equiv 0$. If $F(t, x)\in C_0(x)$ and if the zero solution of* (18.2) *is uniform-asymptotically stable, there exists a Liapunov function $V(t, x)$ defined on $I\times S_{H'}$ which satisfies the conditions in Theorem* 8.3 *and*

$$(19.17)\qquad |V(t, x)-V(t, x')|\leqq f(t)\|x-x'\|,$$

where $f(t)$ is a suitable function. In particular, the condition on $V'_{(18.2)}(t, x)$ can be replaced by $V'_{(18.2)}(t, x)\leqq -cV(t, x)$, where $c>0$ is a constant. Moreover, if $F(t, x)\in \bar{C}_0(x)$, condition (19.17) *is replaced by*

$$(19.18)\qquad |V(t, x)-V(t, x')|\leqq K\|x-x'\|,$$

where $K>0$ is a constant.

REMARK. In case $F(t, x)\in C_0(t, x)$, we can find a Liapunov function which has continuous partial derivatives of all orders (cf. [64], [94], [96]).

Corresponding to Theorems 19.7 and 19.8, we have the following theorem.

THEOREM 19.10. *Suppose that $F(t, x)$ of* (18.2) *is defined and continuous on $-\infty<t<\infty$, $x\in S_H$, $H>0$, and $F(t, 0)\equiv 0$, $F(t, x)\in \bar{C}_0(x)$. Moreover, suppose that $F(t, x)$ is almost periodic in t uniformly with respect to $x\in S_H$. If the zero solution of* (18.2) *is uniform-asymptotically stable for $t_0\geqq 0$, then it is perfectly uniform-asymptotically stable and there exists a Liapunov function $V(t, x)$ defined on $-\infty<t<\infty$, $x\in S_{H'}$, where $H'<H$ is a suitable constant, which is almost periodic in t uniformly with respect to $x\in S_{H'}$ and which satisfies* (19.18) *for all t and the following conditions;*

(i) *$a(\|x\|)\leqq V(t, x)\leqq b(\|x\|)$, where $a(r)$ and $b(r)$ are the same as in Theorem* 8.2,

(ii) *$V'_{(18.2)}(t, x)\leqq -cV(t, x)$, where $c>0$ is a constant.*

In particular, if $F(t, x)$ is periodic in t of period ω, so is $V(t, x)$. In case F is autonomous, we can find an autonomous Liapunov function.

In the case of exponential-asymptotic stability in an almost periodic system, we have the following property.

THEOREM 19.11. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* $(-\infty, \infty)\times R^n$, $F(t, x)\in \bar{C}_0(x)$ *and that* $F(t, x)$ *is almost periodic in* t *uniformly with respect to* $x\in R^n$. *If there exist a* $K(\alpha)\geqq 1$ *and a* $c>0$ *such that for any* $t_0\geqq 0$, *if* $x_0\in S_\alpha$,

$$\| x(t\,;\, x_0, t_0)\| \leqq K(\alpha)e^{-c(t-t_0)}\| x_0\| \,, \qquad t\geqq t_0\,, \tag{19.19}$$

then for any $t_0\in(-\infty, \infty)$, *if* $x\in S_\alpha$, *we have also* (19.19).

For the proof, see [149].

§ 20. Converse theorems on boundedness.

Throughout this section, $F(t, x)$ of the system (18.2) will be assumed to be defined and continuous on $I\times R^n$.

THEOREM 20.1. *If* $F(t, x)\in C_0(x)$ *and if the solutions of* (18.2) *are uniform-bounded, there exists a Liapunov function* $V(t, x)$ *defined on* $I\times R^n$ *which satisfies the conditions in Theorem* 10.2.

PROOF. The argument is the same as in the proof of Theorem 18.4. Define a Liapunov function $V(t, x)$ by

$$V(t, x)=\min_{\tau}\,\{\| x(\tau\,;\, x, t)\|\,;\ \tau\in[0, t]\cap I^*\}\,, \tag{20.1}$$

where I^* is the largest interval to the left of t on which $x(\tau\,;\, x, t)$ is defined. By uniform-boundedness, for any $\alpha>0$ there exists a $\beta(\alpha)>0$ such that if $x_0\in S_\alpha$, then $\| x(t\,;\, x_0, t_0)\| <\beta(\alpha)$ for all $t\geqq t_0$, and we can assume $\beta(\alpha)$ to be continuous, strictly increasing.

In this theorem, for a solution $x(\tau\,;\, x_0, t_0)$ such that $\| x(\tau\,;\, x_0, t_0)\| \to\infty$ as $\tau\to\sigma+0$ for some $\sigma\geqq 0$, t^* in the proof of Theorem 18.4 is determined to be the value of t where $\| x(t^*\,;\, x_0, t_0)\| =\beta(2\| x_0\|)+1$ at the first time to the left of t_0, $t^*<t_0$.

Thus, this theorem can be proved by the same idea used in the proof of Theorem 18.4. For the details, see [137].

REMARK. If $F(t, x)\in C_0(t, x)$, we can find a Liapunov function which has continuous partial derivatives of all orders [137].

Under only the assumption that $F(t, x)$ is continuous, we can prove necessary and sufficient conditions for boundedness, by using the section of solutions by hyperplanes (cf. [123], [137]).

THEOREM 20.2. *In order that for any point* $(t_0, x_0) \in I \times R^n$ *there exists a* $\beta(t_0, x_0) > 0$ *such that* $\| x(t; x_0, t_0) \| < \beta(t_0, x_0)$ *for all* $t \geqq t_0$, *it is necessary and sufficient that there exists a scalar function* $V(t, x) > 0$ *on* $I \times R^n$ *which satisfies the conditions that* (i) $V(t, x) \to \infty$ *uniformly for* t *as* $\| x \| \to \infty$, (ii) *for any solution* $x(t)$ *of* (18.2), $V(t, x(t))$ *is nonincreasing in* t. *For equi-boundedness, we require, in addition, the condition that* (iii) *there exists a* $\beta(t, \alpha) > 0$ *such that* $V(t, x) \leqq \beta(t, \alpha)$ *for* $x \in S_\alpha$. *Moreover, for uniform-boundedness,* $V(t, x)$ *may be defined on* $t \in I$, $\| x \| \geqq R$, *where* R *may be large, and* β *in* (iii) *depends only on* α.

In case $F(t, x) \in C_0(x)$, converse theorems on equiultimate and uniform-ultimate boundedness will be obtained by the same argument as in Theorem 19.3 (cf. [137]).

THEOREM 20.3. *If* $F(t, x) \in C_0(x)$ *and if the solutions of* (18.2) *are equiultimately bounded for bound* B', *there exists a Liapunov function* $V(t, x)$ *defined on* $I \times R^n$ *which satisfies the conditions in Theorem* 10.5 *on* $t \in I$, $\| x \| \geqq B$, $B > B'$.

This theorem and Theorem 20.4 follow immediately from converse theorems on quasi-asymptotic stability of a set in the large which will be seen in Section 22. However, we shall give here an outline of a direct proof of Theorem 20.3.

Let $\Omega_{\sigma,\alpha}$ be the same domain as in Theorem 19.3. By equi-boundedness, there exists a $\beta(\sigma, \alpha) > 0$ for which $\| x(t; x_0, t_0) \| < \beta$ for $(t_0, x_0) \in \Omega_{\sigma,\alpha}$. Moreover, there exists a $T(\sigma, \alpha) > 0$ such that if $(t_0, x_0) \in \Omega_{\sigma,\alpha}$, then $\| x(t; x_0, t_0) \| < B'$ for all $t \geqq t_0 + T(\sigma, \alpha)$. Let $L(\sigma, \alpha) > 0$ be a constant such that $\| F(t, x) - F(t, x') \| \leqq L(\sigma, \alpha) \| x - x' \|$ for $t \in [0, \sigma + T]$, $x \in S_\beta$ and $x' \in S_\beta$, and let $F^*(\sigma, \alpha)$ be $1 + \max \| F(t, x) \|$ for $t \in [0, \sigma + T]$, $x \in S_\beta$. Further, β, T, L and F^* can be assumed to be continuous.

Letting $G(z)$ be a function such that $G(z)=z-B'$ for $z \geqq B'$, $G(z)=0$ for $0 \leqq z < B'$,

$$V(t, x)=\sup_{\tau \geqq 0} G(\| x(t+\tau\,;\, x, t) \|)e^{c\tau}.$$

This $V(t, x)$ can be defined for all $(t, x) \in I \times R^n$. For example, the Lipschitz condition on $V(t, x)$ will be

(20.2) $\quad | V(t, x)-V(t', x') | \leqq A(\sigma, \alpha) | t-t' | + e^{(c+L(\sigma,\alpha))T(\sigma,\alpha)} \| x-x' \|$,

where $(t, x) \in \Omega_{\sigma,\alpha}$, $(t', x') \in \Omega_{\sigma,\alpha}$ and $A(\sigma, \alpha)=2F^*(\sigma, \alpha) \exp \{(c+L(\sigma, \alpha))T(\sigma, \alpha)\}$.

THEOREM 20.4. *If $F(t, x) \in C_0(x)$ and if the solutions of* (18.2) *are uniform-bounded and uniform-ultimately bounded, there exists a Liapunov function $V(t, x)$ defined on $t \in I$, $\| x \| \geqq R$, $R > 0$, which satisfies the conditions in Theorem* 10.4. *More precisely, if the solutions are uniform-ultimately bounded for bound B', $V(t, x)$ can be defined on $I \times R^n$, satisfies the conditions in Theorem* 10.5, *and $V(t, x) \leqq b(\| x \|)$ on $t \in I$, $\| x \| \geqq B$, $B > B'$, where $b(r) \in CI$.*

COROLLARY. *In Theorem* 20.4, *if $F(t, x) \in \bar{C}_0(x)$, for $x \in S_\alpha$, $x' \in S_\alpha$ and a suitable function $h(\alpha) > 0$,*

(20.3) $$| V(t, x)-V(t, x') | \leqq h(\alpha) \| x-x' \| .$$

Corresponding to Theorems 19.7 and 19.8, by the same idea, we have the following theorem concerned with an almost periodic system.

THEOREM 20.5. *Suppose that $F(t, x)$ of* (18.2) *is defined and continuous on $-\infty < t < \infty$, $x \in R^n$ and $F(t, x) \in \bar{C}_0(x)$ and that $F(t, x)$ is almost periodic in t uniformly with respect to $x \in R^n$. If the solutions of* (18.2) *are uniform-bounded and uniform-ultimately bounded for $t_0 \geqq 0$, then they are perfectly uniform-bounded and perfectly uniform-ultimately bounded, and there exists a Liapunov function $V(t, x)$ defined on $(-\infty, \infty) \times R^n$ which is almost periodic in t uniformly with respect to $x \in R^n$ and which satisfies* (20.3) *and the conditions in Theorem* 20.4 *for all t. In particular, if $F(t, x)$ is periodic in t of period ω, so is $V(t, x)$. In case F is autonomous,* *we can find an autonomous Liapunov function.*

In the case where $F(t, x)$ of (18.2) has continuous partial derivatives of the first order with respect to x and $F(t, x)$ is bounded for bounded x, it is known that if the solutions of (18.2) are uniform-bounded and uniform-ultimately bounded, there exists a Liapunov function which satisfies the conditions in Theorem 10.4 and which has continuous partial derivatives of the first order with respect to all variables.

§ 21. Converse theorems on extreme stability.

Consider the system (18.2) and its associated system

$$\frac{dx}{dt} = F(t, x), \qquad \frac{dy}{dt} = F(t, y). \tag{21.1}$$

First of all, we consider the case where (18.2) is uniform-asymptotically stable with respect to (∞, ∞). The following assumptions will be made:

(a) $F(t, x)$ of (18.2) is defined and continuous on $I \times R^n$ and $F(t, x) \in \bar{C}_0(x)$.

(b) The solutions of (18.2) are uniform-bounded.

(c) The system (18.2) is uniform-asymptotically stable with respect to (∞, ∞).

THEOREM 21.1. *Under the assumptions above, there exists a Liapunov function $V(t, x, y)$ defined on $I \times R^n \times R^n$ which satisfies the following conditions;*

(i) $a(\| x-y \|) \leqq V(t, x, y) \leqq b(\| x-y \|)$, *where* $a(r) \in CIP$ *and* $b(r) \in CIP$,

(ii) *for any* $\alpha > 0$, *if* $x_i \in S_\alpha$ *and* $y_i \in S_\alpha$, $i = 1, 2$,

$$| V(t, x_1, y_1) - V(t, x_2, y_2) | \leqq h(\alpha) \{ \| x_1 - x_2 \| + \| y_1 - y_2 \| \},$$

where $h(\alpha) > 0$ *is a continuous function,*

(iii) $$V'_{(21.1)}(t, x, y) \leqq -V(t, x, y).$$

PROOF. By (b), for $x_0 \in S_\alpha$ and $y_0 \in S_\alpha$, there exists a $\beta(\alpha) > 0$ for which $\| x(t; x_0, t_0) \| \leqq \beta(\alpha)$, $\| x(t; y_0, t_0) \| \leqq \beta(\alpha)$ for all $t \geqq t_0$. By (c), corresponding to each $\varepsilon>0$, there exists a $T(\varepsilon, \alpha)>0$ such that if $t \geqq t_0+T(\varepsilon, \alpha)$, then $\| x(t; x_0, t_0)-x(t; y_0, t_0)\| < \varepsilon$ for $x_0 \in S_\alpha$, $y_0 \in S_\alpha$. We assume that if $\varepsilon>1$, $T(\varepsilon, \alpha)=T(1, \alpha)$. Moreover, there exists an $L(\beta(\alpha))>0$ such that $\| F(t, x)-F(t, x')\| \leqq L(\beta(\alpha))\| x-x'\|$ for $x \in S_\beta$, $x' \in S_\beta$, because $F(t, x) \in \bar{C}_0(x)$. We can assume that β, T and L are continuous. Let $A(\varepsilon, \alpha)$ be a positive continuous function for $\varepsilon>0$, $\alpha \geqq 0$ such that

$$A(\varepsilon, \alpha)=\exp\{L(\beta(\alpha))+1)T(\varepsilon, \alpha)\}+2\beta(\alpha)\exp\{T(\varepsilon, \alpha)\}\,.$$

Then there exist two continuous functions $g(\varepsilon)$, $h(\alpha)$ such that $g(\varepsilon)>0$ for $\varepsilon \neq 0$, $g(0)=0$, $h(\alpha)>0$ and that $g(\varepsilon)A(\varepsilon, \alpha) \leqq h(\alpha)$, by Lemma 19.1.

For $k=1, 2, \cdots$, we define $V_k(t, x, y)$ to be

$$V_k(t, x, y)=g\Big(\frac{1}{k}\Big)\sup_{\tau \geqq 0} G_k(\| x(t+\tau; x, t)-x(t+\tau; y, t)\|)e^\tau\,,$$

where $G_k(z)=z-\dfrac{1}{k}$ for $z \geqq \dfrac{1}{k}$ and $G_k(z)=0$ for $0 \leqq z<\dfrac{1}{k}$, and if $V(t, x, y)$ is defined by $V(t, x, y)=\sum\limits_{k=1}^{\infty}\dfrac{1}{2^k}V_k(t, x, y)$, this $V(t, x, y)$ can be verified to be the desired Liapunov function.

In the case where $H^*<\infty$, the following assumptions will be made:

(d) $F(t, x)$ of (18.2) is continuous on $I \times S_H$, $H>0$.

(e) $\| F(t, x)-F(t, x')\| \leqq L\| x-x'\|$, where $L>0$ is a constant.

(f) The system (18.2) is uniform-asymptotically stable with respect to (H^*, H).

THEOREM 21.2. *Under the assumptions above, there exists a Liapunov function $V(t, x, y)$ defined on $I \times S_{H^*} \times S_{H^*}$ which satisfies the conditions* (i), (iii) *in Theorem* 21.1 *and*

$$\text{(ii)}' \qquad | V(t, x_1, y_1)-V(t, x_2, y_2)| \leqq K\{\| x_1-x_2\|+\| y_1-y_2\|\}\,,$$

where $K>0$ is a constant.

This theorem can be proved by setting $\alpha=H^*$ in the proof of Theorem 21.1.

THEOREM 21.3. *Suppose that $F(t, x)$ of* (18.2) *is continuous on $I\times R^n$ and that $F(t, x)\in C_0(x)$. If the system* (18.2) *is uniform-distance-bounded and uniform-ultimately distance-bounded, there exists a Liapunov function $V(t, x, y)$ satisfying the conditions in Theorem* 15.7.

REMARK. In Theorem 21.3, we can find a Liapunov function which satisfies the condition $V'_{(21.1)}(t, x, y)\leqq -cV(t, x, y)$, where $c>0$ is a constant.

For other converse theorems, see [139]. Here, it is noticed that if F of (18.2) is autonomous or periodic in t, we can find a Liapunov function which is autonomous or periodic in t. This is clear from the construction of the Liapunov function.

§ 22. Converse theorems on stability of a set.

We shall establish relationships between the stability of a set and Liapunov functions. Considering the system (18.2) and a set M in $I\times R^n$, we use the notations in Section 16 and assume that M satisfies the conditions given in Section 16.

LEMMA 22.1. *If M satisfies condition* (16.1), *the β in Definition* 16.4 *can be chosen as a continuous function of t_0.*

LEMMA 22.2. *If* (16.1) *is satisfied, for any $\varepsilon>0$*

$$|d(x, M(t, \varepsilon))-d(x, M(t', \varepsilon))|\leqq K|t-t'|.$$

LEMMA 22.3. *Let $G(t, z, \varepsilon)$ be a function such that $G(t, z, \varepsilon)=d(z, M(t, \varepsilon))$. Then,*

$$|G(t, z, \varepsilon)-G(t', z', \varepsilon')|\leqq K|t-t'|+\|z-z'\|+|\varepsilon-\varepsilon'|, \tag{22.1}$$

where K is a positive constant when (t, z), (t', z') belong to a compact set in $I\times R^n$.

These lemmas can be proved easily, see [143], [146]. Now we have the following theorem.

THEOREM 22.1. *Suppose that $F(t, x)$ of* (18.2) *is defined and continuous on $I\times R^n$ and that $F(t, x)\in C_0(x)$. If a set M is an equiasymptotically stable set of* (18.2) *in the large, there exists a Liapunov function $V(t, x)$ defined on $I\times R^n$ which satisfies the following conditions;*

(a) $V(t, x)=0$ *if* $(t, x)\in M$,

(b) $a(d(x, M(t)))\leqq V(t, x)\leqq b(t, d(x, M(t)), \|x\|)$, *where* $a(r)\in CIP$, $a(r)\to\infty$ *as* $r\to\infty$ *and* $b(t, r, s)$ *is continuous,* $b(t, r, s)\to 0$ *as* $r\to 0$,

(c) $V(t, x)\in C_0(t, x)$, *and if* $x\in S_\alpha$, $x'\in S_\alpha$, $d(x, M(t))\leqq\eta$ *and* $d(x', M(t))\leqq\eta$,

$$|V(t, x)-V(t, x')|\leqq h(\eta)m(\alpha)f(t)\|x-x'\|, \tag{22.2}$$

where $h(\eta)$, $m(\alpha)$ and $f(t)$ are positive continuous functions,

(d) $V'_{(18.2)}(t, x)\leqq -cV(t, x)$, *where $c>0$ is a constant.*

Moreover, if M is bounded, then M is a uniform-asymptotically stable set in the large with respect to α and there exists a Liapunov function $V(t, x)$ defined on $I\times R^n$ which satisfies the conditions (a), (d) *and*

(b$_1$) $a(d(x, M(t)))\leqq V(t, x)\leqq b(t, d(x, M(t)))$, *where* $a(r)\in CIP$, $a(r)\to\infty$ *as* $r\to\infty$ *and* $b(t, r)$ *is continuous,* $b(t, r)\to 0$ *as* $r\to 0$,

(c$_1$) $V(t, x)\in C_0(t, x)$, *and if* $d(x, M(t))\leqq\eta$ *and* $d(x', M(t))\leqq\eta$,

$$|V(t, x)-V(t, x')|\leqq h(\eta)f(t)\|x-x'\|, \tag{22.3}$$

where $h(\eta)$, $f(t)$ are functions similar to those in (c).

PROOF. Let $\Omega_{\sigma,\eta,\alpha}$ be a domain such that $t\in[0, \sigma]$, $d(x, M(t))\leqq\eta$ and $x\in S_\alpha$, and denote it by Ω to make the notation simple. If $(t_0, x_0)\in\Omega$, there exist a $\beta(\sigma, \eta, \alpha)$ and a $T(\sigma, \varepsilon, \eta, \alpha)$ such that $d(x(t; x_0, t_0), M(t))\leqq\beta$ for all $t\geqq t_0$ and that $d(x(t; x_0, t_0), M(t))<\varepsilon$ for all $t\geqq t_0+T$. By Lemma 22.1, β can be assumed to be continuous. If $\varepsilon>1$, we set $T(\sigma, \varepsilon, \eta, \alpha)=T(\sigma, 1, \eta, \alpha)$. Since every solution $x(t; x_0, t_0)$, $(t_0, x_0)\in\Omega$, is continuable to $t=\sigma+T(\sigma, \varepsilon, \eta, \alpha)$, there exists a $\gamma(\sigma, \varepsilon, \eta, \alpha)>0$ for which $\|x(t; x_0. t_0)\|\leqq\gamma$ for $t\in[t_0, \sigma+T]$, by Theorem 3.7. Let $L(\sigma, \varepsilon, \eta, \alpha)$, $F^*(\sigma, \varepsilon, \eta, \alpha)$ and $K(\sigma, \varepsilon, \eta, \alpha)$ be the Lipschitz constant of $F(t, x)$, $1+\max\|F(t, x)\|$ and K in (16.1), respectively, in the domain $0\leqq t\leqq\sigma+T(\sigma, \varepsilon, \eta, \alpha)$,

$\|x\| \leqq \gamma(\sigma, \varepsilon, \eta, \alpha)$. Further, T, γ, L, F^* and K can be assumed to be continuous.

For $c>0$, let $A(\sigma, \varepsilon, \eta, \alpha)$ be such that

$$A(\sigma, \varepsilon, \eta, \alpha)=e^{cT(\sigma,\varepsilon,\eta,\alpha)}\{K(\sigma, \varepsilon, \eta, \alpha)$$
$$+2F^*(\sigma, \varepsilon, \eta, \alpha)e^{L(\sigma,\varepsilon,\eta,\alpha)T(\sigma,\varepsilon,\eta,\alpha)}+\beta(\sigma, \eta, \alpha)\},$$

where four continuous functions $g(\varepsilon)$, $h(\eta)>0$, $m(\alpha)>0$ and $f(\sigma)>0$ are such that $g(\varepsilon)>0$ for $\varepsilon \neq 0$, $g(0)=0$, and $g(\varepsilon)A(\sigma, \varepsilon, \eta, \alpha) \leqq h(\eta)m(\alpha)f(\sigma)$ by Lemma 19.1. For $k=1, 2, \cdots$ with the function $G_k(t, z)=d\left(z, M\left(t, \frac{1}{k}\right)\right)$, if we define $V_k(t, x)$ and $V(t, x)$ by

$$V_k(t, x)=g\left(\frac{1}{k}\right)\sup_{\tau\geqq 0} G_k(t+\tau, x(t+\tau\,;\, x, t))e^{c\tau},$$

$$V(t, x)=\sum_{k=1}^{\infty}\frac{1}{2^k}V_k(t, x),$$

it is clear that $g\left(\frac{1}{k}\right)G_k(t, x) \leqq V_k(t, x)$, $V_k(t, x)=0$ for $(t, x)\in M$ and for $(t, x)\in \Omega$,

$$V_k(t, x)\leqq g\left(\frac{1}{k}\right)e^{cT\left(\sigma,\frac{1}{k},\eta,\alpha\right)}\beta(\sigma, \eta, \alpha)\leqq h(\eta)m(\alpha)f(\sigma).$$

For $(t, x)\in \Omega$ and $(t', x')\in \Omega$, we have, by Lemma 22.3,

$$\begin{aligned}
&|V_k(t, x)-V_k(t', x')| \\
&\quad \leqq g\left(\frac{1}{k}\right)\sup_{\tau\geqq 0}|G_k(t+\tau, x(t+\tau\,;\, x, t)) \\
&\qquad -G_k(t'+\tau, x(t'+\tau\,;\, x', t'))|e^{c\tau} \\
&\quad \leqq g\left(\frac{1}{k}\right)\sup_{T\left(\sigma,\frac{1}{k},\eta,\alpha\right)\geqq\tau\geqq 0} e^{c\tau}\left\{K\left(\sigma, \frac{1}{k}, \eta, \alpha\right)|t-t'|\right. \\
&\qquad \left.+\|x(t+\tau\,;\, x, t)-x(t'+\tau\,;\, x', t')\|\right\},
\end{aligned}$$

and hence, in the same manner as in the proof of Theorem 19.3, it can be seen that

$$|V_k(t, x)-V_k(t', x')|\leqq h(\eta)m(\alpha)f(\sigma)\{|t-t'|+\|x-x'\|\}.$$

The remainder of the proof can be verified by the same argument as in the proof of Theorem 19.3.

THEOREM 22.2. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* $I \times R^n$ *and that* $F(t, x) \in C_0(x)$. *If a set* M *is a quasi-equiasymptotically stable set of* (18.2) *in the large, there exists a non-negative Liapunov function* $V(t, x)$ *defined on* $I \times R^n$ *which satisfies the conditions* (c), (d) *in Theorem* 22.1 *and*

(b_2) $a(d(x, M(t))) \leqq V(t, x)$, *where* $a(r) \in CI$, $a(r) > 0$ *for* $r \neq 0$ *and* $a(r) \to \infty$ *as* $r \to \infty$.

Moreover, if M *is bounded,* $V(t, x)$ *satisfies* (b_2), (c_1) *and* (d).

Since we can see that if M satisfies (16.1) and M is a quasi-equiasymptotically stable set in the large, the solutions of (18.2) are equi-bounded with respect to M (cf. [143]), this theorem follows immediately from the proof of Theorem 22.1. In this case, $V_k(t, x)$ and consequently $V(t, x)$ is not necessarily zero for $(t, x) \in M$.

From Theorem 16.2 and Theorem 22.2, we can also obtain Theorem 20.3.

THEOREM 22.3. *Suppose that* $F(t, x)$ *of* (18.2) *is continuous on* $I \times R^n$ *and that* $F(t, x) \in C_0(x)$. *If a set* M *is a uniform-asymptotically stable set of* (18.2) *in the large, there exists a Liapunov function* $V(t, x)$ *defined on* $I \times R^n$ *which satisfies the conditions* (a), (c), (d) *in Theorem* 22.1 *and*

(b_3) $a(d(x, M(t))) \leqq V(t, x) \leqq b(d(x, M(t)))$, *where* $a(r)$ *is the same as in* (b) *and* $b(r)$ *is continuous,* $b(r) \to 0$ *as* $r \to 0$.

Moreover, if $F(t, x) \in \bar{C}_0(x)$ *and the solutions of* (18.2) *are uniform-bounded,* $V(t, x)$ *satisfies* (a), (b_3), (d) *and*

(c_2) $V(t, x) \in C_0(t, x)$, *and if* $x \in S_\alpha$, $x' \in S_\alpha$, $d(x, M(t)) \leqq \eta$ *and* $d(x', M(t)) \leqq \eta$,

$$(22.4) \qquad |V(t, x) - V(t, x')| \leqq h(\eta) m(\alpha) \| x - x' \|,$$

where $h(\eta)$, $m(\alpha)$ *are functions similar to those in* (c).

We can prove easily the following theorem which implies Theorem 20.4 by Theorem 16.3.

THEOREM 22.4. *Suppose that $F(t, x)$ of* (18.2) *is continuous on $I\times R^n$ and that $F(t, x)\in C_0(x)$. Moreover, suppose that the solutions of* (18.2) *are uniform-bounded. If M is a quasi-uniform-asymptotically stable set of* (18.2) *in the large and if the solutions are uniform-bounded with respect to M, there exists a Liapunov function $V(t, x)$ defined on $I\times R^n$ satisfying* (c), (d) *in Theorem* 22.1 *and*

(b_4) $a(d(x, M(t)))\leqq V(t, x)\leqq b(d(x, M(t)))$, *where $a(r)$ is the same as in* (b_2),*and $b(r)$ is continuous.*

Moreover, if M is bounded, $V(t, x)$ satisfies (b_4), (c_1) *and* (d). *In case M is bounded and $F(t, x)\in \bar{C}_0(x)$, $V(t, x)$ satisfies* (b_4), (d) *and*

(c_3) $V(t, x)\in C_0(t, x)$, *and if $d(x, M(t))\leqq \eta$ and $d(x', M(t))\leqq \eta$,*

$$|V(t, x)-V(t, x')|\leqq h(\eta)\|x-x'\| . \tag{22.5}$$

Finally, we shall consider the case where $F(t, x)$ of (18.2) is defined and continuous on $D: 0\leqq t<\infty$, $d(x, M(t))\leqq H$, $H>0$.

THEOREM 22.5. *Suppose that $F(t, x)$ of* (18.2) *is continuous on D and that there exists a $B>0$ such that $\|F(t, x)-F(t, x')\|\leqq B\|x-x'\|$. If M is a uniform-asymptotically stable set of* (18.2), *there exists a Liapunov function $V(t, x)$ defined on $D': 0\leqq t<\infty$, $d(x, M(t))\leqq H'$, $H'<H$, which satisfies the following conditions;*

(i) $a(d(x, M(t)))\leqq V(t, x)\leqq b(d(x, M(t)))$, *where $a(r)\in CIP$ and $b(r)\in CIP$,*

(ii) $V(t, x)\in C_0(t, x)$, *and for a constant $L>0$,*

$$|V(t, x)-V(t, x')|\leqq L\|x-x'\| , \tag{22.6}$$

(iii) $V'_{(18.2)}(t, x)\leqq -cV(t, x)$, *where $c>0$ is a constant.*

By uniform-asymptotic stability, there is a $\delta_0=H'$ such that if $d(x_0, M(t_0))\leqq\delta_0$, then $x(t; x_0, t_0)\to M$ as $t\to\infty$. To prove this theorem, consider the domain $\Omega_{\sigma,\alpha}$ such that $0\leqq t\leqq\sigma$, $d(x, M(t))\leqq H'$, $\|x\|\leqq\alpha$. Then, this theorem can be proved by following the proof of Theorem 22.1.

REMARK. In Theorems 22.3, 22.4 and 22.5, when $F(t, x)$ and $M(t)$ are periodic in t with the same period ω, $V(t, x)$ is also periodic in t of period ω, and if F is autonomous and $M(t)=M(0)$ for all $t \geqq 0$, then V is independent of t.

Here, we shall add some investigations in dynamical systems. Auslander and Seibert [9], [10] have presented a unified theory of stability and boundedness in dynamical systems by means of generalized Liapunov functions and prolongations. The notion of prolongation was first used by Poincaré and Bendixson in a very special sense (cf. [25], [77]). In a much more general sense, prolongations were considered by Ura [133]. The concept of prolongation leads to a whole hierarchy of new types of stability, all of which occupy an intermediate place between stability in the sense of Liapunov and asymptotic stability. Ura developed a method of extending orbits of dynamical systems beyond their positive limiting sets in a way which is closely related to the concept of stability. The most significant among new types of stability, called "absolute stability", is characterized by the existence of a continuous generalized Liapunov function.

On the other hand, it has been observed [137] that there exists a kind of duality between the concepts of stability in the sense of Liapunov and boundedness. Auslander and Seibert have formalized this duality by compactifying the phase space, and in this way they have obtained a corresponding boundedness theorem from every stability theorem.

We mention here only the concepts of prolongation and absolute stability due to Ura, in a rather simple form. Let X be a locally compact separable metric space and let R be the real line. A dynamical system or continuous flow on X is a continuous map $\pi : X \times R \to X$ satisfying

(a) $\pi(x, 0)=x,\ x \in X,$

(b) $\pi(\pi(x, t_1), t_2)=\pi(x, t_1+t_2),\ x \in X,\ t_1, t_2 \in R.$

Typically, dynamical systems arise from the solution curves of autonomous systems of differential equations. The positive semi-orbit $\gamma^+(x)$ is the set $\{\pi(x, t)\,;\, t \geqq 0\}$. A subset M of X is called a positively invariant set if $\pi(x, t) \in M$ whenever $x \in M$ and $t \geqq 0$. We define two operators G and S on the class of maps from X to

2^X (the set of all subsets of X). If $Q: X \to 2^X$ and $M \subset X$, then $Q(M) = \cup\{Q(x); x \in M\}$. GQ is defined to be such that $y \in GQ(x)$ if and only if there are sequences $\{x_n\}$, $\{y_n\}$ with $y_n \in Q(x_n)$ such that $x_n \to x$, $y_n \to y$. $SQ(x)$ is defined by $SQ(x) = \bigcup_{n=1}^{\infty} Q^n(x)$.

DEFINITION 22.1. A prolongation is a map $Q: X \to 2^X$ satisfying

(i) $\gamma^+(x) \subset Q(x)$, if $x \in X$,

(ii) $GQ = Q$,

(iii) if M is a compact subset of X and $x \in M$, then either $Q(x) \subset M$ or $Q(x)$ meets the boundary of M.

If $x \in X$, let $E_0(x)$ be $\gamma^+(x)$ and let $D_1 = GE_0$. We define $E_1 = SD_1$ and $D_2 = GE_1$. For any ordinal number α, we define D_α inductively. Suppose for every ordinal number $\beta < \alpha$, D_β has been defined. Let $E_\beta = SD_\beta$ and $E_\alpha^* = \cup\{E_\beta; \beta < \alpha\}$. D_α is defined by $D_\alpha = GE_\alpha^*$. These D_α can be verified to be prolongations. Here, it is noticed that $y \in D_\alpha(x)$ if and only if there are sequences $\{x_n\}$, $\{y_n\}$ in X and $\{\beta_n\}$, $\{k_n \geqq 1\}$, where β_n is an ordinal number less than α and k_n is an integer, such that $x_n \to x$, $y_n \to y$ and $y_n \in D_{\beta_n}^{k_n}(x_n)$.

DEFINITION 22.2. Let M be a compact positively invariant set and let α be an ordinal number. M is said to be stable of order α, if $D_\alpha(M) = M$. If M is stable of order α for every ordinal number α, then M is said to be absolutely stable.

The stability of order 1 is the stability in the sense of Liapunov. The study of stability in dynamical systems has been facilitated by the use of generalized Liapunov function (cf. [76], [152]).

DEFINITION 22.3. Let M be a compact positively invariant set. A generalized Liapunov function for M is a non-negative function V defined in a positively invariant neighborhood W of M and satisfying

(i) for any $\varepsilon > 0$, there exists a $\lambda(\varepsilon) > 0$ such that $V(x) > \lambda$ for $x \notin N(\varepsilon, M)$,

(ii) for any $\lambda > 0$, there exists an $\eta(\lambda) > 0$ such that $V(x) < \lambda$

for $x \in N(\eta, M)$,

(iii) if $x \in W$ and $t \geqq 0$, then $V(\pi(x, t)) \leqq V(x)$.

THEOREM 22.6. *The compact set M is stable in the sense of Liapunov if and only if there exists a generalized Liapunov function for M.*

For the proof, see [76] or [152].

Auslander and Seibert proved the following theorem which shows the relation between generalized Liapunov function and absolute stability.

THEOREM 22.7. *Let M be a compact subset of X. Then M is absolutely stable if and only if there exists a generalized Liapunov function V for M which is continuous in some neighborhood W of M.*

For the details, see [9], [10], [124].

CHAPTER VI. PERTURBED SYSTEMS

By utilizing results obtained in the previous chapter, the properties of solutions of perturbed systems and the asymptotic behavior of solutions near integral manifolds will be discussed.

§ 23. Total stability and total boundedness.

Goršin [36] and Malkin [86] proved that, under fairly general assumptions, asymptotic stability implies total stability. First of all, we shall give the definition of total stability of the zero solution of

$$\frac{dx}{dt} = F(t, x). \tag{23.1}$$

Here, $F(t, x)$ is assumed to be defined and continuous on $I \times S_H$, $H > 0$, and we assume that $F(t, 0) \equiv 0$.

DEFINITION 23.1. The zero solution of (23.1) is totally stable (in the Soviet terminology, stable under constantly acting perturbations), if given $\varepsilon > 0$, there exists a $\delta(\varepsilon) > 0$ such that for any equation

$$\frac{dy}{dt} = F(t, y) + G(t, y), \tag{23.2}$$

where $\| G(t, y) \| < \delta$, the solution $y(t; y_0, t_0)$ of (23.2) satisfies $\| y(t; y_0, t_0) \| < \varepsilon$ for any initial value such that $\| y_0 \| < \delta$, $t \geqq t_0 \geqq 0$.

THEOREM 23.1. *Suppose that $F(t, x)$ of (23.1) is continuous on $I \times S_H$ and that $F(t, 0) \equiv 0$. If $F(t, x) \in \overline{C}_0(x)$ and if the zero solution of (23.1) is uniform-asymptotically stable, then it is totally stable.*

PROOF. By Theorem 19.9, there exists a Liapunov function $V(t, x)$ which satisfies the conditions in Theorem 8.3 with $c = 1$ and (19.18). For any $\varepsilon > 0$, choose a $\delta_1(\varepsilon) > 0$ so that $b(\delta_1) < a(\varepsilon)$, where $a(r)$, $b(r)$ are the functions in (ii)′ in Theorem 8.2. Choose $\delta > 0$ so small that $a(\delta_1) - K\delta > 0$, $\delta < \delta_1$, where K is the Lipschitz constant in (19.18).

Suppose that a solution $y(t; y_0, t_0)$ of (23.2), where $\| G(t, y) \| < \delta$ and $\| y_0 \| < \delta$, satisfies $\| y(t; y_0, t_0) \| = \varepsilon$ at some t. Then, there are t_1, t_2 such that $\| y(t_1; y_0, t_0) \| = \delta_1$, $\| y(t_2; y_0, t_0) \| = \varepsilon$ and that $\delta_1 < \| y(t; y_0, t_0) \| < \varepsilon$ for $t \in (t_1, t_2)$. On the other hand, for $t \in [t_1, t_2]$

$$V'(t, y(t; y_0, t_0)) \leqq -V(t, y(t; y_0, t_0)) + K \| G(t, y(t; y_0, t_0)) \|$$
$$\leqq -a(\delta_1) + K\delta \leqq 0 .$$

Thus, $a(\varepsilon) \leqq V(t_2, y(t_2; y_0, t_0)) \leqq V(t_1, y(t_1; y_0, t_0)) \leqq b(\delta_1)$, which contradicts the choice of δ_1. Therefore, $\| y(t; y_0, t_0) \| < \varepsilon$ for all $t \geqq t_0$, which shows that the zero solution of (23.1) is totally stable.

The following is a partial reciprocal of this result.

THEOREM 23.2. *If the zero solution of a linear system $\dot{x} = A(t)x$ is totally stable, it is uniform-asymptotically stable.*

PROOF. If the solution $x(t) \equiv 0$ is totally stable, there exists a $\delta > 0$ such that if $\| y_0 \| < \delta$, the solution $y(t; y_0, t_0)$ of $\dot{y} = A(t)y + \delta y$, where $\| y \| < 1$, satisfies $\| y(t; y_0, t_0) \| < 1$. But the solutions of both equations are related by $y(t; y_0, t_0) = x(t; y_0, t_0)e^{\delta(t-t_0)}$, which proves the theorem.

Here, it should be noticed that total stability does not necessarily imply uniform-asymptotic stability, and Massera [95] gave such an example for an autonomous system.

Corresponding to total stability, we shall consider total boundedness. In this case, F and G are assumed to be continuous on $I \times R^n$.

DEFINITION 23.2. The solutions of (23.1) are totally bounded, if given $\alpha > 0$, there exist two numbers $\beta(\alpha) > 0$, $\gamma(\alpha) > 0$ such that if $y_0 \in S_\alpha$, then $\| y(t; y_0, t_0) \| < \beta(\alpha)$ for all $t \geqq t_0$, where $y(t; y_0, t_0)$ is the solution of (23.2) in which $\| G(t, y) \| < \gamma(\alpha)$, provided $\alpha < \| y \| < \beta$.

DEFINITION 23.3. For a given function $f(r) > 0$, the solutions of (23.1) are ultimately bounded under constantly acting perturbations of order $f(r)$, if there exist two constants $B>0$, $\alpha>0$ such that $\|G(t, y)\| < \alpha f(\|y\|)$ for $\|y\| \geqq B$ and that $\overline{\lim}_{t\to\infty} \|y(t; y_0, t_0)\| < B$, where $y(t; y_0, t_0)$ is the solution of (23.2).

THEOREM 23.3. *Suppose that $F(t, x)$ of (23.1) is continuous on $I \times R^n$ and that there exists a Liapunov function $V(t, x) > 0$ defined on $D: 0 \leqq t < \infty$, $\|x\| \geqq R$, where R may be large, which satisfies the following conditions;*

(i) $a(\|x\|) \leqq V(t, x) \leqq b(\|x\|)$, *where $a(r) \in CI$, $a(r) \to \infty$ as $r \to \infty$ and $b(r) \in CI$,*

(ii) $V(t, x) \in \bar{C}_0(x)$, *and $V'_{(23.1)}(t, x) \leqq -c(\|x\|)$, where $c(r) > 0$ is continuous.*

Then, the solutions are totally bounded.

PROOF. For an $\alpha > 0$, choose a $\beta(\alpha) > 0$ so that $a(\beta) > b(\alpha)$. By (ii), for x such that $\alpha \leqq \|x\| \leqq \beta$, there are $K(\alpha) > 0$ and $\lambda(\alpha) > 0$ such that $|V(t, x) - V(t, x')| \leqq K(\alpha) \|x - x'\|$ and $V'_{(23.1)}(t, x) \leqq -\lambda(\alpha)$. Thus, in the domain $0 \leqq t < \infty$, $\alpha \leqq \|x\| \leqq \beta$,

$$V'_{(23.2)}(t, y) \leqq V'_{(23.1)}(t, y) + K\|G(t, y)\| \leqq -\lambda + K\|G(t, y)\| .$$

Therefore, if we choose $\gamma(\alpha) > 0$ so that $\gamma(\alpha) \leqq \dfrac{K(\alpha)}{\lambda(\alpha)}$, we have $V'_{(23.2)}(t, y) \leqq 0$ for $\|G(t, y)\| < \gamma(\alpha)$ in the domain $0 \leqq t < \infty$, $\alpha \leqq \|x\| \leqq \beta$, which implies that $\|y(t; y_0, t_0)\| < \beta$ if $y_0 \in S_\alpha$.

COROLLARY. *Suppose that $F(t, x)$ of (23.1) is continuous on $I \times R^n$ and that $F(t, x) \in \bar{C}_0(x)$. If the solutions of (23.1) are uniform-bounded and uniform-ultimately bounded, they are totally bounded.*

This follows immediately from Corollary of Theorem 20.4 and Theorem 23.3.

THEOREM 23.4. *Under the assumptions of Theorem 23.3, let $L(\|x\|)$ be the Lipschitz constant of $V(t, x)$. If $L(r)f(r) = 0(c(r))$ as $r \to \infty$, the solutions of (23.1) are ultimately bounded under constantly acting perturbations of order $f(r)$.*

PROOF. By the assumption, we can choose an $\alpha > 0$ such that

$\alpha \leqq \dfrac{c(r)}{2L(r)f(r)}$. For this α, if $\|G(t, y)\| < \alpha f(\|y\|)$, then $V'_{(23.2)}(t, y) \leqq V'_{(23.1)}(t, y) + L(\|y\|)\,\|G(t, y)\| \leqq -c(\|y\|) + \alpha L(\|y\|)f(\|y\|) \leqq -\dfrac{1}{2}c(\|y\|)$. Therefore, by Theorem 20.4, the solutions of (23.2) are ultimately bounded, which proves the theorem.

§ 24. Behavior of solutions of perturbed systems.

There are many results concerned with relationships between solutions of an unperturbed system and solutions of a perturbed system, especially concerning the stability of solutions of the perturbed system (cf. [4], [23], [25], [47], [77], [123], [143]).

For the system (23.1), consider its perturbed system

$$\frac{dx}{dt} = F(t, x) + G_1(t, x) + G_2(t, x)\,. \tag{24.1}$$

THEOREM 24.1. *Suppose that* $F(t, x)$ *of* (23.1) *and* $G_1(t, x)$, $G_2(t, x)$ *of* (24.1) *are defined and continuous on* $I \times S_H$, $H > 0$, *and that* $F(t, 0) \equiv 0$. *Moreover, suppose that* $F(t, x) \in \bar{C}_0(x)$ *and that there exist a* $\delta_0 > 0$, *a* $c > 0$ *and a* $K > 0$ *such that if* $x_0 \in S_{\delta_0}$,

$$\|x(t\,;\, x_0, t_0)\| \leqq Ke^{-c(t-t_0)}\|x_0\| \qquad \textit{for all} \quad t \geqq t_0\,,$$

where $x(t\,;\, x_0, t_0)$ *is a solution of* (23.1). *Then, if* $\|G_1(t, x)\| = o(\|x\|)$ $(\|x\| \to 0)$ *and* $\|G_2(t, x)\| \leqq g(t)\|x\|$ *in* $I \times S_H$ *and if* $\int_0^\infty g(t)dt < \infty$, *the zero solution of* (24.1) *is exponential-asymptotically stable, more precisely, there exist* $\delta_0^* > 0$, $c^* > 0$ *and* $K^* > 0$ *such that for the solution* $x^*(t\,;\, x_0, t_0)$ *of* (24.1), *if* $x_0 \in S_{\delta_0^*}$,

$$\|x^*(t\,;\, x_0, t_0)\| \leqq K^* e^{-c^*(t-t_0)}\|x_0\| \qquad \textit{for all} \quad t \geqq t_0\,. \tag{24.2}$$

PROOF. By assumption, there exists a Liapunov function $V(t, x)$ defined on $I \times S_{\delta_0}$ which satisfies the conditions in Corollary of Theorem 19.2. Using the same notations as in the corollary, choose an $\varepsilon > 0$ such that $\varepsilon \leqq \dfrac{qc}{2L}$. By the condition on $G_1(t, x)$, there is a δ_1, $\delta_1 \leqq \delta_0$, such that $x \in S_{\delta_1}$ implies $\|G_1(t, x)\| \leqq \varepsilon\|x\|$.

Let $\delta_2\ (\leqq \delta_0)$ be such that if $x_0 \in S_{\delta_2}$, we have $K\|x_0\| \exp\left\{L\int_0^\infty g(t)dt\right\} < \delta_1$.

Now, consider a function $W(t, x)$ such that $W(t, x) = V(t, x) \exp\left\{-L\int_0^t g(s)ds\right\}$, which is defined on $I \times S_{\delta_0}$. In the domain $I \times S_{\delta_1}$, we have

$$
\begin{aligned}
&W'_{(24.1)}(t, x) \\
&\quad \leqq e^{-L\int_0^t g(s)ds}\{-Lg(t)V(t, x) - qcV(t, x) + L\|G_1(t, x)\| + L\|G_2(t, x)\|\} \\
&\quad \leqq e^{-L\int_0^t g(s)ds}\{-Lg(t)V(t, x) - qcV(t, x) + L\varepsilon\|x\| + Lg(t)\|x\|\} \\
&\quad \leqq e^{-L\int_0^t g(s)ds}\{-Lg(t)V(t, x) - qcV(t, x) + L\varepsilon V(t, x) + Lg(t)V(t, x)\} \\
&\quad \leqq W(t, x)(L\varepsilon - qc) \leqq -\frac{qc}{2}W(t, x),
\end{aligned}
$$

and hence, as long as a solution $x^*(t; x_0, t_0)$ stays in the domain $\|x\| \leqq \delta_1$,

$$W(t, x^*(t; x_0, t_0)) \leqq W(t_0, x_0)e^{-\frac{qc}{2}(t-t_0)}.$$

This implies that $\|x^*(t; x_0, t_0)\| \leqq K\|x_0\| \exp\left\{L\int_0^\infty g(t)dt\right\} \exp\left\{-\frac{qc}{2}(t-t_0)\right\}$. On the other hand, if $x_0 \in S_{\delta_2}$, we have $\|x^*(t; x_0, t_0)\| < \delta_1$. Thus, if $x_0 \in S_{\delta_2}$, we have (24.2), where $c^* = \frac{qc}{2}$ and $K^* = K\exp\left\{L\int_0^\infty g(t)dt\right\}$.

COROLLARY 1. *Under the same assumptions on G_1, G_2 as in Theorem 24.1, if all characteristic roots of the $n \times n$ matrix A have negative real parts, the zero solution of*

$$\frac{dx}{dt} = Ax + G_1(t, x) + G_2(t, x) \tag{24.3}$$

is exponential-asymptotically stable.

Now consider a system

$$\frac{dx}{dt} = A(t)x + G_1(t, x) + G_2(t, x)\,, \tag{24.4}$$

where $A(t)$ is a real continuous periodic matrix. By Floquet's theorem (cf. [25], [77]), there exists a non-singular periodic matrix $P(t)$ such that by transformation $x = P(t)y$, the periodic system $\dot{x} = A(t)x$ can be reduced to a linear system with constant coefficients $\dot{y} = By$. Then, by $x = P(t)y$, the system (24.4) is transformed into

$$\frac{dy}{dt} = By + P^{-1}(t)G_1(t, P(t)y) + P^{-1}(t)G_2(t, P(t)y)\,. \tag{24.5}$$

Since $\| P(t) \|$, $\| P^{-1}(t) \|$ are bounded, if $G_1(t, x)$ and $G_2(t, x)$ satisfy the conditions required above, $P^{-1}(t)G_1(t, P(t)y)$ and $P^{-1}(t)G_2(t, P(t)y)$ also satisfy the same conditions. Moreover, the transformation $x = P(t)y$ does not affect stability properties and hence, if all characteristic roots of B, in other words, all characteristic exponents of $\dot{x} = A(t)x$ have negative real parts, Corollary 1 of Theorem 24.1 applies to (24.4).

COROLLARY 2. *Under the same assumptions on G_1, G_2 as in Theorem 24.1, if $\dot{x} = A(t)x$, where $A(t)$ is a real continuous periodic matrix, has all characteristic exponents with negative real parts, the zero solution of (24.4) is exponential-asymptotically stable.*

Let us consider more general cases where the stability of a set M is considered. We assume that the set M satisfies the conditions given in Section 16.

LEMMA 24.1. *Consider a perturbed system*

$$\frac{dx}{dt} = F(t, x) + G(t, x)\,, \tag{24.6}$$

where F and G are defined and continuous on $D: 0 \leqq t < \infty$, $d(x, M(t)) \leqq H$, $H > 0$. Suppose that there exists a continuous scalar function $\omega(t, r)$ such that if $x \in S_\alpha$, then $\| G(t, x) \| \leqq \omega(t, \alpha)$. Moreover, suppose that there exists a non-negative continuous function $V(t, x)$ satisfying the following conditions in a domain $0 \leqq t < \infty$,

$d(x, M(t)) \leqq H'$, $0 < H' \leqq H$;

(i) $a(d(x, M(t))) \leqq V(t, x)$, *where* $a(r) \in CI$,

(ii) *for any* $\alpha > 0$, *if* $x \in S_\alpha$ *and* $x' \in S_\alpha$,

$$|V(t, x) - V(t, x')| \leqq \lambda(t, \alpha) \| x - x' \| , \tag{24.7}$$

where $\lambda(t, \alpha)$ *is a continuous function,*

(iii) $V'_{(23.1)}(t, x) \leqq -cV(t, x)$, *where* $c > 0$ *is a constant.*

If $u(t; u_0, t_0)$ *is the solution of the equation of the first order*

$$\frac{du}{dt} = -cu + \lambda(t, \alpha)\omega(t, \alpha), \qquad u_0 = V(t_0, x_0) \tag{24.8}$$

and if $u(t; u_0, t_0) < a(H')$ *for* $t \geqq t_0$, *then the solution* $x^*(t; x_0, t_0)$ *of* (24.6) *such that* $\| x^*(t; x_0, t_0) \| \leqq \alpha$ *for* $t \geqq t_0$ *satisfies the relation*

$$a(d(x^*(t; x_0, t_0), M(t))) \leqq u(t; u_0, t_0) \quad \text{for} \quad t \geqq t_0 . \tag{24.9}$$

PROOF. By the conditions (ii), (iii), if $d(x^*(t; x_0, t_0), M(t)) \leqq H'$,

$$\begin{aligned} V'(t, x^*(t; x_0, t_0)) &\leqq -cV(t, x^*(t; x_0, t_0)) + \lambda(t, \alpha) \| G(t, x^*(t; x_0, t_0)) \| \\ &\leqq -cV(t, x^*(t; x_0, t_0)) + \lambda(t, \alpha)\omega(t, \alpha) . \end{aligned}$$

Therefore, by the comparison principle, we have $V(t, x^*(t; x_0, t_0)) \leqq u(t; u_0, t_0)$, which implies (24.9), because of (i).

LEMMA 24.2. *Suppose that* $F(t, x)$, $G(t, x)$ *and* $V(t, x)$ *in Lemma* 24.1 *are defined on* $I \times R^n$. *Then, if* $u(t; u_0, t_0)$ *is the solution of* (24.8), *the solution* $x^*(t; x_0, t_0)$ *of* (24.6) *such that* $\| x^*(t; x_0, t_0) \| \leqq \alpha$ *for* $t \geqq t_0$ *satisfies* (24.9).

LEMMA 24.3. *In a scalar differential equation*

$$\frac{du}{dt} = -cu + \lambda\{g_1(t, \gamma) + g_2(t, \gamma)\} , \tag{24.10}$$

suppose that $g_1(t, \gamma)$ *and* $g_2(t, \gamma)$ *are non-negative continuous functions on* I, *where* $c > 0$ *is a constant and* λ, γ *are parameters. If we assume that*

$$g_1(t, \gamma) \to 0 \ \text{as} \ t \to \infty , \tag{24.11}$$

$$\int_0^\infty g_2(t, \gamma) dt < \infty , \tag{24.12}$$

then corresponding to any $\rho > 0$, *there exists a* $T(\rho, \lambda, \gamma, u_0) > 0$ *such that for any* $t_0 \in I$ *and* $t \geqq t_0 + T(\rho, \lambda, \gamma, u_0)$, *we have* $u(t; u_0, t_0) < \rho$, *where* $u(t; u_0, t_0)$ *is the solution of* (24.10).

PROOF. The proof presented follows an argument due to Hale [43]. The solution of (24.10) has the form

$$u(t; u_0, t_0) = e^{-c(t-t_0)}u_0 + \lambda e^{-ct}\int_{t_0}^{t} e^{cs}g_1(s, \gamma)ds + \lambda e^{-ct}\int_{t_0}^{t} e^{cs}g_2(s, \gamma)ds.$$

By (24.11), there is a $p(\gamma) > 0$ such that $g_1(t, \gamma) \leqq p(\gamma)$. For any $\rho > 0$, there are $T_i(\rho, \lambda, \gamma)$, $i = 1, 2$, such that if $t \geqq T_1$, we have $e^{-\frac{ct}{2}} < \dfrac{c\rho}{6\lambda p(\gamma)}$ and that if $t \geqq T_2$, we have $\max\limits_{\frac{t}{2} \leqq s \leqq t} g_1(s, \gamma) < \dfrac{c\rho}{6\lambda}$. Therefore, if $t \geqq \max(T_1, T_2)$,

$$\lambda e^{-ct}\int_{t_0}^{t} e^{cs}g_1(s, \gamma)ds \leqq \frac{1}{c}\lambda\{p(\gamma)e^{-\frac{ct}{2}} + \max_{\frac{t}{2} \leqq s \leqq t} g_1(s, \gamma)\} < \frac{\rho}{3}.$$

Choose $T_3(\rho, \lambda, \gamma) \geqq 0$ so large that

$$\int_{T_3}^{\infty} g_2(s, \gamma)ds < \frac{\rho}{6\lambda} \tag{24.13}$$

and choose $T_4(\rho, \lambda, \gamma)$ so that

$$e^{-c(T_4 - T_3)}\int_{0}^{\infty} g_2(s, \gamma)ds < \frac{\rho}{6\lambda}. \tag{24.14}$$

Then, for any $t_0 \geqq 0$ and $t \geqq t_0 + T_4$, by (24.13) and (24.14),

$$\begin{aligned}\lambda e^{-ct}\int_{t_0}^{t} e^{cs}g_2(s, \gamma)ds &\leqq \lambda\int_{0}^{T_3} e^{-c(t-s)}g_2(s, \gamma)ds + \lambda\int_{T_3}^{t} e^{-c(t-s)}g_2(s, \gamma)ds \\ &\leqq \lambda e^{-c(t-T_3)}\int_{0}^{T_3} g_2(s, \gamma)ds + \lambda\int_{T_3}^{t} g_2(s, \gamma)ds \\ &\leqq \lambda e^{-c(T_4-T_3)}\int_{0}^{\infty} g_2(s, \gamma)ds + \lambda\int_{T_3}^{\infty} g_2(s, \gamma)ds < \frac{\rho}{3}.\end{aligned}$$

Moreover, there is a $T_5(\rho, \lambda, \gamma, u_0)$ such that if $t \geqq t_0 + T_5$, we have $u_0 e^{-c(t-t_0)} < \dfrac{\rho}{3}$. Thus, for any $t_0 \geqq 0$ and $t \geqq t_0 + T(\rho, \lambda, \gamma, u_0)$, $T = \max(T_1, T_2, T_3, T_4, T_5)$, we have $u(t; u_0, t_0) < \rho$.

THEOREM 24.2. *Suppose that* $F(t, x)$ *of the system* (23.1) *and* $G_1(t, x)$, $G_2(t, x)$ *of the system* (24.1) *are defined and continuous on* $I\times R^n$. *Moreover, suppose that there exists a Liapunov function* $V(t, x)$ *defined on* $I\times R^n$ *which satisfies the following conditions;*

(i) $a(d(x, M(t)))\leqq V(t, x)\leqq b(d(x, M(t)))$, *where* $a(r)\in CI$, $a(r)>0$ *for* $r\neq 0$, $a(r)\to\infty$ *as* $r\to\infty$ *and* $b(r)$ *is continuous,*

(ii) *if* $d(x, M(t))\leqq\eta$ *and* $d(x', M(t))\leqq\eta$, $\eta>0$,

$$|V(t, x)-V(t, x')|\leqq h(\eta)f(t)\|x-x'\|, \tag{24.15}$$

where $h(\eta)$ *and* $f(t)$ *are continuous function,*

(iii) $V'_{(23.1)}(t, x)\leqq -cV(t, x)$, *where* $c>0$ *is a constant. Moreover, suppose that there are continuous functions* $g_i(t, \alpha)$, $i=1, 2$, *for any* $\alpha>0$ *such that* $\|G_i(t, x)\|\leqq g_i(t, \alpha)$, $i=1, 2$, *for* x *such that* $d(x, M(t))\leqq\alpha$. *Then, if all solutions of* (24.1) *exist in the future and if we have*

$$f(t)g_1(t, \alpha)\to 0 \text{ as } t\to\infty, \tag{24.16}$$

$$\int_0^\infty f(t)g_2(t, \alpha)dt<\infty, \tag{24.17}$$

the solutions of (24.1) *are eventually uniform-bounded with respect to* M, *and* M *is an eventually quasi-uniform-asymptotically stable set of* (24.1) *in the large.*

PROOF. For each $\eta>0$, if $d(x_0, M(t_0))\leqq\eta$, there exists a $B(\eta)>0$ such that $V(t_0, x_0)\leqq B(\eta)$. Choose a $\beta(\eta)>0$ so that $3B(\eta)\leqq a(\beta)$, $\eta<\beta$. Let $x^*(t; x_0, t_0)$ be a solution of (24.1). As long as $d(x^*(t; x_0, t_0), M(t))\leqq\beta(\eta)$, we have

$$\begin{aligned} V'(t, x^*(t; x_0, t_0))\leqq & -cV(t, x^*(t; x_0, t_0)) \\ & +h(\beta)f(t)\{g_1(t, \beta)+g_2(t, \beta)\}. \end{aligned} \tag{24.18}$$

Let $u(t; u_0, t_0)$ be the solution of

$$\frac{du}{dt}=-cu+h(\beta)f(t)\{g_1(t, \beta)+g_2(t, \beta)\}, \qquad u_0=V(t_0, x_0).$$

Then as long as $d(x^*(t; x_0, t_0), M(t))\leqq\beta(\eta)$, we have

$$a(d(x^*(t; x_0, t_0), M(t)))\leqq u(t; u_0, t_0). \tag{24.19}$$

The solution $u(t; u_0, t_0)$ is of the form

$$u(t; u_0, t_0) = e^{-c(t-t_0)} V(t_0, x_0) \\ + h(\beta) e^{-ct} \int_{t_0}^{t} \{ e^{cs} f(s) g_1(s, \beta) + e^{cs} f(s) g_2(s, \beta) \} ds .$$

By (24.16), there is a $T_1(\eta)$ such that $f(t) g_1(t, \beta) < \dfrac{cB(\eta)}{h(\beta)}$ for $t \geqq T_1$, and by (24.17), we can choose $T_2(\eta)$ so large that $\int_{t_0}^{t} f(s) g_2(s, \beta) ds < \dfrac{B(\eta)}{h(\beta)}$ for $t \geqq t_0 \geqq T_2$. From this, it follows that $u(t; u_0, t_0) \leqq 3B(\eta)$ for $t \geqq t_0 \geqq S(\eta)$, where $S(\eta) = \max(T_1, T_2)$. Therefore, by (24.19) and $3B(\eta) \leqq a(\beta)$, we have $d(x^*(t; x_0, t_0), M(t)) \leqq \beta(\eta)$ for all $t \geqq t_0 \geqq S(\eta)$, because $d(x_0, M(t_0)) \leqq \eta < \beta$. This shows that the solutions of (24.1) are eventually uniform-bounded with respect to M.

For the solution $x^*(t; x_0, t_0)$ of (24.1) such that $d(x_0, M(t_0)) \leqq \eta$, if $t_0 \geqq S(\eta)$, we have (24.18) for all $t \geqq t_0$ and consequently, we have (24.19) for all $t \geqq t_0$. Therefore, applying Lemma 24.3 to $u(t; u_0, t_0)$, we can see that for any $\varepsilon > 0$, $t_0 \geqq S(\eta)$, there exists a $T(\varepsilon, \eta) > 0$ such that $d(x^*(t; x_0, t_0), M(t)) < \varepsilon$ for all $t \geqq t_0 + T(\varepsilon, \eta)$, which shows eventually quasi-uniform-asymptotic stability of M.

REMARK. When M is bounded, we see from the proof that we do not require the assumption that all solutions of (24.1) exist in the future. This condition is verified automatically.

From this theorem and Theorem 22.4, the following corollaries follow immediately.

COROLLARY 1. *Suppose that F, G_1 and G_2 are continuous on $I \times R^n$, $F(t, x) \in \bar{C}_0(x)$ and that G_1, G_2 satisfy*

$$\text{(24.20)} \qquad G_1(t, x(t)) \to 0 \text{ as } t \to \infty \text{ and } \int_{t_0}^{\infty} \| G_2(t, x(t)) \| dt < \infty ,$$

where $x(t)$ is an arbitrary continuous bounded function defined for $t \geqq t_0$, $t_0 \in I$. Moreover, suppose that the solutions of (23.1) are uniform-bounded and that M is bounded and is a quasi-uniform-asymptotically stable set of (23.1) in the large. Then, the solutions *of* (24.1) *are eventually uniform-bounded and* M *is an eventually quasi-uniform-asymptotically stable set of* (24.1) *in the large.*

COROLLARY 2. *Suppose that* F, G_1 *and* G_2 *are continuous on* $I\times R^n$, $F(t, x)\in \bar{C}_0(x)$ *and that the solutions of* (23.1) *are uniform-bounded and uniform-ultimately bounded. If* G_1 *and* G_2 *satisfy condition* (24.20), *then the solutions of* (24.1) *are eventually uniform-bounded and eventually uniform-ultimately bounded.*

THEOREM 24.3. *Suppose that* F, G_1 *and* G_2 *are continuous on* $I\times R^n$ *and that there exists a Liapunov function* $V(t, x)$ *defined on* $I\times R^n$ *which satisfies the conditions* (i), (iii) *in Theorem* 24.2 *and*

(ii)$'$ *if* $x\in S_\alpha$ *and* $x'\in S_\alpha$, *for continuous functions* $m(\alpha)$ *and* $f(t)$

$$|V(t, x)-V(t, x')|\leqq m(\alpha)f(t)\|x-x'\|.$$

Moreover, suppose that

$$f(t)\|G_1(t, x(t))\|\to 0 \text{ as } t\to\infty \text{ and } \int_{t_0}^{\infty} f(t)\|G_2(t, x(t))\|\,dt<\infty, \tag{24.21}$$

where $x(t)$ *is an arbitrary continuous bounded function for all* $t\geqq t_0$. *Then the bounded solution* $x^*(t\,; x_0, t_0)$ *of* (24.1) *approaches* M *as* $t\to\infty$.

PROOF. Let $x^*(t\,; x_0, t_0)$ be bounded by $\gamma>0$. Then we have

$$V'(t, x^*(t\,; x_0, t_0))\leqq -cV(t, x^*(t\,; x_0, t_0)) + m(\gamma)f(t)\{g_1(t, \gamma)+g_2(t, \gamma)\},$$

where $g_i(t, \gamma)=\max\|G_i(t, x)\|$ for $x\in S_\gamma$, $i=1, 2$. Let $u(t\,; u_0, t_0)$ be the solution of

$$\frac{du}{dt}=-cu+m(\gamma)f(t)\{g_1(t, \gamma)+g_2(t, \gamma)\}, \qquad u_0=V(t_0, x_0).$$

Then, by Lemma 24.2, $a(d(x^*(t\,; x_0, t_0), M(t))\leqq u(t\,; u_0, t_0)$ and moreover, by Lemma 24.3, for each $\varepsilon>0$, there exists a $T(\varepsilon)>0$ such that if $t\geqq T(\varepsilon)$, then $u(t\,; u_0, t_0)<a(\varepsilon)$, which implies that $d(x^*(t\,; x_0, t_0), M(t))<\varepsilon$ for $t\geqq T(\varepsilon)$. This shows that $x^*(t\,; x_0, t_0)\to M$ as $t\to\infty$.

COROLLARY 1. *Under the assumption in Corollary* 1 *of Theorem* 24.2, *if the solutions of* (24.1) *are uniform-bounded, then M is a quasi-uniform-asymptotically stable set of* (24.1) *in the large. If the solutions of* (24.1) *are equi-bounded, then M is a quasi-equi-asymptotically stable set of* (24.1) *in the large. Moreover, if every solution of* (24.1) *is bounded, the set M is a quasi-asymptotically stable set of* (24.1) *in the large.*

Since by Theorem 22.4, there exists a Liapunov function which satisfies the conditions in Theorem 24.3, this corollary can be easily proved.

COROLLARY 2. *Under the assumption in Corollary* 2 *of Theorem* 24.2, *suppose that G_1 and G_2 satisfy condition* (24.20). *If the solutions of* (24.1) *are uniform-bounded, they are uniform-ultimately bounded. If the solutions of* (24.1) *are equi-bounded, they are equi-ultimately bounded. Moreover, if every solution of* (24.1) *is bounded, the solutions of* (24.1) *are ultimately bounded.*

In the following two theorems, we assume that $F(t, x)$ of the system (23.1) and $G_1(t, x)$, $G_2(t, x)$ of the system (24.1) are defined and continuous on $D: 0 \leqq t < \infty,\ d(x, M(t)) \leqq H,\ H > 0$.

THEOREM 24.4. *Suppose that there exists a Liapunov function $V(t, x)$ defined on a domain $D': 0 \leqq t < \infty,\ d(x, M(t)) \leqq H',\ H' \leqq H$, which satisfies the following conditions;*

(i) $a(d(x, M(t))) \leqq V(t, x) \leqq b(d(x, M(t)), \|x\|)$, *where* $a(r) \in CIP$ *and $b(r, s)$ is continuous, $b(r, s) \to 0$ as $r \to 0$,*

(ii) *for any $\alpha > 0$, if $x \in S_\alpha$ and $x' \in S_\alpha$, we have* $|V(t, x) - V(t, x')| \leqq m(\alpha) f(t) \|x - x'\|$, *where $m(\alpha)$, $f(t)$ are continuous functions,*

(iii) $V'_{(23.1)}(t, x) \leqq -cV(t, x)$, *where $c > 0$ is a constant.*

If G_1 and G_2 satisfy condition (24.21) *and if the solutions of* (24.1) *are uniform-bounded, the set M is an eventually uniform-stable set of* (24.1) *with respect to t and moreover, M is an eventually uniform-asymptotically stable set of* (24.1) *with respect to t.*

PROOF. Let $x^*(t; x_0, t_0)$ be a solution of (24.1) such that $x_0 \in S_\alpha$. Then there is a $\gamma(\alpha) > 0$ for which $\|x^*(t; x_0, t_0)\| \leqq \gamma(\alpha)$ for all

$t \geqq t_0$. By (24.21), we have

$$f(t)g_1(t, \gamma) \to 0 \text{ as } t \to \infty \text{ and } \int_{t_0}^{\infty} f(t)g_2(t, \gamma)dt < \infty ,$$

where $g_i(t, \gamma) = \max \| G_i(t, x) \|$ for $x \in S_\gamma$, $i = 1, 2$. Since $| V(t, x) - V(t, x') | \leqq m(\gamma)f(t) \| x - x' \|$ and $\| G_1(t, x) + G_2(t, x) \| \leqq g_1(t, \gamma) + g_2(t, \gamma)$ for $t \in I$, $x \in S_\gamma$, the equation corresponding to (24.8) is

$$\text{(24.22)} \qquad \frac{du}{dt} = -cu + m(\gamma)f(t)\{g_1(t, \gamma) + g_2(t, \gamma)\} , \qquad u_0 = V(t_0, x_0) .$$

The solution $u(t; u_0, t_0)$ of (24.22) is of the form

$$u(t; u_0, t_0) = e^{-c(t-t_0)} V(t_0, x_0) + m(\gamma)e^{-ct}\int_{t_0}^{t} e^{cs} f(s)\{g_1(s, \gamma) + g_2(s, \gamma)\} ds ,$$

and hence,

$$u(t; u_0, t_0) \leqq e^{-c(t-t_0)} b(d(x_0, M(t_0)), \| x_0 \|) + m(\gamma)e^{-ct}\int_{t_0}^{t} e^{cs} f(s)\{g_1(s, \gamma) + g_2(s, \gamma)\} ds .$$

Corresponding to each $\varepsilon > 0$, choose $\delta(\varepsilon, \alpha)$ and $S(\varepsilon, \alpha)$ so that if $r < \delta(\varepsilon, \alpha)$, $\max\limits_{\| x_0 \| \leqq \alpha} b(r, \| x_0 \|) < \frac{a(\varepsilon)}{3}$, that $\sup\limits_{S(\varepsilon, \alpha) \leqq t} f(t)g_1(t, \gamma) < \frac{ca(\varepsilon)}{3m(\gamma)}$, and that $\int_{S(\varepsilon, \alpha)}^{\infty} f(t)g_2(t, \gamma)dt < \frac{a(\varepsilon)}{3m(\gamma)}$. Then, if $t \geqq t_0 > S(\varepsilon, \alpha)$, $d(x_0, M(t_0)) < \delta(\varepsilon, \alpha)$ and $x_0 \in S_\alpha$, then $u(t; u_0, t_0) < a(\varepsilon)$. From Lemma 24.1, it follows that $a(d(x^*(t; x_0, t_0), M(t)) \leqq u(t; u_0, t_0) < a(\varepsilon)$ for all $t \geqq t_0$, which implies $d(x^*(t; x_0, t_0), M(t)) < \varepsilon$. This shows that M is an eventually uniform-stable set of (24.1) with respect to t.

On the other hand, by Lemma 24.3, for any $\varepsilon > 0$ there is a $T(\varepsilon, \alpha)$ such that $u(t; u_0, t_0) < a(\varepsilon)$ for all $t \geqq t_0 + T(\varepsilon, \alpha)$. For a fixed ε_0, set $S(\varepsilon_0, \alpha) = S_1(\alpha)$ and $\delta(\varepsilon_0, \alpha) = \delta_0(\alpha)$. Then, if $t_0 \geqq S_1(\alpha)$, $t \geqq t_0 + T(\varepsilon, \alpha)$ and $d(x_0, M(t_0)) < \delta_0(\alpha)$, then $d(x^*(t; x_0, t_0), M(t)) < \varepsilon$. Thus, the theorem is proved.

THEOREM 24.5. *Suppose that* $F(t, x) \in \bar{C}_0(x)$ *and* M *is bounded and that* M *is a uniform-asymptotically stable set of* (23.1) *and is an invariant set of* (24.1). *If* G_1 *and* G_2 *satisfy condition* (24.20), *then M is a uniform-asymptotically stable set of* (24.1).

PROOF. By assumption, there exists a Liapunov function $V(t, x)$ defined on D' which satisfies the conditions in Theorem 22.5. Since M is bounded, by Theorem 24.4, M is an eventually uniform-stable set of (24.1) and hence, by Theorem 17.2, M is a uniform-stable set of (24.1). Thus, there exists a $\delta_0 > 0$ such that if $d(x_0, M(t_0)) < \delta_0$, then $d(x^*(t; x_0, t_0), M(t)) < H'$, where $x^*(t; x_0, t_0)$ is a solution of (24.1). For this solution,

$$V'(t, x^*(t; x_0, t_0)) \leqq -cV(t, x^*(t; x_0, t_0)) + L\{g_1(t) + g_2(t)\} ,$$

where $g_i(t) = \max \| G_i(t, x) \|$, $i = 1, 2$. Using Lemmas 24.1 and 24.3, for any $\varepsilon > 0$ we can find a $T(\varepsilon)$ such that if $t_0 \in I$, $t \geqq t_0 + T(\varepsilon)$ and $d(x_0, M(t_0)) < \delta_0$, then $d(x^*(t; x_0, t_0), M(t)) < \varepsilon$, which implies quasi-uniform-asymptotic stability of M. This completes the proof.

EXAMPLE 24.1. Consider the system (24.3) and suppose that $G_i(t, 0) \equiv 0$, $i = 1, 2$, and G_1, G_2 satisfy condition (24.20). If all characteristic roots of A have negative real parts and if the 0-set is an invariant set of (24.3), then the zero solution of (24.3) is uniform-asymptotically stable.

In [43], [143], [146], we can find several relationships between solutions of an unperturbed system and of its perturbed system.

§ 25. Perturbed system with small parameters.

Consider the system (23.1) and its perturbed system

$$\text{(25.1)} \qquad \frac{dx}{dt} = F(t, x) + G(t, x, \varepsilon) + H(t, x, \varepsilon) ,$$

where ε is an m-vector. Let M be a set in $I \times R^n$ and assume that M satisfies the conditions required in Section 16. The following assumptions will be made:

(a) $F(t, x)$ is defined and continuous on $D: 0 \leqq t < \infty$, $d(x, M(t)) \leqq A$, $A > 0$.

(b) $G(t, x, \varepsilon)$, $H(t, x, \varepsilon)$ are defined and continuous in (t, x, ε) for $(t, x) \in D$ and $\| \varepsilon \| \leqq \varepsilon_0$ small.

(c) M is a uniform-asymptotically stable set of (23.1).

(d) There exists a constant $B>0$ such that $\|F(t, x)-F(t, x')\| \leqq B\|x-x'\|$ in D.

(e) M is bounded and $\|G(t, x, \varepsilon)\| \leqq g(t, \varepsilon)$, $\|H(t, x, \varepsilon)\| \leqq h(t, \varepsilon)$ for $(t, x) \in D-M$ and $g(t, \varepsilon)=0(\varepsilon)$, $\int_0^\infty h(t, \varepsilon)dt=0(\varepsilon)$ as $\varepsilon \to 0$, or

(e′) $\|G(t, x, \varepsilon)\| \leqq g(t, \varepsilon)$, $\|H(t, x, \varepsilon)\| \leqq h(t, \varepsilon)$ for $(t, x) \in D$ and $g(t, \varepsilon)=0(\varepsilon)$, $\int_0^\infty h(t, \varepsilon)dt=0(\varepsilon)$ as $\varepsilon \to 0$.

THEOREM 25.1. *Under the assumptions* (a), (b), (c), (d) *and* (e) *or* (e′), *for each* ρ, $0<\rho \leqq A$, *there exist a* $\delta(\rho)>0$ *and an* $\varepsilon_0(\rho)$ *such that if* $d(x_0, M(t_0)) \leqq \delta(\rho)$ *and* $\|\varepsilon\| \leqq \varepsilon_0(\rho)$, *then* $d(x(t; x_0, t_0, \varepsilon), M(t)) < \rho$ *for all* $t \geqq t_0$, *where* $x(t; x_0, t_0, \varepsilon)$ *is a solution of* (25.1) *through the point* (t_0, x_0).

PROOF. By the assumptions and Theorem 22.5, in the domain $D': 0 \leqq t < \infty$, $d(x, M(t)) \leqq A'$, $0<A' \leqq A$, there exists a Liapunov function $V(t, x)$ which satisfies the conditions in Theorem 22.5. Using the same notation, choose $\delta>0$ so that $b(\delta)<\dfrac{a(\rho)}{3}$. Then, if $d(x_0, M(t_0)) \leqq \delta$, we have $V(t_0, x_0)<\dfrac{a(\rho)}{3}$. Choose $\varepsilon_0(\rho)$ so small that, under the assumption (e), for $(t, x) \in D-M$ and $\|\varepsilon\| \leqq \varepsilon_0(\rho)$ or, under the assumption (e′), for $(t, x) \in D$ and $\|\varepsilon\| \leqq \varepsilon_0(\rho)$

$$g(t, \varepsilon)<\frac{a(\rho)}{3L} \quad \text{and} \quad \int_0^\infty h(t, \varepsilon)dt<\frac{a(\rho)}{3L}.$$

As long as $x(t; x_0, t_0, \varepsilon)$ exists and $0<d(x(t; x_0, t_0, \varepsilon), M(t))<A'$, if $d(x_0, M(t_0)) \leqq \delta(\rho)$ and $\|\varepsilon\| \leqq \varepsilon_0(\rho)$, setting $c=1$,

$$V(t, x(t; x_0, t_0, \varepsilon)) \leqq e^{-(t-t_0)}V(t_0, x_0)$$
$$+Le^{-t}\left\{\int_{t_0}^t e^s g(s, \varepsilon)ds+\int_{t_0}^t e^s h(s, \varepsilon)ds\right\}<a(\rho).$$

Therefore, as long as $x(t; x_0, t_0, \varepsilon)$ exists, we have $d(x(t; x_0, t_0, \varepsilon), M(t))<\rho$. Thus, if M is bounded, the proof is completed.

In case M is unbounded, suppose that for some $\sigma>t_0$,

$\|x(t;x_0,t_0,\varepsilon)\|\to\infty$ as $t\to\sigma-0$. On the interval $[t_0,\sigma)$, if $d(x_0,M(t_0))\leqq\delta(\rho)$ and $\|\varepsilon\|\leqq\varepsilon_0(\rho)$, then $d(x(t;x_0,t_0,\varepsilon),M(t))<\rho$ and $d(x(t;x_0,t_0),M(t))<\rho$, where $x(t;x_0,t_0)$ is the solution of (23.1). Since we have

$$\|x(t;x_0,t_0,\varepsilon)-x(t;x_0.t_0)\|\leqq B\int_{t_0}^{t}\|x(s;x_0,t_0,\varepsilon)-x(s;x_0,t_0)\|\,ds$$
$$+\int_{t_0}^{t}\|G(s,x(s;x_0,t_0,\varepsilon),\varepsilon)\|\,ds+\int_{t_0}^{t}\|H(s,x(s;x_0,t_0,\varepsilon),\varepsilon)\|\,ds$$

and for $\|\varepsilon\|\leqq\varepsilon_0(\rho)$ and $t_0\leqq t<\sigma$, $\|G(t,x(t;x_0,t_0,\varepsilon),\varepsilon)\|<\dfrac{a(\rho)}{3L}$, $\displaystyle\int_{t_0}^{t}\|H(s,x(s;x_0,t_0,\varepsilon),\varepsilon)\|\,ds<\frac{a(\rho)}{3L}$, we obtain

$$\|x(t;x_0,t_0,\varepsilon)-x(t;x_0,t_0)\|\leqq\left\{\frac{a(\rho)}{3L}(\sigma-t_0)+\frac{a(\rho)}{3L}\right\}e^{B(\sigma-t_0)}.$$

However, $\|x(t;x_0,t_0)\|$ is bounded on $[t_0,\sigma]$ and hence, there arises a contradiction. Thus, $x(t;x_0,t_0,\varepsilon)$ exists for all $t\geqq t_0$ and $d(x(t;x_0,t_0,\varepsilon),M(t))<\rho$ for all $t\geqq t_0$, if $d(x_0,M(t_0))\leqq\delta(\rho)$ and $\|\varepsilon\|\leqq\varepsilon_0(\rho)$.

By setting $\rho=A'$ in Theorem 25.1 and applying Lemma 24.3, we have the following theorem.

THEOREM 25.2. *Under the assumptions* (a), (b), (c), (d) *and* (e′), *if* $g(t,\varepsilon)\to 0$ *as* $t\to\infty$, *the set* M *is a quasi-uniform-asymptotically stable set of* (25.1) *with small* $\|\varepsilon\|$.

EXAMPLE 25.1. Consider two systems

$$\frac{dx}{dt}=F(x), \tag{25.2}$$

$$\frac{dx}{dt}=F(x)+\varepsilon G(t,x,\varepsilon)+\varepsilon H(t,x,\varepsilon), \tag{25.3}$$

where ε is a scalar parameter. Suppose that (25.2) has a periodic solution $x=p(t)$ of period ω. Let P be the set in R^n such that $x=p(t)$, $0\leqq t<\omega$, and let D^* be the domain in R^n such that $d(x,P)\leqq A$. Assume that $F(x)$ has continuous partial derivatives F_x in D^* and that $G(t,x,\varepsilon)$, $H(t,x,\varepsilon)$ are continuous in (t,x,ε) for $(t,x)\in I\times D^*$, ε small. If $n-1$ characteristic exponents of

$\dot{y}=F_x(p(t))y$ have negative real parts, the set $I\times P$ is a uniform-asymptotically stable set of (25.2). For the proof, see [25]. This is a special case of the result which will be discussed in Section 26 concerning integral manifolds. Thus, if there exist two functions $g(t)$ and $h(t)$ such that $\|G(t, x, \varepsilon)\| \leqq g(t)$, $g(t)\to 0$ as $t\to\infty$ and that $\|H(t, x, \varepsilon)\| \leqq h(t)$, $\int_0^\infty h(t)dt<\infty$, and if ε is small enough, the set $I\times P$ is a quasi-uniform-asymptotically stable set of (25.3).

In the following, to make the investigation simple, the set M is assumed to be bounded and hence, assumptions (a), (b), (c), (d) and (e) will be made. For the case where M is not bounded, the same argument is valid under assumptions (a), (b), (c), (d) and (e$'$). Here, we consider a perturbed system

$$\frac{dx}{dt}=F(t, x)+G(t, x, \varepsilon)\,. \tag{25.4}$$

Let $V(t, x)$ be the Liapunov function defined in Theorem 22.5. Corresponding to each $\lambda>0$, let M_λ be the set of points (t, x) where $V(t, x)\leqq\lambda$. Choosing λ small, we can assume that M_λ is contained in D' in which $V(t, x)$ is defined and that there exists a neighborhood $N(M)$ of M containing M_λ in it. Moreover, by Theorem 25.1 with $\rho=A'$ and by choosing λ such that $\lambda<a(\delta(A'))$, we can assume that the solutions of (25.4) with $\|\varepsilon\|$ small starting in $N(M)$ stay in D'.

LEMMA 25.1. *$M\subset M_\lambda$ and M_λ is a closed set in D'.*

THEOREM 25.3. *Under the assumptions* (a), (b), (c), (d), (e) *and $G(t, x, \varepsilon)\in\bar{C}_0(x)$, the system* (25.4) *with $\|\varepsilon\|$ small has an invariant set containing M which approaches M as $\varepsilon\to 0$ and is M for $\varepsilon=0$ and which is a uniform-asymptotically stable set of* (25.4).

PROOF. By assumptions (c), (d), there exists a Liapunov function $V(t, x)$ used in Theorem 25.1. For $\|\varepsilon\|$ small, consider the set M_λ mentioned above. Let $\delta_1=b^{-1}(\lambda)$, where $b^{-1}(r)$ is the inverse function of $b(r)$. Then, if $d(x, M(t))\leqq\delta_1$ and $t\in I$, then $V(t, x)$

$\leqq b(\delta_1)=\lambda$ and hence, $\overline{N(\delta_1, M)}\subset M_\lambda$, where $N(\delta_1, M)$ is the δ_1-neighborhood of M. Choose an $\varepsilon_1(\lambda)$ so small that for $\|\varepsilon\|\leqq\varepsilon_1(\lambda)$, $L\|G(t, x, \varepsilon)\|\leqq\frac{a(\delta_1)}{2}$ for $(t, x)\in D'-M$. Thus, for $(t, x)\in D'-D_{\delta_1}$, where D_{δ_1} is the domain $0\leqq t<\infty$, $d(x, M(t))<\delta_1$, and for the system (25.4) with $\|\varepsilon\|\leqq\varepsilon_1(\lambda)$, we have

$$(25.5)\qquad V'_{(25.4)}(t, x)\leqq -V(t, x)+L\|G(t, x, \varepsilon)\|$$
$$\leqq -a(\delta_1)+\frac{a(\delta_1)}{2}=-\frac{a(\delta_1)}{2}\,.$$

Since $\overline{N(\delta_1, M)}\subset M_\lambda$, (25.5) implies that any solution of (25.4) starting in M_λ cannot leave M_λ, because clearly $V(t, x)>\lambda$ for $(t, x)\notin M_\lambda$. This shows that M_λ is an invariant set of (25.4) with $\|\varepsilon\|\leqq\varepsilon_1(\lambda)$.

By Lemma 25.1, $M\subset M_\lambda$, and clearly if $\lambda=0$, $M_0=M$ and $\varepsilon_1(0)=0$. Since $V(t, x)\leqq\lambda$ implies $a(d(x, M(t)))\leqq\lambda$, we have $d(x, M(t))\leqq a^{-1}(\lambda)$, where $a^{-1}(r)$ is the inverse function of $a(r)$. From this, it follows that $M_\lambda\subset\overline{N(a^{-1}(\lambda), M)}$, which implies that $M_\lambda\to M$ as $\lambda\to 0$. By the properties of $a(r)$, $b(r)$, δ_1 and consequently $a(\delta_1)$ are continuous increasing in λ and vanish at $\lambda=0$. Therefore, we can choose $\varepsilon_1(\lambda)$ to be continuous increasing and zero at $\lambda=0$. Let $\lambda^*(\varepsilon_1)$ be the inverse function of $\varepsilon_1(\lambda)$. Then, if we consider the set M_λ for $\lambda=\lambda^*(\|\varepsilon\|)$, we see that $M_\lambda\to M$ as $\varepsilon\to 0$.

Next, we shall show that M_λ is a uniform-asymptotically stable set of (25.4) with $\|\varepsilon\|$ small. Consider a function $W(t, x)$ defined by $W(t, x)=V(t, x)-\lambda$ for $(t, x)\in D'-M_\lambda$. Clearly $W(t, x)$ is positive, continuous and $W(t, x)\in C_0(x)$. By (25.5), if $\|\varepsilon\|\leqq\varepsilon_1(\lambda)$, we have $W'_{(25.4)}(t, x)\leqq-\frac{a(\delta_1)}{2}$ in the domain $D'-M_\lambda$. Therefore, it can be seen that every solution $x(t\,;x_0, t_0, \varepsilon)$ of (25.4) such that $d(x_0, M(t_0))<\delta(A')$ remains in M_λ for all $t\geqq t_0+T$, $T=2(b(\delta(A'))-\lambda)/a(\delta_1)$. This shows that M_λ is a quasi-uniform-asymptotically stable set of (25.4).

Finally, we shall see that M_λ is a uniform-stable set of (25.4). Let B_1 be the Lipschitz constant of $G(t, x, \varepsilon)$. For each $\rho>0$, choose $\gamma(\rho)>0$ so small that $\gamma<\rho e^{-(B+B_1)T}$, where B is the Lipschitz constant of $F(t, x)$ and T is chosen as above. Let y_0 be a point in $M_\lambda(t_0)$ such that $d(x_0, M(t_0))=d(x_0, y_0)$. Since M_λ is an invariant set of (25.4), $x(t; y_0, t_0, \varepsilon)\in M_\lambda(t)$. Thus, on the interval $[t_0, t_0+T]$, if $d(x_0, M_\lambda(t_0))<\gamma$,

$$\begin{aligned} d(x(t; x_0, t_0, \varepsilon), M_\lambda(t)) &\leqq d(x(t; x_0, t_0, \varepsilon), x(t; y_0, t_0, \varepsilon)) \\ &\leqq \| x(t; x_0, t_0, \varepsilon)-x(t; y_0, t_0, \varepsilon)\| \\ &\leqq e^{(B+B_1)T}\| x_0-y_0\| <\rho\,. \end{aligned}$$

Cleary, for $t\geqq t_0+T$, $d(x(t; x_0, t_0, \varepsilon), M_\lambda(t))<\rho$, and thus M_λ is a uniform-stable set of (25.4).

REMARK. If $F(t, x)$ and $M(t)$ have the same period ω, $V(t, x)$ has the same period ω and hence, the set $M_\lambda\cap\pi_t$ is periodic in t of period ω. Moreover, if F is autonomous and $M(t)=M(0)$ for all $t\in I$, then V also is independent of t and hence, $M_\lambda=I\times M_\lambda^*$, where M_λ^* is the set in x space such that $V(x)\leqq\lambda$.

§ 26. Asymptotic behavior near integral manifolds.

Let $p(t)$ be a periodic solution of period ω of a periodic system

(26.1) $$\frac{dx}{dt}=F(t, x)\,,\qquad F(t+\omega, x)=F(t, x)$$

for $t\in I$, where $F(t, x)$ is continuous and has continuous partial derivatives of the first order with respect to x in a region of $I\times R^n$ which contains the solution curve $(t, p(t))$, $0\leqq t<\infty$. Let $x=z+p(t)$ and let the matrix with columns $\dfrac{\partial F}{\partial x_i}(t, p(t))$ be denoted by $F_x(t, p(t))$. Then, we have

(26.2) $$\frac{dz}{dt}=F_x(t, p(t))z+f(t, z)\,,$$

where $f(t, z)=o(\| z\|)$ for small $\| z\|$ and $f(t, z)$ is periodic in t. Corresponding to (26.1), consider the linear variational equation of (26.1) with respect to $p(t)$

(26.3) $$\frac{dy}{dt}=F_x(t, p(t))y,$$

which has a periodic coefficient matrix of period ω. The system (26.2) is of the form considered in Corollary 2 of Theorem 24.1. Therefore, the following theorem is an immediate consequence of Corollary 2 of Theorem 24.1.

THEOREM 26.1. *Under the assumptions above, if the characteristic exponents of* (26.3) *all have negative real parts, then the periodic solution $p(t)$ of* (26.1) *is exponential-asymptotically stable, i.e., there exist a $K>0$ and a $\lambda>0$ such that*

$$\| x(t)-p(t)\| \leqq Ke^{-\lambda(t-t_0)}\| x(t_0)-p(t_0)\| .$$

When the right member of (26.1) does not depend on t, that is, when the system is autonomous, the linear variational equation

(26.4) $$\frac{dy}{dt}=F_x(p(t))y$$

has a periodic solution $y=\dot{p}(t)$. In this case, (26.4) can have at most $n-1$ characteristic exponents with negative real parts and thus the hypothesis of Theorem 26.1 cannot be satisfied. Indeed, the conclusion of Theorem 26.1 does not hold in this case, and we have another kind of stability.

THEOREM 26.2. *Let $n-1$ characteristic exponents of* (26.4) *have negative real parts. Then there exists an $\varepsilon>0$ such that if a solution $x(t)$ of $\dot{x}=F(x)$ satisfies $\| x(t_1)-p(t_0)\| <\varepsilon$ for some t_0 and t_1, there exists a constant c such that* $\lim\limits_{t\to\infty}\| x(t)-p(t+c)\| =0$.

This is called asymptotic orbital stability with asymptotic phase. In this case, $p(t+\varphi)$, where φ is an arbitrary constant, is also a solution, and hence, the function $p(t+\varphi)$ defines a cylinder C in $I\times R^n$. Thus, we see that the cylinder C is asymptotically stable.

Here, we shall prove more general results due to Hale and Stokes [51] in the case where an autonomous system has an $(n+1)$-parameter family of periodic solutions, by an idea applied by Kato

[56] in consideration of stability of solutions with respect to a family of functions.

Consider an autonomous system

$$\frac{d\xi}{dt}=F(\xi)\,, \tag{26.5}$$

where ξ is an $n+m+1$-vector and $F(\xi)$ is defined and twice continuously differentiable on R^{n+m+1}. Suppose that (26.5) has a continuous $(n+1)$-parameter family of periodic solutions

$$\xi=\xi_0(\omega(b)t+\theta,\, b)\,, \tag{26.6}$$

where $b=(b_1, \cdots, b_n)$ and θ are real constants. In the (t, ξ) space, such a family of solutions defines an n-parameter family of cylinder C_b or an $(n+2)$-dimensional manifold M of solutions. It will be shown that for any solution $\xi(t)$ of (26.5) with $\xi(0)$ sufficiently close to M, there exist θ_0, b_0 such that $\xi(t)-\xi_0(\omega(b_0)t+\theta_0, b_0)\to 0$ as $t\to\infty$, i.e., $\xi(t)$ approaches the cylinder C_{b_0} with asymptotic phase. For the integral manifold, see [16], [47], [51].

Now, the following assumptions on (26.5) will be made:

(a) $b\in U$ for an open subset U of R^n, $\theta\in R^1$ and $\omega(b)>0$ is scalar and continuous in b and $\xi_0(z+\pi, b)=\xi_0(z, b)$.

(b) m of the characteristic exponents $\lambda_1(b), \cdots, \lambda_m(b)$ of the linear variational equation of (26.6),

$$\frac{dw}{dt}=\frac{\partial F(\xi_0(\omega(b)t+\theta, b))}{\partial\xi}w\,,$$

have negative real parts for all $b\in U$.

(c) For all $b\in U$, $z\in R^1$

$$\operatorname{rank}\left(\frac{\partial\xi_0(z, b)}{\partial z}, \frac{\partial\xi_0(z, b)}{\partial b}\right)=n+1\,.$$

To discuss the problem above, local coordinates (x, y, z) are introduced, where x is an n-vector, y is an m-vector and z is a scalar, in a neighborhood of the manifold of solutions of (26.5) in such a way that the manifold is given by $x=\text{const.}$, $y=0$, $z=\omega(x)t+\theta$. Thus, the result that we want to obtain is that a solution of the transformed system has the property that $x\to c_a$, where

c_a is a constant, $y \to 0$, $z - [\omega(c_a)t + \theta] \to 0$, as $t \to \infty$, where θ is a constant.

Now consider a general system

$$
(26.7) \qquad \begin{cases} \dfrac{dx}{dt} = X(t, x, y, z) \\[2ex] \dfrac{dy}{dt} = F(t, x, z)y + Y(t, x, y, z) \\[2ex] \dfrac{dz}{dt} = G(t, x, y) + Z(t, x, y, z), \end{cases}
$$

where x, y, z are n, m and l-dimensional vectors, respectively. Let $S \subset R^n$ and $H \subset R^m$ be bounded open spheres with center at the origin, and let H_1 be the radius of S. Suppose that the functions in (26.7) satisfy the following conditions on $D = I \times S \times H \times R^l$:

(i) F, G, X, Y and Z are continuous on D and belong to the class $C_0(x, y, z)$ with Lipschitz constant L on D.

(ii) $X(t, x, 0, z) = 0$, $Z(t, x, 0, z) = 0$ for $t \in I$, $x \in S$, $z \in R^l$, and thus $\| X(t, x, y, z) \| \leqq L \| y \|$, $\| Z(t, x, y, z) \| \leqq L \| y \|$.

(iii) Given $\varepsilon > 0$, there exists a $\delta(\varepsilon) > 0$ such that $\| Y(t, x, y, z) \| \leqq \varepsilon \| y \|$ for $\| y \| \leqq \delta$, $x \in S$, $z \in R^l$, $t \in I$.

(iv) Let $x_0 \in S$ be a constant and let $u(t)$ be the solution of $\dot{z} = G(t, x_0, 0)$, $u(0) = z_0$, for $t \in I$. If $W(t; x_0, z_0)$ is the matrix solution of

$$
\frac{dW}{dt} = F(t, x_0, u(t))W, \; W(0; x_0, z_0) = E \qquad (E\text{: unit matrix}),
$$

then there exist constants $B > 0$ and $\alpha(\rho) > 0$, continuous and decreasing in ρ, such that for $t \geqq s \geqq 0$, $z_0 \in R^l$ and $\| x_0 \| \leqq \rho$,

$$
\| W(t; x_0, z_0) W^{-1}(s; x_0, z_0) \| \leqq B e^{-\alpha(\rho)(t-s)} .
$$

Hale and Stokes showed that under the assumptions (a), (b), (c) on (26.5), there exists a real matrix $Q(z, b) = Q(z + 2\pi, b)$ such that in a neighborhood of any periodic solution (26.6), say $b = b_0$, the transformation $\xi = \xi_0(z, b) + Q(z, b)y$ transforms (26.5) into an equivalent system

$$\frac{db}{dt}=B(b,y,z)\,,\qquad \frac{dy}{dt}=A^*(b)y+Y^*(b,y,z)\,,$$

$$\frac{dz}{dt}=\omega(b)+Z^*(b,y,z)\,,$$

where the eigenvalues of $A^*(b)$ are $\lambda_1(b),\cdots,\lambda_m(b)$, provided that $\|b-b_0\|$ is sufficiently small. If we set $b=b_0+x$, we have

$$\frac{dx}{dt}=X(x,y,z)\,,\qquad \frac{dy}{dt}=A(x)y+Y(x,y,z)\,,$$

$$\frac{dz}{dt}=\omega_1(x)+Z(x,y,z)\,,$$

which is a special case of (26.7), and all functions satisfy the conditions (i) through (iv). For the transformation, see [51]. Therefore, we shall prove the following theorem.

THEOREM 26.3. *Under the assumptions on the system* (26.7), *given $\rho>0$, $\rho<H_1$, there exist $\eta(\rho)>0$ and $\sigma(\rho)>0$ such that for all $z_0\in R^l$, $\|x_0\|\leqq\rho$, $\|y_0\|\leqq\eta(\rho)$, the solution $\{x(t),y(t),z(t)\}$ of* (26.7) *through (x_0,y_0,z_0) at $t=t_0$ has the following properties;*

(v) $\|y(t)\|\leqq Be^{-\sigma(\rho)(t-t_0)}\|y_0\|$ *for* $t\geqq t_0$,

(vi) $\lim_{t\to\infty}x(t)=x(\infty)$ *exists and* $\|x(t)-x(\infty)\|\leqq B'e^{-\sigma(\rho)(t-t_0)}\|y_0\|$ *for some $B'(\rho)>0$ and all $t\geqq t_0$,*

(vii) $\|z(t)-u(t)\|\leqq B''e^{-\sigma(\rho)(t-t_0)}\|y_0\|$ *for some $B''(\rho)>0$ and all $t\geqq t_0$, where $u(t)$ is the solution of*

$$\frac{dz}{dt}=G(t,x(\infty),0)\,,\qquad u(t_0)=\xi_0 \tag{26.8}$$

for $t\geqq t_0$ and some $\xi_0(t_0,x_0,y_0,z_0)$.

PROOF. First of all, let $\{x(t),y(t),z(t)\}$ be a solution of

$$\frac{dx}{dt}=0\,,\qquad \frac{dy}{dt}=F(t,x,z)y\,,\qquad \frac{dz}{dt}=G(t,x,0) \tag{26.9}$$

through $(t_0,x_0,y_0,z_0)\in D$. Then clearly $z(0)=z_0-\int_0^{t_0}G(s,x_0,0)ds$ and $y(t)$ satisfies

(26.10) $$\frac{dy}{dt}=F(t, x_0, z(t))y\,.$$

Let $W(t\,; x_0, z_0, t_0)$ denote $W\left(t\,; x_0, z_0-\int_0^{t_0} G(s, x_0, 0)ds\right)$ in (iv). This is a fundamental matrix of (26.10), and hence, $y(t)=W(t\,; x_0, z_0, t_0)W^{-1}(t_0\,; x_0, z_0, t_0)y_0$. Since (iv) implies that $\| W(t\,; x_0, z_0, t_0)W^{-1}(s\,; x_0, z_0, t_0)\| \leqq Be^{-\alpha(\|x_0\|)(t-s)}$, we have

(26.11) $$\| y(t)\| \leqq Be^{-\alpha(\|x_0\|)(t-t_0)}\| y_0\|$$

for all $t\geqq t_0$, $x_0\in S$, $z_0\in R^l$. For a point $(t, x, y, z)\in D$, consider a Liapunov function $V(t, x, y, z)$ defined by

(26.12) $$V(t, x, y, z)=\sup_{\tau\geqq 0}\| y(t+\tau\,; x, y, z, t)\| e^{c(\|x\|)\tau}\,,$$

where $c(\rho)\in C_0(\rho)$ is decreasing and $0<c(\rho)<\alpha(\rho)$. Then, it can be easily seen that

(26.13) $$V(t, x, 0, z)=0 \qquad \text{for all} \quad t\in I,\ x\in S,\ z\in R^l\,,$$

(26.14) $$\| y\| \leqq V(t, x, y, z)\leqq B\| y\|\,,$$

(26.15) $$V'_{(26.9)}(t, x, y, z)\leqq -c(\| x\|)V(t, x, y, z)\,.$$

Moreover, there exists a continuous increasing $L_1(\rho)>0$ such that if $t\in I$, $\| x\| \leqq\rho$, $\| x'\| \leqq\rho$, $\| y\| \leqq\eta$, $\| y'\| \leqq\eta$, $z\in R^l$ and $z'\in R^l$,

(26.16) $$| V(t, x, y, z)-V(t, x', y', z')| \leqq B\| y-y'\| +L_1(\rho)\eta[\| x-x'\| +\| z-z'\|]\,.$$

For the proof of (26.16), see [56].

Now consider the system (26.7), which has perturbation terms $X(t, x, y, z)$, $Y(t, x, y, z)$ and $G(t, x, y)-G(t, x, 0)+Z(t, x, y, z)$ for the system (26.9). Then,

(26.17) $$V'_{(26.7)}(t, x, y, z)\leqq -c(\| x\|)V(t, x, y, z) +B\| Y(t, x, y, z)\| +3LL_1(\| x\|)\| y\|^2$$

by (i), (ii) and (26.16). Set $r(\rho)=\dfrac{H_1+\rho}{2}$ and let $H_0(r)$ be $\min\left\{\dfrac{c(r)}{9LL_1(r)},\ \delta\left(\dfrac{c(r)}{3B}\right),\ \text{radius of } H\right\}$, where δ is the one in (iii). Then, if $\| x\| \leqq r$, $\| y\| \leqq H_0(r)$, $z\in R^l$, $t\in I$, we have

(26.18) $$B\|Y(t,x,y,z)\| \leqq \frac{1}{3}c(r)\|y\|,$$

(26.19) $$3LL_1(\|x\|)\|y\|^2 \leqq \frac{1}{3}c(r)\|y\|.$$

Moreover, let $\eta(\rho)$ be such that $\eta(\rho) < \min\left\{\frac{H_0(r)}{B}, \frac{(r-\rho)c(r)}{3BL}\right\}$.

Now, let $\{x(t), y(t), z(t)\}$ be a solution of (26.7) through (t_0, x_0, y_0, z_0) such that $t_0 \in I$, $\|x_0\| \leqq \rho$, $\|y_0\| \leqq \eta(\rho)$ and $z_0 \in R^l$. We assume that $\|y(t)\| < H_0(r)$ and $z(t)$ exists on $[t_0, \tau)$. Suppose that there is a τ_1, $t_0 < \tau_1 < \tau$, such that $\|x(t)\| < r$ on $[t_0, \tau_1)$ and $\|x(\tau_1)\| = r$. Then, on $[t_0, \tau_1]$

$$\begin{aligned}\|y(t)\| &\leqq V(t, x(t), y(t), z(t)) \\ &\leqq V(t_0, x_0, y_0, z_0)e^{-\frac{c(r)}{3}(t-t_0)} \\ &\leqq B\eta(\rho)e^{-\frac{c(r)}{3}(t-t_0)} < H_0(r),\end{aligned}$$

and hence, $\|x(t)\| \leqq \|x_0\| + L\int_{t_0}^t \|y(s)\|\,ds \leqq \|x_0\| + \frac{3}{c(r)}LB\eta(\rho) < r$ on $[t_0, \tau_1]$, which contradicts $\|x(\tau_1)\| = r$. Thus, $\|x(t)\| < r$, $\|y(t)\| < H_0(r)$ on $[t_0, \tau)$. Next, suppose that there exists a finite number $\sigma > t_0$ such that $z(t)$ exists on $[t_0, \sigma)$ and that $\|z(t)\| \to \infty$ as $t \to \sigma - 0$. Since $\|x(t)\| < r$, $\|y(t)\| < H_0(r)$ on $[t_0, \sigma)$,

$$\|z(t)\| \leqq \|z_0\| + \int_{t_0}^t \|G(s, x(s), y(s))\|\,ds + L\int_{t_0}^t \|y(s)\|\,ds.$$

The right-hand side is bounded on $[t_0. \sigma]$, which contradicts the assumption, and hence, $z(t)$ exists in the future. Thus, $\|x(t)\| \leqq r$, $\|y(t)\| \leqq H_0(r)$ for all $t \geqq t_0$ and $z(t)$ exists in the future.

If $t \in I$, $\|x\| \leqq r$, $\|y\| \leqq H_0(r)$ and $z \in R^l$, we have $V'_{(26.7)}(t,x,y,z) \leqq -\sigma(\rho)V(t,x,y,z)$, where $\sigma(\rho) = \frac{1}{3}c(r)$, and hence the property (v) can be easily verified. By (ii) and (v),

$$\int_{t_0}^t \|X(s, x(s), y(s), z(s))\|\,ds \leqq L\int_{t_0}^t \|y(s)\|\,ds < \infty,$$

which implies that $x(\infty)$ exists. On the other hand,

$$\| x(t)-x(\infty) \| \leqq \int_t^\infty \| X(s, x(s), y(s), z(s)) \| ds \leqq L\int_t^\infty \| y(s) \| ds ,$$

and this implies (vi) with $B'(\rho)=\dfrac{LB}{\sigma(\rho)}$.

Since $\int_{t_0}^t \| G(s, x(s), y(s))-G(s, x(\infty), 0) \| ds < \infty$ and $\int_{t_0}^t \| Z(s, x(s), y(s), z(s)) \| ds < \infty$, there exists a constant ξ_0 such that

$$\xi_0 = z_0 + \int_{t_0}^\infty \{G(s, x(s), y(s))-G(s, x(\infty), 0)+Z(s, x(s), y(s), z(s))\} ds .$$

Setting $u(t)=\xi_0+\int_{t_0}^t G(s, x(\infty), 0)ds$, we have

$$\begin{aligned} \| z(t)-u(t) \| &\leqq \int_t^\infty \{\| G(s, x(s), y(s))-G(s, x(\infty), 0) \| \\ &\qquad + \| Z(s, x(s), y(s), z(s)) \|\} ds \\ &\leqq \int_t^\infty \{L\| x(s)-x(\infty) \| + L\| y(s) \| + L\| y(s) \|\} ds , \end{aligned}$$

which shows (vii) with $B''=LB\left(2+\dfrac{L}{\sigma(\rho)}\right)/\sigma(\rho)$.

REMARK. A special case of this theorem was discussed by Malkin [87].

Now we shall consider a system

$$(26.20) \qquad \begin{cases} \dfrac{dx}{dt} = X(t, x, y, z)+\Phi_1(t, x, y, z) \\[2ex] \dfrac{dy}{dt} = F(t, x, z)y+Y(t, x, y, z)+\Phi_2(t, x, y, z) \\[2ex] \dfrac{dz}{dt} = G(t, x, y)+Z(t, x, y, z)+\Phi_3(t, x, y, z) . \end{cases}$$

Correspoding to the system (26.5), its perturbed system

$$(26.21) \qquad \frac{d\xi}{dt} = F(\xi)+F^*(t, \xi)$$

is transformed into a system which is a special case of (26.20). In addition to (i) through (iv), we assume that

(viii) Φ_i, $i=1, 2, 3$, are continuous on D, and there exists a continuous function $\lambda(t)$ such that $\|\Phi_i(t, x, y, z)\| \leqq \lambda(t)$ on D and that $\int_0^\infty \lambda(t)dt < \infty$.

THEOREM 26.4. *Under the assumptions* (i) *through* (iv) *and* (viii), *for any* $\rho > 0$, $\rho < H_1$, *there exist* $\eta(\rho) > 0$ *and* $\gamma(\rho) \geqq 0$ *such that if* $t_0 \geqq \gamma(\rho)$ *and if* $\{x(t), y(t), z(t)\}$ *is the solution of* (26.20) *through* (t_0, x_0, y_0, z_0), *where* $\|x_0\| \leqq \rho$, $\|y_0\| \leqq \eta(\rho)$, $z_0 \in R^l$, *then* $z(t)$ *is defined for* $t \geqq t_0$, $\lim_{t\to\infty} x(t) = x(\infty) \in S$ *exists and* $y(t) \to 0$ *as* $t \to \infty$.

Moreover, in addition, if $\int_0^\infty \int_t^\infty \lambda(u)dudt < \infty$, *then* $\|z(t) - u(t)\| \to 0$ *as* $t \to \infty$, *where* $\dfrac{du}{dt} = G(t, x(\infty), 0)$, $u(t_0) = \xi_0$, $t \geqq t_0$, $\xi_0 = \xi_0(t_0, x_0, y_0, z_0)$.

PROOF. As was seen in the proof of Theorem 26.3, there exists a Liapunov function $V(t, x, y, z)$ defined on D, which satisfies (26.13), (26.14), (26.15) and (26.16). For any $\rho > 0$, $\rho < H_1$, and $r(\rho) = \dfrac{H_1 + \rho}{2}$, choose $\eta(\rho) > 0$, $\gamma(\rho) \geqq 0$ so that $B\eta(\rho) < \dfrac{H_0(r(\rho))}{2}$, $\dfrac{3LB\eta(\rho)}{c(r(\rho))} < \dfrac{r(\rho) - \rho}{2}$, $B^*(r(\rho))\int_{\gamma(\rho)}^\infty \lambda(t)dt < \dfrac{H_0(r(\rho))}{2}$ and $\left\{1 + \dfrac{3LB^*(r(\rho))}{c(r(\rho))}\right\} \int_{\gamma(\rho)}^\infty \lambda(t)dt < \dfrac{r(\rho) - \rho}{2}$, where $H_0(r)$ is the one in the proof of Theorem 26.3 and $B^*(r(\rho)) = 2L_1(r(\rho))H_0(r(\rho)) + B$.

Suppose that for $t_0 \geqq \gamma(\rho)$, $\|x_0\| \leqq \rho$, $\|y_0\| \leqq \eta(\rho)$, $z_0 \in R^l$, $z(t)$ exists on $[t_0, \tau)$. If there is a τ_1, $t_0 < \tau_1 < \tau$, such that $\|x(t)\| < r$ on $[t_0, \tau_1)$ and $\|x(\tau_1)\| = r$, then as long as $\|y(t)\| \leqq H_0(r)$,

$$V'(t, x(t), y(t), z(t)) \leqq -\sigma(\rho) V(t, x(t), y(t), z(t)) + B^*(r(\rho))\lambda(t), \tag{26.22}$$

where $\sigma(\rho) = \dfrac{c(r(\rho))}{3}$, and hence,

$$\|y(t)\| \leqq V(t_0, x_0, y_0, z_0)e^{-\sigma(\rho)(t-t_0)} + B^*(r(\rho))e^{-\sigma(\rho)t}\int_{t_0}^t e^{\sigma(\rho)s}\lambda(s)ds, \tag{26.23}$$

which implies that $\| y(t) \| < H_0(r(\rho))$ on $[t_0, \tau_1]$. Thus, we have (26.22) on $[t_0, \tau_1]$. The right-hand side of (26.23) is the solution $w(t)$, $w(t_0) = V(t_0, x_0, y_0, z_0)$, of

$$\frac{dw}{dt} = -\sigma(\rho)w + B^*(r(\rho))\lambda(t)\,. \tag{26.24}$$

Let $W(t, t_0, \rho)$ be the solution of (26.24) such that $W(t_0, t_0, \rho) = B\eta(\rho)$. Then clearly

$$\| y(t) \| \leqq w(t) \leqq W(t, t_0, \rho) \leqq H_0(r(\rho))\,. \tag{26.25}$$

From (26.24), it follows that

$$\int_{t_0}^{t} w(s)ds = \frac{1}{\sigma(\rho)} \{w(t_0) - w(t)\} + \frac{B^*}{\sigma(\rho)} \int_{t_0}^{t} \lambda(s)ds\,, \tag{26.26}$$

which implies that

$$\int_{t_0}^{t} W(s, t_0, \rho)ds \leqq \frac{1}{\sigma(\rho)} \left\{B\eta(\rho) + B^* \int_{t_0}^{t} \lambda(s)ds\right\}, \tag{26.27}$$

$$\int_{s}^{\infty} W(u, t_0, \rho)du = \frac{1}{\sigma(\rho)} \left\{W(s, t_0, \rho) + B^* \int_{s}^{\infty} \lambda(u)du\right\}. \tag{26.28}$$

From (26.25) and (26.27),

$$\begin{aligned} \| x(t) \| &\leqq \| x_0 \| + L\int_{t_0}^{t} \| y(s) \| ds + \int_{t_0}^{t} \lambda(s)ds \\ &\leqq \| x_0 \| + \frac{LB\eta(\rho)}{\sigma(\rho)} + \left\{1 + \frac{LB^*}{\sigma(\rho)}\right\} \int_{t_0}^{t} \lambda(s)ds < r(\rho) \end{aligned}$$

on $[t_0, \tau_1]$ and this contradicts $\| x(\tau_1) \| = r$. Therefore, $\| x(t) \| < r$ on $[t_0, \tau)$.

Next, suppose that there exists a τ, $\tau > t_0$, such that $z(t)$ exists on $[t_0, \tau)$ and $\| z(t) \| \to \infty$ as $t \to \tau - 0$. Then $\| x(t) \| < r$, $\| y(t) \| < H_0(r)$ and $\int_{t_0}^{t} \| y(s) \| ds < \frac{r-\rho}{L}$ on $[t_0, \tau)$, and hence

$$\| z(t) \| \leqq \| z_0 \| + \int_{t_0}^{t} \| G(s, x(s), y(s)) \| ds + L\int_{t_0}^{t} \| y(s) \| ds + \int_{t_0}^{t} \lambda(s)ds\,.$$

Since the right-hand side is bounded on $[t_0, \tau]$, there arises a contradiction, which shows that $z(t)$ exists in the future and consequently $\| x(t) \| < r(\rho)$, $\| y(t) \| < H_0(r(\rho))$ for all $t \geqq t_0$.

Thus, we have the inequality (26.22) for all $t \geqq t_0$. By Lemma 24.3, the solution $W(t, t_0, \rho)$ of (26.24) tends to zero as $t \to \infty$, and hence $y(t) \to 0$ as $t \to \infty$. The existence of $x(\infty)$ is clear. This proves the first part of the theorem.

Under the additional condition $\int_0^\infty \int_t^\infty \lambda(u)dudt < \infty$, we shall prove that $\| z(t) - u(t) \| \to 0$ as $t \to \infty$. From (26.28), it follows that

$$(26.29) \quad \int_t^\infty \int_s^\infty W(u, t_0, \rho)duds \leqq \frac{1}{\sigma^2(\rho)} \left\{ W(t, t_0, \rho) + B^* \int_t^\infty \lambda(s)ds \right\} + \frac{B^*}{\sigma(\rho)} \int_t^\infty \int_s^\infty \lambda(u)duds ,$$

which implies that

$$(26.30) \qquad \int_{t_0}^\infty \int_t^\infty \| y(s) \| dsdt < \infty .$$

Moreover, we have

$$(26.31) \qquad \| x(t) - x(\infty) \| \leqq L \int_t^\infty \| y(s) \| ds + \int_t^\infty \lambda(s)ds ,$$

and hence we have $\int_{t_0}^\infty \| G(s, x(s), y(s)) - G(s, x(\infty), 0) \| ds < \infty$. Thus, it can be seen that there exists a constant ξ_0 such that

$$\xi_0 = z_0 + \int_{t_0}^\infty \{ G(s, x(s), y(s)) - G(s, x(\infty), 0) + Z(s, x(s), y(s), z(s)) + \Phi_3(s, x(s), y(s), z(s)) \} ds .$$

Setting $u(t) = \xi_0 + \int_{t_0}^t G(s, x(\infty), 0)ds$, we have

$$\| z(t) - u(t) \| \leqq \int_t^\infty \{ 2L \| y(s) \| + L \| x(s) - x(\infty) \| + \lambda(s) \} ds ,$$

and hence, by (26.25), (26.28), (26.30) and (26.31), we have $\| z(t) - u(t) \| \to 0$.

To prove the next theorem which is a type of converse of the preceding one, first of all, we shall prove a lemma which is concerned with a special n-point boundary value problem which Hukuhara [53] applied in studying the behavior of solutions in a neighborhood of a singular point. Here, we shall state a simple form. For more general cases, see [102].

LEMMA 26.1. *In a system*

$$\frac{dx}{dt}=f(t, x, y), \qquad \frac{dy}{dt}=g(t, x, y), \tag{26.32}$$

where x, y are n-vector and m-vector, suppose that f, g are continuous and bounded on $[a, b]\times R^n\times R^m$. Then, for any $(x_0, y_0)\in R^n\times R^m$, there exists at least one solution $\{x(t), y(t)\}$ of (26.32) which is defined on $[a, b]$ and satisfies $x(b)=x_0$, $y(a)=y_0$.

PROOF. At first, we assume that f and g are continuously differentiable with respect to (x, y). For any $\xi\in R^n$, let $\{x(t;\xi), y(t;\xi)\}$ be a solution of (26.32) through (a, ξ, y_0), where obviously $x(t;\xi)$ is continuous in ξ and clearly $\{x(t;\xi), y(t;\xi)\}$ exists on $[a, b]$. Consider a continuous function $F(\xi)$ such that $F(\xi)=\xi-x(b;\xi)+x_0$. If B is a bound of $f(t, x, y)$ for $t\in[a, b]$, $x\in R^n$, $y\in R^m$, we have $\|F(\xi)\|\leqq B(b-a)+\|x_0\|$. Clearly, the set $S=\{\xi;\ \|\xi\|\leqq B(b-a)+\|x_0\|\}$ is compact and convex. Since the mapping F maps S into itself, by Brouwer's fixed point theorem, we can find a point $\xi_0\in S$ such that $F(\xi_0)=\xi_0$, which implies $x(b;\xi_0)=x_0$. Thus, $\{x(t;\xi_0), y(t;\xi_0)\}$ is the required solution of (26.32).

On the other hand, corresponding to continuous bounded functions f, g, define $f_k(t, x, y)$ and $g_k(t, x, y)$ in the following way: Let $S_k(z)$ be a domain in $R^n\times R^m$ such that $S_k(z)=\left\{z';\ \|z-z'\|\leqq\frac{1}{k}\right\}$, $z=(x, y)$, and let d_k is the measure of $S_k(z)$. If we set $f_k(t, x, y)=\frac{1}{d_k}\int_{S_k(z)} f(t, x, y)dxdy$ and $g_k(t, x, y)=\frac{1}{d_k}\int_{S_k(z)} g(t, x, y)dxdy$, $\{f_k, g_k\}$ tends to $\{f, g\}$ uniformly on $[a, b]\times R^n\times R^m$ as $k\to\infty$, and it is bounded and continuously differentiable with respect to (x, y).

Let $\{x_k(t), y_k(t)\}$ be a solution of $\dot{x}=f_k(t, x, y)$, $\dot{y}=g_k(t, x, y)$ such that $x_k(b)=x_0$, $y_k(a)=y_0$. Since f_k, g_k are uniformly bounded on $[a, b]\times R^n\times R^m$, $\{x_k(t), y_k(t)\}$ is uniformly bounded and equicontinuous. From this, it follows that $\{x_k(t), y_k(t)\}$ has a subsequence which converges uniformly to a function $\{x(t), y(t)\}$. Then, $\{x(t), y(t)\}$ is a solution of (26.32) such that $x(b)=x_0$, $y(a)=y_0$.

THEOREM 26.5. *Under the assumptions* (i) *through* (iv) *and* (viii), *if* $\int_0^\infty \int_t^\infty \lambda(u)dudt < \infty$, *for any* $\rho > 0$, $\rho < H_1$, *there exist* $\zeta(\rho) > 0$ *and* $T(\rho) \geqq 0$ *such that for any* (t_0, x_0, y_0, z_0), $t_0 \geqq T(\rho)$, $\|x_0\| \leqq \rho$, $\|y_0\| \leqq \zeta(\rho)$, $z_0 \in R^l$, *there exists a solution of the system* (26.20) *which starts at* t_0 *and tends to a given solution of the system* (26.9) *through* (t_0, x_0, y_0, z_0) *as* $t \to \infty$.

PROOF. Let $\zeta(\rho)$ be such that $\zeta(\rho) < \eta(r(\rho))$ and let $\{\bar{x}(t), \bar{y}(t), \bar{z}(t)\}$ be a given solution of (26.9) through (t_0, x_0, y_0, z_0) such that $t_0 \geqq T(\rho) = \gamma(r(\rho))$, $\|x_0\| \leqq \rho < r(\rho)$, $\|y_0\| \leqq \zeta(\rho)$, $z_0 \in R^l$, where $r(\rho)$, $\eta(\rho)$, $\gamma(\rho)$ are the same as in Theorem 26.4. Then, as was seen in Theorem 26.4, this solution exists in the future and $\|\bar{x}(t)\| = \|x_0\| < r(r(\rho))$, $\|\bar{y}(t)\| < W(t, t_0, r(\rho))$. Define the functions F^*, G^*, X^*, Y^*, Z^* and Φ_i^*, $i = 1, 2, 3$, by replacing t, x, y, z in Fy, G, X, Y, Z and Φ_i by t, $\min\left(1, \frac{r(r(\rho))}{\|x\|}\right)x$, $\min\left(1, \frac{W(t, t_0, r(\rho))}{\|y\|}\right)y$, $\min\left(1, \frac{A(t)}{\|z\|}\right)z$, respectively, where $A(t) = A_1(t_0, t, \|z_0\| + A_2(t_0, t_0, \rho), r(\rho))$ with $M(\rho) = \frac{LB^*(r(\rho))}{\sigma(\rho)}$ and

$$A_1(t_0, t, \beta, \rho) = \beta + Lr(\rho)(t - t_0) + \frac{2L}{\sigma(\rho)} B\eta(\rho) + \{2M(\rho) + 1\}\int_{t_0}^t \lambda(s)ds + \int_{t_0}^t \|G(s, 0, 0)\|ds\,,$$

$$A_2(t_0, t, \rho) = \left\{\frac{L}{\sigma(r(\rho))} + 2\right\}\frac{L}{\sigma(r(\rho))} W(t, t_0, r(\rho)) + \left\{1 + M(r(\rho))\left(\frac{L}{\sigma(r(\rho))} + 2\right)\right\}\int_t^\infty \lambda(s)ds + \{M(r(\rho)) + 1\}L\int_t^\infty \int_s^\infty \lambda(u)duds\,.$$

Clearly, these functions are continuous on $[t_0, \infty) \times R^n \times R^m \times R^l$ and are bounded on $I' \times R^n \times R^m \times R^l$ for any compact subinterval $I' \subset [t_0, \infty)$. Moreover, we have

$$\begin{aligned} &\|X^*(t, x, y, z)\| \leqq LW(t, t_0, r(\rho))\,, \\ (26.33) \qquad &\|Z^*(t, x, y, z)\| \leqq LW(t, t_0, r(\rho))\,, \\ &\|G^*(t, x, y) - G^*(t, x, 0)\| \leqq LW(t, t_0, r(\rho))\,, \end{aligned}$$

(26.34) $$\|\Phi_i^*(t, x, y, z)\| \leqq \lambda(t), \qquad i=1, 2, 3,$$

on $[t_0, \infty)\times R^n\times R^m\times R^l$.

Consider the system

(26.35)
$$\begin{cases} \dfrac{dx}{dt} = X^*(t, x, y, z)+\Phi_1^*(t, x, y, z) \\[2ex] \dfrac{dy}{dt} = F^*(t, x, y, z)+Y^*(t, x, y, z)+\Phi_2^*(t, x, y, z) \\[2ex] \dfrac{dz}{dt} = G^*(t, x, y)+Z^*(t, x, y, z)+\Phi_3^*(t, x, y, z), \end{cases}$$

which is the same as (26.20) in the domain $t_0 \leqq t < \infty$, $\|x\| \leqq r(r(\rho))$, $\|y\| \leqq W(t, t_0, r(\rho))$, $\|z\| \leqq A(t)$. For a divergent sequence $\{t_k\}$, $t_k > t_0$, and a fixed y_0' such that $\|y_0'\| \leqq \zeta(\rho)$, by Lemma 26.1, there exists a solution $\{x_k(t), y_k(t), z_k(t)\}$ of (26.35) which satisfies $x_k(t_k) = \bar{x}(t_k)$, $y_k(t_0) = y_0'$, $z_k(t_k) = \bar{z}(t_k)$.

Clearly, $\|\bar{x}(t)\| = \|\bar{x}(t_k)\| \leqq \rho < r(\rho)$, and we have for $t \in [t_0, t_k]$

$$\begin{aligned} \|x_k(t)\| &\leqq \|x_k(t_k)\| + L\int_t^{t_k} W(s, t_0, r(\rho))ds + \int_t^{t_k} \lambda(s)ds \\ &\leqq \rho + \frac{LB\eta(r(\rho))}{\sigma(r(\rho))} + \left\{1 + \frac{LB^*(r(r(\rho)))}{\sigma(r(\rho))}\right\} \int_t^{t_k} \lambda(s)ds, \end{aligned}$$

which implies that

(26.36) $$\|x_k(t)\| < r(\rho) \quad \text{for all } t \in [t_0, t_k].$$

Moreover, we have

$$\begin{aligned} \|z_k(t) - \bar{z}(t)\| \leqq &\int_t^{t_k} \|G^*(s, x_k(s), y_k(s)) - G^*(s, x_k(s), 0)\| ds \\ &+ \int_t^{t_k} \|G^*(s, x_k(s), 0) - G(s, \bar{x}(s), 0)\| ds \\ &+ \int_t^{t_k} \|Z^*(s, x_k(s), y_k(s), z_k(s))\| ds \\ &+ \int_t^{t_k} \|\Phi_3^*(s, x_k(s), y_k(s), z_k(s))\| ds. \end{aligned}$$

Since $\|x_k(s)\| < r(\rho) < r(r(\rho))$, $\|G^*(s, x_k(s), 0) - G(s, \bar{x}(s), 0)\| = \|G(s, x_k(s), 0) - G(s, \bar{x}(s), 0)\| \leqq L\|x_k(s) - \bar{x}(s)\|$. On the other hand, by

(26.33) and (26.34)

$$\int_t^{t_k} \| x_k(s)-\bar{x}(s) \| ds \leqq L\int_t^{t_k}\int_s^{t_k} W(u, t_0, r(\rho))duds+\int_t^{t_k}\int_s^{t_k} \lambda(u)duds .$$

Thus, by (26.28), (26.29), (26.33) and (26.34)

(26.37) $\quad \| z_k(t)-\bar{z}(t) \| \leqq A_2(t_0, t, \rho)$ for all $t \in [t_0, t_k]$.

From (26.36) and (26.37), it follows that

(26.38) $\quad \| x_k(t_0) \| < r(\rho)$ and $\| z_k(t_0) \| \leqq \| z_0 \| + A_2(t_0, t_0, \rho)$.

By estimating $\| z_k(t)-\int_{t_0}^t G(s, 0, 0)ds \|$, we obtain $\| z_k(t) \| < A_1(t_0, t, \| z_k(t_0) \|, r(\rho))$, and hence, by (26.38),

(26.39) $\quad \| z_k(t) \| < A_1(t_0, t, \| z_0 \| + A_2(t_0, t_0, \rho), r(\rho)) = A(t)$.

Now it will be proved that $\| y_k(t) \| < W(t, t_0, r(\rho))$ on $[t_0, t_k]$. Since $\| y_k(t_0) \| = \| y_0' \| < \eta(r(\rho))$, $\| y_k(t_0) \| < W(t_0, t_0, r(\rho))$. Suppose that $\| y_k(\tau) \| = W(\tau, t_0, r(\rho))$ at some τ, $t_0 < \tau \leqq t_k$, and that $\| y_k(t) \| < W(t, t_0, r(\rho))$ for $t \in [t_0, \tau)$. Then, on $[t_0, \tau]$, $\{x_k(t), y_k(t), z_k(t)\}$ is a solution of (26.20) such that $\| x_k(t_0) \| \leqq r(\rho)$, $\| y_k(t_0) \| < \eta(r(\rho))$, $z_k(t_0) \in R^l$, and hence, by Theorem 26.4, $\| y_k(t) \| < W(t, t_0, r(\rho))$, which is a contradiction. Thus, $\| y_k(t) \| < W(t, t_0, r(\rho))$ for $t \in [t_0, t_k]$. From this and (26.36), (26.39), it follows that $\{x_k(t), y_k(t), z_k(t)\}$ is a solution of (26.20).

Since $t_0 \geqq \gamma(r(\rho))$, $\| x_k(t_0) \| \leqq r(\rho)$, $\| y_k(t_0) \| \leqq \eta(r(\rho))$, $z_k(t_0) \in R^l$, we can assume that $\{x_k(t), y_k(t), z_k(t)\}$ is a solution of (26.20) defined on $[t_0, \infty)$, which satisfies $\| x_k(t) \| \leqq r(r(\rho))$, $\| y_k(t) \| \leqq W(t, t_0, r(\rho))$, $\| z_k(t) \| \leqq A(t)$ and $x_k(t_k) = \bar{x}(t_k)$, $y_k(t_0) = y_0'$, $z_k(t_k) = \bar{z}(t_k)$. Thus, we can find a subsequence which converges to a function $\{x(t), y(t), z(t)\}$ defined on $[t_0, \infty)$ uniformly on any compact subinterval of $[t_0, \infty)$. Clearly, $\{x(t), y(t), z(t)\}$ is a solution of (26.20). We shall also denote the subsequence by $\{x_k(t), y_k(t), z_k(t)\}$.

Since $\| x_k(t)-\bar{x}(t) \| \leqq L\int_t^{t_k} W(u, t_0, r(\rho))du+\int_t^{t_k} \lambda(u)du$ for $t \in [t_0, t_k]$, if we choose $\tau(\varepsilon, \rho) \geqq 0$ so that

$$\int_{\tau(\varepsilon,\rho)}^{\infty} W(u, t_0, r(\rho))du < \frac{\varepsilon}{2L} \quad \text{and} \quad \int_{\tau(\varepsilon,\rho)}^{\infty} \lambda(u)du < \frac{\varepsilon}{2},$$

then $\| x_k(t)-\bar{x}(t) \| < \varepsilon$ for any $t \in [\tau(\varepsilon, \rho), t_k]$, which implies that $\| x(t)-\bar{x}(t) \| \leqq \varepsilon$ for all $t \geqq \tau(\varepsilon, \rho)$. Moreover, if we choose $\tau(\varepsilon, \rho)$ so that $W(t, t_0, r(\rho)) < \varepsilon$, $A_2(t_0, t, \rho) < \varepsilon$ for all $t \geqq \tau(\varepsilon, \rho)$, then we have $\| z_k(t)-\bar{z}(t) \| < \varepsilon$ for $t \in [\tau(\varepsilon, \rho), t_k]$ by (26.37) and $\| y_k(t) \| < \varepsilon$ for all $t \geqq \tau(\varepsilon, \rho)$. Thus, $\| z(t)-\bar{z}(t) \| \leqq \varepsilon$ and $\| y(t) \| \leqq \varepsilon$ for all $t \geqq \tau(\varepsilon, \rho)$. Therefore, $x(t) \to \bar{x}(t)$, $y(t) \to \bar{y}(t)$ and $z(t) \to \bar{z}(t)$ as $t \to \infty$. This completes the proof.

For the system (26.5), the following theorem follows from Theorem 26.3 (cf. [51]).

THEOREM 26.6. *Suppose that the system* (26.5) *has a continuous* $(n+1)$*-parameter family of periodic solutions* (26.6). *If the assumptions* (a), (b) *and* (c) *are satisfied, then the manifold* M *defined by* (26.6) *is asymptotically stable with asymptotic phase and amplitude.*

Applying Theorems 26.4 and 26.5, the following theorem is obtained (cf. [51], [59], [151]).

THEOREM 26.7. *Consider the system* (26.21), *where the system* (26.5) *satisfies the conditions of Theorem* 26.6 *and* F^* *is continuous on* $I \times R^{n+m+1}$, *and suppose that there exist continuous functions* $h(u)$, $g(t)$ *and a* $T \geqq 0$ *such that*

$$\| F^*(t, \xi) \| \leqq g(t)h(\| \xi \|)$$

for all ξ, $t \geqq T$. *If* $\int_T^\infty g(t)dt < \infty$, *then the manifold of solutions* (26.6) *of* (26.5) *is an asymptotically stable manifold of* (26.21) *with asymptotic amplitude. Moreover, if* $\int_T^\infty \int_t^\infty g(u)dudt < \infty$, *then this asymptotically stable manifold of* (26.21) *has asymptotic amplitude and phase, and there exists a solution of* (26.21) *which has a given asymptotic amplitude and phase, that is, there exists a solution of* (26.21) *which tends to a given periodic solution of* (26.5).

§ 27. Asymptotic equivalence.

There are several cases of asymptotic equivalence of two systems and there are many results concerning such problems.

Here, we shall discuss a simple case. For other cases and references, see [52], [57], [58], [123].

Consider a linear system and its perturbed system

$$\frac{dx}{dt} = Ax + p(t)\,, \tag{27.1}$$

$$\frac{dx}{dt} = Ax + p(t) + f(t, x)\,, \tag{27.2}$$

where A is an $n \times n$ constant matrix, $p(t)$ is continuous on I and $f(t, x)$ is continuous on $I \times R^n$. The following assumptions will be made;

(i) all solutions of (27.1) are bounded,

(ii) $\|f(t, x)\| \leqq \lambda(t)\varphi(\|x\|)$, where $\lambda(t) \geqq 0$ is continuous, $\int_0^\infty \lambda(t)dt < \infty$ and $\varphi(r) > 0$ for $r \geqq 0$, $\varphi(r)$ is continuous, increasing, $\int^\infty \frac{dr}{\varphi(r)} = \infty$.

THEOREM 27.1. *Under the assumptions* (i), (ii), *the systems* (27.1) *and* (27.2) *are asymptotically equivalent, that is, for any given solution of* (27.1) *there exists a solution of* (27.2) *which approaches the given solution of* (27.1) *as* $t \to \infty$, *and conversely.*

This theorem can be proved by using a Liapunov function and by a slight modification of the argument used in the proof of Theorem 26.5. To prove the theorem, we shall transform (27.1) and (27.2) into

$$\frac{dy}{dt} = Ay\,, \tag{27.3}$$

$$\frac{dy}{dt} = Ay + g(t, x) \tag{27.4}$$

by $x = y + \varphi(t)$, where $\varphi(t)$ is a solution of (27.1) starting at $t = 0$ and $g(t, y) = f(t, y + \varphi(t))$. Since all solutions of (27.1) are bounded, there exists a $b > 0$ such that $\|\varphi(t)\| \leqq b$ for all $t \geqq 0$ and all characteristic roots of A have nonpositive real parts and the elementary divisors of characteristic roots with zero real parts are linear. Therefore, by a suitable transformation [57], the systems (27.3) and (27.4) can be transformed into

$$\frac{d\xi}{dt} = B\xi , \qquad \frac{d\eta}{dt} = 0 , \tag{27.5}$$

$$\frac{d\xi}{dt} = B\xi + F_1(t, \xi, \eta) , \qquad \frac{d\eta}{dt} = F_2(t, \xi, \eta) , \tag{27.6}$$

respectively, where all characteristic roots of B have negative real parts and F_1, F_2 will satisfy an inequality

$$\| F_i(t, \xi, \eta) \| \leqq K\lambda(t)\psi(K\sqrt{\| \xi \|^2 + \| \eta \|^2}) , \qquad i = 1, 2 ,$$

for some constant $K > 0$ and $\psi(r) = \varphi(r+b)$. Therefore, it is sufficient to see the asymptotic equivalence of the systems (27.5) and (27.6).

Since the solutions of (27.5) are uniform-bounded, by Theorem 19.1 with $c=0$, there exists a Liapunov function $V(t, \xi, \eta)$. By using this Liapunov function, it can be shown that a solution of (27.6), $\{\xi(t), \eta(t)\}$, is bounded. Moreover, since $\xi(t) \equiv 0$ of $\dot{\xi} = B\xi$ is exponential-asymptotically stable, there exists a Liapunov function, by which we can see that $\xi(t) \to 0$ as $t \to \infty$ and $\lim_{t\to\infty} \eta(t)$ exists. Thus, for any given solution of (27.6), there exists a solution of (27.5) which tends to the given solution of (27.6).

For a given solution $\{\bar{\xi}(t), \bar{\eta}(t)\}$ of (27.5), by a similar argument to the one used in the proof of Theorem 26.5 and by applying Lemma 26.1, we can find a solution $\{\xi(t), \eta(t)\}$ of (27.6) such that $\xi(t) \to \bar{\xi}(t)$, $\eta(t) \to \bar{\eta}(t)$ as $t \to \infty$. The initial time of $\{\xi(t), \eta(t)\}$ may be different from the initial time of $\{\bar{\xi}(t), \bar{\eta}(t)\}$. However, under our assumptions, every solution $y(t; y_0, t_0)$ of (27.4) is continuable to the left, and hence, we can take the same initial time. For the details of the proof, see [59].

CHAPTER VII. EXISTENCE THEOREMS FOR PERIODIC SOLUTIONS AND ALMOST PERIODIC SOLUTIONS

Using fixed point theorems and Liapunov functions, the existence of periodic solutions will be discussed. Moreover, under uniform-asymptotic stability of a bounded solution, we shall obtain existence theorems for almost periodic solutions in almost periodic systems.

§ 28. Fixed point theorems.

In this section, we shall state some fixed point theorems without proofs. The following theorem is due to Brouwer. For the proof, see [20].

THEOREM 28.1. *Let G be a simply connected plane open domain and T a topological mapping of G into itself. If T is sense-preserving and there exist a point $x_0 \in G$ and a subsequence of its successive images $x_1 = Tx_0$, $x_2 = Tx_1$, $\cdots$ which converges to a point in G, then T has a fixed point in G.*

The following theorem due to Browder [21] is very useful in obtaining existence theorems for periodic solutions (cf. [42], [54], [55], [145], [148]).

THEOREM 28.2. *Let S and S_1 be open convex subsets of the Banach space X, S_0 a closed convex subset of X, $S_0 \subset S_1 \subset S$, f a compact mapping of S into X. Suppose that for a positive integer m, f^m is well-defined on S_1, $\bigcup_{0 \leqq j \leqq m} f^j(S_0) \subset S_1$, while $f^m(S_1) \subset S_0$. Then f has a fixed point in S_0.*

Here, f is said to be a compact mapping of S, if f is a continuous mapping and $f(S)$ is contained in a compact set of X.

For the proof, see [21]. This theorem is a generalization of Schauder's fixed point theorem which is described in the following way.

THEOREM 28.3. *A continuous mapping of a compact convex set S in the Banach space X into itself has at least one fixed point.*

As an application of Schauder's Theorem, we shall prove the following theorem which is concerned with a boundary value problem of an equation of the second order.

THEOREM 28.4. *In an equation of the second order*

$$\frac{d^2x}{dt^2}=f\left(t, x, \frac{dx}{dt}\right), \tag{28.1}$$

suppose that $f(t, x, y)$ is defined and continuous on $a\leqq t\leqq b$, $|x|<\infty$, $|y|<\infty$ and that $|f(t, x, y)|\leqq L$ for some constant $L>0$. Then for any pair of constants A, B, there exists a solution $x(t)$ of (28.1) *which satisfies the conditions*

$$x(a)=A, \qquad x(b)=B. \tag{28.2}$$

PROOF. It is seen easily that $x(t)$ satisfying the following equations is the desired, that is,

$$\begin{cases} x(t)=\displaystyle\int_a^b G(t, \xi)f(\xi, x(\xi), \dot{x}(\xi))d\xi+\frac{B(t-a)+A(b-t)}{b-a} \\ \dot{x}(t)=\displaystyle\int_a^b G_t(t, \xi)f(\xi, x(\xi), \dot{x}(\xi))d\xi+\frac{B-A}{b-a}, \end{cases}$$

where

$$G(t, \xi)=\begin{cases} \dfrac{(t-b)(\xi-a)}{b-a} & (a\leqq\xi\leqq t\leqq b) \\ \dfrac{(t-a)(\xi-b)}{b-a} & (a\leqq t\leqq\xi\leqq b). \end{cases}$$

Let z be a pair of continuous functions $x(t), y(t)$ defined on $[a, b]$. Clearly the set X of z is a complete normed linear space, if the norm of z is defined by $\|z\|=\max\limits_{a\leqq t\leqq b}\{|x(t)|, |y(t)|\}$. Let T be a transformation defined by

$$\begin{cases} x^*(t)=\int_a^b G(t,\xi)f(\xi,x(\xi),y(\xi))d\xi+\dfrac{B(t-a)+A(b-t)}{b-a} \\ y^*(t)=\int_a^b G_t(t,\xi)f(\xi,x(\xi),y(\xi))d\xi+\dfrac{B-A}{b-a}, \end{cases}$$

where T is clearly a continuous mapping such that $T(X)\subset X$. Let K be a subset of X such that for $z\in K$, $|x(t)|\leqq\frac{(b-a)^2}{8}L+\max(|A|,|B|)$ and $|y(t)|\leqq\frac{b-a}{2}L+\frac{|B-A|}{b-a}$. Then, K is a convex set and T transforms K into a compact subset in K. Thus, T has a fixed point and hence, there exists a solution of (28.1) which satisfies (28.2).

§ 29. Existence theorems for periodic solutions

There are many results (cf. [22], [24], [75], [77], [84], [118], [123]) concerning the existence of periodic solutions, especially for equations of second order, and many good references can be found in [118], [123]. Here, for the equation of second order, we apply Massera's Theorem which can be proved by Theorem 28.1.

Consider a system of differential equations

$$\frac{dx}{dt}=F(t,x),\qquad F(t+\omega,x)=F(t,x), \tag{29.1}$$

where $F(t,x)$ is continuous on $I\times R^n$ and ω is the period of $F(t,x)$. Throughout this section, we assume the uniqueness of solutions of (29.1).

Massera [93] proved the following theorem for $n=1$.

THEOREM 29.1. *If* (29.1) *is a scalar equation, the existence of a solution which exists and remains bounded in the future implies the existence of a periodic solution of period ω.*

He showed that if $n=2$ in the system (29.1), this theorem does not hold, and he gave the following theorem.

THEOREM 29.2. *If $n=2$ in the system* (29.1) *and if all solutions exist in the future and one of them is bounded, then there exists a periodic solution of period ω.*

PROOF. Let T be a mapping such that $x_0 \to x(\omega; x_0, 0)$. Since every solution of (29.1) is unique and all solutions starting at $t=0$ can reach $t=\omega$, T is a topological mapping of a plane into itself, and T is sense-preserving. If x_0 is the starting point of the bounded solution $x(t)$, its successive images are $x(\omega)$, $x(2\omega)$, $\cdots$, which form a bounded sequence and hence have a convergent subsequence. Apply Theorem 28.1 to G, where G is the whole plane. Thus, T has a fixed point $x^* \in G$. Namely, $x(\omega; x^*, 0)=x^*$ and hence, we can see that $x(t; x^*, 0)$ is a periodic solution of period ω.

COROLLARY. *If $n=2$ in the system* (29.1) *and if all solutions of* (29.1) *are bounded, then there exists a periodic solution of period ω.*

REMARK. One result due to Levinson [84] is a special case, because he assumes uniform-boundedness. In case $n>2$, Theorem 29.2 is not necessarily true (cf. [93]). Deysach and Sell [31] showed that if there exists a solution which is bounded and is uniform-stable, there exists an almost periodic solution. Massera [93] proved that if (29.1) is linear, the existence of a bounded solution implies the existence of a periodic solution of period ω.

EXAMPLE 29.1. Consider the equation (10.4) in Example 10.3, and suppose that $g(x) \in C_0(x)$, that $p(t)$ is periodic of period ω and that $\int_0^\omega p(t)dt=0$. As was seen, every solution of (10.5) is bounded, and hence, by the corollary above, the system (10.5) has a periodic solution of period ω, because $P(t)$ also is periodic of period ω by the condition $\int_0^\omega p(t)dt=0$. Thus, (10.4) has a periodic solution of period ω.

Next, the case where n is an arbitrary positive integer will be considered. By applying Theorem 28.2, we obtain the following theorem which corresponds to Cartwright's Theorem for $n=2$ [22].

THEOREM 29.3. *If the solutions of* (29.1) *are equi-ultimately bounded for bound* B, *then there exists a periodic solution* $x(t)$ *of period* ω *such that* $\| x(0) \| \leqq B$.

PROOF. Let f be a mapping such that $f(x_0)=x(\omega ; x_0, 0)$. Since equi-ultimate boundedness implies equi-boundedness of solutions and (29.1) is periodic, the solutions of (29.1) are uniform-bounded by Theorem 9.2. Therefore, there exists a $\beta(B)>0$ such that if $t_0 \in I$ and $x_0 \in S_B$, then $\| x(t ; x_0, t_0)\| < \beta$ for all $t \geqq t_0$. Moreover, there are γ, γ^* such that if $t_0 \in I$ and $x_0 \in S_\beta$, then $\| x(t ; x_0, t_0)\| < \gamma$ for all $t \geqq t_0$ and that $t_0 \in I$ and $x_0 \in S_\gamma$ imply $\| x(t ; x_0, t_0)\| < \gamma^*$ for all $t \geqq t_0$. Let S be the set in R^n such that $\| x \| < \gamma$, then $f(S)$ is contained in a compact set S_{γ^*}, which shows that f is a compact mapping.

From equi-ultimate boundedness for bound B, it follows that there exists a $T>0$ such that if $t \geqq T$ and $\| x_0 \| < \beta$, then $\| x(t ; x_0, 0)\| < B$, and hence, there exists a positive integer m for which $\| x(m\omega ; x_0, 0)\| < B$ if $\| x_0 \| < \beta$. Let S_B be S_0 in Theorem 28.2 and let S_1 be the set such that $\| x \| < \beta$. Then, these convex sets satisfy the assumptions in Theorem 28.2. Therefore, there exists a fixed point x_0 in S_B, which implies the existence of a periodic solution of period ω.

EXAMPLE 29.2. Consider an equation

$$\ddot{x}+f(\dot{x})+g(x)=p(t)\,, \tag{29.2}$$

where $f(y) \in C_0(y)$, $g(x) \in C_0(x)$, and $p(t)$ is continuous and periodic of period ω. Suppose that $g(x)\operatorname{sgn} x \to \infty$ as $|x| \to \infty$ and that $f(y)\operatorname{sgn} y \to \infty$ as $|y| \to \infty$. Then, (29.2) has at least one periodic solution of period ω. To see this, consider the system

$$\dot{x}=y, \qquad \dot{y}=-f(y)-g(x)+p(t) \tag{29.3}$$

and a Liapunov function $V(t, x)$ such that

$$V(x, y)=\begin{cases} G(x)+\dfrac{y^2}{2} & (|x|<\infty,\ y\geqq b) \\ G(x)+\dfrac{y^2}{2}+y-b & (x\geqq a,\ |y|\leqq b) \\ G(x)+\dfrac{y^2}{2}-2b & (x\geqq a,\ y\leqq -b) \\ G(x)+\dfrac{y^2}{2}-\dfrac{2b}{a}x & (|x|\leqq a,\ y\leqq -b) \\ G(x)+\dfrac{y^2}{2}+2b & (x\leqq -a,\ y\leqq -b) \\ G(x)+\dfrac{y^2}{2}-y+b & (x\leqq -a,\ |y|\leqq b)\,, \end{cases}$$

where $G(x)=\int_0^x g(x)dx$ and a, b are sufficiently large numbers which are chosen suitably. This $V(x, y)$ satisfies the conditions in Theorem 10.4, and hence, the solutions of (29.3) are uniform-ultimately bounded. Thus, (29.3) and consequently (29.2) has a periodic solution of period ω.

As an application of Theorem 29.3, let us consider an equation of third order

$$\dddot{x}+\varphi(\dot{x})\ddot{x}+b\dot{x}+f(x)=p(t)\,, \tag{29.4}$$

the boundedness of which was discussed by Ezeilo [32] and Reissig [117], where $b>0$ is a constant, $\varphi(y), f(x)$ and $p(t)$ are continuous. First of all, following Reissig, we shall show the uniform-ultimate boundedness of solutions of (29.4).

Consider the equivalent system

$$\dot{x}=y, \qquad \dot{y}=z-\Phi(y)+P(t), \qquad \dot{z}=-f(x)-by\,, \tag{29.5}$$

where $\Phi(y)=\int_0^y \varphi(u)du$ and $P(t)=\int_0^t p(s)ds$. The following assumptions will be made;

(i) $|f(x)|\leqq F$ for all x, and $f(x)\operatorname{sgn} x>0$ for $|x|\geqq h$,

(ii) $\Phi(y)\operatorname{sgn} y>0$ for $|y|\geqq k$, and $|\Phi(y)|\to\infty$ as $|y|\to\infty$,

(iii) $|P(t)|\leqq m$ for all $t\geqq 0$.

Now we shall prove that the solutions of (29.5) are uniform-ultimately bounded by using Liapunov functions. Let $K>0$ be a constant such that $K \geqq 1+2m+\frac{\alpha}{F}+4F\left(1+b+\frac{1}{b}\right)+k$, where $\alpha = \max\{b|y|m+2F|\Phi(y)|-by\Phi(y)\} \geqq 0$, and that for $|y| \geqq K$, $|\Phi(y)| \geqq 2m+\frac{F}{b}(1+2b+2F+4m)$. On the domain $0 \leqq t < \infty$, $\max(|y|-K, |z|-K) \geqq 0$, consider a Liapunov function $V(y, z)=W(y, z)+U(y, z)$, where $W(y, z)=\frac{1}{2}(by^2+z^2)$ and

$$U(y, z)=\begin{cases} -2Fy \operatorname{sgn} z & \text{for } |y| \leqq |z| \\ -2Fz \operatorname{sgn} y & \text{for } |y| \geqq |z|. \end{cases}$$

Clearly $V(y, z)$ is continuous, positive and $V(y, z) \to \infty$ as $y^2+z^2 \to \infty$, because

$$\frac{1}{2}(by^2+z^2)-2F|y| \geqq \frac{1}{2}by^2+\frac{1}{2}|z|(|z|-4F) > \frac{1}{2}by^2$$
$$\text{for } |y| \leqq |z|,$$

$$\frac{1}{2}(by^2+z^2)-2F|z| \geqq \frac{1}{2}z^2+\frac{1}{2}|y|(b|y|-4F) > \frac{1}{2}z^2$$
$$\text{for } |y| \geqq |z|.$$

On the other hand, $W'_{(29.5)}(y, z) \leqq -by\Phi(y)+b|y|m+|z|F$ and for $|y| \leqq |z|$,

$$\begin{aligned} U'_{(29.5)}(y, z) &= -2F|z|+2F\Phi(y) \operatorname{sgn} z-2FP(t) \operatorname{sgn} z \\ &\leqq -2F|z|+2F|\Phi(y)|+2Fm \end{aligned}$$

and for $|y| \geqq |z|$,

$$U'_{(29.5)}(y, z)=2Ff(x) \operatorname{sgn} y+2Fb|y| \leqq 2F^2+2Fb|y|.$$

Therefore, for $|y| \geqq |z|$,

$$V'_{(29.5)}(y, z) \leqq -|y|\{b|\Phi(y)|-bm-F-2Fb-2F^2\} < 0,$$

because $|y| \geqq K$, and for $|y| \leqq |z|$, $|y| \geqq K$

$$V'_{(29.5)}(y,z) \leqq -\frac{b}{2}|y|\{|\Phi(y)|-2m\}-\frac{1}{2}|\Phi(y)|\{b|y|-4F\}$$
$$-F(|y|-2m)<0$$

and for $|y| \leqq |z|$, $|y| \leqq K$

$$V'_{(29.5)}(y,z) \leqq 2Fm+\alpha-F|z|<0 \qquad (\text{by } |z| \geqq K).$$

Thus, by Theorem 10.2, as long as $\{x(t), y(t), z(t)\}$ exists, $y^2(t)+z^2(t)$ is bounded, which implies the existence of $x(t)$ in the future, because of $\dot{x}=y$. Therefore, for any $\alpha>0$, if $x_0^2+y_0^2+z_0^2 \leqq \alpha^2$, $t_0 \in I$, $\{x(t\,;x_0,y_0,z_0,t_0),\ y(t\,;x_0,y_0,z_0,t_0),\ z(t\,;x_0,y_0,z_0,t_0)\}$ exists in the future and there exist $\beta(\alpha)$, $T(\alpha)$ and $B>0$ such that

$$y^2(t\,;x_0,y_0,z_0,t_0)+z^2(t\,;x_0,y_0,z_0,t_0)<\beta^2 \qquad \text{for all } t \geqq t_0,$$
$$y^2(t\,;x_0,y_0,z_0,t_0)+z^2(t\,;x_0,y_0,z_0,t_0)<B^2 \qquad \text{for all } t \geqq t_0+T(\alpha),$$

by Theorem 10.4, where B is independent of particular solutions.

Let $v(x,z)$ be a continuous function such that $v(x,z)= \frac{b}{2}\left(x+\frac{z}{b}\right)^2$. Then, on the domain $|z| \leqq B$, $|x| \geqq H=\max\left(h, \frac{2B}{b}\right)$,

$$\text{(29.6)} \qquad \begin{cases} \dfrac{b}{8}x^2 \leqq v(x,z) \leqq \dfrac{9b}{8}x^2, \\[2ex] v'_{(29.5)}(x,z) \leqq -\dfrac{1}{2}xf(x)<0. \end{cases}$$

Since $|x(t_0+T(\alpha)\,;x_0,y_0,z_0,t_0)| \leqq \alpha+\beta(\alpha)T(\alpha)$ and we have (29.6) for $t \geqq t_0+T(\alpha)$, $|x| \geqq H$ and $|z| \leqq B$, we can find a $T_1(\alpha)>0$ and a constant $A>0$ such that if $t \geqq t_0+T(\alpha)+T_1(\alpha)$, then $|x(t\,;x_0,y_0,z_0,t_0)| <A$, where actually $\frac{9b}{8}H^2<\frac{b}{8}A^2$, that is, $3H<A$.

Thus, if $x_0^2+y_0^2+z_0^2 \leqq \alpha^2$, $t_0 \in I$, there exist a constant $C>0$ and a $T_2(\alpha)>0$ for which $x^2(t\,;x_0,y_0,z_0,t_0)+y^2(t\,;x_0,y_0,z_0,t_0)+z^2(t\,;x_0,y_0,z_0,t_0)<C$ for all $t \geqq t_0+T_2(\alpha)$. Namely, the solutions of (29.5) are uniform-ultimately bounded. Therefore, if $f(x) \in C_0(x)$ and $P(t)$ is periodic of period ω, by Theorem 29.3, (29.5) and consequently (29.4) has a periodic solution of period ω.

Now consider a perturbed system with a small periodic perturbation

$$\frac{dx}{dt} = F(t, x) + G(t, x, \varepsilon), \tag{29.7}$$

where $G(t, x, \varepsilon)$ is continuous in (t, x, ε) for $(t, x) \in I \times R^n$ and for $\|\varepsilon\|$ small, $G(t, x, \varepsilon) \in C_0(x)$ and $G(t, x, \varepsilon)$ is periodic in t with the same period ω as the one of $F(t, x)$. Moreover, the following assumptions will be made;

(i) M is a bounded set in $I \times R^n$ and satisfies the conditions given in Section 16,

(ii) $M(t)$ is periodic of period ω and $M(0)$ is convex,

(iii) M is a quasi-equiasymptotically stable set of the system (29.1) in the large,

(iv) $F(t, x) \in C_0(x)$,

(v) for bounded $\|x\|$, $\|G(t, x, \varepsilon)\| = 0(\|\varepsilon\|)$ as $\|\varepsilon\| \to 0$ for $(t, x) \in (I \times R^n) - M$.

THEOREM 29.4. *Under the assumptions above, if $\|\varepsilon\|$ is small enough, the perturbed system* (29.7) *has a periodic solution of period ω.*

PROOF. Since $F(t, x)$ is periodic in t, M is a quasi-uniform-asymptotically stable set of (29.1) in the large and $F(t, x) \in \bar{C}_0(x)$. By Theorem 22.4, there exists a Liapunov function $V(t, x)$ defined on $I \times R^n$ which satisfies the conditions;

(vi) $a(d(x, M(t))) \leqq V(t, x) \leqq b(d(x, M(t)))$, where $a(r) \in CI$ (see, Section 1), $a(r) > 0$ for $r \neq 0$, $a(r) \to \infty$ as $r \to \infty$ and $b(r) \in CI$,

(vii) if $d(x, M(t)) \leqq \eta$ and $d(x', M(t)) \leqq \eta$, for a positive continuous function $h(\eta)$

$$|V(t, x) - V(t, x')| \leqq h(\eta) \|x - x'\|, \tag{29.8}$$

(viii) $V'_{(29.1)}(t, x) \leqq -V(t, x)$.

For a $\delta > 0$, choose α, β and γ so that $a(\alpha) > b(\delta)$, $a(\beta) > b(\alpha)$ and $a(\gamma) > b(\beta)$. Let γ' be such that $a(\gamma') > b(\gamma)$. By (29.8), if $d(x, M(t)) \leqq \gamma'$ and $d(x', M(t)) \leqq \gamma'$,

$$|V(t, x) - V(t, x')| \leqq h(\gamma') \|x - x'\|. \tag{29.9}$$

Choose ε_0 so small that if $0 < d(x, M(t)) \leqq \gamma'$ and $\|\varepsilon\| \leqq \varepsilon_0(\gamma')$, then $h(\gamma')\|G(t, x, \varepsilon)\| \leqq \frac{a(\delta)}{2}$. Consider a solution $x(t; x_0, t_0, \varepsilon)$ of (29.7) with $\|\varepsilon\| \leqq \varepsilon_0(\gamma')$ such that $d(x_0, M(t_0)) \leqq \gamma$. Since $V'_{(29.7)}(t, x) \leqq -\frac{a(\delta)}{2}$ for $t \in I$, $\delta < d(x, M(t)) < \gamma'$ by (viii) and (29.9) and $h(\gamma')\|G(t, x, \varepsilon)\| \leqq \frac{a(\delta)}{2}$, we can see that $d(x(t; x_0, t_0, \varepsilon), M(t)) < \gamma'$ for all $t \geqq t_0$. By the same argument, $d(x_0, M(t_0)) \leqq \delta$ implies $d(x(t; x_0, t_0, \varepsilon), M(t)) < \alpha$ and $d(x_0, M(t_0)) \leqq \alpha$ implies $d(x(t; x_0, t_0, \varepsilon), M(t)) < \beta$. Moreover, if $d(x_0, M(t_0)) \leqq \beta$, then $d(x(t; x_0, t_0, \varepsilon), M(t)) < \gamma$ for all $t \geqq t_0$. Since $V'_{(29.7)}(t, x) \leqq -\frac{a(\delta)}{2}$ for $t \in I$, $\delta < d(x, M(t)) < \gamma'$, we can find a $T > 0$ such that if $d(x_0, M(0)) < \beta$, then $d(x(t; x_0, 0, \varepsilon), M(t)) < \alpha$ for all $t \geqq T$. Therefore, there exists a positive integer m for which, if $d(x_0, M(0)) < \beta$, $d(x(m\omega; x_0, 0, \varepsilon), M(m\omega)) < \alpha$. The sets $S_0 = \{x; d(x, M(0)) \leqq \alpha\}$ and $S_1 = \{x; d(x, M(0)) < \beta\}$ are convex, because $M(0)$ is convex. Thus, Theorem 28.2 can be applied to S_0 and S_1, and hence, (29.7) has a periodic solution $x(t)$ of period ω such that $d(x(0), M(0)) \leqq \alpha$.

COROLLARY. *Under the assumption in Theorem* 29.3 *and* (iv), (v) *above, if* $\|\varepsilon\|$ *is sufficiently small, the system* (29.7) *has a periodic solution of period* ω.

EXAMPLE 29.3. Consider the equation (29.2) again under the same conditions as in Example 29.2. As was seen, the solutions of (29.3) are uniform-ultimately bounded. Now let $q(t, x, y)$ be a continuous periodic function of period ω such that $q(t, x, y) \in C_0(x, y)$. Then, by the corollary above, the equation

$$\ddot{x} + f(\dot{x}) + g(x) = p(t) + \varepsilon q(t, x, \dot{x})$$

has a periodic solution of period ω for $|\varepsilon|$ small.

Now consider the system (29.1) and assume that $F(t, x)$ is defined and continuous on $D: 0 \leqq t < \infty$, $d(x, M(t)) \leqq A$, where M is a bounded set in $I \times R^n$. Moreover, suppose that $M(0)$ is convex and $M(t)$ is periodic of period ω and that M is an asymptotically stable set of (29.1). Corresponding to (29.1), consider the system

(29.7), in which we assume that $G(t, x, \varepsilon)$ is continuous in (t, x, ε) for $(t, x) \in D$ and $\| \varepsilon \| \leqq \varepsilon^*$, $G(t, x, \varepsilon) \in C_0(x)$ and $G(t, x, \varepsilon)$ is periodic in t of period ω and that $G(t, x, \varepsilon)$ satisfies the condition $\| G(t, x, \varepsilon)\| \leqq g(t, \varepsilon)$ for $(t, x) \in D-M$, where $g(t, \varepsilon) = 0(\| \varepsilon \|)$ as $\| \varepsilon \| \to 0$.

THEOREM 29.5. *Under the assumptions above and $F(t, x) \in C_0(x)$, if $\| \varepsilon \|$ is sufficiently small, the system* (29.7) *has a periodic solution of period ω.*

PROOF. Referring to the proof of Theorem 25.3, corresponding to any small $\eta > 0$, choose λ such that $a^{-1}(\lambda) = \delta(\eta)$, where $\delta(\eta)$ is the one in Theorem 25.1. If $\| \varepsilon \|$ is small, there exists a $T > 0$ such that if $d(x_0, M(0)) < \eta$, then $d(x(t; x_0, 0, \varepsilon), M(t)) \leqq a^{-1}(\lambda)$ for all $t \geqq T$ and hence, there exists a positive integer m such that $d(x(m\omega; x_0, 0, \varepsilon), M(m\omega)) \leqq a^{-1}(\lambda)$. Further, if $d(x_0, M(t_0)) \leqq a^{-1}(\lambda)$, then $d(x(t; x_0, t_0, \varepsilon), M(t)) < \eta$ for all $t \geqq t_0$ by Theorem 25.1. Therefore, by choosing η so that $\eta < \delta(A')$ and applying Theorem 28.2, it is seen that (29.7) has a periodic solution of period ω.

REMARK. As can be seen from the proof, a periodic solution of (29.7) stays in the domain $0 \leqq t < \infty$, $d(x, M(t)) \leqq \eta$. Therefore, we obtain a periodic solution of (29.7) as close to M as desired, if $\| \varepsilon \|$ is sufficiently small.

As a corollary, we obtain the following result which is a special case of a result due to Halanay [42] and Jones [54].

COROLLARY. *Suppose that the system* (29.1) *has a periodic solution of period ω, say $x = p(t)$, and that in the domain $0 \leqq t < \infty$, $\| x - p(t) \| \leqq A$, the periodic solution $p(t)$ is asymptotically stable, where $F(t, x) = F(t+\omega, x)$, $F(t, x) \in C_0(x)$. Then, if $\| \varepsilon \|$ is sufficiently small, the system* (29.7) *which satisfies the same assumptions as in Theorem* 29.5 *has a periodic solution $p(t, \varepsilon)$ of period ω, and, if necessary, we may select one such solution $p(t, \varepsilon)$ which tends to $p(t)$ as $\varepsilon \to 0$.*

§ 30. Existence of a bounded solution.

Consider a differential equation of second order

$$\frac{d^2x}{dt^2} = F\left(t, x, \frac{dx}{dt}\right), \tag{30.1}$$

where $F(t, x, y)$ is periodic in t of period ω. To apply Theorem 29.2, it is sufficient to show that every solution is continuable to any T to the right and there exists a bounded solution. For example, concerning the continuation of solutions, the existence of a Liapunov function $V(t, x)$ satisfying the conditions in Theorem 3.4 for each T guarantees that every solution is continuable to any t.

In some cases, the following theorem is more convenient, and it can be proved by the same argument used in the proof of Theorem 10.3, or see [140].

THEOREM 30.1. *Consider a system*

$$\frac{dx}{dt} = F(t, x, y), \qquad \frac{dy}{dt} = G(t, x, y), \tag{30.2}$$

where x, y are n, m-vectors, respectively, and F, G are continuous on $I \times R^n \times R^m$. Suppose that for each T, there exists a positive continuous function $W(t, x, y)$ satisfying the following conditions in the domain $0 \leqq t \leqq T$, $\|x\|^2 + \|y\|^2 \geqq R^2$, where R may be large;

(i) *$W(t, x, y) \to \infty$ uniformly as $\|y\| \to \infty$,*

(ii) *$W(t, x, y) \in C_0(x, y)$ and $W'_{(30.2)}(t, x, y) \leqq 0$.*

Moreover, suppose that for each K and each T, there exists a continuous function $U(t, x, y) > 0$ satisfying the following conditions in the domain $0 \leqq t \leqq T$, $\|x\| \geqq R^*$, $\|y\| \leqq K$, where R^* may be large;

(iii) *$U(t, x, y) \to \infty$ uniformly as $\|x\| \to \infty$,*

(iv) *$U(t, x, y) \in C_0(x, y)$ and $U'_{(30.2)}(t, x, y) \leqq 0$.*

Then, all solutions of (30.2) *exist in the future.*

Here, an existence theorem for a bounded solution will be shown, by considering a boundary value problem, to which Liapunov's second method also can be applied.

Now suppose that $F(t, x, y)$ in (30.1) is defined and continuous on $I\times R^1\times R^1$. For the following theorem, the periodicity of F is not necessary.

THEOREM 30.2. *Suppose that two functions $\varphi(t)$, $\psi(t)$ are defined on I, twice differentiable and bounded on I with their derivatives and that $\psi(t)\leqq\varphi(t)$, $\ddot{\varphi}(t)\leqq F(t, \varphi(t), \dot{\varphi}(t))$ and $\ddot{\psi}(t)\geqq F(t, \psi(t), \dot{\psi}(t))$. Let D be the domain $0\leqq t<\infty$, $\psi(t)\leqq x\leqq\varphi(t)$, and let D_1 be the domain such that $(t, x)\in D$, $y\geqq K$ and D_2 be the domain such that $(t, x)\in D$, $y\leqq -K$, where $K>0$ may be large. Moreover, suppose that there exist two positive Liapunov functions $V_i(t, x, y)$ defined on D_i, $i=1, 2$, respectively, which satisfy the following conditions;*

(i) *$V_i(t, x, y)\leqq b(|y|)$, $i=1, 2$, where $b(r)>0$ is continuous,*

(ii) *$V_i(t, x, y)\to\infty$ uniformly as $|y|\to\infty$,*

(iii) *in the interiors of D_1 and D_2,*

$$V_1'(t, x, y)=\varliminf_{h\to 0+}\frac{1}{h}\{V_1(t+h, x+hy, y+hF(t, x, y))-V_1(t, x, y)\}\geqq 0\,,$$

$$V_2'(t, x, y)=\varlimsup_{h\to 0+}\frac{1}{h}\{V_2(t+h, x+hy, y+hF(t, x, y))-V_2(t, x, y)\}\leqq 0\,.$$

Then, the equation (30.1) has a solution $x(t)$ such that $x(t)^2+\dot{x}(t)^2$ is bounded for all $t\geqq 0$.

PROOF. Let n be a positive integer and let B_n be the domain such that $0\leqq t\leqq n$, $\psi(t)\leqq x\leqq\varphi(t)$, $|y|<\infty$. First of all, it will be shown that there exists a solution $x_n(t)$ of (30.1) which satisfies the conditions $x_n(0)=\psi(0)$, $x_n(n)=\psi(n)$. Choosing a large $K>0$, we can assume that $|\dot{\varphi}(t)|<K$ and $|\dot{\psi}(t)|<K$. By (i), we have

$$V_1(t, x, K)\leqq b(K),\qquad V_2(t, x, -K)\leqq b(K) \tag{30.3}$$

and by (ii), there exists a constant $M>0$ such that for $(t, x)\in D$

$$b(K)<V_1(t, x, M),\ b(K)<V_2(t, x, -M)\ \text{and}\ M\geqq K. \tag{30.4}$$

Letting $H(t, x, y)$ be such that

$$H(t, x, y)=\begin{cases} F(t, x, M) & (y>M)\\ F(t, x, y) & (|y|\leqq M)\\ F(t, x, -M) & (y<-M)\,,\end{cases}$$

define a function $F^*(t, x, y)$ as follows;

$$F^*(t, x, y)=\begin{cases} H(t, \varphi(t), y)+\dfrac{x-\varphi(t)}{x-\varphi(t)+1} & (x>\varphi(t)) \\ H(t, x, y) & (\psi(t)\leqq x\leqq\varphi(t)) \\ H(t, \psi(t), y)-\dfrac{\psi(t)-x}{\psi(t)-x+1} & (x<\psi(t)). \end{cases}$$

Then, $F^*(t, x, y)$ is defined on $0\leqq t\leqq n$, $|x|<\infty$, $|y|<\infty$ and is continuous, bounded. Therefore, by Theorem 28.4, the equation

$$\ddot{x}=F^*(t, x, \dot{x}) \tag{30.5}$$

has a solution $x_n(t)$ such that $x_n(0)=\psi(0)$ and $x_n(n)=\psi(n)$.

Suppose that at some t, $x_n(t)>\varphi(t)$. Then there exists a ξ such that $x_n(\xi)>\varphi(\xi)$, $\dot{x}_n(\xi)=\dot{\varphi}(\xi)$ and $\ddot{x}_n(\xi)\leqq\ddot{\varphi}(\xi)$. Thus, $F^*(\xi, x_n(\xi), \dot{x}_n(\xi))>H(\xi, \varphi(\xi), \dot{x}_n(\xi))$, and $\dot{x}_n(\xi)=\dot{\varphi}(\xi)$ and $|\dot{\varphi}(\xi)|<K\leqq M$ imply $H(\xi, \varphi(\xi), \dot{x}_n(\xi))=F(\xi, \varphi(\xi), \dot{\varphi}(\xi))$. On the other hand, $\ddot{x}_n(\xi)=F^*(\xi, x_n(\xi), \dot{x}_n(\xi))$ and hence, $\ddot{x}_n(\xi)>F(\xi, \varphi(\xi), \dot{\varphi}(\xi))$. Since $\ddot{\varphi}(\xi)\leqq F(\xi, \varphi(\xi), \dot{\varphi}(\xi))$, we have $\ddot{x}_n(\xi)>\ddot{\varphi}(\xi)$, which contradicts $\ddot{x}_n(\xi)\leqq\ddot{\varphi}(\xi)$. Thus, $x_n(t)\leqq\varphi(t)$ on $[0, n]$, and the same argument shows that $x_n(t)\geqq\psi(t)$ on $[0, n]$.

Next, it will be seen that $|\dot{x}_n(t)|<M$. Since $x_n(0)=\psi(0)$ and $\psi(t)\leqq x_n(t)\leqq\varphi(t)$, we have $-K<\dot{x}_n(0)$. Suppose that at some t_1, $\dot{x}_n(t_1)\leqq -M$. Then, there are t_2, t_3, $t_2<t_3$, such that $\dot{x}_n(t_2)=-K$, $\dot{x}_n(t_3)=-M$ and that $-M<\dot{x}_n(t)<-K$ for $t\in(t_2, t_3)$. Considering $V_2(t, x_n(t), \dot{x}_n(t))$, we have $b(K)<V_2(t_3, x_n(t_3), \dot{x}_n(t_3))\leqq V_2(t_2, x_n(t_2), \dot{x}_n(t_2))\leqq b(K)$ by (iii), (30.3) and (30.4), which is a contradiction. Therefore, $\dot{x}_n(t)>-M$. By using $V_1(t, x, y)$, the same argument proves that $\dot{x}_n(t)<M$.

As $\psi(t)\leqq x_n(t)\leqq\varphi(t)$ and $|\dot{x}_n(t)|<M$, we have $F^*(t, x_n(t), \dot{x}_n(t))=F(t, x_n(t), \dot{x}_n(t))$, which implies that $x_n(t)$ is a solutions of (30.1) such that $x_n(0)=\psi(0)$ and $x_n(n)=\psi(n)$. Let $\{x_n^*(t)\}$ be a sequence such that

$$x_n^*(t)=\begin{cases} x_n(t) & (0\leqq t\leqq n) \\ \psi(t) & (n<t<\infty). \end{cases}$$

Then, $\{x_n^*(t)\}$ is uniformly bounded and equicontinuous, and hence, there exists a uniformly convergent subsequence which tends to its limit function $x(t)$. Clearly, $\psi(t) \leqq x(t) \leqq \varphi(t)$ for $t \in I$ and $|\dot{x}(t)| \leqq M$, and $x(t)$ is a bounded solution of (30.1).

REMARK. If $\psi(0) = \varphi(0)$, it is sufficient that $V_2'(t, x, y) \leqq 0$ and $\overline{\lim_{h \to 0^+}} \frac{1}{h} \{V_1(t+h, x+hy, y+hF(t, x, y)) - V_1(t, x, y)\} \leqq 0$ in place of (iii).

In the case where $F(t, x, y)$ is periodic in t of period ω, if all solutions of (30.1) exist in the future, by applying Theorem 30.2, we obtain an existence theorem for a periodic solution of (30.1) of period ω.

As an example, consider an equation

$$\ddot{x} + f(x, \dot{x}) + g(t, x) = p(t)\,, \tag{30.6}$$

where $f(x, y)$, $g(t, x)$ are continuous and $f(x, y) \in C_0(x, y)$, $g(t, x) \in C_0(x)$, and $p(t)$ is continuous and $p(t)$, $g(t, x)$ are periodic in t of period ω. Suppose that $f(x, y)y \geqq 0$ and $-C < G(t, x) = \int_0^x g(t, u)\,du$ for all t, x, where $C > 0$ is a constant, and that $\left|\frac{\partial G}{\partial t}\right| \Big/ \sqrt{G(t, x) + C}$ is bounded.

A special type of (30.6)

$$\ddot{x} + g(x) = p(t) \tag{30.7}$$

with a small period ω was considered by Seifert [126] who proved the existence of a periodic solution by using the index of the boundary curve of a simply connected region relative to a vector field induced by the mapping.

Here, we apply Theorem 30.2 to the case where the period ω is not necessarily small. To do so, we assume that there exist two constants a, b, $a < b$, such that

$$\begin{cases} 0 \leqq f(a, 0) + g(t, a) - p(t) \\ 0 \geqq f(b, 0) + g(t, b) - p(t)\,, \end{cases} \tag{30.8}$$

and moreover, we assume that there exists a continuous function

$\lambda(u)>0$ for $-\infty<u<\infty$ such that $|f(x,y)|\leqq\lambda(y)$, $\int^{\infty}\frac{u}{\lambda(u)+c}du=\infty$ and $\int^{-\infty}\frac{u}{\lambda(u)+c}du=\infty$, where $|g(t,x)|+|p(t)|\leqq c$ for $t\in I, x\in[a,b]$.

For an arbitrary $T>0$, in the domain $0\leqq t\leqq T$, $|x|<\infty$, $|y|<\infty$, consider a system

$$\dot{x}=y, \qquad \dot{y}=-f(x,y)-g(t,x)+p(t)\,. \tag{30.9}$$

On the domain $0\leqq t\leqq T$, $x^2+y^2\geqq R^2$, consider a function

$$W(t,x,y)=\exp\left\{\sqrt{2(G(t,x)+C)+y^2}-\int_0^t|p(s)|ds-kt\right\},$$

where $\left|\frac{\partial G}{\partial t}\right|\Big/\sqrt{2(G+C)}\leqq k$, and we have $W'_{(30.9)}(t,x,y)\leqq 0$, By using $W(t,x,y)$, we can show the boundedness of $|y(t)|$, which implies the boundedness of $|x(t)|$. Thus, it can be seen that all solutions of (30.9) are continuable to $t=T$. As T is arbitrary, all solutions exist in the future.

Setting $\varphi(t)=b$ and $\psi(t)=a$, if $V_1(t,x,y)$ and $V_2(t,x,y)$ are defined by

$$V_1(t,x,y)=\exp\left\{x+\int_K^y\frac{u}{\lambda(u)+c}du\right\},$$

$$V_2(t,x,y)=\exp\left\{x+\int_{-K}^y\frac{u}{\lambda(u)+c}du\right\},$$

all conditions in Theorem 30.2 are satisfied. Thus, applying Theorem 29.2, we can see the existence of a periodic solution of (30.6).

In the special equation

$$\ddot{x}+k\sin x=p(t),\ k>0\,, \tag{30.10}$$

if $p(t)$ is periodic of period ω and $|p(t)|\leqq k$, (30.10) has a periodic solution of period ω.

§ 31. Existence theorems for almost periodic solutions.

In this section, the existence of almost periodic solutions will be discussed under the assumption that a bounded solution is uniform-asymptotically stable, and we shall obtain some results which correspond to steady-state oscillation theorems for periodic solutions (cf. [68], [74]).

Consider a system of differential equations

$$\frac{dx}{dt}=F(t, x) \tag{31.1}$$

and the associated product system

$$\frac{dx}{dt}=F(t, x), \qquad \frac{dy}{dt}=F(t, y), \tag{31.2}$$

where $F(t, x)$ is defined on $-\infty<t<\infty$, $\|x\|\leqq H$ (or $\|x\|<\infty$) and is continuous in (t, x).

When $F(t, x)$ is almost periodic in t uniformly with respect to $x\in S$ for any compact set S in R^n, we shall say that $F(t, x)$ is almost periodic in t uniformly with respect to $x\in R^n$. It is clear that if $F(t, x)$ is almost periodic in t uniformly with respect to $x\in S_H$, there exists an $L(H)>0$ such that if $x\in S_H$, then $\|F(t, x)\| \leqq L(H)$.

First of all, we shall prove the following theorem by following an idea of Hale, see Theorem II. 2 in [49].

Throughout this section, when a Liapunov function $V(t, x)$ is defined for $x\in S_H$, it is understood that the conditions on $V'(t, x)$ for $\|x\|=H$ is satisfied if V' can be defined.

THEOREM 31.1. *Suppose that there exists a Liapunov function $V(t, x, y)$ defined for $t\in I$, $x\in S_H$, $y\in S_H$ (or $x\in R^n$, $y\in R^n$) which satisfies the following conditions;*

(i) *$a(\|x-y\|)\leqq V(t, x, y)\leqq b(\|x-y\|)$, where $a(r)\in CIP$, $b(r)\in CI$ and $b(r)\to 0$ as $r\to 0$ (see, section 1),*

(ii) *$|V(t, x_1, y_1)-V(t, x_2, y_2)|\leqq K\{\|x_1-x_2\|+\|y_1-y_2\|\}$, where $K>0$ is a constant (in case $x, y\in R^n$, K may depend on η for $\|x_i\| \leqq\eta$, $\|y_i\|\leqq\eta$, $i=1, 2$),*

(iii) *$V'_{(31.2)}(t, x, y)\leqq -\alpha V(t, x, y)$, where $\alpha>0$ is a constant. Moreover, suppose that there exists a solution $x(t; x_0, t_0)$ of (31.1) such that $\|x(t; x_0, t_0)\|\leqq H_1$, where $t\geqq t_0\geqq 0$, $H_1<H$. Then, if $F(t, x)$ is almost periodic in t uniformly with respect to $x\in S_H$ (or $x\in R^n$),*

in the region S_H (or R^n) there exists a unique perfectly uniform-asymptotically stable almost periodic solution of (31.1) *which is bounded by H_1, and its module is contained in the module of $F(t, x)$. In particular, if $F(t, x)$ is periodic in t of period T, then there exists a unique perfectly uniform-asymptotically stable periodic solution of* (31.1) *of period T.*

PROOF. Let S be the set S_{H_1}, then S is a compact set in R^n. Let $\{\tau_k\}$ be any sequence of real numbers such that $\tau_k \to \infty$ as $k \to \infty$ and $F(t+\tau_k, x) - F(t, x) \to 0$ uniformly for $t \in (-\infty, \infty)$, $x \in S$. For any real number β, let $k_0(\beta)$ be the smallest value of k such that $\tau_{k_0} + \beta \geqq t_0$. Since $\| x(t\,; x_0, t_0) \| \leqq H_1$ for all $t \geqq t_0$, $x(t+\tau_k\,; x_0, t_0) \in S$ for $t \geqq \beta$, $k \geqq k_0(\beta)$.

Letting $x(t)$ denote $x(t\,; x_0, t_0)$, we prove that the sequence of functions $\{x(t+\tau_k)\}$, $k \geqq k_0(\beta)$, converges to a continuous bounded function $\omega(t)$ defined on $[\beta, \infty)$ with the bound independent of β and the convergence being uniform on all compact subsets of $[\beta, \infty)$. Since $\| x(t) \| \leqq H_1$, $t \geqq t_0$, and H_1 does not depend on β, it is sufficient to prove that the sequence $\{x(t+\tau_k)\}$, $k \geqq k_0(\beta)$ forms a Cauchy sequence on any compact subsets of $[\beta, \infty)$. For any compact subset U of $[\beta, \infty)$ and for any $\varepsilon > 0$, choose an integer $n_0(\varepsilon, \beta) \geqq k_0(\beta)$ so large that for $l \geqq k \geqq n_0(\varepsilon, \beta)$ and all $t \in (-\infty, \infty)$

$$b(2H_1)e^{-\alpha(\beta+\tau_k-t_0)} < \frac{a(\varepsilon)}{2}, \tag{31.3}$$

$$\| F(t+\tau_l, x) - F(t+\tau_k, x) \| < \frac{a(\varepsilon)\alpha}{2K}. \tag{31.4}$$

From the conditions (ii), (iii), it follows that

$$\begin{aligned} V'(s, x(s), x(s+\tau_l-\tau_k)) \leqq & -\alpha V(s, x(s), x(s+\tau_l-\tau_k)) \\ & + K \| F(s+\tau_l-\tau_k, x(s+\tau_l-\tau_k)) - F(s, x(s+\tau_l-\tau_k)) \|. \end{aligned} \tag{31.5}$$

If we set $s - \tau_k = \sigma$, then $F(s+\tau_l-\tau_k, x(s+\tau_l-\tau_k)) - F(s, x(s+\tau_l-\tau_k)) = F(\sigma+\tau_l, x(\sigma+\tau_l)) - F(\sigma+\tau_k, x(\sigma+\tau_l))$, and hence, by (31.4),

$$V'(s, x(s), x(s+\tau_l-\tau_k)) \leqq -\alpha V(s, x(s), x(s+\tau_l-\tau_k)) + \frac{a(\varepsilon)\alpha}{2},$$

which implies that

$$V(t+\tau_k,\ x(t+\tau_k),\ x(t+\tau_l))$$
$$\leqq e^{-\alpha(t+\tau_k-t_0)}V(t_0,\ x(t_0),\ x(t_0+\tau_l-\tau_k))+\frac{a(\varepsilon)}{2}<a(\varepsilon)$$

by (31.3) for $l\geqq k\geqq n_0(\varepsilon,\beta)$ and all $t\in U$. Thus, by (i), $\|x(t+\tau_k)-x(t+\tau_l)\|<\varepsilon$ for $l\geqq k\geqq n_0(\varepsilon,\beta)$ and all $t\in U$. This proves the existence of a function $\omega(t)$ defined on $[\beta,\infty)$ and bounded by H_1. Since β is arbitrary, $\omega(t)$ is defined on $(-\infty,\infty)$ and $x(t+\tau_k)-\omega(t)\to 0$ as $k\to\infty$ uniformly on compact subsets of $(-\infty,\infty)$.

Next we shall show that $\omega(t)$ is differentiable and satisfies the system (31.1). Since $x(t)$ satisfies (31.1), we have

$$\|\dot{x}(t+\tau_k)-\dot{x}(t+\tau_l)\|\leqq\|F(t+\tau_k,\ x(t+\tau_k))-F(t+\tau_l,\ x(t+\tau_k))\|$$
$$+\|F(t+\tau_l,\ x(t+\tau_k))-F(t+\tau_l,\ x(t+\tau_l))\|\,.$$

As $x(t+\tau_k)\in S$ for large τ_k, for each compact subset of $(-\infty,\infty)$ there exists an $n_1(\varepsilon)>0$ such that if $l\geqq k\geqq n_1(\varepsilon)$, then $\|F(t+\tau_k,\ x(t+\tau_k))-F(t+\tau_l,\ x(t+\tau_k))\|<\frac{\varepsilon}{2}$. Since $x(t+\tau_k)\in S$, $x(t+\tau_l)\in S$ for large τ_k, τ_l and $F(t,x)$ is uniformly continuous by Theorem 6.2, there exists an $n_2(\varepsilon)$ such that if $l\geqq k\geqq n_2(\varepsilon)$, then $\|F(t+\tau_l,\ x(t+\tau_k))-F(t+\tau_l,\ x(t+\tau_l))\|<\frac{\varepsilon}{2}$. Thus, if $l\geqq k\geqq n_0(\varepsilon)=\max(n_1,n_2)$, we have $\|\dot{x}(t+\tau_k)-\dot{x}(t+\tau_l)\|<\varepsilon$, which shows that $\lim_{k\to\infty}\dot{x}(t+\tau_k)$ exists uniformly on all compact subsets of $(-\infty,\infty)$. Thus, we have $\lim_{k\to\infty}\dot{x}(t+\tau_k)=\dot{\omega}(t)$, and

$$\dot{\omega}(t)=\lim_{k\to\infty}\{F(t+\tau_k,\ x(t+\tau_k))-F(t+\tau_k,\ \omega(t))+F(t+\tau_k,\ \omega(t))\}$$
$$=F(t,\ \omega(t))\,,$$

because $\omega(t)\in S$, or $\omega(t)$ satisfies (31.1).

To show that $\omega(t)$ is an almost periodic function with the module contained in the module of $F(t,x)$, by Theorems 6.3 and 6.6, it is sufficient to show that for any sequence of real numbers $\{\tau_k\}$ for which $\{F(t+\tau_k,\ x)\}$ converges uniformly on $(-\infty,\infty)$ and all $x\in S$, the sequence $\{\omega(t+\tau_k)\}$ converges uniformly on $(-\infty,\infty)$, where τ_k tends to a finite number or infinity. In case τ_k tends to a finite number τ, clearly $\omega(t+\tau_k)\to\omega(t+\tau)$ uniformly on $(-\infty,\infty)$ as $k\to\infty$. Now suppose that $\tau_k\to\infty$ as $k\to\infty$. For any $\varepsilon>0$, there exists an $n_0(\varepsilon)>0$ such that if $l\geqq k\geqq n_0(\varepsilon)$, then

$$e^{-\alpha\tau_k}b(2H_1)<\frac{a(\varepsilon)}{4},$$

$$\|F(\sigma+\tau_k,x)-F(\sigma+\tau_l,x)\|<\frac{a(\varepsilon)\alpha}{4K}\quad\text{for } \sigma\in(-\infty,\infty),\ x\in S.$$

For each fixed $t\in(-\infty,\infty)$, let c be an $\frac{a(\varepsilon)\alpha}{4K}$—translation number of F such that $t+c\geqq 0$, i.e., $\|F(\sigma,x)-F(\sigma+c,x)\|<\frac{a(\varepsilon)\alpha}{4K}$ for all $\sigma\in(-\infty,\infty)$ and all $x\in S$.

Consider the function $V(c+s,\omega(s),\omega(s+\tau_l-\tau_k))$ on $t\leqq s\leqq t+\tau_k$. Then

$$\begin{aligned}
V'(c+s,\omega(s),\omega(s+\tau_l-\tau_k)) &\leqq -\alpha V(c+s,\omega(s),\omega(s+\tau_l-\tau_k))\\
&\quad +K\|F(s,\omega(s))-F(c+s,\omega(s))\|\\
&\quad +K\|F(s+\tau_l-\tau_k,\omega(s+\tau_l-\tau_k))-F(c+s,\omega(s+\tau_l-\tau_k))\|\\
&\leqq -\alpha V(c+s,\omega(s),\omega(s+\tau_l-\tau_k))+K\|F(s,\omega(s))-F(c+s,\omega(s))\|\\
&\quad +K\|F(\sigma+\tau_l,\omega(\sigma+\tau_l))-F(\sigma+\tau_k,\omega(\sigma+\tau_l))\|\\
&\quad +K\|F(\sigma+\tau_k,\omega(\sigma+\tau_l))-F(c+\sigma+\tau_k,\omega(\sigma+\tau_l))\|,
\end{aligned}$$

where $\sigma=s-\tau_k$, and hence,

$$V'(c+s,\omega(s),\omega(s+\tau_l-\tau_k))\leqq-\alpha V(c+s,\omega(s),\omega(s+\tau_l-\tau_k))+\frac{3a(\varepsilon)\alpha}{4},$$

which implies $V(c+t+\tau_k,\omega(t+\tau_k),\omega(t+\tau_l))\leqq e^{-\alpha\tau_k}V(c+t,\omega(t),\omega(t+\tau_l-\tau_k))+\frac{3a(\varepsilon)}{4}<a(\varepsilon)$. Thus, $\|\omega(t+\tau_k)-\omega(t+\tau_l)\|<\varepsilon$ for $l\geqq k\geqq n_0(\varepsilon)$ and all t, which shows that $\{\omega(t+\tau_k)\}$ converges uniformly on $(-\infty,\infty)$. In the case where $\tau_k\to-\infty$, the same argument proves that $\{\omega(t+\tau_k)\}$ also converges uniformly on $(-\infty,\infty)$.

By use of the Liapunov function $V(t,x,y)$, we can see easily that $\omega(t)$ is uniform-asymptotically stable and that every solution staying in S_H approaches $\omega(t)$ as $t \to \infty$, which implies the uniqueness of $\omega(t)$.

To prove the perfect uniform-asymptotic stability, setting $x = u + \omega(t)$ and $G(t, u) = F(t, u+\omega(t)) - F(t, \omega(t))$, we have

$$\frac{du}{dt} = G(t, u), \qquad G(t, 0) \equiv 0$$

and $u(t) \equiv 0$ is uniform-asymptotically stable for $t_0 \geqq 0$. Clearly, $G(t, u)$ also is almost periodic in t uniformly with respect to $u \in S_{H-H_1}$ (or $u \in R^n$). If we set

$$W(t, u) = V(t, \omega(t), \omega(t)+u),$$

$W(t, u)$ satisfies the conditions in Theorem 11.9 (or Theorem 11.8). Thus, we can see the perfect uniform-asymptotic stability of $u(t) \equiv 0$ or $\omega(t)$.

In the case where $F(t, x)$ is periodic in t of period T, by setting $\tau_k = kT$, $x(t+kT) \to \omega(t)$ as $k \to \infty$ and hence, $x(t+T+kT) \to \omega(t+T)$. On the other hand, clearly $x(t+(k+1)T) \to \omega(t)$, which implies $\omega(t+T) = \omega(t)$. This completes the proof.

From Theorems 21.2 and 31.1, the following theorem follows immediately.

THEOREM 31.2. *Suppose that $F(t, x)$ of* (31.1) *is continuous and almost periodic in t uniformly with respect to $x \in S_H$. Moreover, suppose that*

(i) *the system* (31.1) *for $t \geqq 0$ is uniform-asymptotically stable with respect to (H^*, H),*

(ii) $\| F(t, x) - F(t, x') \| \leqq L \| x - x' \|$, *where $L > 0$ is a constant and $x \in S_H$, $x' \in S_H$,*

(iii) *there exists a solution $x(t; x_0, t_0)$ of* (31.1) *such that $\| x(t; x_0, t_0) \| \leqq H_1$, where $t \geqq t_0 \geqq 0$, $H_1 < H^*$.*

Then, in the region S_{H^*} there exists a unique perfectly uniform-asymptotically stable almost periodic solution of (31.1) *which is bounded by H_1 and the module of which is contained in the module of $F(t, x)$. In particular, if $F(t, x)$ is periodic in t of period T, then so is the above solution.*

COROLLARY. *Suppose that $F(t, x)$ of* (31.1) *is continuous on $t \in (-\infty, \infty)$, $x \in R^n$ and is almost periodic in t uniformly with respect to $x \in R^n$. Moreover, suppose that $F(t, x) \in \bar{C}_0(x)$. If there exists a solution $x(t\,; x_0, 0)$ of* (31.1) *for $t \geqq 0$ which is bounded by H_1 for $t \geqq 0$ and is uniform-asymptotically stable in the large, then there exists a unique almost periodic solution of* (31.1), *which is perfectly uniform-asymptotically stable in the large and which is bounded by H_1. Further, the module of this solution is contained in the module of $F(t, x)$. In particular, if $F(t, x)$ is periodic in t of period T, then so is the above solution.*

PROOF. By Theorem 15.2, the system (31.1) is uniform-asymptotically stable with respect to (H, H'), $H_1 < H$, where $H' > 0$ is a constant depending on H and H_1. Therefore, by Theorem 31.2, in the region S_H there exists a unique perfectly uniform-asymptotically stable almost periodic solution of (31.1) bounded by H_1. Since H is arbitrary, it can be concluded that this solution is unique and perfectly uniform-asymptotically stable in the large.

THEOREM 31.3. *Suppose that there exists a Liapunov function $V(t, x, y)$ defined on $I \times S_H \times S_H$ (or $I \times R^n \times R^n$) which satisfies the conditions* (i), (iii) *in Theorem* 31.1 *and*

(ii)′ $|V(t, x_1, y_1) - V(t, x_2, y_2)| \leqq K\|(x_1 - x_2) - (y_1 - y_2)\|$, *where $K > 0$ is a constant (in case $x, y \in R^n$, K may depend on η for $\|x_i\| \leqq \eta$, $\|y_i\| \leqq \eta$, $i = 1, 2$).*

Moreover, suppose that (31.1) *has a solution $x^0(t)$ such that $\|x^0(t)\| \leqq c$ for $t \geqq t_0 \geqq 0$ and some constant $c > 0$. Consider a system*

$$\frac{dx}{dt} = F(t, x) + h(t)\,, \tag{31.6}$$

where $F(t, x)$ is almost periodic in t uniformly with respect to $x \in S_H$ (or $x \in R^n$) and $h(t)$ is almost periodic in t and $\|h(t)\| \leqq R$ for all t. Then, if $a^{-1}\left(\frac{KR}{\alpha}\right) + c \leqq H_1 < H$ (or for some H and $K(H)$), where $a^{-1}(r)$ is the inverse function of $a(r)$, in the region S_{H^*}, $H_1 < H^* < H$, (or R^n) the system (31.6) *has a unique perfectly uniform-asymptoti-*

cally stable almost periodic solution bounded by H_1 and its module is contained in the module associated with the vector field. In particular, if $F(t, x)$ and $h(t)$ are periodic in t of period T, then so is the above solution. In case V' is well defined for all $\|x\|=H$, $\|y\|=H$, the conclusion is true in S_H.

For any H^*, $H_1<H^*<H$, and for the associated system

$$\text{(31.7)} \qquad \begin{cases} \dfrac{dx}{dt}=F(t, x)+h(t) \\[2ex] \dfrac{dy}{dt}=F(t, y)+h(t), \end{cases}$$

$$\begin{aligned} V'_{(31.7)}(t, x, y) &= \overline{\lim_{\delta\to 0+}} \frac{1}{\delta}\{V(t+\delta, x+\delta F(t, x)+\delta h(t), y+\delta F(t, y) \\ &\qquad\qquad +\delta h(t))-V(t, x, y)\} \\ &\leqq \overline{\lim_{\delta\to 0+}} \frac{1}{\delta}\{V(t+\delta, x+\delta F(t, x), y+\delta F(t, y))-V(t, x, y)\} \\ &\leqq -\alpha V(t, x, y), \end{aligned}$$

and hence, Theorem 31.3 can be proved by Theorem 31.1 and the following lemma which shows the existence of a solution $x(t; x_0, t_0)$ of (31.6) bounded by H_1.

LEMMA 31.1. *Under the same assumptions as in Theorem* 31.3, *if* $\|x_0-x^0(t_0)\|\leqq b^{-1}\left(\dfrac{KR}{\alpha}\right)$ *and* $\|x_0\|<H$, *we have*

$$\text{(31.8)} \qquad \|x(t; x_0, t_0)-x^0(t)\|\leqq a^{-1}\left(\frac{KR}{\alpha}\right) \quad \text{for all } t\geqq t_0,$$

where $x(t; x_0, t_0)$ is a solution of (31.6).

PROOF. Let $x(t; x_0, t_0)$ be a solution of (31.6) such that $\|x_0-x^0(t_0)\|\leqq b^{-1}\left(\dfrac{KR}{\alpha}\right)$ and $\|x_0\|<H$. By (ii)$'$ and (iii), as long as $\|x(t; x_0, t_0)\|\leqq H^*$, where $H_1<H^*<H$ and $\|x_0\|<H^*$, we have

$$V'(t, x^0(t), x(t; x_0, t_0))\leqq -\alpha V(t, x^0(t), x(t; x_0, t_0))+KR,$$

which implies that

$$V(t, x^0(t), x(t; x_0, t_0)) \leqq e^{-\alpha(t-t_0)}\left\{V(t_0, x^0(t_0), x_0)-\frac{KR}{\alpha}\right\}+\frac{KR}{\alpha}$$

$$\leqq \frac{KR}{\alpha}.$$

Thus, by (i), $\| x^0(t)-x(t; x_0, t_0)\| \leqq a^{-1}\left(\frac{KR}{\alpha}\right)$ as long as $\|x(t; x_0, t_0)\| \leqq H^*$. On the other hand, $\| x(t; x_0, t_0)\| \leqq a^{-1}\left(\frac{KR}{\alpha}\right)+c \leqq H_1 < H^*$. Therefore, (31.8) is valid for all $t \geqq t_0$. In particular, if $x_0 = x^0(t_0)$, clearly $x(t; x_0, t_0)$ is a solution of (31.6) such that $\| x(t; x_0, t_0)\| \leqq H_1$.

In the case where the system (31.1) is a linear system, i. e., $F(t, x) = A(t)x$, where $A(t)$ is a continuous $n \times n$ matrix on I, if the zero solution is uniform-asymptotically stable in the large, there exist constants $K \geqq 1$, $\alpha > 0$ and a Liapunov function $V(t, x)$ defined on $I \times R^n$ such that $\| x\| \leqq V(t, x) \leqq K\| x\|$, $| V(t, x) - V(t, x')| \leqq K\| x-x'\|$ and $V'_{(31.1)}(t, x) \leqq -\alpha V(t, x)$, by Theorem 19.1. Setting $W(t, x, y) = V(t, x-y)$, for all $t \geqq 0$, $x \in R^n$, $x_i \in R^n$, $y_i \in R^n$, $i=1, 2$,

$$\| x-y\| \leqq W(t, x, y) \leqq K\| x-y\| ,$$

$$| W(t, x_1, y_1) - W(t, x_2, y_2)| \leqq K\| (x_1-x_2)-(y_1-y_2)\| .$$

Moreover, we have

$$W'_{(31.2)}(t, x, y) = \overline{\lim_{\delta\to 0+}} \frac{1}{\delta}\{ V(t+\delta, x+hA(t)x-y-hA(t)y) - V(t, x-y)\}$$

$$= \overline{\lim_{\delta\to 0+}} \frac{1}{\delta}\{ V(t+\delta, x-y+hA(t)(x-y)) - V(t, x-y)\}$$

$$= V'_{(31.1)}(t, x-y) \leqq -\alpha V(t, x-y) = -\alpha W(t, x, y).$$

Thus, $W(t, x, y)$ satisfies the conditions in Theorem 31.3 for $t \in I$, $x \in R^n$, $y \in R^n$. Since $x(t) \equiv 0$ is a solution of (31.1), c in Theorem 31.3 can be zero. By Lemma 31.1, any solution $x(t; x_0, t_0)$ of (31.6) such that $\| x_0\| \leqq \frac{R}{\alpha}$ satisfies $\| x(t; x_0, t_0)\| \leqq \frac{KR}{\alpha}$ for all $t \geqq t_0$. Therefore, if $F(t, x) = A(t)x$ is almost periodic in t uniformly with respect to $x \in R^n$ and $h(t)$ is almost periodic in t, Theorem 31.3 gives the following theorem due to Hale [49], which was essentially known by Bochner and Favard.

THEOREM 31.4. *Consider two systems*

$$\frac{dx}{dt}=A(t)x\,, \tag{31.9}$$

$$\frac{dx}{dt}=A(t)x+h(t)\,, \tag{31.10}$$

where $A(t)$ is a continuous $n\times n$ matrix defined for $t\in(-\infty,\infty)$ and $h(t)$ is a continuous function defined for $t\in(-\infty,\infty)$ which is bounded by R for all t. If the zero solution of (31.9) *is uniform-asymptotically stable in the large for $t_0\geqq 0$, then the solutions of* (31.10) *satisfy the following properties:*

(i) *All solutions of* (31.10) *are bounded for $t\geqq t_0$. In fact, there exist constants $K\geqq 1$, $\alpha>0$ such that $\|x_0\|\leqq\dfrac{R}{\alpha}$ implies $\|x(t;x_0,t_0)\|\leqq\dfrac{KR}{\alpha}$ for all $t\geqq t_0$, where $x(t;x_0,t_0)$ is a solution of* (31.10).

(ii) *If $A(t)$, $h(t)$ are almost periodic in t, then there is a unique almost periodic solution of* (31.10), *which is bounded by $\dfrac{KR}{\alpha}$ and which is perfectly uniform-asymptotically stable in the large.*

Now we consider a system

$$\begin{cases}\dfrac{dx}{dt}=f(t)x+F(t,y)+X(t,x,y,\mu)+\mu h(t)\\[2ex] \dfrac{dy}{dt}=G(t,x,y)+\dfrac{1}{\mu}g(t)y+Y(t,x,y,\mu)+H(t)\,,\end{cases} \tag{31.11}$$

where $\mu>0$ is a real parameter and x, y are m, n-vectors, respectively, and f, g are $m\times m$, $n\times n$ matrices, respectively. The following assumptions will be made:

(i) All functions in (31.11) are continuous in all their variables for all t, $x\in S_H$, $y\in S_H$, μ, and are almost periodic in t uniformly with respect to the other variables.

(ii) There exist $K\geqq 1$ and $\alpha>0$ such that $\|x(t;x_0,t_0)\|\leqq Ke^{-\alpha(t-t_0)}\|x_0\|$ and $\|y(t;y_0,t_0)\|\leqq Ke^{-\frac{\alpha}{\mu}(t-t_0)}\|y_0\|$ for $t\geqq t_0\geqq 0$, where $x(t;x_0,t_0)$, $y(t;y_0,t_0)$ are solutions of

$$\frac{dx}{dt}=f(t)x\,, \tag{31.12}$$

$$\frac{dy}{dt}=\frac{1}{\mu}g(t)y\,, \tag{31.13}$$

respectively.

(iii) $F(t, 0)\equiv X(t, 0, 0, 0)\equiv 0$, $G(t, 0, 0)\equiv Y(t, 0, 0, 0)\equiv 0$.

(iv) There exists an $L>0$ such that $\|F(t, y_1)-F(t, y_2)\|\leqq L\|y_1-y_2\|$ and that $\|G(t, x_1, y_1)-G(t, x_2, y_2)\|\leqq L\{\|x_1-x_2\|+\|y_1-y_2\|\}$.

(v) There exists a function $\eta(\rho, \mu)$, continuous and non-decreasing in ρ, μ for $0\leqq\rho\leqq H$, $0\leqq\mu\leqq\mu_0$ such that $\eta(0, 0)=0$ and that

$$\|X(t, x_1, y_1, \mu)-X(t, x_2, y_2, \mu)\|\leqq\eta(\rho, \mu)\{\|x_1-x_2\|+\|y_1-y_2\|\}\,, \tag{31.14}$$

$$\|Y(t, x_1, y_1, \mu)-Y(t, x_2, y_2, \mu)\|\leqq\eta(\rho, \mu)\{\|x_1-x_2\|+\|y_1-y_2\|\} \tag{31.15}$$

for all t, $\|x_i\|\leqq\rho$, $\|y_i\|\leqq\rho$, $i=1, 2$, $\rho\leqq H$, $0\leqq\mu\leqq\mu_0$.

THEOREM 31.5. *Under the assumptions* (i) *through* (v), *there exist* $\mu_0^*>0$, $M>0$ *such that there is a unique perfectly uniform-asymptotically stable almost periodic solution of* (31.11) *in the region* $\|x\|\leqq M$, $\|y\|\leqq M$, $0<\mu\leqq\mu_0^*$. *Furthermore, this almost periodic solution approaches zero as* $\mu\to 0$ *and its module is contained in the module associated with the vector field. In particular, if all functions in* (31.11) *are periodic in t of period T, then so is the above solution.*

PROOF. Here, the associated systems of (31.11), (31.12) and (31.13) will be denoted by (31.11)*, (31.12)* and (31.13)*, respectively. By (ii), there exist two Liapunov functions $V_1(t, x, u)$ and $V_2(t, y, v)$ which satisfy the conditions

$$\|x-u\|\leqq V_1(t, x, u)\leqq K\|x-u\|,\ \|y-v\|\leqq V_2(t, y, v)\leqq K\|y-v\|\,,$$

$$V'_{1\,(31.12)^*}(t, x, u)\leqq -\alpha V_1(t, x, u),\ V'_{2\,(31.13)^*}(t, y, v)\leqq -\frac{\alpha}{\mu}V_2(t, y, v)\,,$$

$$|V_1(t, x_1, u_1)-V_1(t, x_2, u_2)|\leqq K\|(x_1-x_2)-(u_1-u_2)\|\,,$$

$$|V_2(t, y_1, v_1)-V_2(t, y_2, v_2)|\leqq K\|(y_1-y_2)-(v_1-v_2)\|\,.$$

Consider a Liapunov function

$$V(t, x, u, y, v)=V_1(t, x, u)+\beta(\mu)V_2(t, y, v).$$

Then, we have

$$\begin{aligned}
&V'_{(31.11)^*}(t, x, u, y, v)\\
&\quad\leqq -\alpha V_1+K\|F(t, y)-F(t, v)\|+K\|X(t, x, y, \mu)-X(t, u, v, \mu)\|\\
&\quad -\frac{\alpha}{\mu}\beta V_2+\beta K\|G(t, x, y)-G(t, u, v)\|\\
&\quad +\beta K\|Y(t, x, y, \mu)-Y(t, u, v, \mu)\|.
\end{aligned}$$

From (iv), (v), $\|x-u\|\leqq V_1$ and $\|y-v\|\leqq V_2$, it follows that for $\|x\|, \|y\|, \|u\|, \|v\|\leqq\rho$,

$$V'_{(31.11)^*}(t, x, u, y, v)\leqq -A_1V_1-A_2V_2,$$

where

$$A_1=\alpha-\beta KL-K\eta(\rho, \mu)-\beta K\eta(\rho, \mu),$$

$$A_2=\frac{\alpha}{\mu}\beta-KL-\beta KL-K\eta(\rho, \mu)-\beta K\eta(\rho, \mu).$$

First of all, choose μ so small that $\alpha-\mu KL\geqq\alpha^*>0$ for a fixed $\alpha^*<\alpha$, and set $\beta(\mu)=\dfrac{\gamma+KL}{\alpha^*}\mu\leqq 1$ for μ small, where γ is a positive fixed number such that $\gamma<\alpha^*$. Choose M and r so small that

$$\gamma-2K\eta(M, r)=\gamma^*>0, \qquad \alpha-KL\frac{\gamma+KL}{\alpha^*}r\geqq\gamma.$$

Then, $V(t, x, u, y, v)$ is defined on $\|x\|, \|y\|, \|u\|, \|v\|\leqq M, 0\leqq\mu\leqq r$ and

$$V'_{(31.11)^*}(t, x, u, y, v)\leqq -\gamma^*V(t, x, u, y, v),$$

because $A_1\geqq\gamma^*$, $A_2\geqq\gamma^*$ for $\rho\leqq M$ and $\beta(\mu)\leqq 1$.

Let $R(\mu)=\max\{\sup_{0\leqq\tau\leqq\mu}\sup_t\|X(t, 0, 0, \tau)\|, \sup_{0\leqq\tau\leqq\mu}\sup_t\|Y(t, 0, 0, \tau)\|\}$. Then $R(\mu)$ is continuous, non-decreasing for $0<\mu\leqq\mu_0$ and $R(0)=0$. Since $h(t)$, $H(t)$ are almost periodic, there exists a $P>0$ such that $\|h(t)\|\leqq P$, $\|H(t)\|\leqq P$. Choose r small again so that if $0\leqq\mu\leqq r$, (31.16) $$M_1=\frac{K}{\gamma^*}\{R(\mu)(1+\beta)+P(\mu+\beta)\}<M,$$

(31.17) $$M_2=\frac{K\mu}{\gamma^*}\{(L+\eta(M, r))M+R(\mu)+P\}<M.$$

Let $x(t)=x(t;x_0,y_0,t_0)$, $y(t)=y(t;x_0,y_0,t_0)$ be a solution of (31.11). As long as $\|x(t)\|\leqq M$, $\|y(t)\|\leqq M$ and $0<\mu\leqq r$, we have

$$V'(t, 0, x(t), 0, y(t))$$
$$\leqq -\gamma^* V(t, 0, x(t), 0, y(t))+KR(\mu)(1+\beta)+KP(\mu+\beta),$$

and hence we have

$$\|x(t)\|\leqq V(t, 0, x(t), 0, y(t))\leqq e^{-\gamma^*(t-t_0)}\{V(t_0, 0, x_0, 0, y_0)-M_1\}+M_1.$$

Thus, if $\|x_0\|+\|y_0\|\leqq\frac{1}{\gamma^*}\{R(\mu)(1+\beta)+P(\mu+\beta)\}$ ($<M$ by (31.16) and $K\geqq 1$), we have $\|x(t)\|\leqq M_1<M$. Now consider $V_2(t, 0, y(t))$. Then, we have

$$V_2'(t, 0, y(t))\leqq -\left(\frac{\alpha}{\mu}-KL-K\eta(M, r)\right)V_2(t, 0, y(t))$$
$$+K\{(L+\eta(M, r))\|x(t)\|+R(\mu)+P\}$$
$$\leqq -\frac{\gamma^*}{\mu}V_2(t, 0, y(t))+K\{(L+\eta(M, r))\|x(t)\|$$
$$+R(\mu)+P\},$$

because

$$\frac{\alpha}{\mu}-KL-K\eta(M, r)\geqq\frac{1}{\mu}(\alpha-KL\mu-K\eta(M, r))$$
$$\geqq\frac{1}{\mu}(\gamma-K\eta(M, r))\geqq\frac{\gamma^*}{\mu}.$$

Thus, we have

$$\|y(t)\|\leqq V_2(t, 0, y(t))\leqq e^{-\frac{\gamma^*}{\mu}(t-t_0)}\{V_2(t_0, 0, y_0)-M_2\}+M_2.$$

Therefore, if $\|y_0\|\leqq\frac{\mu}{\gamma^*}\{(L+\eta(M, r))M+R(\mu)+P\}$ ($<M$ by (31.17) and $K\geqq 1$), we have $\|y(t)\|\leqq M_2$. Thus, it can be seen that there exists a solution $\{x(t), y(t)\}$ of (31.11) such that

$$\| x(t) \| \leqq M_1 < M, \ \| y(t) \| \leqq M_2 < M.$$

Therefore, by Theorem 31.1, there exists a unique perfectly uniform-asymptotically stable almost periodic solution of the system (31.11) which tends to zero as $\mu \to 0$.

CHAPTER VIII. FUNCTIONAL-DIFFERENTIAL EQUATIONS

The concept of a functional-differential equations is a generalization of differential-difference equations. To apply Liapunov's second method to functional-differential equations, one must actually use a Liapunov functional. By using Liapunov functional, one can extend to functional-differential equations most of the well-known results for ordinary differential equations. Here, we shall give some results for functional-differential equations.

In this chapter, the following notations will be used: For $x \in R^n$, $|x|$ is the norm of x. For a given $h > 0$, C denotes the space of continuous functions mapping the interval $[-h, 0]$ into R^n and for $\varphi \in C$, $\|\varphi\| = \sup_{-h \leqq \theta \leqq 0} |\varphi(\theta)|$. C_H will denote the set of $\varphi \in C$ such that $\|\varphi\| \leqq H$. For any continuous function $x(u)$ defined on $-h \leqq u < A$, $A > 0$, and any fixed t, $0 \leqq t < A$, the symbol x_t will denote the restriction of $x(u)$ to the interval $[t-h, t]$, i. e., x_t is an element of C defined by $x_t(\theta) = x(t+\theta)$, $-h \leqq \theta \leqq 0$.

§ 32. Existence theorem and uniqueness.

Let $\dot{x}(t)$ denote the right-hand derivative of $x(u)$ at $u = t$, and consider the functional-differential equation

$$\dot{x}(t) = F(t, x_t), \tag{32.1}$$

where $F(t, \varphi) \in R^n$ is defined on $[0, c] \times C_H$.

In the case where F is independent of t and linear, if we replace F by L and if $L(\varphi)$ is continuous on C, we have

$$L(\varphi) = \int_{-h}^{0} [d\eta(\theta)]\varphi(\theta) \qquad \text{for all} \quad \varphi \in C, \tag{32.2}$$

in which $\eta(\theta)$ is a matrix whose elements are functions of bounded variation and the integral is in the sense of Stieltjes, see [122].

As an example, consider the equation

$$\dot{x}(t)=ax(t)+bx(t-h),$$

where a, b are constants and x is a scalar. If we define η by the relation

$$\eta(\theta)=\begin{cases} 0 & (\theta=-h) \\ b & (-h<\theta<0) \\ a+b & (\theta=0), \end{cases}$$

then this equation is equivalent to $\dot{x}(t)=L(x_t)$, where $L(x_t) =\int_{-h}^{0} x(t+\theta)d\eta(\theta)$.

DEFINITION 32.1. A function $x(t_0, \varphi)$ is said to be a solution of (32.1) with initial condition $\varphi \in C_H$ at $t=t_0$, $t_0 \geqq 0$, if there is an $A>0$ such that $x(t_0, \varphi)$ is a function from $[t_0-h, t_0+A)$ into R^n with the properties

(i) $x_t(t_0, \varphi) \in C_H$ for $t_0 \leqq t < t_0+A$,

(ii) $x_{t_0}(t_0, \varphi)=\varphi$,

(iii) $x(t_0, \varphi)$ satisfies (32.1) for $t_0 \leqq t < t_0+A$.

In this chapter, we shall denote by $x(t\,; t_0, \varphi)$ the value of $x(t_0, \varphi)$ at t.

THEOREM 32.1. *If $F(t, \varphi)$ in (32.1) is continuous in t, φ, for every $\varphi \in C_{H_1}$, $H_1<H$, and t_0, $0 \leqq t_0 < c$, there exists a solution of (32.1) with initial value φ at $t=t_0$, and this solution has a continuous derivative for $t>t_0$.*

PROOF. As $F(t, \varphi)$ is continuous in (t, φ), there exist a $T_0>0$ and a $\delta>0$ such that if $t_0 \leqq t \leqq t_0+T_0$ and $\|\varphi-\psi\| \leqq \delta$, then $|F(t_0, \varphi)-F(t, \psi)| \leqq 1$ or $|F(t, \psi)| \leqq |F(t_0, \varphi)|+1$. Let a be such that $|\varphi(t_0)|<a$ and let b be such that $|F(t_0, \varphi)|+1<b$. Since $\varphi(t_0+\theta)$ is continuous on $-h \leqq \theta \leqq 0$, there is a $\delta_0>0$ for which if $|\tau| \leqq \delta_0$, then $|\varphi(t_0+\theta+\tau)-\varphi(t_0+\theta)|<\dfrac{\delta}{2}$. Let $A>0$ be a number such that $A \leqq T_0$, $A \leqq \delta_0$, $bA \leqq \dfrac{\delta}{2}$, $A-h<0$ and that $|\varphi(t_0)| +bA<a$.

Let $S(a, b, A)$ be the set of all functions $x(t)$ which are continuous and satisfy

$$|x(t)| \leqq a \quad \text{for} \quad t_0 \leqq t \leqq t_0+A\,,$$
$$|x(t_1)-x(t_2)| \leqq b\,|t_1-t_2| \quad \text{for} \quad t_0 \leqq t_1,\, t_2 \leqq t_0+A\,,$$
$$x(t)=\varphi(t) \quad \text{for} \quad t_0-h \leqq t \leqq t_0\,.$$

Then $S(a, b, A)$ is a compact convex set in the space of continuous functions on $[t_0-h, t_0+A]$. Consider a mapping T of $S(a, b, A)$ defined by

$$\begin{cases} Tx(t)=\varphi(t) & \text{for} \quad t \leqq t_0 \\ Tx(t)=\varphi(t_0)+\displaystyle\int_{t_0}^{t} F(s, x_s)ds & \text{for} \quad t>t_0\,, \end{cases}$$

where T is a continuous mapping. Consider the function $x(s+\theta)-\varphi(t_0+\theta)$ on $-h \leqq \theta \leqq 0$, where $t_0 \leqq s \leqq t_0+A$. If $s+\theta \leqq t_0$, then $x(s+\theta)-\varphi(t_0+\theta)=\varphi(s+\theta)-\varphi(t_0+\theta)$, and hence, $|x(s+\theta)-\varphi(t_0+\theta)| < \frac{\delta}{2}$, because $0 \leqq s-t_0 \leqq A \leqq \delta_0$. If $s+\theta>t_0$, we have

$$\begin{aligned} |x(s+\theta)-\varphi(t_0+\theta)| &\leqq |x(s+\theta)-\varphi(t_0)|+|\varphi(t_0)-\varphi(t_0+\theta)| \\ &\leqq b(s+\theta-t_0)+|\varphi(t_0)-\varphi(t_0+\theta)| \\ &\leqq bA+\frac{\delta}{2}<\delta\,, \end{aligned}$$

because $bA \leqq \frac{\delta}{2}$ and $0 \leqq -\theta \leqq s-t_0 \leqq A \leqq \delta_0$. Thus, for any $s \in [t_0, t_0+A]$, $\|\varphi-x_s\| \leqq \delta$. Therefore, $|Tx(t)| \leqq |\varphi(t_0)|+b(t-t_0) \leqq |\varphi(t_0)| +bA<a$ for $t \in [t_0, t_0+A]$, and for t_1, $t_2 \in [t_0, t_0+A]$ we have $|Tx(t_1)-Tx(t_2)| \leqq b\,|t_1-t_2|$, which implies that $TS \subset S$. Thus, by Theorem 28.3, there exists a fixed point $x(t) \in S$, that is,

$$\begin{cases} x(t)=\varphi(t) & \text{for} \quad t \leqq t_0 \\ x(t)=\varphi(t_0)+\displaystyle\int_{t_0}^{t} F(s, x_s)ds & \text{for} \quad t>t_0\,. \end{cases}$$

This shows that $x(t)$ is a solution of (32.1) with initial value φ at t_0. Clearly $x(t)$ has a continuous derivative for $t>t_0$.

Next we shall discuss the uniqueness of solutions of (32.1) by a Liapunov functional $V(t, \varphi)$.

DEFINITION 32.2. Let $V(t, \varphi)$ be a continuous functional defined for $t \geqq 0$, $\varphi \in C_H$. The upper right-hand derivative of V along the solutions of (32.1) will be denoted by $V'_{(32.1)}$ and is defined to be

$$V'_{(32.1)}(t, x_t(t_0, \varphi)) = \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t_0, \varphi)) - V(t, x_t(t_0, \varphi))\}, \tag{32.3}$$

where $x(t_0, \varphi)$ is the solution of (32.1) with $x_{t_0}(t_0, \varphi) = \varphi$. We shall also sometimes write

$$V'_{(32.1)}(t, \varphi) = \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t, \varphi)) - V(t, \varphi)\}, \tag{32.4}$$

where it is understood in this notation that φ denote the segment for the interval $[t-h, t]$ of the solution of (32.1). Therefore, in general, it depends on the solution considered. In other words, $V'_{(32.1)}(t, \varphi)$ is not necessarily defined uniquely for given t, φ, and it depends on which solution φ belongs to. However, clearly, if the solution is unique, $V'_{(32.1)}(t, \varphi)$ is determined uniquely by t, φ. Furthermore, in the case where $V(t, \varphi) \in C_0(\varphi)$, $V'_{(32.1)}(t, \varphi)$ is defined only by giving t, φ. Suppose that there are two solutions $x(t, \varphi)$, $y(t, \varphi)$ such that $x_t = y_t = \varphi$. For small δ, there exists a compact set Q in (t, φ) space such that $(s, x_s) \in Q$, $(s, y_s) \in Q$ for $t \leqq s \leqq t+\delta$. Let L be a Lipschitz constant of $V(t, \varphi)$ on Q. Then, along the solution $x(t, \varphi)$,

$$\begin{aligned} V'_{(32.1)}(t, \varphi) &= \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t, \varphi)) - V(t, \varphi)\} \\ &\leqq \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, y_{t+\delta}(t, \varphi)) - V(t, \varphi)\} \\ &\quad + \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t, \varphi)) - V(t+\delta, y_{t+\delta}(t, \varphi))\} \\ &\leqq \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta}\{V(t+\delta, y_{t+\delta}(t, \varphi)) - V(t, \varphi)\} \\ &\quad + \overline{\lim_{\delta \to 0^+}} \frac{1}{\delta} L \, \| x_{t+\delta}(t, \varphi) - y_{t+\delta}(t, \varphi) \| \end{aligned}$$

$$\leqq \overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\{V(t+\delta, y_{t+\delta}(t, \varphi)) - V(t, \varphi)\} + L|\dot{x}(t) - \dot{y}(t)|$$

$$\leqq \overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\{V(t+\delta, y_{t+\delta}(t, \varphi)) - V(t, \varphi)\},$$

because $\dot{x}(t) = \dot{y}(t)$. In the same way, along the solution $y(t, \varphi)$, we have

$$V'_{(32.1)}(t, \varphi) \leqq \overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t, \varphi)) - V(t, \varphi)\}.$$

Thus, it is seen that $V'_{(32.1)}(t, \varphi)$ can be defined uniquely.

From the mentioned above, we can see the following: Let $V(t, \varphi) \in C_0(\varphi)$ and let $x(t, \varphi)$, $y(t, \varphi)$ be continuous functions with the right-hand derivatives such that $x_t = y_t = \varphi$. Then, we have

$$V'(t, y_t) \leqq V'(t, x_t) + L|\dot{y}(t) - \dot{x}(t)|,$$

where L is a Lipschitz constant of $V(t, \varphi)$ and

$$V'(t, x_t) = \overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t, \varphi)) - V(t, x_t)\},$$

$$V'(t, y_t) = \overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\{V(t+\delta, y_{t+\delta}(t, \varphi)) - V(t, y_t)\}.$$

In the above, we used the relation

$$\overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\| x_{t+\delta}(t, \varphi) - y_{t+\delta}(t, \varphi)\| = |\dot{x}(t) - \dot{y}(t)|.$$

In general, for two continuous functions $x(t, \varphi)$, $y(\tau, \varphi)$ with the right-hand derivatives such that $x_t = y_\tau = \varphi$, we have

$$\overline{\lim_{\delta\to 0^+}} \frac{1}{\delta}\| x_{t+\delta}(t, \varphi) - y_{\tau+\delta}(\tau, \varphi)\| = |\dot{x}(t) - \dot{y}(\tau)|.$$

This relation is proved in the following way.

Since $\| x_{t+\delta}(t, \varphi) - y_{\tau+\delta}(\tau, \varphi)\| = \sup_{-h\leqq\theta\leqq 0} |x(t+\delta+\theta) - y(\tau+\delta+\theta)|$, the supremum is taken at some θ such that $0 \leqq \delta+\theta \leqq \delta$, because the supremum is zero for θ such that $\delta+\theta \leqq 0$. Thus,

$$\sup_{-h\leqq\theta\leqq 0} |x(t+\delta+\theta) - y(\tau+\delta+\theta)| = \sup_{-\delta\leqq\theta\leqq 0} |x(t+\delta+\theta) - y(\tau+\delta+\theta)|,$$

and hence, we have

$$\overline{\lim_{\delta\to 0^+}}\frac{1}{\delta}\|x_{t+\delta}(t,\varphi)-y_{\tau+\delta}(\tau,\varphi)\|=\overline{\lim_{\delta\to 0^+}}\frac{1}{\delta}|x(t+\delta+\theta_\delta)-y(\tau+\delta+\theta_\delta)|,$$

where $-\delta\leqq\theta_\delta\leqq 0$. Since $0\leqq\dfrac{\delta+\theta_\delta}{\delta}\leqq 1$ and we have

$$\frac{|x(t+\delta+\theta_\delta)-y(\tau+\delta+\theta_\delta)|}{\delta}=\frac{|x(t+\delta+\theta_\delta)-y(\tau+\delta+\theta_\delta)|}{\delta+\theta_\delta}\cdot\frac{\delta+\theta_\delta}{\delta},$$

by setting $\delta+\theta_\delta=\bar{\delta}$, we have

$$\begin{aligned}\overline{\lim_{\delta\to 0^+}}\frac{1}{\delta}|x(t+\delta+\theta_\delta)-y(\tau+\delta+\theta_\delta)|&\leqq\overline{\lim_{\bar\delta\to 0^+}}\frac{1}{\bar\delta}|x(t+\bar\delta)-y(\tau+\bar\delta)|\\&\leqq\overline{\lim_{\bar\delta\to 0^+}}\frac{1}{\bar\delta}|x(t+\bar\delta)-x(t)-(y(\tau+\bar\delta)-y(\tau))|\qquad(x(t)=y(\tau))\\&\leqq|\dot{x}(t)-\dot{y}(\tau)|.\end{aligned}$$

On the other hand, we have

$$\begin{aligned}&\overline{\lim_{\delta\to 0^+}}\frac{1}{\delta}|x(t+\delta+\theta_\delta)-y(\tau+\delta+\theta_\delta)|\\&\qquad\geqq\overline{\lim_{\delta\to 0^+}}\frac{1}{\delta}|x(t+\delta)-y(\tau+\delta)|\geqq|\dot{x}(t)-\dot{y}(\tau)|.\end{aligned}$$

Therefore, we have the relation above.

THEOREM 32.2. *Suppose that $F(t,\varphi)$ is defined and continuous on $0\leqq t\leqq c$, $\varphi\in C_H$ and that there exists a continuous Liapunov functional $V(t,\varphi,\psi)$ defined on $0\leqq t\leqq c$, $\varphi\in C_H$, $\psi\in C_H$, which satisfies the following conditions;*

(i) *$V(t,\varphi,\psi)=0$ if $\varphi=\psi$,*

(ii) *$V(t,\varphi,\psi)>0$ if $\varphi\neq\psi$,*

(iii) *for the associated system*

$$\dot{x}(t)=F(t,x_t),\qquad \dot{y}(t)=F(t,y_t),\tag{32.5}$$

we have $V'_{(32.5)}(t,\varphi,\psi)\leqq 0$, where for $\|\varphi\|=H$ or $\|\psi\|=H$, we understand that the condition $V'_{(32.5)}\leqq 0$ is satisfied in case V' can be defined.

Then, for given initial value $\varphi\in C_{H_1}$, $H_1<H$, there exists a unique solution of (32.1).

PROOF. Suppose that there are two solutions $x(t_0, \varphi)$, $y(t_0, \varphi)$ on $[t_0, t_0+A)$. Then there exists a t_1, $t_0 < t_1 < t_0+A$, such that $\| x_{t_1} - y_{t_1} \| > 0$. From this and (iii), it follows that $0 < V(t_1, x_{t_1}, y_{t_1}) \leqq V(t_0, x_{t_0}, y_{t_0}) = 0$, which is a contradiction. Thus, $x(t_0, \varphi) = y(t_0, \varphi)$.

EXAMPLE 32.1. If there exists an $L > 0$ such that $|F(t, \varphi) - F(t, \psi)| \leqq L \| \varphi - \psi \|$, $\varphi \in C_H$, $\psi \in C_H$, then $V(t, \varphi, \psi) = e^{-Lt} \| \varphi - \psi \|$ is the desired Liapunov functional.

It is clear that $V(t, \varphi, \psi)$ satisfies (i), (ii). Let $x(t_0, \varphi)$ and $y(t_0, \psi)$ be two solutions of (32.1). Then we have

$$
\begin{aligned}
V'_{(32.5)}(t, x_t, y_t) &= \overline{\lim_{\delta \to 0+}} \frac{1}{\delta} \{ e^{-L(t+\delta)} \| x_{t+\delta} - y_{t+\delta} \| - e^{-Lt} \| x_t - y_t \| \} \\
&= \overline{\lim_{\delta \to 0+}} \frac{1}{\delta} \{ (e^{-L(t+\delta)} - e^{-Lt}) \| x_t - y_t \| \\
&\quad + e^{-L(t+\delta)} (\| x_{t+\delta} - y_{t+\delta} \| - \| x_t - y_t \|) \} \\
&\leqq -e^{-Lt} L \| x_t - y_t \| \\
&\quad + e^{-Lt} \overline{\lim_{\delta \to 0+}} \frac{1}{\delta} \int_t^{t+\delta} | F(s, x_s) - F(s, y_s) | \, ds \\
&\leqq e^{-Lt} \{ -L \| x_t - y_t \| + L \| x_t - y_t \| \} = 0 ,
\end{aligned}
$$

and hence, $V(t, \varphi, \psi)$ satisfies (iii).

For functional-differential equations, the following should be noticed. Since C_α is not a compact set, there is, in general, no $L > 0$ such that $|F(t, \varphi)| \leqq L$ for $t \in [0, c]$, $\varphi \in C_\alpha$. Therefore, the fact that $\| x_t(t_0, \varphi) \| \leqq \alpha$ as long as $x(t_0, \varphi)$ exists, where $\alpha > 0$ is a constant, does not necessarily imply the continuation of $x(t_0, \varphi)$ to the right. If there exists an $L > 0$, the continuation of $x(t_0, \varphi)$ can be verified.

§ 33. Stability of solutions.

The definitions of stability and boundedness can be given in the same manner as for ordinary differential equations, that is, by replacing the initial value x_0 and the solution $x(t; x_0, t_0)$ by φ and $x_t(t_0, \varphi)$, respectively, and we can extend to functional-differential equations most of results obtained in the previous sections (cf. [39]-[42], [43]-[46], [48], [49], [62], [130]-[132], [147]-[149]). Here, we shall discuss some of these results and give some conditions for stability of functional-differential equations.

Consider the system of functional-differential equations (32.1) and suppose that $F(t, \varphi)$ is defined and continuous on $I \times C_H$ and that $F(t, 0) \equiv 0$. Moreover, suppose that for any $\alpha > 0$, there exists an $L(t, \alpha) > 0$ such that if $\varphi \in C_\alpha$, then $|F(t, \varphi)| \leqq L(t, \alpha)$, where $L(t, \alpha)$ is continuous in t. This condition is satisfied, if $F(t, \varphi) \in \bar{C}_0(\varphi)$, i.e., for any $\alpha > 0$ there exists a $K(\alpha) > 0$ such that $|F(t, \varphi) - F(t, \psi)| \leqq K(\alpha) \|\varphi - \psi\|$ for $\varphi, \psi \in C_\alpha$.

By the same arguments used for ordinary differential equations, the following theorems can be proved.

THEOREM 33.1. *Suppose that there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\|\varphi\| < H_1$, $0 < H_1 < H$, which satisfies the following conditions;*

(i) *$a(\|\varphi\|) \leqq V(t, \varphi) \leqq b(\|\varphi\|)$, where $a(r) \in CIP$ and $b(r) \in CIP$,*

(ii) *$V'_{(32.1)}(t, \varphi) \leqq -c(\|\varphi\|)$, where $c(r)$ is continuous and positive for $r > 0$.*

Then, the zero solution of (32.1) *is uniform-asymptotically stable.*

THEOREM 33.2. *If $F(t, \varphi) \in \bar{C}_0(\varphi)$ and if the zero solution of* (32.1) *is uniform-asymptotically stable, there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\|\varphi\| \leqq \delta_0$, $0 < \delta_0 < H$, which satisfies the conditions* (i), (ii) *in Theorem* 33.1 *and*

(iii) *$|V(t, \varphi) - V(t, \psi)| \leqq K \|\varphi - \psi\|$, where $K > 0$ is a constant.*

The condition (ii) *can be replaced by $V'_{(32.1)}(t, \varphi) \leqq -cV(t, \varphi)$, where $c > 0$ is an arbitrary constant. Moreover, if $F(t, \varphi)$ is defined on $-\infty < t < \infty$, $\varphi \in C_H$ and is almost periodic in t uniformly with respect to $\varphi \in S$ for any compact set $S \subset C_H$, there exists a continuous Liapunov functional which is almost periodic in t uniformly with respect to $\varphi \in S^*$ for any compact set $S^* \subset C_{\delta_0}$.*

Now we shall prove the following theorem. For a function $\varphi \in C$ defined on $[-h, 0]$, we introduce a norm and denote it by

$\|\varphi\|_{(a,b)}$. This norm is defined by $\|\varphi\|_{(a,b)} = \sup_{-a \leqq \theta \leqq -b} |\varphi(\theta)|$, where $-h \leqq -a \leqq -b \leqq 0$.

THEOREM 33.3. *Suppose that* $|F(t,\varphi)| \leqq L$, *where* $L>0$ *is a constant, and that there exists a continuous Liapunov functional* $V(t,\varphi)$ *defined on* $0 \leqq t < \infty$, $\|\varphi\| < H$ *which satisfies the following conditions;*

(i) $a(|\varphi(0)|) \leqq V(t,\varphi) \leqq b(\|\varphi\|)$, *where* $a(r)$ *is positive continuous for* $r \neq 0$ *and* $b(r)$ *is continuous, non-decreasing and* $b(0)=0$,

(ii) $V'_{(32.1)}(t,\varphi) \leqq -c(\|\varphi\|_{(h_2,h_1)})$, *where* $0 \leqq h_1 \leqq h_2 \leqq h$ *and* $c(r)$ *is a non-negative continuous function.*

Then the zero solution of (32.1) *is uniform-stable. If, in addition,* $c(r)$ *is positive, increasing for* $r>0$, *then zero solution of* (32.1) *is uniform-asymptotically stable.*

PROOF. For a given $\varepsilon>0$, choose a δ so small that $b(\delta)<a(\varepsilon)$. Then, $\sup_{\|\varphi\|\leqq\delta} V(t,\varphi) < \inf_{|\varphi(0)|=\varepsilon} V(t,\varphi)$. Since $V(t,\varphi)$ is non-increasing along the solutions of (32.1), the same arguments as used in ordinary differential equations show that if $\|\varphi\|<\delta(\varepsilon)$, then $|x(t;t_0,\varphi)|<\varepsilon$, which implies that $\|x_t(t_0,\varphi)\|<\varepsilon$ for all $t \geqq t_0$. This proves the uniform stability.

By uniform stability, there exists a $\delta_0>0$ such that if $\|\varphi\|<\delta_0$, then $\|x_t(t_0,\varphi)\|<H$ for all $t\geqq t_0$. Suppose that a solution $x(t_0,\varphi)$, $\|\varphi\|<\delta_0$, satisfies $\|x_t(t_0,\varphi)\| \geqq \delta$ for all $t \geqq t_0$, where δ is the one used above for uniform stability. Then there exists a sequence $\{t_k\}$ such that $t_0+(2k-1)h \leqq t_k \leqq t_0+2kh$, $k=1,2,\cdots$, and that $|x(t_k;t_0,\varphi)| \geqq \delta$. Therefore, on the interval $t_k - \frac{\delta}{2L} \leqq t \leqq t_k + \frac{\delta}{2L}$, we have $|x(t;t_0,\varphi)| > \frac{\delta}{2}$ and hence, $\|x_t(t_0,\varphi)\|_{(h_2,h_1)} > \frac{\delta}{2}$ for t such that $t_k - \frac{\delta}{2L} + h_1 \leqq t \leqq t_k + \frac{\delta}{2L} + h_1$. Therefore, we have

$$V'_{(32.1)}(t, x_t(t_0,\varphi)) \leqq -c\left(\frac{\delta}{2}\right), \qquad t_k - \frac{\delta}{2L} + h_1 \leqq t \leqq t_k + \frac{\delta}{2L} + h_1 .$$

By taking a large L, if necessary, we can assume that these intervals do not overlap, and hence $V(t_k, x_{t_k}(t_0,\varphi)) - V(t_0,\varphi) \leqq$

$-c\left(\frac{\delta}{2}\right)\frac{\delta}{L}(k-1)$. Let $n_0(\delta)$ be the smallest integer $\geqq \dfrac{b(\delta_0)}{c\left(\frac{\delta}{2}\right)\frac{\delta}{L}}$.

If $k>1+n_0(\delta)$, then $V(t_k, x_{t_k}(t_0, \varphi)) < b(\delta_0) - c\left(\frac{\delta}{2}\right)\frac{\delta}{L}\cdot\dfrac{b(\delta_0)}{c\left(\frac{\delta}{2}\right)\frac{\delta}{L}} = 0$,

which is a contradiction. Therefore, at some t_1 such that $t_0 \leqq t_1 \leqq t_0 + 2h(1+n_0(\delta))$, we have $\| x_t(t_0, \varphi)\| < \delta$. Thus, if $t \geqq t_0 + T(\varepsilon)$, $T(\varepsilon) = 2h(1+n_0(\delta))$, then $\| x_t(t_0, \varphi)\| < \varepsilon$. This completes the proof.

COROLLARY. *Suppose that* $|F(t, \varphi)| \leqq L$, $L>0$, *and that there exists a continuous Liapunov functional* $V(t, \varphi)$ *defined on* $0 \leqq t < \infty$, $\|\varphi\| < H$ *which satisfies the following conditions;*

(i) $a(|\varphi(0)|) \leqq V(t, \varphi) \leqq b_1(|\varphi(0)|) + b_2(\|\varphi\|_2)$, *where* $\|\varphi\|_2 = \left(\int_{-h}^{0}\left[\sum_{i=1}^{n}\varphi_i^2(\theta)\right]d\theta\right)^{\frac{1}{2}}$, $a(r)$ *is continuous, positive for* $r \neq 0$, $b_1(r)$ *and* $b_2(r)$ *are continuous, monotone for* $r \geqq 0$ *and* $b_1(0) = b_2(0) = 0$,

(ii) $V'_{(32.1)}(t, \varphi) \leqq -c(|\varphi(0)|)$, *where* $c(r)$ *is continuous and positive, increasing for* $r \neq 0$.

Then the zero solution of (32.1) *is uniform-asymptotically stable.*

Since $|\varphi(0)| \leqq \|\varphi\|$ and $\|\varphi\|_2 \leqq (n\|\varphi\|^2 h)^{\frac{1}{2}}$, this corollary follows immediately from Theorem 33.3.

EXAMPLE 33.1. Consider a system

$$\dot{x}_i(t) = \sum_{j=1}^{n} a_{ij}x_j(t) + f_i(x_1(t), \cdots, x_n(t);\ x_1(t-h_{i1}), \cdots, x_n(t-h_{in}), t), \tag{33.1}$$

$$i=1, 2, \cdots, n; \qquad a_{ij} \text{ constants, } h_{ij} \geqq 0 \text{ constants.}$$

Suppose that f_i satisfy $|f_i(x_1, \cdots, x_n;\ y_1, \cdots, y_n, t)| \leqq \beta(|x|+|y|)$, $|x| = \sup_i |x_i|$, and that the characteristic roots of $A=(a_{ij})$ have negative real parts. By Theorem 18.1, there exists a continuous Liapunov function $v(x)$ which is a quadratic form

$$v(x) = \sum_{i,j=1}^{n} b_{ij}x_ix_j, \qquad b_{ij} = b_{ji}$$

such that $\sum_{i=1}^{n}\frac{\partial v}{\partial x_i}(\sum_{j=1}^{n}a_{ij}x_j)=-\sum_{i=1}^{n}x_i^2$. Now consider a Liapunov functional

$$V(\varphi)=v(\varphi(0))+\sum_{i,j=1}^{n}\frac{1}{8n^2}\int_{-h_{ij}}^{0}\varphi_j^2(\theta)d\theta\,.$$

Then, if β is sufficiently small,

$$\begin{aligned}V'_{(33.1)}(x_t)&=\sum_{i=1}^{n}\frac{\partial v}{\partial x_i}(\sum_{j=1}^{n}a_{ij}x_j)\\&\quad+\sum_{i=1}^{n}\frac{\partial v}{\partial x_i}f_i+\sum_{i,j=1}^{n}\frac{1}{8n^2}x_j^2-\sum_{i,j=1}^{n}\frac{1}{8n^2}x_j^2(t-h_{ij})\\&\leqq-\sum_{i=1}^{n}x_i^2+\sum_{i=1}^{n}\{|\sum_{j=1}^{n}b_{ij}x_j|\beta(\sum_{j=1}^{n}|x_j|+\sum_{j=1}^{n}|x_j(t-h_{ij})|)\}\\&\quad+\frac{1}{4n^2}\sum_{i,j=1}^{n}x_j^2-\frac{1}{8n^2}\sum_{i,j=1}^{n}x_j^2-\frac{1}{8n^2}\sum_{i,j=1}^{n}x_j^2(t-h_{ij})\\&\leqq-\frac{1}{2}\sum_{i=1}^{n}x_i^2(t)\,,\end{aligned}$$

or $V'_{(33.1)}(\varphi)\leqq-\frac{1}{2}|\varphi(0)|^2$. Thus, if β is sufficiently small, the zero solution of (33.1) is uniform-asymptotically stable.

Next, following Hale [43], we shall consider a linear system

$$\dot{x}(t)=f(t,x_t)\,, \tag{33.2}$$

where $f(t,\varphi)$ is linear in φ and is continuous in t, φ for all $t\geqq 0$ and $\varphi\in C$.

THEOREM 33.4. *If there exist a continuous function $\alpha(t)$ possessing a continuous derivative for $t\geqq 0$ and a positive continuous function $K(t)$ for $t\geqq 0$ such that for every $t_0\geqq 0$, every $t\geqq t_0$ and $\varphi\in C$ the solution $x(t_0,\varphi)$ of (33.2) satisfies*

$$\|x_t(t_0,\varphi)\|\leqq K(t_0)e^{-[\alpha(t)-\alpha(t_0)]}\|\varphi\|\,, \tag{33.3}$$

then there exists a continuous Liapunov functional $V(t,\varphi)$ defined on $I\times C$ which satisfies the following conditions;

(i) $\|\varphi\|\leqq V(t,\varphi)\leqq K(t)\|\varphi\|$,

(ii) $|V(t,\varphi_1)-V(t,\varphi_2)|\leqq K(t)\|\varphi_1-\varphi_2\|$, $\varphi_1,\varphi_2\in C$,

(iii) $V'_{(33.2)}(t, \varphi) \leqq -\alpha'(t) V(t, \varphi)$, *where* $\alpha'(t)$ *is the derivative of* $\alpha(t)$.

PROOF. Let $V(t, \varphi)$ be defined by

$$V(t, \varphi) = \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi) \| e^{\alpha(t+\tau)-\alpha(t)} .$$

Then, clearly $\|\varphi\| \leqq V(t, \varphi) \leqq K(t) \|\varphi\|$ by (33.3). If $\varphi_1, \varphi_2 \in C$, then

$$\begin{aligned} | V(t, \varphi_1) - V(t, \varphi_2) | &= | \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi_1) \| e^{\alpha(t+\tau)-\alpha(t)} \\ &\qquad - \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi_2) \| e^{\alpha(t+\tau)-\alpha(t)} | \\ &\leqq \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi_1) - x_{t+\tau}(t, \varphi_2) \| e^{\alpha(t+\tau)-\alpha(t)} \\ &\leqq \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi_1 - \varphi_2) \| e^{\alpha(t+\tau)-\alpha(t)} \\ &\leqq K(t) \| \varphi_1 - \varphi_2 \| , \end{aligned}$$

because $x_{t+\tau}(t, \varphi_1) - x_{t+\tau}(t, \varphi_2) = x_{t+\tau}(t, \varphi_1 - \varphi_2)$ by the linearity of $f(t, \varphi)$.

Now the continuity of $V(t, \varphi)$ will be established. Let $\eta > 0$, $\varphi \in C$, and $\psi \in C$. Then we have

$$\begin{aligned} | V(t+\eta, \psi) - V(t, \varphi) | \leqq & | V(t+\eta, \psi) - V(t+\eta, \varphi) | \\ & + | V(t+\eta, \varphi) - V(t+\eta, x_{t+\eta}(t, \varphi)) | \\ & + | V(t+\eta, x_{t+\eta}(t, \varphi)) - V(t, \varphi) | . \end{aligned}$$

Since $V(t, \varphi)$ is Lipschitzian in φ and $x_{t+\eta}(t, \varphi)$ is continuous in η, the first two terms are small if η and $\|\varphi - \psi\|$ are small. Therefore, it is sufficient to see that the third term approaches zero as $\eta \to 0$. Since $x_{t+\eta+\tau}(t+\eta, x_{t+\eta}(t, \varphi)) = x_{t+\eta+\tau}(t, \varphi)$,

$$\begin{aligned} & | V(t+\eta, x_{t+\eta}(t, \varphi)) - V(t, \varphi) | \\ &= | \sup_{\tau \geqq 0} \| x_{t+\eta+\tau}(t, \varphi) \| e^{\alpha(t+\eta+\tau)-\alpha(t+\eta)} - \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi) \| e^{\alpha(t+\tau)-\alpha(t)} | \\ &= | \sup_{\tau \geqq \eta} \| x_{t+\tau}(t, \varphi) \| e^{\alpha(t+\tau)-\alpha(t)+\alpha(t)-\alpha(t+\eta)} - \sup_{\tau \geqq 0} \| x_{t+\tau}(t, \varphi) \| e^{\alpha(t+\tau)-\alpha(t)} | . \end{aligned}$$

Set $a(\eta) = \sup_{\tau \geqq \eta} \| x_{t+\tau}(t, \varphi) \| e^{\alpha(t+\tau)-\alpha(t)}$. Then we have

$$| V(t+\eta, x_{t+\eta}(t, \varphi)) - V(t, \varphi) | = | a(\eta) e^{\alpha(t)-\alpha(t+\eta)} - a(0) | .$$

On the other hand, $\|x_{t+\tau}(t,\varphi)\|e^{\alpha(t+\tau)-\alpha(t)}$ is a bounded continuous function for all $\tau\geqq 0$, and hence we can see that $a(\eta)$ is non-increasing and $a(\eta)\to a(0)$, that is, $a(\eta)$ is continuous at $\eta=0$.

Finally, we have

$$\begin{aligned}
V'_{(33.2)}(t,\varphi)&=\overline{\lim_{\delta\to 0+}}\frac{1}{\delta}\{V(t+\delta, x_{t+\delta}(t,\varphi))-V(t, x_t(t,\varphi))\}\\
&=\overline{\lim_{\delta\to 0+}}\frac{1}{\delta}\{\sup_{\tau\geqq 0}\|x_{t+\delta+\tau}(t+\delta, x_{t+\delta}(t,\varphi))\|e^{\alpha(t+\delta+\tau)-\alpha(t+\delta)}\\
&\quad-\sup_{\tau\geqq 0}\|x_{t+\tau}(t,\varphi)\|e^{\alpha(t+\tau)-\alpha(t)}\}\\
&=\overline{\lim_{\delta\to 0+}}\frac{1}{\delta}\{\sup_{\tau\geqq\delta}\|x_{t+\tau}(t,\varphi)\|e^{\alpha(t+\tau)-\alpha(t+\delta)}\\
&\quad-\sup_{\tau\geqq 0}\|x_{t+\tau}(t,\varphi)\|e^{\alpha(t+\tau)-\alpha(t)}\}\\
&\leqq\overline{\lim_{\delta\to 0+}}\frac{1}{\delta}\{\sup_{\tau\geqq 0}\|x_{t+\tau}(t,\varphi)\|e^{\alpha(t+\tau)-\alpha(t)}(e^{\alpha(t)-\alpha(t+\delta)}-1)\}\\
&\leqq -\alpha'(t)V(t,\varphi)\,.
\end{aligned}$$

§ 34. Asymptotic behavior of solutions of an autonomous system.

In this section, we shall discuss some results due to Hale [48], [50] which are extensions to functional-differential equations of results due to LaSalle [69], [70] for ordinary differential equations.

Consider the system

$$\dot{x}(t)=F(x_t)\,, \tag{34.1}$$

where $F(\varphi)$ is continuous and $F(\varphi)\in\bar{C}_0(\varphi)$ on S, $S=\{\varphi\in C;\ \|\varphi\|<H\}$, where H is some positive number (or ∞). Since the system (34.1) is autonomous, the initial time t_0 will be taken as 0 and solutions of (34.1) will be denoted by $x(\varphi)$. We can define a motion or a path in C through φ as the set of functions in C given by $\bigcup_{0\leqq t<A} x_t(\varphi)$, where the interval $[-h, A)$ is the region of definition of $x(\varphi)$.

An element ψ of C is said to be in $\Gamma^+(\varphi)$, the positive limiting set of φ, if $x(\varphi)$ is defined on $[-h,\infty)$ and there is a sequence of non-negative real numbers $\{t_k\}$ such that $t_k \to \infty$ as $k \to \infty$ and $\| x_{t_k}(\varphi) - \psi \| \to 0$ as $k \to \infty$. A set M in C is called an invariant set if for any $\varphi \in M$, the solution $x(\varphi)$ is defined on $(-\infty, \infty)$ and $x_t(\varphi)$ is in M for $t \in (-\infty, \infty)$.

For any $H_1 < H$, there is a constant L such that $|F(\varphi)| \leqq L$ for all $\varphi \in C_{H_1}$, because $F \in \bar{C}_0(\varphi)$. From this, we obtain the following result.

LEMMA 34.1. *If $x(\varphi)$ is a solution of* (34.1) *with initial function φ at $t=0$ which is defined on $[-h, \infty)$ and satisfies $\| x_t(\varphi) \| \leqq H_1 < H$ for all $t \in I$, then the family of functions $\{x_t(\varphi) ; t \geqq 0\}$ belongs to a compact subset of C, that is, the motion through φ belongs to a compact subset of C.*

LEMMA 34.2. *If the solution $x(\varphi)$ of* (34.1) *is defined on $[-h, \infty)$ and $\| x_t(\varphi) \| \leqq H_1 < H$ for all $t \in I$, then $\Gamma^+(\varphi)$ is a non-empty, compact, connected, invariant set and* $\operatorname{dist}(x_t(\varphi), \Gamma^+(\varphi)) \to 0$ *as $t \to \infty$.*

This lemma can be proved by Lemma 34.1 and the standard type of argument used in the theory of dynamical systems, for the proof, see [50].

THEOREM 34.1. *Let $V(\varphi)$ be a continuous Liapunov functional on S and let U_l denote the region such that $V(\varphi) < l$. Suppose that $V(\varphi) \geqq 0$ and $V'_{(34.1)}(\varphi) \leqq 0$ for all $\varphi \in U_l$ and that there exists a constant $K \geqq 0$ such that $|\varphi(0)| \leqq K$ for $\varphi \in U_l$. If E is the set of all points in U_l where $V'_{(34.1)}(\varphi) = 0$ and M is the largest invariant set in E, then every solution of* (34.1) *with initial value in U_l approaches M as $t \to \infty$.*

PROOF. Since $V'_{(34.1)}(\varphi) \leqq 0$, $V(x_t(\varphi))$ is non-increasing in t within U_l. Therefore, $\varphi \in U_l$ implies $x_t(\varphi) \in U_l$ and $|x(t ; \varphi)| \leqq K$ for all $t \geqq 0$, which implies $\| x_t(\varphi) \| \leqq K$ for all $t \geqq h$, that is, $x_t(\varphi)$ is bounded. By Lemma 34.2, $\Gamma^+(\varphi)$ is an invariant set. Since $V(x_t(\varphi))$ is non-increasing and bounded below, $V(x_t(\varphi))$ has a limit $l_0 < l$ as $t \to \infty$, and $V = l_0$ on $\Gamma^+(\varphi)$. Hence, $\Gamma^+(\varphi)$ is in U_l and $V'_{(34.1)} = 0$

on $\Gamma^+(\varphi)$. Consequently, $\Gamma^+(\varphi)$ is in M and Lemma 34.2 implies that $x_t(\varphi)\to M$ as $t\to\infty$.

COROLLARY. *If the conditions of Theorem* 34.1 *are satisfied and if $V'_{(34.1)}(\varphi)<0$ for all $\varphi\neq 0$ in U_l, then every solution of* (34.1) *with initial value in U_l approaches* 0 *as $t\to\infty$.*

PROOF. As is seen from the proof of Theorem 34.1, $\varphi\in U_l$ implies $x_t(\varphi)\in U_l$, $\Gamma^+(\varphi)$ is in U_l, and $V'_{(34.1)}=0$ on $\Gamma^+(\varphi)$. Since $V'_{(34.1)}(\varphi)<0$ for all $\varphi\neq 0$ in U_l and $\Gamma^+(\varphi)$ is non-empty, we can see that $\Gamma^+(\varphi)$ must be 0. Thus, $x_t(\varphi)\to 0$ as $t\to\infty$.

REMARK. Since $\Gamma^+(\varphi)$ is in U_l and $\Gamma^+(\varphi)=0$, 0 must be in U_l, and we can see that $F(0)=0$, because $\Gamma^+(\varphi)$ is an invariant set. Namely, the conditions of the corollary above imply that $F(0)=0$ and 0 is in U_l.

THEOREM 34.2. *Suppose that there exists a continuous Liapunov functional $V(\varphi)$ defined on S which satisfies the following conditions;*

(i) $a(|\varphi(0)|)\leqq V(\varphi)\leqq b(\|\varphi\|)$, *where $a(r), b(r)\in CIP$ for $0\leqq r<H$,*

(ii) $V'_{(34.1)}(\varphi)<0$, $\varphi\neq 0$, $V'_{(34.1)}(0)=0$.

Then, $x\equiv 0$ is the solution of (34.1) *and it is asymptotically stable and every solution of* (34.1) *approaches* 0 *as $t\to\infty$, provided the initial value φ satisfies $V(\varphi)<l_0$, $l_0=\lim_{r\to H} a(r)$.*

PROOF. Since $a(r)$ is increasing, the set U_l of φ such that $V(\varphi)<l$ satisfies the conditions of Theorem 34.1 if $l<l_0$. Thus, by the corollary of Theorem 34.1, every solution in U_l approaches zero as $t\to\infty$. The stability of the zero solution follows from Theorem 33.3.

THEOREM 34.3. *Let $S=C$ and $V(\varphi)$ be a continuous Liapunov functional on C. If $V(\varphi)\geqq 0$, $V'_{(34.1)}(\varphi)\leqq 0$ for all $\varphi\in C$ and E is the set of $\varphi\in C$ for which $V'_{(34.1)}(\varphi)=0$ and M is the largest invariant set in E, then all bounded solutions of* (34.1) *approach M as $t\to\infty$.*

If, in addition, there exists a continuous function $a(r) \geqq 0$, $0 \leqq r < \infty$, *such that* $a(r) \to \infty$ *as* $r \to \infty$ *and that* $a(|\varphi(0)|) \leqq V(\varphi)$ *for all* $\varphi \in C$, *then all solutions of* (34.1) *are bounded for all* $t \geqq 0$.

PROOF. The first part of the theorem can be proved in the same manner as in the proof of Theorem 34.1. For any $\varphi_0 \in C$, there is an $m > 0$ such that $V(\varphi) > V(\varphi_0)$ for $|\varphi(0)| \geqq m$. Since $V(x_t(\varphi))$ is non-increasing in t, it follows that $|x(t\,;\varphi_0)| < m$ for all $t \geqq 0$ which implies $\|x_t(\varphi_0)\| < m$ for all $t \geqq h$.

COROLLARY. *If the conditions of Theorem* 34.3 *are satisfied and* $V(0) = 0$, $V'_{(34.1)}(\varphi) < 0$ *for* $\varphi \neq 0$, *then all solutions of* (34.1) *approach zero as* $t \to \infty$ *and the zero solution is asymptotically stable in the large.*

EXAMPLE 34.1. Suppose $n = 1$ and consider

$$f(\varphi) = -\frac{1}{h}\int_{-h}^{0} g(\varphi(\theta))(h+\theta)d\theta\,,$$

where $g(x)$ is a real function defined for all real x, locally Lipschitzian in x and $xg(x) > 0$, $G(x) = \int_0^x g(u)du \to \infty$ as $|x| \to \infty$. The system (34.1) is then given by

$$\dot{x}(t) = -\frac{1}{h}\int_{-h}^{0}(h+\theta)g(x(t+\theta))d\theta\,, \qquad t \geqq 0\,. \tag{34.2}$$

We shall observe a relationship between the solutions of (34.2) and the second order ordinary differential equation

$$\ddot{x} + g(x) = 0\,, \tag{34.3}$$

along the same lines as Levin and Nohel [83]. Consider a Liapunov functional $V(\varphi)$ defined by $V(\varphi) = G(\varphi(0)) + \frac{1}{2h}\int_{-h}^{0}\Big[\int_{s}^{0} g(\varphi(\theta))d\theta\Big]^2 ds$. Then, clearly $V(\varphi)$ is continuous on C and $V(\varphi) \geqq 0$. $V'_{(34.2)}(\varphi)$ can be calculated in the following way.

$$V'_{(34.2)}(x_t) = g(x(t))\Big(-\frac{1}{h}\int_{-h}^{0}(h+\theta)g(x(t+\theta))d\theta\Big)$$

$$+\varlimsup_{\delta\to 0^+}\frac{1}{\delta}\frac{1}{2h}\Big\{\int_{-h}^{0}\Big[\int_{s}^{0}g(x(t+\theta+\delta))d\theta\Big]^2 ds$$

$$-\int_{-h}^{0}\Big[\int_{s}^{0}g(x(t+\theta))d\theta\Big]^2 ds\Big\}$$

$$=-\frac{1}{h}g(x(t))\int_{-h}^{0}(h+\theta)g(x(t+\theta))d\theta$$

$$+\frac{1}{h}\int_{-h}^{0}\Big[\int_{s}^{0}g(x(t+\theta))(g(x(t))-g(x(t+s)))d\theta\Big]ds\,.$$

Since

$$\int_{-h}^{0}\Big(\int_{s}^{0}g(x(t+\theta))d\theta\Big)ds=\int_{-h}^{0}(h+\theta)g(x(t+\theta))d\theta$$

and

$$\int_{-h}^{0}\Big(\int_{s}^{0}g(x(t+\theta))g(x(t+s))d\theta\Big)ds=\frac{1}{2}\Big\{\int_{-h}^{0}g(x(t+\theta))d\theta\Big\}^2,$$

we have

$$V'_{(34.2)}(\varphi)=-\frac{1}{2h}\Big\{\int_{-h}^{0}g(\varphi(\theta))d\theta\Big\}^2. \tag{34.4}$$

The conditions of Theorem 34.3 are satisfied and the set E is the set of φ such that $\int_{-h}^{0}g(\varphi(\theta))d\theta=0$ and the set M consists of points $x_t(\varphi)$ where $x(\varphi)$ is a solution of (34.2) such that

$$\int_{-h}^{0}g(x(t+\theta\,;\,\varphi))d\theta=0\,. \tag{34.5}$$

If $x(\varphi)$ is a solution of (34.2), then from $\dot{x}(t)=-\frac{1}{h}\int_{t-h}^{t}(h-t+u)$ $g(x(u))du$, it follows that $\ddot{x}(t)=-g(x(t))+\frac{1}{h}\int_{t-h}^{t}g(x(u))du$ or $\ddot{x}(t)$ $+g(x(t))=\frac{1}{h}\int_{-h}^{0}g(x(t+\theta))d\theta$, and hence, M is generated by periodic solutions of $\ddot{x}(t)+g(x(t))=0$ which satisfy (34.5). Thus, from Theorem 34.3, each solution of (34.2) approaches M as $t\to\infty$.

§ 35. Boundedness of solutions.

Consider the system (32.1), where $F(t,\varphi)$ of (32.1) is defined and continuous on $I\times C$, and suppose that for any $\alpha>0$, there exists an $L(t, \alpha) > 0$ such that if $\varphi \in C_\alpha$, then $|F(t, \varphi)| \leqq L(t, \alpha)$, where $L(t, \alpha)$ is continuous in t.

The following theorems can be proved by the same arguments used for ordinary differential equations. Let S be the set of $\varphi \in C$ such that $\|\varphi\| \geqq H$, where $H > 0$ is a constant and it may be large (cf. [148]).

THEOREM 35.1. *Suppose that there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\varphi \in S$ which satisfies the following conditions;*

(i) $a(\|\varphi\|) \leqq V(t, \varphi) \leqq b(\|\varphi\|)$, *where $a(r) \in CI$, positive for $r > H$, $a(r) \to \infty$ as $r \to \infty$ and $b(r) \in CI$,*

(ii) $V'_{(32.1)}(t, \varphi) \leqq -c(\|\varphi\|)$, *where $c(r)$ is continuous and positive for $r > H$.*

Then the solutions of (32.1) *are uniform-bounded and uniform-ultimately bounded.*

THEOREM 35.2. *If the solutions of* (32.1) *are uniform-bounded and uniform-ultimately bounded and if $F(t, \varphi) \in \bar{C}_0(\varphi)$, there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\varphi \in S$ which satisfies the conditions* (i), (ii) *in Theorem* 35.1 *and*

(iii) $V(t, \varphi) \in \bar{C}_0(\varphi)$.

Moreover, condition (ii) *can be replaced by $V'_{(32.1)}(t, \varphi) \leqq -cV(t, \varphi)$, where $c > 0$ is a constant.*

PROOF. Let the solutions of (32.1) be uniform-ultimately bounded for bound H. Then, for any t_0 and any $\alpha > 0$ there exists a $T(\alpha) > 0$ such that if $\varphi \in C_\alpha$, $\|x_t(t_0, \varphi)\| < H$ for all $t \geqq t_0 + T(\alpha)$, where $x(t_0, \varphi)$ is a solution of (32.1). The uniform-boundedness implies that corresponding to each $\alpha > 0$, there exists a $\beta(\alpha) > 0$ such that if $\varphi \in C_\alpha$, then $\|x_t(t_0, \varphi)\| \leqq \beta(\alpha)$ for all $t \geqq t_0$. Moreover, there exists an $L(\beta(\alpha)) > 0$ such that $\varphi_1 \in C_\beta$ and $\varphi_2 \in C_\beta$ imply $|F(t, \varphi_1) - F(t, \varphi_2)| \leqq L(\beta(\alpha)) \|\varphi_1 - \varphi_2\|$. Further, $T(\alpha)$, $\beta(\alpha)$ and $L(\beta(\alpha))$ can be assumed to be continuous and increasing in α. Define $V(t, \varphi)$ by

$$V(t, \varphi)=\sup_{\tau\geqq 0} G(\| x_{t+\tau}(t, \varphi)\|)e^{\tau}$$

for $t \in I$ and $\varphi \in C$, where

$$G(z)=\begin{cases} z-H & (z\geqq H) \\ 0 & (0\leqq z<H). \end{cases}$$

Then, in the same way as in ordinary differential equations, it can be seen that this is the desired $V(t, \varphi)$.

THEOREM 35.3. *Suppose that there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t\in I$, $\varphi\in S$ which satisfies the following conditions;*

(i) $a(|\varphi(0)|)\leqq V(t, \varphi)\leqq b(\|\varphi\|)$, *where $a(r)\in CI$ for $r\geqq 0$, positive for $r>H$, $a(r)\to\infty$ as $r\to\infty$ and $b(r)\in CI$ for $r>H$,*

(ii) $V'_{(32.1)}(t, \varphi)\leqq -c(\|\varphi\|_{(h_2,h_1)})$, *where $0\leqq h_1\leqq h_2\leqq h$ and $c(r)\geqq 0$ is continuous for $r\geqq 0$.*

Then the solutions of (32.1) *are uniform-bounded. If, in addition, for any $\alpha>0$ there exists an $L(\alpha)>0$ such that $|F(t, \varphi)|\leqq L(\alpha)$ for all $t\geqq 0$, $\varphi\in C_\alpha$ and if $c(r)$ is positive for $r>H$, then the solutions of* (32.1) *are uniform-ultimately bounded.*

PROOF. For a given α, where we may assume that $\alpha>H$, choose a $\beta(\alpha)>0$ such that $a(\beta)>b(\alpha)$, $\beta>\alpha$. For a solution $x(t_0, \varphi)$ such that $\varphi\in C_\alpha$, suppose that at some t_1 we have $\|x_{t_1}(t_0, \varphi)\| =\beta(\alpha)$. Since $\| x_t(t_0, \varphi)\|$ is continuous in t and $\| x_{t_0}(t_0, \varphi)\|\leqq\alpha$, there exist t_2, t_3, $t_0\leqq t_2<t_3\leqq t_1$, such that $\| x_{t_2}(t_0, \varphi)\|=\alpha$, $|x(t_3; t_0, \varphi)|=\beta$ and that $\alpha<\| x_t(t_0, \varphi)\|<\beta$ for $t\in(t_2, t_3)$. For $t\in[t_2, t_3]$, $\| x_t(t_0, \varphi)\|\geqq\alpha>H$, and hence $x_t(t_0, \varphi)\in S$. By (ii), $V(t_3, x_{t_3}(t_0, \varphi)) \leqq V(t_2, x_{t_2}(t_0, \varphi))\leqq b(\alpha)$. Since $|x(t_3; t_0, \varphi)|=\beta$ implies $a(\beta)\leqq V(t_3, x_{t_3}(t_0, \varphi))$, we have $a(\beta)\leqq b(\alpha)$, which contradicts $a(\beta)>b(\alpha)$. Thus, if $\varphi\in C_\alpha$, then $\| x_t(t_0, \varphi)\|<\beta(\alpha)$ for all $t\geqq t_0$. This proves uniform-boundedness of solutions.

From the above, it follows that for a fixed H_1, $H_1>H$, there exists an $H^*>0$ such that if $\varphi\in C_{H_1}$, then $\| x_t(t_0, \varphi)\|<H^*$ for all $t\geqq t_0$. Suppose that a solution $x(t_0, \varphi)$, $\varphi\in C_\alpha$, satisfies $\| x_t(t_0, \varphi)\|$

$\geqq H_1$ for all $t \geqq t_0$. Then there exists a sequence $\{t_k\}$ such that $t_0+(2k-1)h \leqq t_k \leqq t_0+2kh$, $k=1, 2, \cdots$ and that $|x(t_k; t_0, \varphi)| \geqq H_1$. Since $\|x_t(t_0, \varphi)\| \leqq \beta(\alpha)$ for all $t \geqq t_0$, we have $|\dot{x}(t; t_0, \varphi)| \leqq L(\beta(\alpha))$ for all $t \geqq t_0$. Therefore, on the intervals $t_k - \frac{H_1-H}{2L(\beta)} \leqq t \leqq t_k + \frac{H_1-H}{2L(\beta)}$, $k=1,2,\cdots$, we have $|x(t; t_0, \varphi)| \geqq \frac{H_1+H}{2}$, which implies $\|x_t(t_0, \varphi)\|_{(h_2,h_1)} \geqq \frac{H_1+H}{2}$ for t such that

$$t_k - \frac{H_1-H}{2L(\beta)} + h_1 \leqq t \leqq t_k + \frac{H_1-H}{2L(\beta)} + h_1 \,. \tag{35.1}$$

Since $\frac{H_1+H}{2} > H$, $x_t(t_0, \varphi) \in S$ and $\|x_t(t_0, \varphi)\|_{(h_2,h_1)} \leqq \beta(\alpha)$, there exists a constant $c>0$ such that $V'_{(32.1)}(t, x_t(t_0, \varphi)) \leqq -c$ for t on (35.1), where c depends only on $\beta(\alpha)$. By taking a large L, if necessary, we can assume that the intervals (35.1) do not overlap, and hence

$$V(t_k, x_{t_k}(t_0, \varphi)) - V(t_0, \varphi) \leqq -c\frac{H_1-H}{L(\beta)}(k-1)\,.$$

Let $n_0(\alpha)$ be the smallest integer $\geqq \frac{b(\alpha)L(\beta)}{(H_1-H)c}$. Then, it can be seen that at some t' such that $t_0 \leqq t' \leqq t_0+2h(1+n_0(\alpha))$, we have $\|x_{t'}(t_0, \varphi)\| < H_1$. Thus, if $t \geqq t_0+T(\alpha)$, $T(\alpha)=2h(1+n_0(\alpha))$, then $\|x_t(t_0, \varphi)\| < H^*$. This proves the theorem.

The following theorem gives a sufficient condition for uniform-boundedness of solutions. We shall denote by S^* the set of $\varphi \in C$ such that $|\varphi(0)| \geqq H$, where H may be large.

THEOREM 35.4. *Suppose that there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\varphi \in S^*$ which satisfies the following conditions;*

(i) $a(|\varphi(0)|) \leqq V(t, \varphi) \leqq b_1(|\varphi(0)|) + b_2(\|\varphi\|)$, *where* $a(r)$, $b_1(r)$, $b_2(r) \in CI$, *positive for* $r > H$ *and* $a(r)-b_2(r) \to \infty$ *as* $r \to \infty$,

(ii) $V'_{(32.1)}(t, \varphi) \leqq 0$.

Then, the solutions of (32.1) *are uniform-bounded.*

PROOF. For a given $\alpha>0$, $\alpha>H$, choose $\beta(\alpha)>0$ so large that $b_1(\alpha)<a(\beta)-b_2(\beta)$. Suppose that a solution $x(t_0, \varphi)$ of (32.1), $\varphi\in C_\alpha$, satisfies $\|x_t(t_0,\varphi)\|=\beta$ at some t. Then there exist t_1, t_2, $t_0\leqq t_1<t_2$, such that $|x(t_1; t_0,\varphi)|=\alpha$, $|x(t_2; t_0,\varphi)|=\beta$ and that $\alpha<|x(t; t_0,\varphi)|<\beta$ for $t\in(t_1, t_2)$, and we can assume that $|x(t; t_0,\varphi)|<\beta$ for $t\in[t_0, t_2)$, and hence $\|x_{t_1}(t_0,\varphi)\|<\beta$. For $t\in[t_1, t_2]$, $x_t(t_0,\varphi)\in S^*$ and hence, $V(t_2, x_{t_2}(t_0,\varphi))\leqq V(t_1, x_{t_1}(t_0,\varphi))$ by (ii). This implies that $a(\beta)\leqq b_1(\alpha)+b_2(\beta)$, which contradicts $b_1(\alpha)<a(\beta)-b_2(\beta)$. Thus, $\|x_t(t_0,\varphi)\|<\beta(\alpha)$ for all $t\geqq t_0$. This proves the uniform-boundedness of solutions of (32.1).

§ 36. Perturbed systems.

Here, we shall consider the linear system (33.2) and its perturbed system

$$\dot{x}(t)=f(t, x_t)+X(t, x_t)\,, \tag{36.1}$$

where $X(t,\varphi)$ is defined and continuous on $I\times C_H$, $H>0$, or on $I\times C$. Following Hale [43], we shall discuss the behavior of solutions of (36.1). In the following, in case $X(t,\varphi)$ is defined on $I\times C$, we understand that $H>0$ is an arbitrary constant.

LEMMA 36.1. *If the solutions of* (33.2) *satisfy* (33.3) *and if* $x^*(t_0,\varphi)$ *is a solution of* (36.1) *with initial function* φ *at* t_0, *then the function* $V(t,\varphi)$ *given in Theorem* 33.4 *satisfies*

$$V'_{(36.1)}(t, x_t^*)\leqq-\alpha'(t)V(t, x_t^*)+K(t)|X(t, x_t^*)| \tag{36.2}$$

as long as $\|x_t^*(t_0,\varphi)\|<H$. *In particular, if there exists a continuous function* $g(t)\geqq 0$ *such that*

$$|X(t,\varphi)|\leqq g(t)\|\varphi\|\,, \tag{36.3}$$

then we have

$$\|x_t^*(t_0,\varphi)\|\leqq K(t_0)e^{-[\alpha(t)-\alpha(t_0)]+\int_{t_0}^t K(s)g(s)ds}\|\varphi\|\,, \tag{36.4}$$

as long as $\|x_t^*(t_0,\varphi)\|<H$.

PROOF. As long as $\|x_t^*\|<H$, we have

$$V'_{(36.1)}(t, x_t^*) = \overline{\lim_{\delta \to 0+}} \frac{1}{\delta} \{ V(t+\delta, x_{t+\delta}^*) - V(t+\delta, x_{t+\delta})$$
$$+ V(t+\delta, x_{t+\delta}) - V(t, x_t^*) \},$$

where x is a solution of (33.2) with initial function x_t^* at t. Since $x_t = x_t^*$ and $|V(t, \varphi_1) - V(t, \varphi_2)| \leqq K(t) \| \varphi_1 - \varphi_2 \|$, we have

$$V'_{(36.1)}(t, x_t^*) \leqq V'_{(33.2)}(t, x_t) + \overline{\lim_{\delta \to 0+}} \frac{K(t+\delta)}{\delta} \| x_{t+\delta}^* - x_{t+\delta} \|$$
$$\leqq -\alpha'(t) V(t, x_t^*) + K(t) | X(t, x_t^*) | .$$

If $X(t, \varphi)$ satisfies (36.3), then $V'_{(36.1)}(t, x_t^*) \leqq -\alpha'(t) V(t, x_t^*) + K(t) g(t) V(t, x_t^*)$, because $\| x_t^* \| \leqq V(t, x_t^*)$. Consequently, by Theorem 4.1,

$$\| x_t^* \| \leqq V(t, x_t^*) \leqq V(t_0, \varphi) e^{-[\alpha(t) - \alpha(t_0)] + \int_{t_0}^t K(s) g(s) ds},$$

which implies (36.4), because $V(t_0, \varphi) \leqq K(t_0) \| \varphi \|$.

THEOREM 36.1. *Suppose that the solutions of* (33.2) *satisfy* (33.3) *and* $X(t, \varphi)$ *of* (36.1) *satisfies* (36.3) *and that there exists a continuous function* $M(t)$, $t \geqq 0$, *such that*

$$m(t, t_0) = K(t_0) \exp \{ -[\alpha(t) - \alpha(t_0)] + \int_{t_0}^t K(s) g(s) ds \} < M(t_0)$$

for $t \geqq t_0$. *If* $\| \varphi \| \leqq \frac{H}{M(t_0)}$, *then the solution* $x^*(t_0, \varphi)$ *of* (36.1) *satisfies*

$$\| x_t^*(t_0, \varphi) \| \leqq m(t, t_0) \| \varphi \| \qquad \text{for all} \quad t \geqq t_0 . \tag{36.5}$$

PROOF. By Lemma 36.1, as long as $\| x_t^*(t_0, \varphi) \| < H$, we have (36.4). Therefore, if $\| \varphi \| \leqq \frac{H}{M(t_0)}$, then $\| x_t^*(t_0, \varphi) \| < H$ and the theorem is proved.

COROLLARY 1. *If the conditions of Theorem* 36.1 *are satisfied and* $m(t, t_0) \to 0$ *as* $t \to \infty$, *then the zero solution of* (36.1) *is asymptotically stable. If* $M(t_0)$ *is independent of* t_0 *and for every* $\varepsilon > 0$ *there exists a* $T(\varepsilon) > 0$ *such that for every* $t_0 \geqq 0$, $m(t, t_0) < \varepsilon$ *for all*

$t \geqq t_0 + T(\varepsilon)$, *then the zero solution of* (36.1) *is uniform-asymptotically stable.*

COROLLARY 2. *If* $X(t, \varphi)$ *is defined on* $I \times C$ *and the conditions of Theorem* 36.1 *are satisfied and if* $M(t_0)$ *can be chosen to be independent of* t_0, *then the solutions of* (36.1) *are uniform-bounded.*

COROLLARY 3. *Suppose that* $X(t, \varphi)$ *is defined on* $I \times C$ *and the conditions of Theorem* 36.1 *are satisfied and that* $\alpha(t)$, $K(t)$ *are constants. If* $g(t)$ *in* (36.3) *satisfies* $\int_0^\infty g(t)dt < \infty$, *then the solutions of* (36.1) *are uniform-bounded.*

PROOF. In this case, $m(t, t_0) = Ke^{K\int_{t_0}^t g(s)ds} \leqq Ke^{K\int_0^\infty g(s)ds}$ for all $t \geqq t_0 \geqq 0$. Thus, $M(t_0)$ can be chosen to be a constant M and we have $\| x_t^*(t_0, \varphi) \| \leqq M \| \varphi \|$, which implies uniform-boundedness of solutions of (36.1).

EXAMPLE 36.1. Consider the system

$$\dot{x}(t) = \sum_{k=1}^{m} (A_k(t) + B_k(t)) x(t - \tau_k), \tag{36.6}$$

where $A_k(t)$, $B_k(t)$ are continuous in t for all $t \geqq 0$ and τ_k, $k = 1, 2, \cdots, m$, are constants, $0 \leqq \tau_k \leqq h$. The system (36.6) clearly can be written in the form (36.1). If all of the solutions of $\dot{x}(t) = \sum_{k=1}^{m} A_k(t) x(t - \tau_k)$ are uniform-bounded for $t \geqq t_0$ for every t_0 and $\varphi \in C$ and if $\int_0^\infty \| B_k(t) \| dt < \infty$, then by Corollary 3 above, all of the solutions of (36.6) are bounded.

COROLLARY 4. *If the zero solution of* (33.2) *is uniform-asymptotically stable and for every* $\varepsilon > 0$ *there is a* $\delta > 0$ *such that* $|X(t, \varphi)| \leqq \varepsilon \| \varphi \|$ *if* $\| \varphi \| < \delta$, *then the zero solution of* (36.1) *is uniform-asymptotically stable.*

For further applications of Theorem 36.1, see [43].

§ 37. Existence theorems for periodic and almost periodic solutions.

In the system (32.1), suppose that $F(t, \varphi)$ is continuous on $I \times C$ and that for any $\alpha > 0$ there exists an $L(t, \alpha) > 0$ such that if $\varphi \in C_\alpha$, then $|F(t, \varphi)| \leqq L(t, \alpha)$, where $L(t, \alpha)$ is continuous in t. Corresponding to Theorem 29.3, we obtain the following theorem by applying Theorem 28.2.

THEOREM 37.1. *Suppose that $F(t, \varphi) \in \bar{C}_0(\varphi)$ and $F(t, \varphi)$ is periodic in t of period ω, $\omega \geqq h$, i.e., $F(t+\omega, \varphi) = F(t, \varphi)$. If the solutions of* (32.1) *are uniform-bounded and uniform-ultimately bounded for bound B, there exists a periodic solution of* (32.1) *of period ω which is bounded by B.*

COROLLARY. *Suppose that $F(t, \varphi) \in \bar{C}_0(\varphi)$ and that $F(t, \varphi)$ is periodic in t of period ω, $\omega \geqq h$. If there exists a continuous Liapunov functional $V(t, \varphi)$ which satisfies the conditions in Theorem* 35.1 *or Theorem* 35.3, *there exists a periodic solution of* (32.1) *of period ω.*

THEOREM 37.2. *Suppose that $F(t, \varphi) \in \bar{C}_0(\varphi)$ and $F(t, \varphi)$ is periodic in t of period ω, $\omega \geqq h$, and consequently for any $\alpha > 0$, there exists an $L(\alpha) > 0$ such that $\varphi \in C_\alpha$ implies $|F(t, \varphi)| \leqq L(\alpha)$. Moreover, suppose that there exists a continuous Liapunov functional $V(t, \varphi)$ defined on $t \in I$, $\varphi \in S^*$, where S^* is the set of $\varphi \in C$ such that $|\varphi(0)| \geqq H$ (H may be large), and that $V(t, \varphi)$ satisfies the following conditions;*

(i) *$a(|\varphi(0)|) \leqq V(t, \varphi) \leqq b(\|\varphi\|)$, where $a(r), b(r) \in CI$, positive for $r \geqq H$ and $a(r) \to \infty$ as $r \to \infty$,*

(ii) *$V'_{(32.1)}(t, \varphi) \leqq -c(|\varphi(0)|)$, where $c(r)$ is continuous and positive for $r \geqq H$.*

Suppose that there exists an $H_1 > 0$, $H_1 > H$, such that

$$hL(\gamma^*) < H_1 - H, \tag{37.1}$$

where $\gamma^* > 0$ is a constant which is determined in the following way:

By the condition on $V(t, \varphi)$, *there exist* $\alpha > 0$, $\beta > 0$ *and* $\gamma > 0$ *such that* $b(H_1) \leqq a(\alpha)$, $b(\alpha) \leqq a(\beta)$ *and* $b(\beta) \leqq a(\gamma)$. γ^* *is defined by* $b(\gamma) \leqq a(\gamma^*)$.

Under the above conditions, there exists a periodic solution of (32.1) *of period* ω. *In particular, the relation* (37.1) *can always be satisfied if* h *is sufficiently small.*

PROOF. Let S be the set of φ such that $\|\varphi\| < \gamma$. We shall show that if $\varphi \in C_\gamma$, then $\|x_t(t_0, \varphi)\| < \gamma^*$ for all $t \geqq t_0$. Suppose a solution $x(t_0, \varphi)$, $t_0 \in I$, $\varphi \in C_\gamma$, satisfies $\|x_{t_1}(t_0, \varphi)\| = \gamma^*$ at some t_1. Then there exist t_2, t_3, $t_0 \leqq t_2 < t_3 \leqq t_1$, such that $\|x_{t_2}(t_0, \varphi)\| = \gamma$, $|x(t_3; t_0, \varphi)| = \gamma^*$ and that $\gamma < \|x_t(t_0, \varphi)\| < \gamma^*$ for $t \in (t_2, t_3)$. Suppose that at some $t \in [t_2, t_3]$, we have $|x(t; t_0, \varphi)| < H$. Then there are t_4, t_5, $t_2 \leqq t_4 < t_5 < t_3$, such that $|x(t_4; t_0, \varphi)| = H$, $|x(t_5; t_0, \varphi)| = H_1$ and that $H < |x(t; t_0, \varphi)| < H_1$ for $t \in (t_4, t_5)$. On the interval $t_4 \leqq t \leqq t_5$, $\|x_t(t_0, \varphi)\| < \gamma^*$ and hence $|\dot{x}(t; t_0, \varphi)| \leqq L(\gamma^*)$. Thus, $|x(t_5; t_0, \varphi)| \leqq H + L(\gamma^*)(t_5 - t_4)$. If $t_5 - t_4 \leqq h$, $|x(t_5; t_0, \varphi)| \leqq H + hL(\gamma^*) < H_1$, and hence, $t_5 - t_4$ must be greater than h. Therefore, $\|x_{t_5}(t_0, \varphi)\| = H_1$. However, $\|x_{t_5}(t_0, \varphi)\| > \gamma$, because $t_2 < t_5 < t_3$. Since $\gamma \geqq H_1$, we have a contradiction. Thus, $|x(t; t_0, \varphi)| \geqq H$ for all $t \in [t_2, t_3]$. Therefore, we have

$$a(\gamma^*) \leqq V(t_3, x_{t_3}(t_0, \varphi)) < V(t_2, x_{t_2}(t_0, \varphi)) \leqq b(\gamma),$$

which contradicts $b(\gamma) \leqq a(\gamma^*)$. Thus, it can be seen that if $\varphi \in C_\gamma$, then $\|x_t(t_0, \varphi)\| < \gamma^*$ for all $t \geqq t_0$.

The same argument shows that $\varphi \in C_{H_1}$ implies $\|x_t(t_0, \varphi)\| < \alpha$ for all $t \geqq t_0$, $\varphi \in C_\alpha$ implies $\|x_t(t_0, \varphi)\| < \beta$ for all $t \geqq t_0$, and $\varphi \in C_\beta$ implies $\|x_t(t_0, \varphi)\| < \gamma$ for all $t \geqq t_0$.

Consider a solution $x(0, \varphi)$ such that $\varphi \in C_\beta$. Then we have $\|x_t(0, \varphi)\| < \gamma$ for all $t \geqq 0$. Suppose that $|x(t; 0, \varphi)| > H$ for all $t \geqq h$. Then, there exists a constant $k > 0$ such that

$$V'_{(32.1)}(t, x_t(0, \varphi)) \leqq -k \qquad \text{for } h \leqq t < \infty, \tag{37.2}$$

because $H < |x(t; 0, \varphi)| < \gamma$ for all $t \geqq h$. From (37.2), it follows that

$$V(t, x_t(0, \varphi)) \leqq V(h, x_h(0, \varphi)) - k(t-h) \leqq b(\gamma) - k(t-h).$$

Therefore, if $t > \frac{1}{k}\{b(\gamma)+kh\}$, we have $V(t, x_t(0, \varphi)) < 0$, which contradicts $V > 0$. Thus, at some t_1 such that $h \leqq t_1 \leqq \frac{1}{k}\{b(\gamma)+kh\}$, we have $|x(t_1; 0, \varphi)| = H$. By (37.1), $|x(t; 0, \varphi)| \leqq H + hL(\gamma^*) < H_1$ for $t \in [t_1 - h, t_1]$, and hence $\| x_{t_1}(0, \varphi) \| < H_1$, which implies that $\| x_t(0, \varphi) \| < \alpha$ for all $t \geqq t_1$.

Let f be the mapping such that $f(\varphi) = x_\omega(0, \varphi)$. Clearly the set S is an open convex subset of C. Since $h \leqq \omega$ and $x(0, \varphi)$, $\varphi \in S$, satisfies $\| x_\omega(0, \varphi) \| < \gamma^*$ and $|\dot{x}(t; 0, \varphi)| \leqq L(\gamma^*)$ for $t \in [0, \omega]$, S is mapped into a compact subset of C, that is, f is a compact mapping. Let $S_1 = \{\varphi \in C;\ \|\varphi\| < \beta\}$ and let $S_0 = \{\varphi \in C;\ \|\varphi\| \leqq \alpha\}$, then S_1 is open and convex while S_0 is closed and convex. As was seen above, $\| x_t(0, \varphi) \| < \alpha$, $\varphi \in S_1$, for all $t \geqq t_1$, and hence, there exists an integer $m > 0$ such that if $\varphi \in S_1$, we have $\| x_{m\omega}(0, \varphi) \| < \alpha$ or $x_{m\omega}(0, \varphi) \in S_0$.

Thus, from Theorem 28.2, it follows that f has a fixed point in S_0, that is, $x_\omega(0, \varphi) = \varphi$. Since $F(t, \varphi)$ is periodic in t of period ω, it is seen that there exists a periodic solution of (32.1) of period ω.

As an example, we consider the system

$$(37.3) \qquad \begin{cases} \dot{x} = y \\ \dot{y} = -\Phi(t, y) - f(x) + p(t) + \int_{-h}^{0} g(x(t+\theta))y(t+\theta)d\theta . \end{cases}$$

If $g(x) = \frac{df}{dx}$, the system is equivalent to

$$\ddot{x} + \Phi(t, \dot{x}) + f(x(t-h)) = p(t).$$

For the system (37.3), the following assumptions will be made;

(i) $\Phi(t, y)$ is continuous in t, y on $0 \leqq t < \infty$, $|y| < \infty$, $\Phi(t, y) \in \bar{C}_0(y)$, $\Phi(t, y)$ is periodic in t of period ω and for some positive constants a, A

$$(37.4) \qquad \frac{\Phi(t, y)}{y} > a > 0 \qquad \text{for } |y| \geqq A,$$

(ii) $f(x)$ is continuous on $|x|<\infty$, $f(x)\in C_0(x)$ and $f(x)\operatorname{sgn} x \to\infty$ as $|x|\to\infty$,

(iii) $p(t)$ is periodic in t of period ω and $|p(t)|\leqq k$,

(iv) $g(x)$ is continuous on $|x|<\infty$, $g(x)\in C_0(x)$ and $|g(x)|\leqq L$.

Then, there exists an $h_0>0$ such that for $0\leqq h\leqq h_0$, the system (37.3) has a periodic solution of period ω.

In case $h=0$, (37.3) is a system of ordinary differential equations, and this result is well-known. Hence, we assume that $h>0$. First of all, consider a continuous functional $W(\varphi,\psi)$ defined by

$$W(\varphi,\psi)=2F(\varphi(0))+\psi^2(0)$$
$$+\frac{a}{2h}\int_{-h}^{0}\Big(\int_s^0\psi^2(\theta)d\theta\Big)ds+L\int_{-h}^{0}\Big(\int_s^0|\psi(\theta)|d\theta\Big)ds,$$

where $F(x)=\int_0^x f(u)du$ and $0<h\leqq h_0<\frac{a}{2L}$. Then, we have

$$W'_{(37.3)}(x_t,y_t)\leqq -2y\Phi(t,y)+2k|y|+2\int_{-h}^{0}L|y(t)y(t+\theta)|d\theta$$
$$+\frac{a}{2h}\int_{-h}^{0}\{y^2(t)-y^2(t+\theta)\}d\theta$$
$$+Lh|y|-L\int_{-h}|y(t+\theta)|d\theta\,.$$

If $|y|\geqq c=\max(A,\frac{4k}{a}+1)$,

$$W'_{(37.3)}(x_t,y_t)\leqq -\frac{1}{2}ay^2-\int_{-h}^{0}\Big\{\frac{a}{2h}y^2(t)-2L|y(t)y(t+\theta)|$$
$$+\frac{a}{2h}y^2(t+\theta)\Big\}d\theta-L\int_{-h}^{0}|y(t+\theta)|d\theta\,,$$

which implies that $W'_{(37.3)}(x_t,y_t)\leqq -\frac{1}{2}ay^2(t)$, because $h<\frac{a}{2L}$. Thus, if $|\psi(0)|\geqq c$,

$$W'_{(37.3)}(\varphi,\psi)\leqq -\frac{1}{2}a\psi^2(0)\,. \tag{37.5}$$

Next we shall consider the case where $|\psi(0)|\leqq c$. Let B be a constant such that if $|y|\leqq c$, then $2|y||\Phi(t,y)|+|\Phi(t,y)|\leqq B$. By (ii), there exists a constant $d>0$ such that $\frac{4}{a}\leqq d$ and

$$B+2ck+\frac{ac}{2}+ac^2+k-f(x)\leqq -1 \qquad \text{for } x\geqq d\,,$$

$$B+2ck+\frac{ac}{2}+ac^2+k+f(x)\leqq -1 \qquad \text{for } x\leqq -d\,.$$

In case $\varphi(0)\geqq d$, if we consider $V(\varphi,\psi)=W(\varphi,\psi)+\psi(0)$,

$$\begin{aligned}V'_{(37.3)}(x_t,y_t)&=W'_{(37.3)}(x_t,y_t)-\Phi(t,y)-f(x)+p(t)\\&\quad+\int_{-h}^{0}g(x(t+\theta))y(t+\theta)d\theta\\&\leqq -2y\Phi(t,y)+2ck+Lhc-\Phi(t,y)+k-f(x)+ay^2\\&\quad-\int_{-h}^{0}\Big\{\frac{a}{2h}y^2(t)-2L|y(t)y(t+\theta)|+\frac{a}{2h}y^2(t+\theta)\Big\}d\theta\\&\leqq -1\,.\end{aligned}$$

If we set $V(\varphi,\psi)=W(\varphi,\psi)-\psi(0)$ for $\varphi(0)\leqq -d$, the same argument gives $V'_{(37.3)}(\varphi,\psi)\leqq -1$.

In case $|\varphi(0)|\leqq d$, $\psi(0)\geqq c$, if we set $V(\varphi,\psi)=W(\varphi,\psi)+\frac{c}{d}\varphi(0)$.

$$V'_{(37.3)}(x_t,y_t)\leqq -\frac{1}{2}ay^2+\frac{c}{d}y\leqq -\frac{1}{4}ay^2\,.$$

In case $|\varphi(0)|\leqq d$, $\psi(0)\leqq -c$, $V(\varphi,\psi)=W(\varphi,\psi)-\frac{c}{d}\varphi(0)$ satisfies $V'_{(37.3)}(x_t,y_t)\leqq -\frac{1}{4}ay^2$.

If $\varphi(0)\geqq d$, $\psi(0)\geqq c$ or $\varphi(0)\leqq -d$, $\psi(0)\leqq -c$, set $V(\varphi,\psi)=W(\varphi,\psi)+c$, and if $\varphi(0)\geqq d$, $\psi(0)\leqq -c$ or $\varphi(0)\leqq -d$, $\psi(0)\geqq c$, set $V(\varphi,\psi)=W(\varphi,\psi)-c$. Then, we obtain a continuous Liapunov functional $V(\varphi,\psi)$ which satisfies condition (ii) in Theorem 37.2.

Since we have

$$2F(\varphi(0))+\psi^2(0)-c\leqq V(\varphi,\psi)\leqq 2F(\varphi(0))+\psi^2(0)+c+\frac{a^2}{8L}(r^2+r)\,,$$

where r is the norm of the vector (φ,ψ), $V(\varphi,\psi)$ satisfies condition (i), if necessary, by adding a positive number so that $V>0$. Easily it is seen that if h_0 is small enough, condition (37.1) is satisfied.

Thus, by Theorem 37.2, if h is small, there exists a periodic solution of (37.3) of period ω.

Now we consider the system (32.1) and its associated system

$$\dot{x}(t)=F(t, x_t), \qquad \dot{y}(t)=F(t, y_t) \tag{37.6}$$

and assume that $F(t, \varphi)$ is defined for $t \in (-\infty, \infty)$, $\varphi \in C_H$, $H>0$, (or $\varphi \in C$) and is continuous in t, φ. Moreover, we assume that for any $\alpha>0$, there exists an $L(\alpha)>0$ such that $\varphi \in C_\alpha$ implies $|F(t, \varphi)| \leqq L(\alpha)$. When $F(t, \varphi)$ is almost periodic in t uniformly with respect to $\varphi \in S$ for any compact set S in C_H (or C), we shall say that $F(t, \varphi)$ is almost periodic in t uniformly with respect to $\varphi \in C_H$ (or $\varphi \in C$).

In the following, we understand that for $\|\varphi\|=H$, a condition on the upper right-hand derivative of V defined for $\varphi \in C_H$ is satisfied in case V' can be defined.

The results in Section 31 can be extended to almost periodic functional-differential equations by the same idea. We shall mention some of these results without proof (cf. [49], [147], [149]).

THEOREM 37.3. *Suppose that there exists a continuous Liapunov functional $V(t, \varphi, \psi)$ for $t \geqq 0$, $\varphi \in C_H$, $\psi \in C_H$ (or $\varphi \in C$, $\psi \in C$) which satisfies the following conditions;*

(i) $a(\|\varphi-\psi\|) \leqq V(t, \varphi, \psi) \leqq b(\|\varphi-\psi\|)$, *where* $a(r) \in CIP$, $b(r) \in CI$ *and* $b(r) \to 0$ *as* $r \to 0$,

(ii) $|V(t, \varphi_1, \psi_1)-V(t, \varphi_2, \psi_2)| \leqq K\{\|\varphi_1-\varphi_2\|+\|\psi_1-\psi_2\|\}$, *where* $K>0$ *is a constant (in case* $\varphi, \psi \in C$, K *may depend on* η *for* $\varphi_i, \psi_i \in C_\eta$, $i=1, 2$),

(iii) $V'_{(37.6)}(t, \varphi, \psi) \leqq -\alpha V(t, \varphi, \psi)$, *where* $\alpha>0$ *is a constant. Moreover, suppose that there exists a solution* $x(t_0, \varphi)$ *of* (32.1) *such that* $\|x_t(t_0, \varphi)\| \leqq H_1$, *where* $t \geqq t_0 \geqq 0$ *and* $H_1<H$. *Then, if* $F(t, \varphi)$ *is almost periodic in* t *uniformly with respect to* $\varphi \in C_H$ *(or* $\varphi \in C$), *in the region* C_H *(or* C) *there exists a unique perfectly uniform-asymptotically stable almost periodic solution of* (32.1) *which is bounded by* H_1, *and its module is contained in the module of* $F(t, \varphi)$. *In particular, if* $F(t, \varphi)$ *is periodic in* t *of period* T, *then there*

exists a unique perfectly uniform-asymptotically stable periodic solution of (32.1) *of period* T.

To prove this theorem, the set S such that $\|x\| \leqq H_1$ in the proof of Theorem 31.1 should be replaced by the following set S. Since $\|x_t(t_0, \varphi)\| \leqq H_1$, $F(t, x_t)$ is bounded and hence, $|\dot{x}(t; t_0, \varphi)|$ is bounded by a constant R for $t \geqq t_0$. Let S be the compact subset of C consisting of all functions which are bounded by H_1 and are Lipschitzian with Lipschitz constant R.

THEOREM 37.4. *Suppose that* $F(t, \varphi)$ *of* (32.1) *is continuous on* $(-\infty, \infty) \times C$ *and is almost periodic in* t *uniformly with respect to* $\varphi \in C$ *and that* $F(t, \varphi) \in \overline{C}_0(\varphi)$. *If there exists a solution* $x(0, \varphi)$ *of* (32.1) *for* $t \geqq 0$ *which is bounded by* H_1 *for* $t \geqq 0$ *and is uniform-asymptotically stable in the large, then there exists a unique almost periodic solution of* (32.1), *which is perfectly uniform-asymptotically stable in the large and which is bounded by* H_1. *Further, the module of this solution is contained in the module of* $F(t, \varphi)$. *In particular, if* $F(t, \varphi)$ *is periodic in* t *of period* T, *then so is the above solution.*

For the proof, see [147], [149].

Corresponding to Theorem 31.3, we have the following theorem, which includes Theorem 37.6 due to Hale [49]. For the proof, see [147], [149].

THEOREM 37.5. *Suppose that there exists a continuous Liapunov functional* $V(t, \varphi, \phi)$ *defined for* $t \in I$, $\varphi \in C_H$, $\phi \in C_H$ (*or* $\varphi \in C$, $\phi \in C$) *which satisfies the conditions* (i), (iii) *in Theorem* 37.3 *and*

(ii)′ $$|V(t, \varphi_1, \phi_1) - V(t, \varphi_2, \phi_2)| \leqq K \|(\varphi_1 - \varphi_2) - (\phi_1 - \phi_2)\|,$$

where $K > 0$ *is a constant* (*in case* $\varphi, \phi \in C$, K *may depend on* η *for* $\varphi_i, \phi_i \in C_\eta$, $i = 1, 2$). *Moreover, suppose that* (32.1) *has a solution* $x^0(t)$ *such that* $\|x_t^0\| \leqq c$ *for* $t \geqq t_0 \geqq 0$ *and some* $c > 0$. *Consider a system*

$$\dot{x}(t) = F(t, x_t) + h(t), \tag{37.7}$$

where $F(t, \varphi)$ *is almost periodic in* t *uniformly with respect to* $\varphi \in C_H$

(or $\varphi \in C$) and $h(t)$ is almost periodic in t and $|h(t)| \leqq R$ for all t. Then, if $a^{-1}\left(\frac{KR}{\alpha}\right)+c \leqq H_1 < H$ (or for some H and $K(H)$), in the region C_{H^*}, $H_1 < H^* < H$, (or C) the system (37.7) *has a unique perfectly uniform-asymptotically stable almost periodic solution which is bounded by H_1, and its module is contained in the module associated with the vector field. In particular, if $F(t, \varphi)$ and $h(t)$ are periodic in t of period T, then so is the above solution. In case $V'_{(37.6)}$ is well defined for all $\|\varphi\| = H$, $\|\psi\| = H$, the conclusion is true in C_H.*

THEOREM 37.6. *Suppose that $F(t, \varphi)$ of* (32.1) *is continuous on $(-\infty, \infty) \times C$ and linear in φ, and that the zero solution of* (32.1) *is uniform-asymptotically stable in the large for $t_0 \geqq 0$. If for each fixed φ, $F(t, \varphi)$ is bounded for $-\infty < t < \infty$ and if $h(t)$ is any bounded continuous function of t, $-\infty < t < \infty$, then the solutions of* (37.7) *satisfy the following properties:*

(i) *All solutions of* (37.7) *are bounded for $t \geqq t_0$. In fact, if $R = \sup_t |h(t)|$, there exist constants $K \geqq 1$, $\alpha > 0$ such that $\|\varphi\| \leqq \frac{R}{\alpha}$ implies $\|x_t(t_0, \varphi)\| \leqq \frac{KR}{\alpha}$ for all $t \geqq t_0$.*

(ii) *If $F(t, \varphi)$, $h(t)$ are almost periodic in t uniformly with respect to $\varphi \in C$, then there exists a unique almost periodic solution of* (37.7), *which is bounded by $\frac{KR}{\alpha}$ and which is perfectly uniform-asymptotically stable in the large.*

For the proof, see [49] or [147]. This theorem has been proved in a different manner and in a less general form by Krasovskii [62], Halanay [40] and Shimanov [130].

For other existence theorems of almost periodic solutions in functional-differential equations, see [39], [41], [49], [62], [129], [147].

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